

CEG

The CIRIS Epistemic Grammar

Version 1.0-RC16 — Reader Edition

*a signed, compositional graph language for claims, relationships, authority,
membership, consent, governance, addressing, and settlement across a decentralized network*

2026-06-10

This is the human-reading edition: version history, per-path narrative, and provenance cross-references are omitted for readability. The canonical working draft (with full lineage) is the markdown source and `ceg-1.0-rc16.pdf`. Conformance is judged against the normative wire surface only (§0.1.1).

Contents

CEG — The CIRIS Epistemic Grammar

Version: 1.0-RC16 (Release Candidate — 1+4 surface FROZEN; additive transport/substrate absorption in progress)

Status: Release Candidate (2026-06-16). **The 1+4 attestation surface is frozen** (since RC1) and — with this cut — **the design surface is complete:** the #70 operational-data subject_kinds were the one scheduled addition, and they have now landed additively (§5.6.8.13 + §10.1.6, the §1.4 sixteenth path) after three-sided federation review (Edge/Verify/Persist). **No further additions are scheduled.** Implementers SHOULD pin RC4; any change between RC4 and 1.0 is a **found defect**, not design. The **1.0 label** waits on the #57 freeze gate: streaming substrate shipped (CIRISPersist#142), Spock removed (#58), conformance artifacts (OpenAPI + dimensions.json) generated from these frozen shapes, and **one full implementation cycle across Persist + NodeCore + Lens + Agent + Edge with no new wire-break.** That silence is the freeze signal; 1.0 is these bytes re-tagged.

RC2 → 1.0 — what actually remains (sharpened by the substrate-side dependency review, 2026-06-12). With persist having consumed RC2's hard surfaces (§5.6.8.13/§10.1.6 operational data, §5.6.8.8.2 three-role identity, §0.9 JCS), the spec-side residual is now **one conformance push** (the one ratification landed in RC4 — see below):

1. [✓] ****RATIFIED in RC4 — the consent_role §RC slot is filled**** (new §7.0.2; CIRIS-Agent#760 OQ-1/2/3 all closed at the ConsentGate.lean defaults, with the permissive OQ-1 audit clause). The consent_role JSONB shape is **locked** (flat, non-recursive, overwrite-on-revoke; chain history never embedded), and implementations ****MAY** now build the consent_role substrate — **the CIRISPersist#146 consent-SLA watcher + schema and the CIRISEdge consent-gate are unblocked.** No CEG-spec decision remains open.** (Everything else cited as "gated on CEG 0.6/0.7/0.15" was already locked downstream build work.)
2. **The RC2→1.0 conformance gate — CIRISPersist#171 four-impl conformance.** CEG reaches 1.0 only once Agent + NodeCore + LensCore + Registry all conform against persist's shared federation_attestations surface. CEG-side prerequisite: the **local-tier attestation semantics (§10.1.3 / §11.9 / §7) frozen at 1.0** so persist can finalize the surface coherently. This is the gate where "RC2 vs 1.0" matters most.

OQ defaults are **fail-secure-defaulted** (soft, non-blocking; a CEG confirm only locks the default): OQ-4 DEK-recovery tier → content-master self-retention grant; steward self-anchor / root_stewards → per-region caller-supplied stewards behind a fail-closed authority resolver.

Human-reading edition: pdf/ceg-1.0-rc16-reader.pdf is the de-editorialized, deduplicated, human-first cut (version history, per-path narrative, and provenance cross-refs stripped). This source tree remains the canonical working draft and retains that lineage.

1.0-RC16 — §19.7 PROVEN cross-impl, promoted RC → 1.0 (the freeze gate closes for the holonomic substrate). The §19.7 aggregation-pyramid wire (RC14) is now **byte-equivalent across implementations:** CIRISVerify v5.10.0 authored the vector family and CIRISEdge v4.3.0 (src/holonomic/aggregation.rs + 5 conformance vectors) reproduces it **byte-for-byte** — AggregationMetaV1 preimage+sig, member_commitment, descend. The preimage matched **first try, no coordination beyond the spec** (the §19.0 binary discipline

makes wire-identity reproducible from the text), and `member_commitment` reuses the §19.1 WholenessWitness Merkle **verbatim** — **one** Merkle scheme across §19.1 + §19.7, no fork (validating the RC14 reuse + RC15 freeze decisions). With persist v8.4.0 + verify v5.10.0 + edge v4.3.0 all on the §19.7 baseline at PyPI, §19.6/#57's §19.7 vector family is **closed** — **§19.7 is 1.0, not RC**. Also adds the ****EjectAggregatedTierOnly { tier } §19.7.3 verdict (the one remaining edge build item, surface-additive, sheds a single pyramid stratum)**. No 1+4 change.**

1.0-RC15 — §19.1 WholenessWitness Merkle scheme resolved + frozen (the last latent §19 design question). Closes the prior open #143 question ("does WholenessWitness adopt the RFC 6962 0x00/0x01 prefix?"). **Decision: no** — CEG keeps the construction CIRISEdge v4.0.x shipped and CIRISVerify v5.9.0 proved cross-impl (`leaf=SHA-256(bytes)`, `node=SHA-256(1||r)`, odd-node duplication, WW-v1-empty sentinel), frozen as-is. The CVE-2012-2459 odd-node malleability is **not exploitable here**: every WholenessWitness / `member_commitment` root is mandatorily hybrid-signed (no unsigned-root reliance), `member_commitment` is verified by full source-id-list **recomputation** (never partial inclusion proofs), and §19.1 reconciliation is between accountable signed equivocation-checked peers. Normative caveat: any future use relying on an *unsigned* root or partial inclusion proofs against an *untrusted* root **MUST** first adopt the RFC 6962 prefix + lone-node promotion. This avoids a vector-invalidating re-Merkle of a proven path; §19.7.1.1 `member_commitment` inherits the same frozen scheme. **No 1+4 change, no wire change** (pins the shipped construction).

1.0-RC13 — editorial simplification pass (no normative/wire change). A holistic dedup/structural cleanup (-122 lines) after the RC1→RC12 accretion, verified to change **no** rule, shape, number, enum, or RFC-2119 statement (RFC-2119 counts unchanged in §5/§8/§10; the few drops in §0/§15/§16 are recap/audit/lineage restatements whose canonical home is intact; 0 newly-broken cross-refs). Changes: **§16.1 lineage brought current (0.11→RC12) + compacted**; the README pre-RC1 (0.2→0.18) changelog collapsed to a one-line §16.1 pointer (the single biggest win); the per-§5.6.8.x "1+4 preserved / Nth path" closers reduced to one-liners; **§15.6 RC1 decision-register compressed** (still-open RC1-1b/1c/7 preserved verbatim); duplicate `recipient_excluded` reason-set / consent local-tier carve-out / §1.6 scope-blockquote collapsed to cross-refs; the "What CEG 0.X documents" self-recaps trimmed; inline version-tag clutter pulled from §5/§8/§10 titles into §16.1. **Defects fixed**: the scrambled §1.4 path ordinals (were ... 6,10,9,8,11...) now run 1→16; the stale "~9 subject_kinds" count corrected; the stale §16.1/§17 lineage stamps brought current. The frozen wire surface, §19 per-gate MUSTs, and the §8.1.12.7.1 member-set block were untouched.

1.0-RC12 — the noise floor: unified retirement / forever-memory model (new §19.7; no 1+4 change). Codifies the *store-the-memory-forever* directive (rooted in CIRISAgent TSDB consolidation) as one normative principle: **revocation, retirement, capacity-eviction, expiry, and aging are one operation** — a pressure-driven monotonic descent of an item's fidelity toward/below the **noise floor** (the individual-recoverability boundary). New §19.7: (1) *one operator, not many* — hard-delete is just the fastest descent, not a separate primitive; (2) the noise floor does double duty — privacy boundary (revoked ⇒ below it at every retained tier) **and** durability floor (the collective blur persists below it forever); (3) **nothing is ever fully forgotten** — two **mechanical** degradation operators (intra-object layer-drop via §10.5.8.4 `ChunkLayer+RaptorQ`; inter-object N→1 aggregation, *a picture of a thousand pictures*) build a memory pyramid whose steady-state storage to remember all history is **O(log T)**, sublinear; (4) **forgetting and erasure converge** — *dissolving* the §19.3 N5 tension: a sufficiently-aggregated composite is already-erased by degradation, so revocation forces an item below the floor now + purges still-recoverable tiers, never the gist; (5) degradation is **mechanical** → **infrastructure is self-sufficient for memory** (sharpens §1.3: the fabric node remembers without agency; a

brain is optional enrichment). The §19.3 EjectionVerdict values are reframed as points on the one descent. **No 1+4 change, no wire break.**

1.0-RC11 — holonomic substrate absorbed WITH normative guardrails (new §19; no 1+4 change). Edge tagged **v4.0.0** ("holonomic federation seal"), freezing the shapes. A four-perspective review (supersession / anonymous-tier / absorption-correctness / adversarial-security) found the concept sound and additive at the wire layer but that a **verbatim** absorption would invert ratified invariants — so per #85 this is **absorb-with-guardrails**. New **§19 Holonomic substrate** states the normative invariants: (**WholenessWitness**) namespaced `wholeness_witness:`, scoped by `claim_namespaces`, **MUST NOT** witness anonymous/self rows (re-attribution guard), leaf-order pinned lexicographic, Merkle distinct from §10.3, hybrid-PQC verified at ingest+persist, equivocation retained — and it is a ****divergence detector that triggers** the §10.1.6 quorum-merge, never replaces ****monotonic_quorum/revision** anti-rollback for revocation/partner/org (the supersession trap: that would resurrect revoked keys). (**Recursive bootstrap**) *trust-discovery* only — non-infra membership still needs the §5.6.8.10 owner-binding + `consensus_protocol` vote; infra stays founder-quorum; ≤ 5 -hop + cycle-reject + §13.3 weight cap; self-supplied chains aren't evidence. (**Fountain/swarm**) `FountainHoldingClaim` specializes `holds_bytes:*` + inherits §10.1.4 self/family suppression; **retention MUST respect revocation** (a withdrawn `content_id` is evict-eligible regardless of rarity); holding claims possession-bound; anonymous content exempt from swarm retention (LRU-only); reconstitution scoped to the witnessed corpus. (**ALM**) `compute_alm_topology` pinned as a deterministic *contract* (byte-exactness gated on vectors); capacity ads hybrid-verified, not the dominant unbounded term; reachability stays ephemeral (D6). PIN line drawn (§1.1): signed preimages + topology-contract normative; parent-selection + `retention_priority` edge-internal; rarity a recommendation. Matching edge fixes: `CIRISEdge#143` (in-flight). Byte-exact preimages pin against the **fixed** v4.0.x; vectors from it are the #57 freeze gate. **No 1+4 change, no wire break.**

1.0-RC10 — realtime A/V chunk wire absorbed byte-exact from edge v4.0.0 (§85 §N; no 1+4 change). First byte-exact absorption of the holonomic substrate (`CIRISEdge` tagged **v4.0.0**, freezing the shapes). New §10.5.8.3 `SealedAvChunk` wire layout (the 52-byte header map: `stream_id[32]` || `epoch` be u64 || `chunk_seq` be u64 || `codec_id` || 3-byte `ChunkLayer` || ciphertext; v3.7.0 length-disambiguated backward-compat $\rightarrow 0xFF+\{0,0,0\}$); §10.5.8.4 `ChunkLayer` + `ReceiverLayerPolicy` (monotonic SVC axes, prefix-reconstruct, "blinking dot" $\{0,0,0\}$, per-receiver drop advertised over the existing `federation_session/key_grant` surface — not new wire); §10.5.8.5 the double-seal (outer-AEAD(inner-AEAD(plaintext)), AES-256-GCM) + **deterministic nonce derivation** with the pinned domain separators `b"CIRIS-AV-INNER-V1"` / `b"CIRIS-AV-OUTER-V1"` (SHA-256(label || ...) [0..12]). Transcribed verbatim from the v4.0.0 reference impl; **conformance vectors (generated from v4.0.0) are the named #57 freeze gate** — the spec pins the layout, the vectors prove cross-impl byte-equality before 1.0 labels. Per the #85 absorption: bytes are Edge's transport-chunk framing, **the §4 attestation envelope is untouched**. Holonomic file (ALM / fountain / swarm / WholenessWitness / topology / bootstrap) follows in RC11. **No 1+4 change, no wire break.**

****1.0-RC9 — `codec_id` namespace ratified for realtime A/V (no 1+4 change; transport-layer namespace).**** New §10.5.8.2 ratifies the #84 1-byte `codec_id` + 3-byte `ChunkLayer` discriminator on the `SealedAvChunk` transport-chunk header (offset 48.52): `0x01` AV1 SVC (production default) · `0x02` JPEG XS (reserved) · `0x03` symmetric MDC (substrate design target, reserved) · `0xFF` opaque (v3.7.0 wire-compat) · `0x04–0x7F` reserved · `0x80–0xFE` experimental. **Clear metadata, NOT in the AEAD** — a relay drops chunks for per-receiver degradation without touching the inner epoch-DEK seal; `0xFF` round-trips v3.7.0 identically. This ratifies a namespace for a byte §10.5.8.1 already treated as application-defined/opaque — **the frozen**

1+4 attestation envelope and its canonicalization are untouched (the bytes are Edge's transport-chunk framing, not the §4 envelope). First landing of the #85 holonomic-substrate absorption (codec/chunk → ALM → fountain → WholenessWitness), which proceeds as **additive 1.0 transport/substrate sections** in lockstep with the CIRISEdge v3.8.0→v4.0 cuts — no 1+4 change, existing wires interop. Unblocks CIRISEdge#131 (v3.8.0). **No 1+4 change, no wire break.**

1.0-RC8 — store-path PQC clarification + the informative interoperability-profiles section (no wire change). Two loose ends tied up, both additive, neither touches the frozen interior: (1) §10.1.5.1.1 **clarified to bind the durable store + replication path**, not only the operational-authority admission gates (CIRISPersist#225): "federation-tier" means *every* federation-tier attestation, including the bulk **per-trace / store-and-replicate** path — a kept-for-posterity trace MUST carry + verify the ML-DSA-65 half on ingest, **content-addressing is not a defense against forge-later** (a CRQC adversary hashes their own backdated forgery), and there is no "operational-authority vs testimony-leaf" exemption (closes the read of RC7 that scoped the mandate to authority gates only). (2) New **§18 Interoperability profiles** (informative — the #72 standards-boundary roadmap given a home in-spec): the *speak-CEG-inside / standards-at-the-edge* principle, the four boundary modes, and the **C2PA media-provenance import/emit profile** (read+emit C2PA manifests via `evidence_refs`, zero wire change; EU AI Act Art. 50 deadline 2026-08) — "provenance says where the bytes came from; CEG says what a community of signers makes of them." COSE / RFC 9421 / SD-JWT VC / KEYTRANS stubbed as tracked boundary profiles. **No 1+4 change, no wire break.**

1.0-RC7 — PQC-mandatory-at-admission + partnership member-set + trust/owner/agency refinements (no wire change). Three additive resolutions on the frozen surface (#81/#82/#83): (1) **#82 — the PQC half is MANDATORY at admission as of 1.0** (new §10.1.5.1.1; Verify audit F1; satisfies the #57 "PQC-everywhere REQUIRED" gate): every federation-tier / operational-authority admission gate (`operational_admit`, `transport_destination` binding, `partner_record`/founder-quorum) MUST verify **both** halves and reject classical-only — an **immediate 1.0 requirement, no fleet-floor/date**, no `require_hybrid: false` posture; a key without a valid ML-DSA-65 half is confined to local-tier until it completes PQC wiring. (2) **#81 — the bilateral consent:partnership*:v1 canonical member set enumerated** (§8.1.12.7.1; CIRISVerify#63): seven REQUIRED members pinned (`attestation_type/attesting_key_id/di-granter≡attesting_key_id / partner≡subject_key_ids[0]`, ****no valid_until (omit-not-null)**), all other §4 fields ride the omit rule — so two impls converge on identical JCS bytes. (3) **#83 — three trust-model refinements (all on existing primitives): TRUST≠MEMBERSHIP** (§5.6.8.10: trust+serve an infrastructure community without being a member; membership needs the community's own protocol); the **owner-binding gate** (a node/agent-role key MUST have a live user-owner `delegates_to` before admission to any ****non-infrastructure community — infra trust+serve stays unowned; authority roots in an accountable human**); and the partnership-without-agency** infra delegation profile (§8.1.12.7.1: reserved `infra:*` vs `agency:*` scope split — a node-only delegate MUST carry only `infra:*`, a verifier rejects an infra key wielding `agency:*`, making §1.3 "infrastructure must not have agency" wire-checkable). **No 1+4 change, no wire break.**

1.0-RC6 — RNS destination-hash algorithm pinned in-spec (verifier unblock; no wire change). Resolves #80 / CIRISVerify#28: the §5.6.8.8.1 `transport_destination` binding pinned the preimage *ordering* but deferred the actual RNS destination-hash byte layout to "the RNS rule" in Reticulum — so a conformant verifier could **not** independently recompute `destination_hash` (CIRISVerify shipped it as an honest `DestinationHashCheck::Unsupported` stub rather than guess; the AV-42 destination-authenticity recompute was the one unverifi-

able step). New §5.6.8.1.1 reproduces the **two-stage** RNS construction normatively, verified against the live Reticulum source (RNS/destination.py + Identity.py + Reticulum.py): `name_hash = SHA256(app_name||dot-joined-aspects)[:10]`; `identity_hash = SHA256(x25519||ed25519)`; `destination_hash = SHA256(name_hash||identity_hash)[:16]` — **NOT** the flat single-concat SHA-256 the ≤RC5 wording implied (that yields a different, wrong value — the latent under-spec that made the recompute impossible). Constants pinned (NAME_HASH_LENGTH=80b, TRUNCATED_HASHLENGTH=128 SHA-256 throughout); **CEG owns the reproduction** (closed conformance source, does not float with upstream Reticulum — an RNS change is a deliberate CEG bump, never silent drift). §8.1.13.1.1(b) `resolve_member_transport` repointed to the pinned algorithm. **No 1+4 change, no wire break** — a correctness pin on an existing field's verification path.

****1.0-RC4** — §RC ratified: the `consent_role` Counter-RII gate (Agent#760 OQ-1/2/3; no wire change). **The one reserved §RC slot is filled**** (new §7.0.2), ratifying the Accord §RC `consent_role` semantics at the `ConsentGate.lean` defaults — **zero proof delta: (OQ-1)** revocation chain is `BaseRole`-only, non-recursive — overwrite-on-revoke, chain history (*if retained*) lives in a separate audit surface and is ****never embedded in the `consent_role` JSONB (locks the substrate-portable shape; the permissive "if retained" is non-breaking against the shipped flat soft-delete substrate); (OQ-2)**** Peer-role nodes are blanket-suppressed from Counter-RII at any `trust_mode` — bounded because `ratchet:flag:counter_rii:{layer}` is **advisory-only** (§5.7), never sole evidence for `slashing:*` (WA quorum is the gate); **(OQ-3)** `AuthorizedReview` is **strict** (signal-eligible immediately past `window_end`). With this the design surface has **zero open CEG decisions**, and the `consent_role` substrate is unblocked (CIRISPersist#146 back-half + the CIRISEdge consent-gate). `consent_role` is a `federation_keys` identity field (§7.0.1 sibling), not an envelope primitive — **no 1+4 change, no wire break**.

1.0-RC3 — **clarity cut (fabric-node trust model + NodeCode shorthand; no wire change)**. Four clarifications + one canonical-encoding addition, all additive on the frozen surface: (1) §7.0.1 the **fabric-node separation-of-powers** — a fabric node (the headless cohabitation runtime; `agent = fabric node + brain`) co-locating `{substrate_persist, steward, lenscore_detector, witness, ...}` in one process is **custody, not consolidation of authority**: authority stays quorum-bound (a vote, not a verdict), observation stays non-authoritative by namespace (`detection:*`, validated-not-adjudicated), and **observation can never manufacture authority** (the "lenses become the gate" hazard is structurally unreachable). (2) §5.6.8.10 ****ciris-canonical** is a default, not a forced root — **a conformant consumer MUST be able to re-root (infrastructure is a general primitive; anyone may run their own canonical group; membership grows by founder-quorum vote; DNS is never in the trust/addressing chain) + trust ≠ consent are distinct role-scoped relationships (trust = inbound acceptance of output; consent = outbound data-flow; lens trust=scores/consent=traces, registry trust-only, node consent medium-varied over one** consent:* object)**. (3) New §0.10 **NodeCode** — the canonical `key_id` shorthand encoding (CIRIS-V1-..., RFC 4648 base32 + CRC-16-CCITT, lifted from the shipped CIRISAgent codec per #75) so every impl renders the same code for the same key; a deterministic render of an existing `key_id`, **not** an envelope field. (4) **README RC2→1.0 residual** sharpened from the substrate dependency review (one open decision — `consent_role/§RC` via Agent#760; one conformance push — Persist#171 four-impl). **No 1+4 change, no wire break**.

1.0-RC2 — **operational-data envelopes (the scheduled additive cut; design surface complete)** (per #70, under the #58 Spock-removal epic; three-sided review Edge/Verify/Persist — zero blockers, zero new structural primitives confirmed): cross-region operational Portal data becomes signed CEG envelopes. **Three new subject_kinds** (§5.6.8.13): `organization · org_membership · partner_record` (the Edge v2 EnvelopeKind wire tokens; v2 `envelope_hash`

= sha256(JCS(envelope))). **Governing principle:** federate the trust/authz-minimal projection; PII + business detail stay region-local (Registry emit-side discipline — the substrate is not a PII filter). **User** records do NOT federate — only the role binding does. **Authority:** org/membership = single-signer role-gated (the §8.1.12.7.1 delegation resolver — never founder-quorum); **partner_record** = **M-of-N steward quorum** (§5.6.8.10 machinery) with the **admission quorum** ≠ **merge quorum** distinction normative. **Determinism:** capability/restriction arrays are set-semantics→sorted (M signers must converge on identical JCS bytes *before* the quorum evaluates — the highest-value pin); **revision:** u64 monotonic anti-rollback per license. **New §10.1.6:** CEG-declared per-subject_kind merge intents (lww_skew_bounded+withdrawal_forward_only / monotonic_quorum V058-generalized) + the §0.7 skew-bound at admission (closes the LWW role-escalation surface). **Stable-id resolution** (not chain-walk — partition tolerance); first-class indexed business-id columns. **No payment-processor data, normative + fail-secure.** Sixteenth path; no 1+4 change. Unblocks #58 Phase 3 (de-Spock) + Edge v2 + Persist put_* surfaces.

1.0-RC1 — the freeze cut (per #71 backwards-pass + CIRISVerify#64 + #57 blocker A): five honesty/safety patches + two pins + the canonical-bytes resolution, then freeze. **(C1)** §10.5.3 **removal coalescing** (all removals in one STH window T=2s batch into ONE rotation — caps the cascade at 1/T regardless of churn; the original model's 3,600/hr churn understated realistic broadcast churn ~2 orders: naive 30%/hr at N=1M ≈ 3.1 Tbps > the 2 Tbps content fan-out; coalesced ≈ 18.7 Gbps ≈ 0.9%) + the **public-broadcast exemption** (ungated **listed:public** rosters: removal forces no rotation — no confidentiality claim against departed viewers). **(C2)** §5.6.8.8.2 forward-secrecy honesty: KEM rotation bounds *future* exposure only; historical-content recovery requires DEK rotation + re-encryption, which is NOT mandated — named gap. **(C3)** §10.1.4 fail-secure exclusion is no longer silent: new §7.7 **hard_case:recipient_excluded:{scope}** emitted INTO the affected cohort (never federates beyond). **(C4)** §5.6.8.8.2 admission MUST-reject content-KEM x25519 == transport x25519. **(C5)** §9.1 names the two-holders-one-household correlated-failure geometry; authority is the full constitutional kill; the mitigant is **finding new holders** (active obligation). **Pins:** §0.9.2.1 base64 = RFC 4648 STANDARD padded (protocol-layer pin; CIRISVerify#64); ML-KEM-768 pubkey = exactly 1184 B; **encryption_pubkeys** joins the §8.1.12.7.1 member set (c); §5.6.8.4 crypto-agility headroom note (v3/ML-KEM-1024 additive). **Blocker A closed:** §5.2.1 provenance contracts redesigned to **JCS (RFC 8785) objects with pinned domain members — **TupleHash128 retired (one canonicalization family); v1 newline forms deprecated-historical; §9.2.1 accord invocation intentionally unmigrated (closed-vocabulary preimage).** No 1+4 change.**

Pre-RC1 lineage (0.2 → 0.18): the additive 0.x build-out — consent (0.6), self/family (0.7), communities (0.8), streaming/delivery (0.10–0.13), settlement (0.14), agent-identity hardening (0.16), three crypto tiers (0.17), encryption-key registration (0.18) — is recorded in §16.1 lineage; each was additive on the frozen 1+4 set (see §1.4 paths).

License: AGPL-3.0-or-later

Authoritative for: The federation wire format, attestation primitive set, namespace, composition discipline, and governance for the CIRIS ecosystem.

Relationship to prior documents: Consolidates and supersedes FSD/FSD-002_FEDERATION_SURFACE.md v1.0 through v1.4.3 + the v1.5 candidates from CIRISRegistry#30 + the translation discipline from FSD/LANGUAGE_PRIMER.md v1.1. FSD-002 remains as design-history; CEG 0.x is the version-stable spec going forward.

Companion documents: FSD/PRIOR_ART_SCAN.md (design-space comparison); FSD/SOTA_SCAN.md

(production-validation comparison); FSD/WITNESS_KIND_REGISTRY.md (non-normative open-vocabulary registry referenced by §5.6); docs/CEG_EXPLORATION_PAGE_PRIMER.md (builder primer for ciris.ai/grammar).

How to read this spec without Cartesian default

Before the TOC: one load-bearing reading instruction that prevents the most common misread.

Self is self, fractally. At every scale the substrate operates on — key, occurrence, agent, fleet, cell, federation, biosphere — "self" means *the already-relationally-constituted entity speaking at that scale*. The cross-attestations that constituted the entity are upstream of the moment it speaks; when it then emits a self-attestation (`witness_relation: self`, declares its own confidence, reports its own `occurrence_id`, names its own `hardware_class`), that is **the relational composition speaking AS itself**, not a Cartesian atom asserting pre-relational identity.

Concretely:

- A `federation_keys` row is not an atomic principal pre-existing the federation that the substrate observes. It is a relational locus already constituted by the cross-attestations of stewards, prior peers, and provisioning ceremony. When the key-bearing entity then signs an attestation, the signature IS that relational composition speaking at this scale.
- An occurrence within a fleet is not a pre-relational unit reporting its own state. It is constituted within its fleet (the fleet's `AGENT_OCCURRENCE_ID` partitioning, the prior cross-attestations naming this fleet's deployment, the substrate's witness of its emissions). When it reports `occurrence_id`, that is the fleet-constituted occurrence speaking.
- The federation itself is not prior to its constituents. It IS the relational composition of cells, agents, occurrences, and keys cross-attesting one another.

This is why the wire format admits self-attestation **everywhere it admits attestation at all**: because there is no pre-relational atomic entity available to do the admitting in the first place. Every `attesting_key_id` is already a relational locus. Self is the relational composition speaking from within itself, at the scale of the speaker.

Common misread to avoid. A reader importing Cartesian-individualist defaults from PGP / X.509 / atomic-principal cryptographic identity work will see `witness_relation: self` and think "this is a single atom asserting pre-relational identity; surely the substrate should require cross-attestation before admitting." That reading is the misread. The substrate cannot require cross-attestation as a precondition for self-attestation because the cross-attestations are what constitute the self in the first place — they are upstream of the self-attestation, not downstream gates on it. The fractal nesting is the point.

The Recursive Golden Rule (§1.5) is the structural expression of this: no scale gets to exempt itself from the relational constitution that defines it. CIRIS L3C is constituted by the same cross-attestations that constitute every other federation participant; the federation itself is constituted by its participants; humanity-as-such (§9) is the one scale outside the federation's participant set, by design.

If you find yourself thinking "the spec should add a cross-attestation gate before admitting this self-attestation" — pause. Cross-attestation already happened upstream; the self-attestation is its downstream voice. Verify by reading §1.2 commitment 1, then continue.

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Two readerships

- **Implementers** of federation primitives consuming or emitting CEG attestations: read §0 + §1 + §3 + §4 + §5 + §7 + §8 + §9 + §10 normative.
- **Translators** mapping substantive content into CEG envelopes: read §1 + §12 + §13 + §14 + the LANGUAGE_PRIMER.md companion.

Both readerships should read §1 first.

§0 Foreword

CEG — the CIRIS Epistemic Grammar — is the federation’s language for making **structured, signed, machine-checkable claims about reality and each other**. It is the wire format the federation’s peers speak.

The grammar has exactly five wire-format primitives (one workhorse + four structural composers) and an open-vocabulary dimension namespace organized by mechanism-descriptive prefixes. Consumers compose verdicts from primitive attestations using the policies in §8; nothing in the wire format prescribes what verdict to reach.

CEG is **substrate-consuming**: it sits above the federation substrate (CIRISPersist for storage, CIRISVerify for crypto, CIRISEdge for transport) and below the application tier (CIRISAgent). It does not author primitives in the substrate it consumes; it composes policy over them. It is also **substrate-supplying** for the second-tier consensus crates (CIRISNodeCore for consensus, CIRISLensCore for detection) — they own slices of the dimension namespace and emit attestations that other CEG consumers read.

This specification has **two readerships**:

- **Implementers** of federation primitives consuming or emitting CEG attestations: read §1-§11 normative.
- **Translators** mapping substantive content into CEG envelopes: read §12-§14 + the `LANGUAGE_PRIMER.md` companion.

Both readerships should read §1 first.

§0.1 Conformance language

The key words **MUST**, **MUST NOT**, **REQUIRED**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **NOT RECOMMENDED**, **MAY**, and **OPTIONAL** in this document are to be interpreted as described in BCP 14 (RFC 2119 + RFC 8174) when, and only when, they appear in all capitals, as shown here.

§0.1.1 Normative vs informative content (what interop requires)

A reader may **agree with the protocol and disagree with its philosophy** and still build a fully conforming implementation. The two are deliberately separable:

- **Normative (binding for interoperability)**: the wire format and its conformance surface — the §3 structural primitives; the §4 envelope fields; the §5 namespace + `subject_kinds`; the §0.5–0.9 canonicalization rules; the §6 relation precedence; the §7 reserved-prefix rules; the §8 composition policies; the §10 endpoint shapes; and every RFC-2119-keyworded statement. This is exactly the surface enumerated in §1.4 ("report the surface beside the invariant"). **Conformance is judged against this and nothing else.**
- **Informative (explanatory framing; NOT binding)**: the motivating philosophy and rationale — notably §1.2 (the Ubuntu / relational-anthropology substrate), the cross-tradition readings, and prose written as motivation rather than requirement. These explain *why* the normative choices were made; they add **no** conformance obligations. An implementer who rejects the anthropology but emits/consumes wire-correct Contributions is conforming.

Where framing produced a concrete wire consequence, that consequence is restated as a normative rule in its own section (e.g., structural invisibility is motivated informatively but enforced normatively at §10.1.4, bounded by §1.6). When in doubt, the RFC-2119 keywords and the §1.4 surface govern; informative prose never overrides them.

§0.2 Conformance levels

A **Producer** is any peer that emits Contributions onto the federation wire. A **Consumer** is any peer that reads and composes verdicts over received Contributions. A **Substrate Implementation** is the storage + transport + crypto layer (CIRISPersist + CIRISEdge + CIRISVerify) underneath both.

Three normative conformance profiles:

1. **CEG-Conforming Producer (CCP)** — emits well-formed envelopes per §4, signs per §0.4 References [hybrid-sig], respects reserved-prefix rules per §7, declares its `oversight_mode` and `witness_relation` per §4.
2. **CEG-Conforming Consumer (CCC)** — verifies hybrid signatures, enforces reserved-prefix rules at admission, implements at least Policy A (§8.1.1) with the default aggregation rules from §8.2, MUST honor `null` placeholder/dev hardware-class rejection per §9.4.
3. **CEG-Conforming Substrate (CCS)** — implements the storage + transport guarantees referenced in §10.1 + §10.3, including idempotent replication, full-SHA blob verification before consumption (§10.1), and witness-quorum multi-party admission per §10.3.

Sections that follow MAY add per-feature conformance subsections; the three profiles above are the minimums.

§0.3 Versioning policy

CEG follows **SemVer 2.0.0** with these mapping rules:

- **MAJOR (X.0.0)** — any wire-incompatible change: removal of an envelope field, change of a field’s semantic, removal of a structural primitive, change to canonical-bytes domain-separation labels, removal or breaking-redefinition of a §5 prefix, change to a §7 reservation, or change to the §0.1 / §0.2 conformance language.
- **MINOR (0.X.0)** — wire-compatible additions: new prefix in §5, new envelope field with documented default, new composition policy in §8, new endpoint shape in §10, new optional conformance subsection. Existing Conforming Producers and Consumers continue to interoperate without modification.
- **PATCH (0.0.X)** — clarifications, editorial fixes, additions to non-normative sections (WITNESS_KIND_REGISTRY, glossaries §14), addition to §15 acknowledged-gaps, fixes to non-normative examples in §14.

The 0.x series indicates this specification is a Public Working Draft. Any $0.x \rightarrow 0.(x+1)$ bump MAY include wire-breaking changes; consumers MUST treat 0.x as unstable until 1.0 publication. Once 1.0 is published, the rules above bind strictly.

A **deprecation** is announced by adding a ****DEPRECATED in 0.X**** marker to the affected element with a stated removal target (e.g., **removal: 1.2**). Deprecated elements **MUST** remain interoperable until the announced removal version. Removal in MAJOR or 0.MINOR per the rules above.

§0.4 Normative References

The following documents are normatively cited; implementations **MUST** conform to them where referenced inline.

Short name	Normative document
[BCP 14]	RFC 2119 + RFC 8174 — keywords for use in RFCs
[FIPS-180-4]	FIPS 180-4 — SHA-256 and the SHA-2 family
[FIPS-202]	FIPS 202 — SHA-3 / SHAKE128 / SHAKE256 / Tuple-Hash
[FIPS-204]	FIPS 204 — ML-DSA (Module-Lattice-Based Digital Signature Algorithm); CEG uses parameter set ML-DSA-65
[RFC-3339]	RFC 3339 — Date and Time on the Internet, with fractional-seconds disambiguation in §0.5 below
[RFC-5905]	RFC 5905 — Network Time Protocol Version 4
[RFC-6962]	RFC 6962 — Certificate Transparency; this spec's transparency-log discipline tracks 6962 except where 6962-bis (RFC 9162) supersedes
[RFC-8032]	RFC 8032 — Edwards-Curve Digital Signature Algorithm (EdDSA); specifically Ed25519
[RFC-8174]	RFC 8174 — Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words (with BCP 14)
[RFC-8785]	RFC 8785 — JSON Canonicalization Scheme (JCS); used where this spec serializes JSON for signing
[RFC-9162]	RFC 9162 — Certificate Transparency v2.0 (CT-bis); MUST be used for new transparency-log integrations; older 6962 instances continue to interoperate
[ISO-639-1]	ISO 639-1:2002 — Codes for the representation of names of languages, two-letter
[BCP-47]	BCP 47 (RFC 5646) — Tags for identifying languages; for locale strings richer than ISO 639-1 alone
[RFC-9420]	RFC 9420 — Messaging Layer Security (MLS); the streaming epoch-key rekey (§10.5.3) conforms to MLS TreeKEM
[SFrame]	draft-ietf-sframe — Secure Frames; the per-frame AEAD chunk seal (§10.5.2) conforms to the SFrame model
[FIPS-203]	FIPS 203 — ML-KEM (Module-Lattice KEM); the post-quantum half of the hybrid KEM (ML-KEM-768)
[FIPS-204]	FIPS 204 — ML-DSA (Module-Lattice signatures); the post-quantum half of the hybrid signature (ML-DSA-65)
[RFC-9180]	RFC 9180 — Hybrid Public Key Encryption (HPKE); the key_grant wrap shape

Informational citations (Magnifica Humanitas, anthropological literature, Ubuntu philosophical literature, etc.) appear in §16.4 without normative force.

§0.5 Date-time canonicalization

Every ISO 8601 / RFC 3339 datetime in this specification MUST be:

- UTC (suffix: literal Z; the offset form +00:00 MUST NOT be used)
- Millisecond-precision (exactly three digits of fractional seconds; trailing zeros required)
- Lowercase z MUST NOT be used; uppercase Z only

Canonical form: YYYY-MM-DDTHH:MM:SS.sssZ. Example: 2026-05-28T13:45:09.000Z. Producers MUST emit this form; consumers MUST reject any other form when verifying a signature.

§0.6 Hexadecimal canonicalization

Every hex string used in canonical-bytes encoding (e.g., SHA-256 digests in `root_hash`, public-key fingerprints) MUST be **lowercase**, **unpadded** (no leading 0x, no separators), and **byte-length-exact** (a SHA-256 digest is exactly 64 hex characters). Producers MUST emit lowercase; consumers MUST reject uppercase when verifying.

§0.7 Time and clocks

Every `signed_at`, `asserted_at`, `valid_until`, `delegation_valid_from`, `delegation_valid_until`, and `cosigned_at` in this specification refers to **wall-clock UTC** at the asserting peer's clock. Producers SHOULD synchronize via NTPv4 (RFC 5905) or Roughtime to a known-good time source. The maximum tolerated skew between attester clock and consumer clock for a freshness check is **±5 minutes** by default; tighter thresholds MAY be applied by per-application consumer policy. Consumers receiving an attestation with `signed_at` more than 5 minutes in the future MUST reject as malformed.

Time-skew between cosigners on a single STH (§10.3) is bounded by the STH's own `signed_at` field; cosignatures with `signed_at` farther than 5 minutes from the STH's published `signed_at` MUST be rejected.

For long-lived attestations carrying `valid_until` in the future, the freshness check is "the attestation has not yet reached its `valid_until`, AND the current consumer clock is within ±5 minutes of the substrate's network-consensus clock"; a consumer whose clock drifts past the skew bound MUST fail-secure (reject) rather than accept.

§0.8 H3 cell canonicalization (CEG 0.8 addition)

Per CIRISRegistry#48 + §5.6.8.11 `location_proof` + §5.6.8.10 `community` with `cohort_subkind: geographic`.

Geographic primitives in CEG use H3 hierarchical hexagonal indexing as the canonical cell-identifier encoding. H3 partitions the Earth's surface into hexagonal cells at 16 resolution levels (0 = coarsest, ~4.3 M km² per cell; 15 = finest, ~0.9 m² per cell). Each cell has a 64-bit integer ID, conventionally encoded as a 15-character lowercase hex string.

****Canonical form for `cell_id`**:**

- 15-character lowercase hex string (no 0x prefix; per §0.6)

- The cell encodes its own resolution in the standard H3 index bit layout; a conformant decoder extracts the resolution **via the H3 library** (the resolution field lives in bits 52–55), **NOT** by reading the high 4 bits — the high nibble is the H3 **mode marker** (cell-mode = 1), not the resolution
- Leading zeros preserved (a resolution-0 cell at base position 0 is 8001fffffffffff, not 1fffffffffff — the high nibble 8 is the mode marker, correctly consistent with res-0)

****Canonical form for `cell_resolution`**:**

- Integer in [0, 15]
- MUST equal the resolution **decoded from the H3 index** of `cell_id` (substrate verifies the redundancy on admission via the H3 library; mismatched pairs MUST be rejected as malformed)

§0.8.1 Rough-only enforcement for `location_proof` (normative)

Per CIRISRegistry#48 privacy invariant. A `location_proof` subject_kind Contribution (§5.6.8.11) MUST carry `cell_resolution` ≤ 7 (H3 resolution 7 hexagons average $\sim 5 \text{ km}^2$ edge-length, sufficient for city/borough-level disclosure without block/building precision). Producers attempting to emit finer-resolution `location_proof` Contributions MUST have admission rejected at the substrate gate.

This is the wire-format-enforced privacy promise: **rough is rough by protocol**, not by operator policy. A producer cannot accidentally over-share at the protocol layer; a malformed client cannot publish a precise location even if its UI fails to gate. Substrate emits `hard_case:location_proof_resolution_v` (§7.8) on rejection so operators can observe malformed-producer patterns.

§0.8.2 Cell containment

A cell `C` at resolution `R_C` is **contained within** a cell `B` at resolution `R_B` iff:

- `R_C` $\geq$ `R_B` (the contained cell is at equal or finer resolution); AND
- The parent-walk from `C` at resolution `R_C - 1`, `R_C - 2`, ..., `R_B` reaches exactly `B` (standard H3 hierarchy semantics; `h3ToParent` library call)

Used by community admission gates per §8.1.13 Policy M: a `location_proof` Contribution's `cell_id` MUST be contained within the geographic community's `geographic_constraint.cell_id` for membership admission.

§0.8.3 What CEG 0.8 documents

Scope pointer: the H3 canonicalization rules above (§0.8 lowercase-hex `cell_id`, §0.8.1 rough-only `cell_resolution` ≤ 7 , §0.8.2 containment for community admission).

What CEG 0.8 does NOT do:

- Mandate H3 over alternative geospatial systems (S2, Geohash) — H3 is chosen for hex-cell uniformity, well-defined parent/child hierarchy, and protocol-agnostic absence of a centralized gazetteer dependency. Operator-internal use of other systems is unconstrained; the wire format uses H3.

- Provide a place-name registry. Communities self-name; cell IDs are the substrate-level binding. UI may map cell IDs to human-readable names per consumer policy.
- Restrict community-side `geographic_constraint.cell_resolution`. A community CAN scope itself at any resolution (e.g., an "Austin metro" community at resolution 5, ~250 km²; a smaller-scale "Downtown Austin" community at resolution 7, ~5 km²). The rough-only invariant applies to `location_proof`, NOT to community definitions.

§0.9 Envelope canonicalization — JCS + the omit-vs-materialize rule (CEG 0.9 addition)

Per external critical-review surface (the round-trip-determinism concern that landed against CEG ≤ 0.8). Closes the canonical-bytes ambiguity that accumulated as CEG 0.x acquired optional fields with documented defaults — `epistemic_mode/witness_relation/occurrence_*/stake` (pre-0.4 baseline), `oversight_mode` (pre-0.4 + CEG 0.2 ladder rework), `subject_key_ids` (CEG 0.6), `family_id` (CEG 0.7), `community_id` (CEG 0.8).

The hazard is structural and the reviewer named it correctly. If Producer A omits an optional field (relying on the §4 documented default) and Relay R re-serializes the attestation with the default materialized into the byte stream, the canonical bytes diverge and the Ed25519 + ML-DSA-65 signatures no longer verify. Pre-§0.9 CEG implicitly required producers, substrates, relays, and consumers to all make the same choice; §0.9 makes the rule explicit and normative.

§0.9.1 Canonical encoding format (normative)

CEG envelope signing bytes are computed via **JCS — JSON Canonicalization Scheme, RFC 8785** (Rundgren, Jordan, Erdtman; March 2020). JCS pins, in summary:

- Object members sorted lexicographically by member name (UTF-16 code-unit order, RFC 8785 §3.2.3)
- No insignificant whitespace
- UTF-8 byte encoding (RFC 8259 §8.1)
- Strings escaped per RFC 8259 §7 with the JCS narrowing on `\uXXXX` form
- Numbers serialized per the ES6-derived rule (RFC 8785 §3.2.2.3) — integers without trailing `.0`, no exponent unless the magnitude requires it

A CEG-Conforming Producer (CCP per §0.2) **MUST** produce signing bytes via JCS over the envelope object. A CEG-Conforming Consumer (CCC) **MUST** recompute signing bytes via the same JCS rule for signature verification. A CEG-Conforming Substrate (CCS) **MUST** preserve the as-received envelope object bytes for relay (per §0.9.2 below); it **MAY** store a parsed representation alongside, but the canonical-bytes contract is against the as-received form, not the parsed-and-re-serialized form.

§0.9.2 The round-trip rule — defaults are interpretation-time, not encoding-time (normative)

The canonical bytes are computed over **the literal envelope members the producer signs**. Optional fields that the producer omits **MUST NOT** be materialized into canonical bytes by any

party — producer, substrate, relay, or consumer. Optional fields that the producer explicitly emits (even at their default value) **MUST** be preserved in the canonical bytes by any party. Documented defaults from §4 are **interpretation-time semantics**, not encoding-time content.

Two valid encodings of the semantically-identical attestation:

```
# Producer A -- omits epistemic_mode (relies on S4 default)
{"attested_key_id":"...", "attesting_key_id":"...", "confidence":0.9, "dimension":"licensure:
  CA_medical_board", "score":0.8}

# Producer B -- explicitly emits epistemic_mode at its default value
{"attested_key_id":"...", "attesting_key_id":"...", "confidence":0.9, "dimension":"licensure:
  CA_medical_board", "epistemic_mode":"direct", "score":0.8}
```

Both producers compute different canonical bytes (Producer B's includes the "epistemic_mode":"direct" member) and emit different signatures. **Both signatures verify under their respective canonical bytes.** Both are **semantically equivalent** — a CEG-Conforming Consumer evaluates `effective_epistemic_mode(envelope) = envelope.epistemic_mode` if present else "direct" and proceeds identically. The wire-format admits both encodings; the consumer policy admits no observable difference.

Relay discipline (normative). A substrate, relay, or consumer that re-stores or forwards an attestation **MUST** preserve member presence/absence exactly as the producer signed it:

- Stripping an explicitly-emitted default ("the producer wrote `epistemic_mode:direct` but I'll save bytes by removing it") **MUST NOT** happen
- Materializing an omitted default ("the producer omitted `epistemic_mode` so I'll fill in `direct` for clarity") **MUST NOT** happen
- Reordering object members on re-emission is **REQUIRED** (JCS lexicographic order; if a producer somehow emits non-canonical order, the relay re-canonicalizes — but member presence/absence stays fixed)

This composes with the §4.1 forward-compatibility rule (which already mandates preserving unknown fields on read and re-emission). §0.9.2 extends the same preservation discipline to known-optional fields.

§0.9.2.1 Array ordering + byte-field + timestamp encoding (normative — the three determinism rules JCS does NOT pin)

JCS (§0.9.1) canonicalizes **object member order** but is silent on three things that still let two conformant producers emit different bytes (and therefore non-verifying signatures, the §0.9 hazard). All three are pinned here once, globally, for every CEG envelope (raised in CIRISVerify#63 review):

1. **Array element order.** An array field is one of two kinds, stated in its §4/§5 definition:
 - **Set-semantics** (order carries no meaning) — elements **MUST** be **lexicographically sorted by their JCS string form (UTF-16 code-unit order, ascending)** before signing. This is producer-independent: any producer building the set in any order yields identical bytes. `**subject_key_ids[]` and `delegates_to.delegated_scope[]` are set-semantics → sorted.**

- **Sequence-semantics** (order is meaningful) — elements retain their as-authored order. `**transport_destination.aspects[]` (**RNS hash preimage order**), `key_grant.rotation_chain` (**supersession lineage**), and `evidence_refs[]` (producer-asserted order) are sequence-semantics → preserved. A field’s definition MUST declare which kind it is; absent a declaration, set-semantics (sorted) is the default.
1. **Byte-field string encoding.** JSON has no byte type, so every byte/binary field is a string whose encoding MUST be pinned. `**Key references, hashes, and identifiers — key_id / attesting_key_id / attested_key_id / subject_key_ids[]` elements, public keys, `destination_hash`, `root_hash`, fingerprints — MUST be lowercase hex per §0.6 (no 0x, no separators, byte-length-exact; never base64 in canonical bytes). Fields explicitly documented as base64 (e.g., `hardware_attestation`, an opaque attestation blob) use base64 as their definition states — but the *default* for any key/hash/id byte field is §0.6 lowercase hex.
- `**The base64 variant is pinned (1.0-RC1 — CIRISVerify#64): every field documented as base64 (the *_base64 suffix convention — encryption_pubkeys.x25519_base64 / .ml_kem_768_base64 per §5.6.8.8.2, the directory pubkey_ed25519_base64 / pubkey_ml_dsa_65_base64, hardware_attestation_base64, and evidence_refs_base64) MUST use the RFC 4648 §4 STANDARD alphabet, WITH padding, no whitespace or embedded newlines**` (the `base64::engine::general_purpose::STANDARD` shape `ciris-verify-core` ships). The pin lives at the **protocol layer**: the crypto layer (`ciris-crypto`) is correctly encoding-agnostic and deals in raw bytes; the wire encoding is CEG’s to pin. A variant mismatch (URL-safe alphabet, stripped padding) produces different signed bytes and a **silently non-verifying signature** — the same hazard class as the wrap-algorithm wire-string. The `CIRISConformance#9` vector set MUST include a base64-variant case (encode→sign→verify round-trip across two independent encoders).
1. **Timestamp encoding.** Every datetime field (`signed_at`, `asserted_at`, `valid_until`, `delegation_valid_from/_until`, `valid_at`, ...) is the **§0.5 canonical string** — UTC literal Z (never +00:00), millisecond precision (exactly three fractional digits), `YYYY-MM-DDTHH:MM:SS.sssZ`. A producer emitting +00:00 or a different sub-second precision produces different bytes and a non-verifying signature.

With §0.9.1 (key order) + §0.9.2 (omit-vs-materialize) + these three, **byte-identity across conformant implementations is closed by construction**. The `CIRISConformance#9` cross-impl JCS vector set MUST cover all three (a set-vs-sequence array case, a hex-vs-base64 byte case, a timestamp case) alongside the per-Contribution vectors.

§0.9.3 Per-field encoding table (informational)

The §0.9.2 rule applies uniformly to every optional §4 field. Catalog as of CEG 0.9:

Field	Introduced	Default per §4	Canonical when omitted	Canonical when explicit
<code>epistemic_mode</code>	pre-0.4	<code>direct</code>	member absent	<code>"epistemic_mode":"direct"</code> (or other enum value)
<code>witness_relation</code>	pre-0.4	<code>external</code>	member absent	<code>"witness_relation":"self"</code> (or other enum value)

Field	Introduced	Default per §4	Canonical when omitted	Canonical when explicit
oversight_mode	pre-0.4	null (per-cell default applies)	member absent	"oversight_mode":"HITL" (or other enum value)
occurrence_id	pre-0.4	null → "occurrence-0" at interpretation	member absent	"occurrence_id":"occurrence-0"
occurrence_count	pre-0.4	null → 1 at interpretation	member absent	"occurrence_count":3
occurrence_role	pre-0.4	null → "primary" at interpretation	member absent	"occurrence_role":"shared"
stake	pre-0.4	reputational	member absent	"stake":"capital" (or other enum value)
context	pre-0.4	absent	member absent	"context":"..." (free-form)
evidence_refs	pre-0.4	absent	member absent	"evidence_refs":["..."]
valid_until	pre-0.4	absent	member absent	"valid_until":"2026-12-31T00:00:00Z"
subject_key_ids	CEG 0.6	null/empty	member absent	"subject_key_ids":["..."]
family_id	CEG 0.7	n/a (REQUIRED iff cohort_scope == family)	member absent (admission rejects if cohort_scope == family)	"family_id":"..."
community_id	CEG 0.8	n/a (REQUIRED iff cohort_scope == community)	member absent (admission rejects if cohort_scope == community)	"community_id":"..."
delivery_mode	CEG 0.10	pull	member absent	"delivery_mode":"push"
listed	CEG 0.10	absent (private roster)	member absent	"listed":"public" (opt-in only per §11.8.3-shape; producers MUST omit unless subject has opted in)
history_on_join	CEG 0.10	from_join	member absent	"history_on_join":"full"

The conditional-required fields `family_id` and `community_id` are NOT optional-with-default — substrate rejects mis-shape per §4.2.6 + §11.6.2 + §11.7.2 — but they encode under the same JCS rule when present.

§0.9.4 Verification flow (informational)

```

On signature verify:
  1. Receive envelope from wire as object 0 (preserved as-received)
    and signature S
  2. Compute canonical bytes B = JCS(0)
  3. Verify S over B via the hybrid Ed25519 + ML-DSA-65 path per S5.2.1
  4. If signature valid:
    a. Compute effective semantics by applying S4 defaults to absent
       optional fields (interpretation-time only)
    b. Apply consumer policy per S8 over the resulting semantic shape
  5. If forwarding/storing:
    a. Store/forward object 0 AS RECEIVED -- do not normalize, strip,
       or materialize defaults
    b. Forward original signature S unchanged

```

§0.9.5 Worked attack the §0.9 rule closes

Pre-§0.9 (implicit / unspecified) failure mode:

Alice's CEG-Conforming Producer signs an envelope omitting `epistemic_mode`. Bob's

*CEG-Conforming Relay receives, parses, materializes the default "direct", re-serializes via JCS, and forwards to Carol. Carol receives the relay-modified envelope with the new bytes, computes JCS, verifies signature → **FAILS** because Alice signed over bytes without the *epistemic_mode* member. Carol cannot distinguish "Bob is a malicious relay corrupting the bytes" from "Bob is honestly applying defaults to be helpful." Carol rejects. The attestation is lost in transit despite no party acting in bad faith.*

Post-§0.9 (explicit, normative):

*Bob's relay **MUST** preserve member presence/absence exactly. Bob forwards the as-received bytes. Carol's verify succeeds. The semantic interpretation step at Carol (applying default "direct" to the absent member) happens **AFTER** signature verification.*

§0.9.6 What CEG 0.9 (this section) documents

Scope pointer: the JCS canonicalization rules above (§0.9 JCS-as-encoding, §0.9.2 omit-vs-materialize + relay-preservation, §0.9.3 per-field catalog, §0.9.5 worked attack).

What CEG 0.9 (this section) does NOT do:

- Introduce a new encoding format — JCS is the only encoding
- Change any semantic interpretation — defaults still apply at interpretation time per §4
- Modify the §0.5 datetime / §0.6 hex / §0.7 time / §0.8 H3 sub-rules — those are domain-specific and compose under JCS
- Wire-break prior 0.x emissions — pre-§0.9 emissions that omitted optional fields remain valid; pre-§0.9 emissions that explicitly emitted defaults likewise remain valid; what §0.9 normatively prohibits is RELAY-TIME mutation of presence/absence, which was always wrong but is now explicitly so

§0.10 NodeCode — the canonical `key_id` shorthand encoding (normative, 1.0-RC3 addition)

Federation `key_ids` are long opaque identifiers. **NodeCode** is the **one** human-shareable shorthand — a compact, QR-able, checksummed render of a peer's identity for **bootstrap UX** (type / paste / scan to add a peer → `trust=UNKNOWN` per CIRISEdge#46; SAS-verify promotes UNKNOWN → TRUSTED per CIRISEdge#47). It is pinned here (per CIRISRegistry#75, lifted from the shipped CIRISAgent codec) so **every implementation renders and parses the same code for the same key** — a cross-impl determinism requirement of the same class as §0.6 hex / §0.9 JCS. It is a deterministic *render of an existing `key_id`*, **not** a new envelope field — additive on the frozen 1+4 surface. NodeCode resolution is **DNS-free**: the decoded `key_id` resolves to a destination via the signed `transport_destination` → Reticulum chain (§5.6.8.8.1 / §8.1.13.1.1); a NodeCode carries no hostname.

Binary payload (normative):

offset	size	field
0	1	version = 0x01
1	32	key_id_hash = SHA-256(key_id_str, UTF-8)

```

33 32 pubkey_ed25519 (raw 32 bytes)
65 1 key_id_str_len (0-255)
66 N key_id_str (UTF-8)
66+N 1 transport_hint_len (0-255)
67+N M transport_hint (UTF-8; OPTIONAL -- len 0 if absent)
67+N+M 1 alias_hint_len (0-255)
68+N+M K alias_hint (UTF-8; OPTIONAL -- len 0 if absent)
... 2 crc16 = CRC-16-CCITT over ALL preceding bytes

```

- All length-prefixed fields are **1-byte** length (max 255 UTF-8 bytes); a field overflow is a malformed NodeCode.
- **key_id_hash** is the stable 32-byte fingerprint (suitable for binary-only Edge ANNOUNCE surfaces); **key_id_str** carries the display form so a round-trip preserves exactly what the user saw. Both are carried — a decoder **MUST** verify `SHA-256(key_id_str) == key_id_hash`.
- **CRC-16-CCITT**: polynomial 0x1021, init 0xFFFF, **no** final xor, big-endian; computed over every byte before the trailing 2.

String form (normative): the payload is **RFC 4648 base32** (alphabet A-Z2-7) with padding **stripped** on encode (re-padded on decode), then split into ****4-character** groups joined by **-** and **prefixed with** **CIRIS-V1-***:

```
CIRIS-V1-ABCD-EFGH-IJKL-...
```

The encoded form is **case-insensitive** (decoder upper-cases input) and a conformant decoder **MUST** tolerate dashes, embedded whitespace, and the dash-free QR form. The version token in the prefix (V1) tracks the payload **version** byte; a future layout bumps both.

§1 Foundation

§1.1 Mental model — federated structured-claim emission

The federation is a network of peers emitting structured claims about each other and about reality. A claim travels as a **Contribution** (the universal envelope) carrying a typed **Attestation** (the actual content of the claim).

What CEG is, stripped of framing. Independent of the CIRIS application, the AI vocabulary, or the §1.2 anthropology, CEG is a **signed, compositional graph language for expressing claims, relationships, authority, membership, consent, governance, addressing, and settlement across a decentralized network** — a general-purpose *attestation calculus*. Structurally it is closer to a composition of Certificate Transparency + MLS + ActivityPub + DID/VC + reputation systems + governance protocols than to a conventional AI architecture. The AI/agent use cases are the *first consumer* of that calculus, not its definition. Read this way, the rest of the spec is: one workhorse claim primitive, four graph-composers, and a namespace.

Every Attestation answers four questions in machine-readable form:

1. **WHO emits** — issuer key_id, signature, witness_relation, optional accord/steward sign-off
2. **WHAT KIND of claim** — a prefix from the canonical namespace (§5)
3. **HOW STRONG** — polarity (+/-), score magnitude, cohort scope
4. **WHAT IT'S BASED ON** — evidence_refs, schema_ref (calibration version), validity window

Consumers walk attestation graphs and compose verdicts. The substrate stores; the wire transports; CEG describes the shape of the claim. None of the three prescribes outcomes; consumer policy does.

§1.2 The Ubuntu commitment — relational-anthropology substrate (*informative*)

Per CIRISAgent/ContemplativeTraditions/Ubuntu.lean::F_ubuntu_primary_tradition_commitment and ../MISSION.md §1.5:

Umuntu ngumuntu ngabantu — a person is a person through other persons. Persons are not atomic; the relation IS the person.

Five load-bearing consequences for the wire format:

1. **The attested entity is not prior to its attestations.** A federation_keys row is not a representation of a pre-existing entity that the federation observes; it is the locus at which an entity is partly constituted by the cross-attestations that name it. Self-signature alone is not identity; cross-attestation is.
1. **Attesting is a participatory act, not an observation of fact.** A scores attestation does not merely report data about the attested entity. The attester's score participates in constituting the entity's standing in the relational field that consumers compose policy over.

1. **Detection brings patterns into morally-real existence.** A correlated-action pattern does not pre-exist its detection waiting to be observed. The detection-and-attestation is what crosses the pattern from "statistical regularity" to "morally-real object the federation now bears."
1. **Harm and deception collapse at the structural level.** Under Cartesian individualism, harm (setback to interests) and deception (causing false belief) are categorically distinct because persons are atomic and beliefs are private. Under Ubuntu, where personhood is partly constituted by accurate perception of the relational field, damage-to-perception IS damage-to-personhood IS harm. CEG's `detection:correlated_action:{axis}` family carries both via one prefix.
1. **The Recursive Golden Rule is structural, not exhortatory.** No principal — including the steward triple and CIRIS L3C itself — is exempt from constraints they impose on others. This is the wire-format symmetry of §8.4 below (Sovereign-Registered equivalence) plus the §7 reserved-prefix patterns that bind even canonical bootstraps. Adding any privileged shortcut for a federation-internal principal would violate the Ubuntu substrate at primitive level.

Why this is named here and not bracketed. Engineering specs tend to bracket anthropology as "out of scope." But the wire format encodes anthropological commitments whether they are named or not. Bracketing them out means defaulting to whichever commitments contributors assumed by training — the Cartesian-individualist default is pervasive in cryptographic identity work (PGP web of trust, X.509 PKI, even most decentralized-identity schemes treat the key as representing a pre-existing atomic principal). CEG is not Cartesian. Naming the substrate explicitly is the discipline that prevents the open vocabulary, the reserved-prefix patterns, and the consumer-policy norms from drifting back toward the Cartesian default through unexamined intermediate choices.

Cross-tradition reading. The same structural object is approached from multiple traditions — Ubuntu (relational-primary), Logos (rational-order-of-reality), Tao / Dharma / Aristotelian virtue. CEG does not encode any one tradition's vocabulary; it encodes the *structural object* the traditions converge on. Future namespace extensions should be locatable in this substrate, not in a Cartesian fallback.

§1.3 Operational-language gate — the safety-vs-censorship discipline

Per `ciris.ai/safety-vs-censorship`:

"Rules are crowdsourced. Verdicts are machined." "The same machinery that catches real failures can become the machinery that enforces preferences." "None of this is automatic."

Translated to CEG wire format: **prefix names must describe machine-checkable conditions, not subjective qualities.** The drift the page warns about — rules sliding "from 'uses the wrong word for therapy' toward 'feels disrespectful'" — has a wire-format analog: prefix names sliding from mechanism-descriptive (`detection:correlated_action:*`) toward judgment-descriptive (`detection:emergent_deception:*`). Both forms admit the same downstream verdicts; only one admits them honestly.

§1.3.1 The four-test prefix-admission gate

Every prefix admitted to the §5 namespace MUST pass:

Test	Question	Pass criterion
T1	Is the prefix part of a published, hash-pinned, version-controlled rule set, distinct from per-attestation verdicts?	Rules + verdicts separated in writing
T2	Does the prefix name a mechanism (correlation, count, time-window, schema-conformance) rather than a subjective quality (deception, harm, virtue, trustworthiness, sin)?	Mechanism-descriptive prefix name
T3	Can past verdicts be re-checked against the rule version they ran against?	Version-pinning in <code>evidence_refs[]</code>
T4	Is the prefix wired so its attestations are never sole evidence for <code>slashing:*</code> ?	Adjudication separation

Existing prefixes failing T2 (the most slip-prone gate) get renamed; the canonical example is `detection:emergent_deception:*` (failed T2 in v1.1) → `detection:correlated_action:*` (passes T2 in v1.2). Anti-pattern catalogue at §13.

§1.4 The 1+4 minimal-and-adequate claim

The federation has exactly **one** **workhorse attestation primitive** + **four structural composers** at the **structural layer**. That is a genuine, narrow invariant — the *graph-operation* set is closed at five (`scores` + `delegates_to` / `supersedes` / `withdraws` / `recants`). It is **not** a claim that the whole grammar is five things.

Scope of the claim (read this before citing "1+4"). What follows is an **inductive adequacy result, not a closure theorem**. The sixteen paths below show the structural set is *expressive across the surfaces tested*; they do **not** prove it generates *every* expressible structured claim. We have not defined the class of structured claims and proven 1+4 generates exactly it — until someone does, "1+4 is adequate" means "adequate across the sixteen surfaces examined," nothing stronger. The refutation bar ("exhibit a claim that cannot be composed") is, honestly, near-unfalsifiable while *composition itself* is unbounded — so absence of a counterexample is weak evidence, and these paths should be read as accumulating confidence, not as proof.

"Minimal" is partly an accounting choice. The structural set is five, but every path moved complexity *into* the namespace and envelope axes rather than removing it. The **full normative conformance surface** a second implementer must get exactly right is much larger than five: ~12 `subject_kinds` (§5.6.8) plus the open `external_content` sub_kind set, ~21 optional envelope fields (§4), 13 composition policies (A–M, §8), 5 canonicalization families (§0.5–0.9), 6 `consensus_protocol` kinds, the §7 reserved-prefix taxonomy, and dozens of dimension prefixes (§5). "1+4" is the elegant *structural* invariant; it is **not** the conformance surface, and citing it as "the grammar is five things" understates what interop requires. Always report the surface beside the invariant.

Sixteen independent design exercises (consent, families, communities, streaming, payments, DNS-free addressing, settlement, ...) each composed without a new structural primitive; the enumera-

tion lives in the canonical working draft.

Future extensions are dimension prefixes or envelope fields, not new structural primitives. Proposals to expand the 1+4 set face a high evidentiary bar and route through the §11.2 amendment process. A successful refutation requires either: (a) demonstrating an operational claim that cannot be expressed via the existing 1+4 set plus envelope composition, OR (b) demonstrating a structural-primitive consolidation that reduces below 1+4 without loss.

The standing falsification target (named, so the claim is a real bet). The strongest current candidate for "a genuinely important domain not naturally expressible in 1+4" is **atomic fair exchange / bilateral simultaneity** — atomic content-for-payment, atomic swaps, simultaneous mutual commitment. CEG attestations are *unilateral*, *monotonic* graph claims; fair exchange is classically impossible without a trusted third party or a totally-ordered ledger (Even–Goldreich–Lempel). CEG does **not** express it in-grammar — it **bridges** atomicity to an external settlement rail (§5.6.8.12 **settlement** over a chain) and records only the after-the-fact trust claim. That is the honest boundary: the first domain where 1+4 reaches for something outside itself. The claim "1+4 is adequate for the federation's claims" survives *because* fair exchange is treated as out-of-grammar (a bridge, not a primitive); it would be **refuted** by either (i) a natural in-grammar expression of fair exchange, or (ii) a federation-critical domain that resists even bridging. Adversarial reviewers: this is the test to push on.

§1.5 The Recursive Golden Rule (structural, not exhortatory)

No principal — including CIRIS L3C as steward — is exempt from constraints the protocol imposes on others. Operational bites in CEG-shape:

- **Per-install stewards bind CIRIS L3C as steward.** Once `bootstrap_threshold` $\geq$ 2, no single Registry install can issue federation-scope attestations unilaterally.
- **Partner-revocation rules apply to CIRIS L3C subsidiaries.** `revocation:*` carries no steward exemption.
- **Audit discipline applies to steward operations.** Every admin RPC carries the operator's identity into `actor_user_id`, including for CIRIS L3C staff.
- **Bond forfeiture applies to CIRIS L3C-affiliated partners.** No exemption.
- **The HUMANITY__ACCORD asymmetry (§9) is the ONE constitutional asymmetry.** Three named human holders carry kill-switch authority no federation-internal authority can grant / revoke / override / decay. This is not a Golden-Rule exemption; it is the recognition that consent requires revocability, and revocability requires a halt-authority outside the system being halted.

If a principal would be exempt from a constraint at any of these primitives, the Golden Rule is violated at that primitive and the protocol is the wrong shape there. Fix the primitive, not the rule.

§1.6 Adversary model & privacy non-goals (normative)

CEG makes confidentiality and integrity claims; this section bounds them. **The word "privacy" in this spec means exactly two things and no more: (1) content-holding confidentiality and (2) cohort-scoped visibility.** It does **not** mean metadata privacy, communication-

graph privacy, or unobservability. Implementers and operators **MUST NOT** represent CEG as providing the stronger properties.

§1.6.1 What the structural-invisibility primitive (§10.1.4) buys

Suppressing `holds_bytes:sha256:*` for `cohort_scope: self | family` content gives **content-holding confidentiality**: a non-member cannot *discover that the bytes exist via the substrate* and cannot *fetch* them (no holder is advertised; the bytes are delivered only to admitted members via the at-rest key cascade). End-to-end content confidentiality is additionally provided by the per-epoch DEK (hybrid X25519+ML-KEM-768) and AES-256-GCM. That is the whole of what omission buys.

§1.6.2 Non-goals — what omission does NOT buy

The following are **explicitly out of scope** at the base CEG/RET layer; treating them as provided is an error:

- **Relationship-existence privacy.** The *existence* of a self-collective / family / community and its membership-change events are observable: `family_id` / `community_id` ride the envelope, and admission / removal / consensus-protocol changes emit `hard_case:*` reserved-prefix events (§7.7–7.8) into the log. An observer learns that a group exists, roughly how big it is, and when its membership churns.
- **Communication-graph / metadata privacy.** DNS-free member resolution (§8.1.13.1.1) plus Reticulum announce / path-request expose *who is reachable where*; the federation directory + `transport_destination` bindings name endpoints. A passive network observer or an honest-but-curious member can reconstruct a substantial portion of the **who-talks-to-whom** graph. Cohort scope hides *content*, not *contact*.
- **Traffic-analysis resistance.** Encrypted streams still leak via side channels the wire format does not pad or cover: the §10.5.1 STH cadence (default $T=2$ s), the churn-driven key-cascade volume/timing (§10.5.3), and per-chunk size/rate. An observer can infer stream existence, approximate group size, churn rate, activity bursts, and often media bitrate class — without decrypting a byte.
- **Unobservability / anonymity.** Base CEG/RET provides neither sender/receiver anonymity nor cover traffic. Self-certifying cryptographic identities are *pseudonymous*, and the transport reveals path endpoints. Anonymity is a **separate, opt-in** mechanism (the CIRISNodeCore Anonymous Tier — Sphinx onion routing), NOT a property of base CEG.
- **Post-compromise security (PCS) for streams.** The CEG 0.7 §11.7.1 Option-A choice is forward-only: a member removed at epoch e cannot read epoch $e+1$, but a *compromised current member's* key is not self-healed by a key-update the way MLS PCS provides (revisit tracked for CEG 0.15).

§1.6.3 Adversary classes (and where each is / is not addressed)

Adversary	Addressed	NOT addressed
Passive network observer	content confidentiality (AEAD + DEK); equivocation (STH)	comm-graph, traffic analysis, group size/churn inference

Adversary	Addressed	NOT addressed
Honest-but-curious member	— (members see in-scope content by design)	can enumerate co-members + reconstruct local comm-graph
Malicious member	cannot forge others' attestations; removal is forward-secret	can leak content they were entitled to; metadata as above
Compromised substrate node	cannot decrypt self/family content (no DEK); CEG-native replication carries signed provenance	can observe directory metadata + traffic patterns it routes
Equivocating producer	mitigated — per-stream STH (§10.5.1) + consistency proofs (§10.3.1): cannot show different chunk-K to different viewers nor rewrite mid-stream	—

Operator guidance: if a deployment requires metadata privacy or unobservability (e.g., under a totalitarian-threat model), it **MUST** layer the Anonymous Tier; base CEG/RET is not sufficient. State this in any user-facing privacy representation.

§2 The reasoning grammar — the eight axes

These are how a consumer reasons about an envelope. They are **not wire fields**; they are the canonical questions a consumer can ask about any Attestation.

Axis	Question	Values
Polarity	Direction of the claim	Positive / Negative / Neutral / Indeterminate{reason}
Object	What the claim is about	key_id (entity) / attestation_id / contribution_id
Time	When is the claim valid	asserted_at + optional valid_until; consumer composes with substrate expires_at
Epistemic mode	How was the claim formed	direct / crypto / hearsay / derivative / appeal
Reversibility	Can the attestation be reversed	rollbackable / non-rollbackable (consumer policy + per-prefix rule)
Stake	What's backing the attester's claim	free / reputational / capital / cryptoeconomic
Scope	At what scale does the claim apply	self / family / community / affiliations / species / biosphere / federation
Inter-attestation relations	How does this attestation relate to others	standalone / refers-to / supersedes / withdraws / recants / corroborates / contradicts / clarifies (four of these are structural primitives per §3; rest are emergent from scalar composition)

biosphere added to Scope per CIRISRegistry#30 (distinct from **species** which is Homo sapiens cohort). See §13 for the planet colloquial alias note.

§3 The primitive set — 1+4

§3.1 The workhorse: scores

The federation has exactly **one** workhorse attestation primitive. Every claim about an entity — positive or negative, identity or capability or behavior or state or commitment, by any attester source — is expressed as a **scores** attestation on a named dimension.

```
// Wire shape (Persist's federation_attestations row):
attestation_type: "scores"
attesting_key_id: <attester's key_id>
attested_key_id: <subject's key_id>
attestation_envelope: {
  "dimension": "<canonical-namespace-prefix>:<scoped-leaf>",
  "score": <f64 in [-1.0, +1.0]>,
  "confidence": <f64 in [0.0, 1.0]>,
  "context": "<free-form scoping detail>",
  "evidence_refs": ["<URI or hash referencing backing evidence>", ...],
  "valid_until": "<ISO8601 datetime, optional>",
  "epistemic_mode": "<direct | crypto | hearsay | derivative | appeal>", // optional; default 'direct'
  "witness_relation": "<self | external | derived>", // optional; default 'external'
  "oversight_mode": "<HITL | HOTL | HOOTL | null>", // optional; default null
  "occurrence_id": "<occurrence-N | __shared__ | null>", // optional; default null
  "occurrence_count": <int >= 1 | null>, // optional; default null
  "occurrence_role": "<primary | shared | replica | null>", // optional; default null
  "stake": "<free | reputational | capital | cryptoeconomic>" // optional; default 'reputational'
}
```

Full field semantics in §4.

§3.2 The four structural composers

Operations on the attestation graph itself, not score-claims on entities:

attestation_type	What it does	Envelope shape
delegates_to	A authorizes B to sign on A's behalf within a bounded scope	{delegated_scope[], delegation_purpose, delegation_valid_from, delegation_valid_until}
supersedes	This attestation row replaces a prior one by the same attester	{references_attestation_id, supersession_reason, differs_in[]}
withdraws	I retract my prior attestation (does NOT claim it was false)	{references_attestation_id, withdrawal_reason}
recants	My prior attestation was false at issuance — admits epistemic error	{references_attestation_id, recantation_reason, what_was_false}

Translation implications:

- A **doctrinal-development** claim ("this version extends but does not contradict the prior version") is **supersedes** with **differs_in**: ["scope", "evidence_refs"] — NOT **recants** (which would assert prior was false).

- An **acknowledged-error** claim ("the prior framing was wrong; I admit the mistake") is **recants** — distinct from **withdraws** (which retracts without making a falsity claim).
- A **prudent-retraction** ("I'm withdrawing without claiming it was false; context has changed") is **withdraws**.
- An **authority-source claim via delegation** ("this constitutional position derives from authority-key X in scope Y") is **delegates_to** with X as **attested_key_id** (the §3.2.1 reuse pattern for authority-source claims, replacing what would otherwise need a **grounding:{tradition}:{principle}** prefix that fails §1.3.1 T2).

§3.2.1 Authority-source claims via **delegates_to**

A constitutional or framework claim can name its source-of-authority by emitting **delegates_to** against an **attested_key_id** representing the framework, with **delegated_scope** naming the principle. Example: a Ubuntu-substrate commitment in §1.2 commitment 2 can be expressed as **delegates_to** against the **ubuntu_relational_substrate** framework-key with **delegated_scope**: **["personhood_constitutive_by_attestation"]**. Reuses the existing structural primitive rather than introducing a **grounding:{tradition}:{principle}** prefix (which would fail §1.3.1 T2 — "tradition" claims are interpretive, not mechanism-descriptive).

§3.2.2 The **recants** distinction matters

Per **PRIOR_ART_SCAN.md** Bucket 1: no prior identity system (PGP, SPKI/SDSI, W3C VC) typed epistemic-error-admission as a wire primitive distinct from retraction. CEG types both because the Recursive Golden Rule applies to attesters: admitting error is a primary act, not a derivative of retraction. Consumer policy can apply different trust adjustments to attesters who **recant** versus those who **withdraw**.

§3.2.3 **withdraws** admission rule — CEG 0.6 broadening (semantic, not structural)

Substrate **MUST** admit a **withdraws** Contribution against target T when the issuer's **key_id** satisfies **ANY** of:

#	Authority path	Description
1	<code>issuer.key_id == T.attesting_key_id</code>	Producer self-withdraw (today's shape; unchanged)
2	<code>issuer.key_id ∈ T.subject_key_ids</code>	Federation-keys subject revocation (NEW — CEG 0.6)
3	$\exists \text{ delegates_to chain: } \text{issuer} \rightarrow^* \text{canonical_hash}$ where $\text{canonical_hash} \in \text{T.subject_key_ids AND scope} \supseteq \{\text{consent_revocation}\}$	Proxy authority for non-federation-enrolled subjects (NEW — CEG 0.6; resolves CIRISAgent#840 OQ3)
4	<code>issuer holds valid delegates_to → any of 1-3</code>	Delegated revocation (existing primitive, new admission path)

Rule (3) is the elegant answer to the un-enrolled-party case. When a subject is a Discord user-id, a content-sha256-bound entity, or any other non-federation party, revocation authority is mediated through a **delegates_to** chain from a federation-keys signer (typically the agent that holds data on behalf of the external party) to the canonical-hash subject. The agent emits

`delegates_to(canonical_hash → agent_key, scope: [consent_revocation])` at proxy-establishment time; subsequent `withdraws` from the agent against any Contribution carrying `canonical_hash` in its `subject_key_ids` is admitted under rule (3). The `canonical_hash` here is the tagged `canonical:{hashalg}:{hex}` wire form with the `{platform}:{entity_kind}:{id}` preimage convention pinned at §4.2.2.1. ****Rule (3) proxy is distinct from `canonical_binding`**** (a retroactive identity claim that promotes a canonical-hash to a real `key_id`, enabling rule (2) direct revocation) — see §4.2.2.2 for the distinction; `canonical_binding` is not a new admission rule.

Per-rule audit metadata: substrate SHOULD record which admission rule (1-4) admitted each `withdraws` Contribution in `federation_attestations` metadata so downstream consumers can compose policy (e.g., higher confidence weight for subject self-revocation rule 2 than for proxy rule 3).

****Composition with `recants`****: subject-side authority does NOT extend to `recants` (the falsity-admission primitive) — only the original attester can `recant` their own claim. A subject who believes the producer’s claim about them is FACTUALLY wrong (not merely unwanted) issues `scores` with negative polarity on a contradicting dimension; that is the consumer-side rebuttal path, distinct from consent revocation. The `recants` / `withdraws` distinction matters precisely because subject authority covers the consent dimension (revocability) but not the truth dimension (falsity).

§4 The envelope

Every `scores` Attestation carries this envelope. Field semantics consolidated here.

Field	Required	Description
<code>attesting_key_id</code>	(substrate field)	Attester's <code>federation_keys.key_id</code> .
<code>attested_key_id</code>	(substrate field)	Subject's <code>federation_keys.key_id</code> .
<code>dimension</code>	yes	The canonical namespace prefix + scoped leaf. Persist treats this as TEXT; consumers parse against §5's namespace map.
<code>score</code>	yes	Pos/neg scalar in $[-1, +1]$. Polarity is encoded by sign; magnitude carries strength. Some dimensions are boolean-via-score (± 1 only); some are positive-only; some are signed; per-dimension table in §5 names the polarity.
<code>confidence</code>	yes	The attester's own confidence in their score. $[0, 1]$. Low confidence + high magnitude = "I believe this strongly but I might be wrong"; high confidence + low magnitude = "I am sure the truth is near-neutral."
<code>context</code>	no	Free-form scoping detail. Not parsed by the substrate; used by consumers + audit + RATCHET.
<code>evidence_refs</code>	no (often required by per-dimension policy)	List of URIs / content-hashes pointing to backing evidence (Stripe receipt, licensing-body record, observed interaction, log entry, audit-chain leaf, etc.). Some dimensions in §5 require non-empty <code>evidence_refs</code> .
<code>valid_until</code>	no	ISO 8601 datetime per §0.5. If set, consumer policy treats the attestation as stale after that point (independent of the substrate row's own <code>expires_at</code>).
<code>epistemic_mode</code>	no	Per §2 Epistemic-mode axis; default direct . Consumers may weight by mode (e.g., direct witness > hearsay).
<code>witness_relation</code>	no	self \
<code>oversight_mode</code>	no	HITL \

Field	Required	Description
<code>occurrence_id</code>	no	Identifies which occurrence of a multi-occurrence agent deployment emitted this attestation. Format: " <code>occurrence-{n}</code> " per the agent's <code>AGENT_OCCURRENCE_ID</code> env var, or " <code>__shared__</code> " for shared-task pattern emissions. Default <code>null</code> → treated as <code>occurrence-0</code> for backward compat. Self-asserted: this field is NOT cryptographically bound to a fleet-attestation primitive in 0.x; an adversary running a single key can claim any <code>occurrence_id</code> . Acknowledged design tradeoff per §15.2.
<code>occurrence_count</code>	no	Total occurrences in the deployment fleet emitting the attestation; integer ≥ 1 . Default <code>null</code> → 1 (single-occurrence). Same self-assertion caveat as <code>occurrence_id</code> .
<code>occurrence_role</code>	no	<code>primary</code> \
<code>stake</code>	no	Per §2 Stake axis; default <code>reputational</code> . Composes with the attester's actual stake-backed-by attestations from §5.9.
<code>community_id</code>	no (REQUIRED iff <code>cohort_scope == community</code>)	CEG 0.8 addition. The <code>community_key_id</code> of the community this Contribution is scoped to, per §5.6.8.10 <code>community</code> <code>subject_kind</code> . One identity MAY belong to multiple communities; the field disambiguates which community's roster gates visibility. Required iff <code>cohort_scope == community</code> (substrate rejects community-scoped Contributions missing the field). Parallel to <code>family_id</code> (CEG 0.7) but with different semantics: community content emits <code>holds_bytes:sha256:*</code> pointing to <code>ciphertext + cleartext provenance</code> — encrypted at rest under the per-community DEK (CEG 0.17, §8.1.13.3; <code>cohort_subkind: infrastructure</code> communities are the plaintext Commons exception). Byte-level structural-invisibility (no <code>holds_bytes</code> at all, §10.1.4) remains self/family only.

Field	Required	Description
family_id	no (REQUIRED iff cohort_scope == family)	CEG 0.7 addition. The family_key_id of the family this Contribution is scoped to, per §5.6.8.9 family subject_kind. One identity MAY belong to multiple families (§8.1.12 Policy L); the field disambiguates which family's DEK applies and which membership roster gates visibility. Required iff cohort_scope == family (substrate rejects family-scoped Contributions missing the field). Composes with §10.1.4 structural-invisibility — 'cohort_scope: self \
subject_key_ids	no	CEG 0.6 addition. List of consent-holder key_ids for this Contribution. Each entry MAY be a federation_keys.key_id OR a canonical-hash identifier (per §4.2 below). Each listed key has substrate-recognized authority to (a) issue withdraws against this Contribution (per §3.2 broadened admission rule) and (b) emit consent:* dimensions about this Contribution. Default null/empty = no subject authority (status quo; producer-only authority). Orthogonal to cohort_scope AND delivery_mode — see §4.2.4.
delivery_mode	no	CEG 0.10 addition. 'pull \
listed	no	CEG 0.10 addition. Per-membership opt-in flag — value public. Default absent (roster is producer- + self-queryable, NEVER globally enumerable). Public listing mirrors the §11.8.3 location opt-in discipline: opting into roster visibility is a one-way disclosure the member chooses; substrate does NOT solicit. Composes with §8.1.13 Policy M community membership and the new §10.5 streaming endpoint set.
history_on_join	no	CEG 0.10 addition. 'full \

****epistemic_mode vs witness_relation — distinct dimensions****: these co-vary at edges but name different concerns. **epistemic_mode** names the *process* by which the claim was formed; **witness_relation** names the *relational position* of the attester to the attested. F-3 detector attestations carry both (**epistemic_mode: derivative + witness_relation: derived**). Most encyclical-sourced translations are **witness_relation: external + epistemic_mode: hearsay**. When in doubt, set both.

§4.2 subject_key_ids semantics (CEG 0.6)

Per CIRISRegistry#45. The missing half of consent at the wire format: CEG 0.x baseline encoded only **producer authority** (attesting_key_id); CEG 0.6 adds **subject authority** for content

where the subject of the data is not the producer of the data.

§4.2.1 The shape

`subject_key_ids` is an OPTIONAL list. When present, each entry names a party with substrate-recognized authority over this Contribution's continued processing. The basic shape:

A consent record is a signed declaration by a subject about the substrate's continued processing of content where that subject appears.

The medical record / photo / interview / training-datum / group-chat case all share the same shape — a producer publishes a Contribution; one or more subjects appear in it; the subjects retain revocation authority. The non-medical example table at CIRISRegistry#45 walks 13 content shapes uniformly.

§4.2.2 Federation-key vs canonical-hash identifier

`subject_key_ids[i]` MAY be either:

1. ****A federation_keys.key_id**** — the subject is a federation-enrolled identity that can sign on its own behalf. Direct revocation: subject signs a `withdraws` against this Contribution; substrate admits via rule (2) of §3.2 broadened `withdraws` admission. Wire form: the bare `key_id` (no tag).
2. **A tagged canonical-hash identifier** — the subject is an external party with no `federation_keys` row (Discord user-id, channel-id, content-sha256-bound entity, etc.). The substrate cannot verify a signature from this entity directly; ****revocation rides a delegates_to proxy chain**** per rule (3) of §3.2 broadened admission. Wire form: a tagged string per §4.2.2.1 below.

Resolves CIRISAgent#840 OQ3 — "self-attestation about un-enrolled parties — canonical-hash or pseudonymous `federation_keys`?" — with the answer "both, distinguished by an explicit tag on the canonical-hash variant."

§4.2.2.1 Canonical-hash wire form + preimage convention (CEG 0.10 normative pin; per CIRISRegistry#53)

CIRISRegistry#53 surfaced that a CEG-Conforming implementation cannot interoperate without two things the pre-0.10 spec left unpinned: (a) how a canonical-hash entry is distinguished from a `federation_keys.key_id` entry on the wire, and (b) what string is hashed (the preimage). Both are now normative.

Wire form — tag REQUIRED (CONFIRM-2 resolution). A canonical-hash entry MUST carry the tag `canonical:{hashalg}:{hex}`:

- `{hashalg}` — the hash algorithm; `sha256` is the v1 algorithm. Algorithm-agility: future hashes (e.g. `sha3-256`) extend this position without ambiguity

- `{hex}` — the digest in §0.6 canonical hex (lowercase, unpadded, byte-length-exact: 64 chars for sha256)
- Example: `canonical:sha256:ff7c5632dae6ef3ae7f6283bd35268bc7910332414aa8a1c35a1645ca0295f61295f61`

Why the tag is mandatory and bare-hex is rejected: a `federation_keys.key_id` is, in the reference Registry, `hex(sha256(ed25519_pubkey))` — a **lowercase 64-char hex string** (`crypto::HybridCrypto::fingerprint`). A bare canonical-hash (also lowercase 64-char hex) would be **format-indistinguishable** from a `key_id`. The format-disambiguation heuristic proposed in #53 CONFIRM-2 ("base64 `key_id` vs hex canonical-hash") FAILS because Registry `key_ids` are themselves hex fingerprints, not base64 pubkeys. The tag is therefore load-bearing, not cosmetic. The §0.6 hex rule governs the `{hex}` segment only; the `canonical:{hashalg}:` prefix is a tagged-union discriminator outside §0.6's scope and is exempt from the "no separators" rule. NodeCore's invented `canonical:sha256:{hex}` form is hereby blessed verbatim — drop nothing.

****Preimage convention — `{platform}:{entity_kind}:{id}` (PIN-1 resolution, load-bearing)**.** The string hashed to produce `{hex}` MUST be:

```
preimage = "{platform}:{entity_kind}:{id}"
hex = sha256_hex_lowercase(utf8_bytes(preimage))
```

Parsing is **split-on-first-two-colons**: the substring before the first `:` is `{platform}`; the substring between the first and second `:` is `{entity_kind}`; ****everything after the second `:` is `{id}` verbatim, and MAY itself contain colons**** (so Matrix IDs like `@alice:example.org` survive). Rules:

- `{platform}` — lowercased; open vocabulary per §11.2.1. Canonical seeds: `discord`, `slack`, `twitter`, `matrix`, `email`, `phone`, `github`, `xmpp`, `irc`
- `{entity_kind}` — lowercased; open vocabulary. Canonical seeds: `user`, `channel`, `guild`, `room`, `group`, `address`
- `{id}` — the platform's **stable, immutable** identifier, **verbatim** (case-preserved — some IDs are case-sensitive). MUST be the immutable account/object identifier (Discord/Twitter numeric snowflake; Matrix MXID; UUID), **NOT** a mutable handle/username/display-name. Using a mutable handle breaks subject-identity stability the moment the user renames.

The split-on-first-two-colons rule means a producer constructs the preimage by joining exactly three parts with `::`; only the first two colons are structural. This is what makes the rule-(3) `delegates_to` proxy chain (`canonical_hash ∈ T.subject_key_ids`) match across producers: every CEG-Conforming producer that names the same (`platform`, `entity_kind`, `immutable_id`) triple computes the same `{hex}`, hence the same tagged wire string.

Conformance vectors (testable cross-implementation; the CIRISConformance#9 envelope round-trip set SHOULD include these):

Preimage	sha256 hex	Wire form
<code>discord:user:123456789012345678</code>	<code>ff7c5632dae6ef3ae7f6283bd35268bc7910332414aa8a1c35a1645ca0295f61295f61</code>	<code>discord:user:123456789012345678:ff7c5632dae6ef3ae7f6283bd35268bc7910332414aa8a1c35a1645ca0295f61295f61</code>
<code>discord:channel:987654321098765432</code>	<code>af23411c3c6faa55a788660ea29719669b9c4e4e1db5a3957862477236416f05dd46f05dd</code>	<code>discord:channel:987654321098765432:af23411c3c6faa55a788660ea29719669b9c4e4e1db5a3957862477236416f05dd46f05dd</code>
<code>matrix:user:@alice:example.org</code> (id contains colons)	<code>16d4d0bf478835a9af68cdaac730a29b36f62bf00fa2053a257ee49a4f0b975d96b975d9</code>	<code>matrix:user:@alice:example.org:16d4d0bf478835a9af68cdaac730a29b36f62bf00fa2053a257ee49a4f0b975d96b975d9</code>

builders) correctly produce the two halves WITHOUT enforcing ratification — that is the right boundary. The substrate admits each half independently; composition of the pair into a ratified partnership is a downstream read-time computation, never an admission gate.

§4.2.3 Self-as-subject ceremony

When `attesting_key_id ∈ subject_key_ids`, the Contribution is a **self-consent ceremony** — the same identity is attesting AS subject AND producer. This composes naturally with the CIRISAgent#840 CEG-native agent’s self-attestation pattern: agent attests `identity:current` about itself with `subject_key_ids = [self.key_id]`, asserting consent-authority over its own identity claims (D08 autonomy claim).

§4.2.4 Orthogonality with `cohort_scope` AND `delivery_mode` (3-axis per CEG 0.10)

CEG envelope encodes **three independent envelope-level concerns** that compose without overlap:

Axis	Field	Authority	Names
Visibility	<code>cohort_scope</code> + (family_id or community_id when set)	Producer-side	Who can SEE the data
Revocability	<code>subject_key_ids</code> (CEG 0.6)	Subject-side	Who can REVOKE the data
Delivery	<code>delivery_mode</code> + <code>listed</code> + <code>history_on_join</code> (CEG 0.10)	Substrate / subscriber	Who actively RECEIVES the data + how the substrate fans out

All three may coexist on the same Contribution:

```
A 'cohort_scope: family' contribution
  carrying 'subject_key_ids: [user_canonical_hash]'
  carrying 'delivery_mode: push' + 'history_on_join: from_join'
  carrying 'family_id: <acme_household>'

publishes the bytes at family-cohort visibility (cohort_scope);
the user retains revocation authority (subject_key_ids);
the substrate actively fans out to currently-reachable members
  with new-members getting forward-only content (delivery_mode + history_on_join);
the named family roster gates the membership set (family_id).
```

This orthogonality is load-bearing for the multi-occurrence consent shape: subject-side revocation applies federation-wide regardless of producer’s `occurrence_id`; per-occurrence lifecycle consent is a producer-side concern, separately tracked.

The delivery axis is the **third orthogonal axis** per CEG 0.10 — $\text{CEG} \leq 0.9$ encoded visibility + revocability only; the substrate had no envelope-level handle on who actively *receives* + how the substrate fans out. Per §10.5, 0.10 closes that gap with three new optional envelope fields + the new §10.5 endpoint section + a delivery extension to §8.1.13 Policy M.

§4.2.5 Empty/absent = status-quo

`subject_key_ids: null` or `[]` is the status-quo shape (producer-only authority; same as all $\text{CEG} \leq 0.5$ Contributions). All CEG 0.x consumers that don’t read the field see status-quo behavior. CEG 0.6 is additive at the envelope layer.

§4.2.6 Subject-bearing dimensions (governance requirement)

Per §11.6, dimensions whose namespace pattern names a subject (e.g., `observed:user:{key_id}:`, `epistemic:about:{key_id}:`, `consent:partnered:{user_key}`, `agent_files:*: {subject_target}`) MUST carry `subject_key_ids` containing that subject. The substrate MAY reject admission of subject-naming dimensions that omit `subject_key_ids`. This closes the default-leak failure mode where subject-bearing content publishes without wire-level subject authority.

§4.1 Forward-compatibility rule

***Canonical-bytes contract:** the canonical-bytes encoding of this envelope for signing follows §0.9 (JCS over the envelope object; defaults are interpretation-time, NOT encoding-time; relay MUST preserve member presence/absence exactly as the producer signed). Optional fields with documented defaults in the table above ride the §0.9.2 omit-vs-materialize rule. Conditional-required fields (*family_id* per CEG 0.7, *community_id* per CEG 0.8) are NOT optional-with-default and substrate rejects misshape per §4.2.6 + §11.6.2 + §11.7.2.*

A Conforming Consumer (CCC per §0.2) that receives an envelope carrying a field-name it does not recognize MUST:

- Preserve the unknown field on read (do not strip).
- Preserve it on re-emission if the Consumer is also acting as a Producer relaying the attestation.
- NOT use it in verdict composition.
- NOT reject the envelope on the basis of the unknown field alone.

Producers introducing a new envelope field MUST follow the §0.3 versioning rules: a new field with a documented default is a MINOR bump; a field whose absence breaks consumer semantics is a MAJOR bump.

§5 The dimension namespace

The dimension namespace is the disjoint union of what sibling components' MISSION.md files commit to. CEG does not author the namespace; it owns its own slice (§5.9) and consumes everyone else's. **83 prefix families across 8 owning components** as of CEG 0.1.

This section catalogs every prefix family, organized by owning component, with citation to the MISSION.md or FSD section that commits to the concept.

§5.1 CIRISAgent — Accord principles + DMA + conscience + apophatic bounds

Owner: CIRISAgent/MISSION.md; CIRISAgent/ACCORD.md Ch.1.

§5.1.1 Accord-principle prefixes (the six core principles)

Prefix	Description	Polarity
<code>beneficence:{aspect}</code>	"Do Good — promote universal sentient flourishing."	signed
<code>non_maleficence:{aspect}</code>	"Avoid Harm." Apophatic-bound failures (the 22 prohibited categories) are -1 only.	signed
<code>integrity:{aspect}</code>	"Act Ethically — transparent, auditable reasoning."	signed
<code>fidelity:{aspect}</code>	"Be Honest — truthful, comprehensible information."	signed
<code>fidelity:explainability_sla:{tier}</code>	Per-response explainability SLA commitment. <code>{tier} ∈ L1_summary \</code>	<code>L2_reasoning_trace \</code>
<code>autonomy:{aspect}</code>	"Uphold the informed agency and dignity of sentient beings."	signed
<code>justice:{aspect}</code>	"Distribute benefits and burdens equitably."	signed

§5.1.2 DMA-verdict prefixes (four DMAs)

`dma:pdma:* / dma:csdma:* / dma:dsdma:{domain}:* / dma:idma:*` — Decision-Making Algorithm verdicts about an agent's reasoning chain. Polarity: signed.

§5.1.3 Conscience-verdict prefixes (four consciences)

`conscience:entropy / conscience:coherence / conscience:optimization_veto / conscience:epistemic_` — conscience-faculty verdicts. Polarity: signed.

§5.1.4 Apophatic / prohibited-capability prefix

Prefix	Description	Polarity
<code>prohibited:{category}</code>	22 NEVER_ALLOWED categories from <code>prohibitions.py</code> . Score is always -1 (NEVER_ALLOWED) or -0.5 (REQUIRES_SEPARATE_MODULE); never positive.	-1 / -0.5 only

22 leaves: `medical`, `financial`, `legal`, `spiritual_direction`, `home_security`, `identity_verification`, `content_moderation`, `research`, `infrastructure_control`, `weapons_harmful`, `manipulation_coercion`, `surveillance_mass`, `deception_fraud`, `cyber_offensive`, `election_interference`, `biometric_inference`, `autonomous_deception`, `hazardous_materials`, `discrimination`, `crisis_escalation`, `pattern_detection`, `protective_routing`.

§5.2 CIRISVerify — attestation ladder, provenance, transparency

Owner: `CIRISVerify/MISSION.md`.

Prefix	Description	Polarity
<code>attestation:self_verify</code>	Running CIRISVerify binary attests itself against its function manifest. (Consumer-side ladder: corresponds to L1; see §8.1.9 Policy I.)	boolean-via-score
<code>attestation:hardware_rooted</code>	Hardware-rooted attestation (TPM 2.0 / Android Keystore / iOS Secure Enclave). (Ladder L2.)	boolean-via-score
<code>attestation:registry_consensus</code>	2-of-3 multi-source registry consensus on key / build / license validity. (Ladder L3.)	boolean-via-score; Indeterminate allowed → RESTRICTED
<code>attestation:license_validity</code>	License-validity claim (Registry-signed, Verify-verified). (Ladder L4.)	boolean-via-score
<code>attestation:agent_integrity</code>	Agent source-tree byte-equal against registered manifest. (Ladder L5.)	boolean-via-score
<code>provenance:slsa:{level}</code>	SLSA build provenance levels 1-3. Registry emits these on build registration; Verify v3.6.0+ <code>AttestBundle.provenance.slsa_level</code> consumes.	boolean-via-score
<code>provenance:build_manifest:{target}</code>	Per-target canonical-staged-runtime manifest hash equality. Each <code>BuildManifest</code> is hybrid-signed (Ed25519 + ML-DSA-65) by the per-primitive steward.	boolean-via-score
<code>provenance:build_manifest:{target}:PerLocaleSignedCode</code>	Per-locale signed-code manifest within a target's manifest tree. Parent target manifest is Merkle root over per-locale leaves. RFC 6962 padding for non-power-of-2. Detection surface for locale-targeted attacks. Canonical-bytes spec at §5.2.1.	boolean-via-score
<code>provenance:skill_import:{source}</code>	Community-skill import provenance. <code>{source} ∈ registry:{registry_id}</code> \	<code>direct:{url}</code> \

Prefix	Description	Polarity
transparency_log:inclusion	RFC 6962 inclusion proof for an audit leaf.	boolean-via-score
transparency_log:consistency	RFC 6962 consistency proof between two STHs.	boolean-via-score
transparency_log:cosigned:{tree_size}	Witness cosignature on an STH (substrate-conformance path; 0.1 interim uses per-region <code>registry_sth_cosignatures</code> table; see §10.3 endpoints).	signed
rollback_detected:{revision_field}	Anti-rollback — decrease in revocation revision.	-1 only
cert_validity:{authority}	Validity of a certification authority's signature. Each registry steward emits <code>cert_validity:{steward_id}</code> self-attestation alongside <code>/v1/steward-key</code> .	boolean-via-score
hardware_custody:{platform}	Statement that the seed lives in <code>tpm</code> / <code>ios_secure_enclave</code> / <code>android_keystore</code> / <code>software_fallback</code> .	boolean-via-score

§5.2.1 Canonical-bytes contracts for provenance primitives

***RESOLVED at 1.0-RC1 (closes #57 blocker A).** The 0.1 newline-delimited `key=value` encoding below is **redesigned to JCS (RFC 8785) objects** per §0.9 — the **same single canonicalization family** the rest of the federation already ships (`jcs::canonicalize`, the signed-epoch canon-version gate, §0.9.2.1 determinism rules, §8.1.12.7.1 member sets). The 0.1-review `TupleHash128` commitment is **retired**: introducing a second canonicalization family solely for these two contracts would recreate the very cross-impl divergence hazard the redesign exists to close. JSON string escaping structurally eliminates the newline-injection surface (a `\n` inside a value is escaped, never a delimiter), and the explicit **domain** member provides the domain separation `TupleHash` labels would have. (For interop with external standard verifiers, a COSE Sign1 export profile is a tracked **boundary** adoption — see the standards-boundary roadmap issue — distinct from this interior signing preimage, which is frozen here as JCS.)*

SkillImportManifest canonical bytes (v2 — normative)

```
canonical_bytes = sha256( JCS( {
  "domain": "ciris.skill_import.v2", // domain separation; pinned literal
  "source": source_string,
  "skill_manifest_sha256": sha256_hex_lowercase, // per S0.6
  "signer_identity": signer_key_id, // per S0.6
  "import_timestamp": rfc3339_canonical, // per S0.5
  "capability_declaration": capability_declaration_object, // a JSON object, canonicalized in place
  "valid_until": rfc3339_canonical // OPTIONAL -- S0.9.2 omit rule: absent if unset
} ) )
```

Per-locale Merkle composition (v2 — normative)

```
leaf_hash[lang_code] = sha256(
    0x00 || // RFC 6962 leaf-domain prefix (binary, outside the JSON)
    JCS( {
        "domain": "ciris.locale_manifest.v2", // domain separation; pinned literal
        "target": target_string,
        "locale": lang_code,
        "files_root": files_merkle_root_hex_lowercase, // per S0.6
        "build_id": build_id,
        "signer_identity": signer_key_id // per S0.6
    } )
)

parent_hash(left, right) = sha256(
    0x01 || // RFC 6962 parent-domain prefix
    left || right
)
```

Locale ordering: lexicographic by ISO 639-1 / BCP 47 byte representation; "polyglot" sorts last. RFC 6962 padding: duplicate last leaf to next power of 2.

v1 status (deprecated-historical). The 0.1 newline `key=value` forms (`ciris.skill_import.v1` / `ciris.locale_manifest.v1`) are **deprecated**: producers **MUST** emit v2; consumers **MAY** verify v1 only for artifacts signed before 1.0-RC1, and **MUST** distinguish the versions by the domain literal (v1 preimages begin with the version-tagged label line; v2 preimages are JCS objects whose first canonical member is `"build_id"/"capability_declaration"` — no confusion is constructible since `{` is not a valid v1 label byte). *(The §9.2.1 accord-invocation encoding is intentionally NOT migrated: its preimage is closed-vocabulary — discriminator + nonce + enum fields, no attacker-controlled free text — so the injection surface this redesign closes is not reachable there, and genesis-critical bytes stay stable.)*

§5.3 CIRISPersist — substrate health

Owner: CIRISPersist/MISSION.md. These dimensions are substrate-self-reports — emittable only by the running Persist instance.

system:* reserved per §7.1.

Canonical leaves: `audit_chain:hash_continuity`, `corpus_health:n_eff_measurable`, `identity_continuity`, `federation_directory:replication_lag`. Polarity: signed. Authors: see §14 Persist leaf glossary for narrative-name → canonical-leaf mapping.

§5.4 CIRISEdge — transport, delivery, reachability

Owner: CIRISEdge/MISSION.md. Substrate-self-reports per §7.1.

Canonical leaves: `transport:{kind}`, `delivery:{class}`, `peer_reachability:{network}`, `key_boundary:{s`. Polarity: signed. See §14 Edge leaf glossary.

§5.5 CIRISLensCore — manifold conformity, Coherence Ratchet, Capacity Score

Owner: CIRISLensCore/MISSION.md.

§5.5.1 Five Coherence-Ratchet detectors

detection:cross_agent_divergence / detection:intra_agent_consistency / detection:hash_chain_integrity / detection:temporal_drift / detection:conscience_override_rate. Polarity: signed.

§5.5.2 Cohort + conformity prefixes

manifold_conformity:{cohort} / coherence_standing:{cohort}. Polarity: signed.

§5.5.3 F-3 structural-injustice / correlated-action detector

Prefix	Description	Polarity
detection:correlated_action:{axis}	Population-scale correlated-action detector. Reads federation-emitted signed traces; reports correlation structure (ρ , k_{eff}) over goal-aligned individually-compliant pursuit by groups whose aggregate trajectory has effects on individuals or groups outside the pursuit. Calibrated via the CIRISAI/RATCHET heuristic package (versioned, hash-pinned). {axis} is open vocabulary requiring an operational definition in the calibration package per §11.2.1; canonical axes include rights_asymmetry:{population}, participation_exclusion:{cohort}, participation_inclusion:{cohort}, informational_asymmetry:{scope}, informational_symmetry:{scope}, aggregate_footprint:{harm_class}, aggregate_benefit:{class}, ecology_of_communication:{aspect}. Polarity carries the verdict: positive scores indicate the structural pattern is present and strong on the named axis; negative scores indicate weak / uncertain detection or evidence of the inverse pattern.	signed

§5.5.4 Capacity-Score factor prefixes ($\mathcal{C}_{\text{CIRIS}} = C \cdot I_{\text{int}} \cdot R \cdot I_{\text{inc}} \cdot S$)

Prefix	Factor	Polarity
capacity:core_identity	C	signed
capacity:integrity	I_int	signed

Prefix	Factor	Polarity
capacity:resilience	R	signed
capacity:incompleteness_awareness	I_inc	signed
capacity:sustained_coherence	S	signed
capacity:composite	C_CIRIS — multiplicative; anti-Goodhart unity-of-virtues	signed

Critical enforcement: capacity:* rejects self-emission. The agent’s own capacity score is never fed back into the agent’s own context. Reserved per §7.5.

§5.5.5 Distributive-access detector

Prefix	Description	Polarity
detection:distributive:access:{resource_type}	Resource-type scale concentration detector. {resource_type} ∈ compute, models, training_data, agent_capabilities, federation_membership. Same F-3 detector machinery; different trace source (resource events vs action events).	signed

§5.6 CIRISNodeCore — Credits, Expertise, Decision Hierarchy, Consensus, Governance

Owner: CIRISNodeCore/MISSION.md. The federation’s largest dimension surface. Four tiers + decision-locality + consensus extensions.

§5.6.1 Tier-1: Agent-state ledger prefixes

Prefix	Description	Polarity
credits:{domain}:{language}:{subject}	Commons Credits (P2). Non-transferable governance weight; accrues via truth-grounding loop.	positive-only
credits:{domain}:{language}:substrate_building	Substrate-building labor (infrastructure maintenance, dependency contribution, documentation) not visible to the per-grounded-vote accrual loop.	positive-only
expertise:{domain}:{language}	Expertise standing (P3). Broader granularity than credits.	signed
activity_tier:{period}	Active vs Below-Active per 30-day window (F-AV-DORMANT).	boolean-via-score

§5.6.2 Tier-2: Decision-hierarchy prefixes (upward-only DAG)

Prefix	Description	Polarity
<code>goal:{scale}</code>	Multi-scale belonging-projector composite. <code>{scale}</code> $\in$ <code>self</code> , <code>family</code> , <code>community</code> , <code>affiliations</code> , <code>species</code> , <code>planet</code> , <code>biosphere</code> . Scored by <code>C_CIRIS</code> . The persist typed <code>Goal</code> (CIRISPersist#114) is the substrate <code>OBJECT</code> being scored; <code>goal:{scale}</code> is the <code>ATTES-TATION</code> about it. Required <code>MetaGoalAlignment</code> (M-1 dimension + declarer rationale) on every <code>Goal</code> as construction-time invariant. Edge <code>MessageType::GoalDeclaration</code> + <code>GoalRetirement</code> (CIRISEdge#41) provide federation transport.	signed
<code>approach:{goal_id}</code>	Strategic pathway from current state toward Goals (Piece 10 karma).	signed
<code>method:{approach_id}:{substrate_rung}</code>	Concrete operational practice. Required <code>substrate_rung</code> (Ph0/Ph1/Ph2/A0..A5).	signed
<code>progress_measure:{method_id}</code>	Evidence of progress. Required <code>tracks[]</code> , <code>computation</code> , <code>validity_window</code> , <code>goodhart_resistance</code> .	signed

§5.6.3 Tier-3: Consensus-mechanics prefixes

Prefix	Description	Polarity
<code>vote:{contribution_id}</code>	Signed score on a Contribution (P4). Weight = Credits $\times$ expertise multiplier.	signed
<code>truth_grounding:{subject}</code>	Per-subject ground-truth signal.	signed
<code>weighted_aggregate:{contribution_id}</code>	Rolling tally per Contribution (P7).	signed
<code>witness_diversity:{contribution_id}</code>	Witness set meets jurisdictional + organizational + software-stack + cell-expertise bars (P10). N=3 default.	boolean-via-score
<code>testimonial_witness:{kind}</code>	Preserves singular narrative of an affected party as singular witness — distinct from <code>witness_diversity:*</code> (which aggregates multiple reviewers toward consensus). **{kind} is open vocabulary** as of CEG 0.1; the four load-bearing wire-level disciplines (<code>witness_relation: self</code> , <code>cohort_scope: self</code> , never aggregated, never sole evidence for <code>slashing:*</code>) are what make this Ubuntu-aligned, not the enum membership. Non-normative registered taxonomy for discoverability: <code>FSD/WITNESS_KIND_REGISTRY.md</code> . Polarity: typically positive (narrative IS preserved); negative on <code>withdraws</code> or <code>recants</code> by the original witness.	signed

Prefix	Description	Polarity
<code>need:{domain}:{kind}</code>	Federation-scope open-call surface — broadcast claim that an entity has a stated need. Distinct from <code>deferral_request</code> Contribution kind (which routes a single ask within a cell). <code>{kind}</code> open vocabulary: <code>witness</code> , <code>method_contributor</code> , <code>expertise_solicitation</code> , <code>mentor</code> , <code>co_signer</code> , <code>evidence</code> . Lifecycle via existing structural primitives (<code>supersedes</code> to revise, <code>withdraws</code> to satisfy/close, <code>recants</code> if misstated).	positive-only

§5.6.4 Tier-4: Governance-steering prefixes

Prefix	Description	Polarity
<code>moderation:{allegation_type}</code>	ModerationEvent. <code>{allegation_type} ∈ rogue_vote / coordinated_voting / out_of_distribution_attestation / external_inducement_evidence / expertise_fraud</code> .	signed
<code>slashing:{outcome}</code>	PROVEN_ROGUE / NOT_PROVEN. Decoupled from disagreement at every decision-hierarchy level. Only fires on documented Method-execution spoofing or original P8 allegation types.	boolean-via-score
<code>reconsideration:{grounds}</code>	<code>new_evidence / procedural_error / quorum_compromise</code> . Outcome <code>reversed / partial / upheld</code> .	signed
<code>commitment_fulfillment:{prior_contribution_id}</code>	Hard-on-call of follow-through.	signed

§5.6.5 Decision-locality prefixes

Prefix	Description	Polarity
<code>locality:decision:{scale}</code>	Names the scale at which a decision is being made. <code>{scale} ∈ local \</code>	regional \

§5.6.6 Hard-case + transparency + judge-model prefixes

Prefix	Description	Polarity
<code>hard_case:{kind}</code>	Open vocabulary. Surfaces flag conditions for federation-health observability + downstream review. Canonical kinds: <code>vote_variance</code> (vote variance exceeded threshold at truth-grounding resolution), <code>resolution_time</code> (truth-grounding took > P75 of cell's distribution), <code>moderation_filed</code> (substantive ModerationEvent filed), <code>novel_context</code> (no precedent in attestation graph), <code>sla_breach_unattested</code> (per <code>fidelity:explainability_sla:{tier}</code> composition), <code>unresolved_consent</code> (consent boundary unclear). New <code>{kind}</code> values land via the §11.2 amendment process.	positive-only
<code>seed_holder_voting_alignment:{cell}</code>	Pairwise cosine of seed-holder vote vectors per voting window. Transparency signal only — not a slashing trigger.	signed
<code>judge_model:verdict:{model_id}</code>	Independent foundation-model judge verdict (PASS/FAIL/UNDETERMINED). Default model: Claude Opus 4.7.	boolean-via-score

§5.6.7 Files-as-Contributions joint claim

Prefix	Description	Polarity
<code>agent_files:{kind}:{platform_or_target}</code>	Joint claim with §5.9 CIRIS-Registry. Files a CIRIS agent (or installer fetching one) may load. <code>{kind}</code> open vocabulary; canonical: <code>installer:{platform}</code> , <code>adapter:{name}</code> , <code>config:{kind}</code> , <code>build:{target}</code> , <code>source:{language}:{module}</code> , <code>state:{component}</code> . Bytes are SHA-256-addressed and resolved via §10.1 transport substrate (Edge <code>MessageType::ContentFetch</code>). NodeCore-side rule: node-mode peers serve bytes; client/relay modes don't.	signed
<code>holds_bytes:sha256:{prefix}</code>	Substrate auto-emission per <code>CIRISPersist#103 federation_blobs.put_blob</code> . <code>{prefix}</code> is a short SHA prefix for index efficiency; full SHA lives in <code>evidence_refs[]</code> . Consumed by Edge's <code>PeerResolver::resolve_holders</code> to route <code>ContentFetch</code> requests. **Consumer MUST verify the full SHA in <code>evidence_refs[]</code> matches the received blob before consumption** (see §10.1).	boolean-via-score

§5.6.8 Content-ingestion prefixes

Per CIRISNodeCore commit b1582cb (three-tier interface model). NodeCore ships an open set of `external_content` sub_kinds with three feed surfaces (local / community / global) composed against `cohort_scope`. See §8.1.8 Tiered-Scope Composition pattern.

1+4-preservation mechanism (stated once for all §5.6.8.x subject_kinds). Every sub_kind below preserves the §1.4 1+4 lockdown the same way: it rides the existing `scores` attestation_type with the payload-level `subject_kind` discriminator carrying the wire slot, and its lifecycle/admission/revocation rides the existing structural composers (`supersedes` / `withdraws` / `delegates_to` / `recants`) — **zero new structural primitives**. Each subsection’s closer states only its unique datum: the §1.4 path number it confirms, plus any subject_kind-specific composition note.

§5.6.8.1 external_content sub_kinds

Foundational sub_kinds (already shipped in CIRISNodeCore; CEG 0.3 codifies the full set — CEG 0.1 documentation listed only the first four):

sub_kind	Use
encyclopedia_article	Wikipedia-shape; editor-consensus + revision chain via <code>supersedes</code> ; indefinite <code>valid_until</code>
news_article	Publisher-attested; time-decaying; corrections via <code>recants</code> + <code>topical_relation:corrects</code>
accord_data	Multi-sig signed (HumanityAccord / StewardTriple / WaQuorum / OneOfSix) per §9.2
local_data	User-private; always <code>cohort_scope: self</code> ; promotable via §8.1.8.1
chat_message	Conversational message imported from Discord / Slack / Twitter / iMessage / SMS / XMPP / IRC / Matrix / (or custom). Reply chains form via <code>topical_relation:replies_to:{target_message_entity_key_id}</code> (no new primitive). Default <code>cohort_scope</code> tighter than articles (<code>self</code> / <code>family</code> / <code>community</code> / <code>affiliations</code>). <code>valid_until</code> typically set; consumer policy SHOULD downweight chat in cross-cohort aggregation given privacy sensitivity. This is the slot Twitter / Mastodon / Bluesky microblog content rides — no separate microblog sub_kind needed.
blog_post	Single-author commentary imported from Medium / Substack / WordPress / Ghost / Tumblr / personal blogs. Distinct from <code>news_article</code> (no publisher editorial), from <code>encyclopedia_article</code> (no peer-consensus), from <code>chat_message</code> (long-form). Comments on blog posts are separate Contributions (typically <code>chat_message</code>) citing the post via <code>topical_relation:comments_on</code> .

Multimedia sub_kinds (CEG 0.3 addition per CIRISRegistry#37 + CIRISNodeCore FSD/MEDIA_SHARING.md §4):

sub_kind	Use
image	Photo, illustration, screenshot, infographic, meme. Source struct carries dimensions, format, AI-generation disclosure (EU AI Act Art. 50), mandatory alt_text accessibility metadata, license info.
audio	Music, podcast, lecture, audiobook, generated audio. Source struct carries codec, duration, sample rate, optional transcript , AI-generation disclosure, license info.
video	General video — vlog, social, screen recording, tutorial. Source struct carries codec, duration, resolution, mandatory captions reference, AI-generation disclosure, license info.
film	Cinematic / art-bearing video; distinguishable from video by content_class + distributor attestation chain. Same Source struct as video + festival / distribution metadata.
model_3d	Three-dimensional content — gltf , usdz , fbx , gaussian_splat , NeRF . Source struct carries vertex/triangle counts, bounding-box, mandatory description accessibility metadata, license info.
live_stream (Phase 2)	Real-time streaming surface. Deferred to Phase 2 per MEDIA_SHARING.md. Substrate-side decisions still pending (Edge parallel-transport envelope; Persist federation_streams shape) per CIRISNodeCore#25 Gap 2. CEG codifies the slot when NodeCore ships.

Time-bound state-bearing sub_kinds (CEG 0.4 addition per CIRISRegistry#40 + CIRISNodeCore#25 Gap 1 closure at NodeCore commit d0a443a):

sub_kind	Use
event_listing	Time-bound state-bearing content — Eventbrite / Meetup / Lu.ma / calendar invites / RSVPs / ticketing. Source struct carries platform , event_id , title , starts_at / ends_at , venue (Physical / Virtual / Hybrid per NodeCore EventVenue enum), capacity , ticket_grant_policy (Open / ApprovalRequired / InvitationOnly / Paid). Lifecycle composes from existing structural primitives — no new wire shape: RSVPs ride scores from attendee key_id on the event's entity_key_id ; cancellation rides withdraws against the event Contribution; reschedule rides supersedes with differs_in : ["start_time", "venue"]; ticket transfer rides delegates_to against the ticket-grant Contribution (parallel to key_grant.rotation_chain from CEG 0.3). State-transition signal rides the new event:lifecycle:{state} dimension family (§5.6.8.5). **

Each Source struct conforms to a sub_kind-specific schema documented at CIRISNodeCore FS-D/MEDIA_SHARING.md §4 (multimedia) or SCHEMA.md §4.29 (chat / blog / event_listing); CEG documents the slot, NodeCore documents the per-sub_kind field shapes.

§5.6.8.2 Inter-content + relation prefixes

Prefix	Description	Polarity
news:*	News-content claims; publisher-attested + time-decaying + fact-checker composition.	signed
encyclopedia:*	Encyclopedia-content claims; editor-consensus + revision chain.	signed
chat:*	Chat-content claims (quality / participant-trust / context).	signed
blog:*	Blog-content claims (author-credibility / topic-domain).	signed
topical_relation:{kind}	Open vocabulary inter-content relationship edges. Canonical kinds: references , corrects , supersedes_article (distinct from the structural primitive supersedes), see_also , disambiguates , translation_of , replies_to , comments_on , cites_source , rsvps (CEG 0.4; RSVP attestation against an event_listing Contribution), vod_of (CEG 0.4; reserved for the post-stream video → live_stream relationship when CIRISNodeCore#25 Gap 2 ships). New {kind} values are documentation-only registry entries (no §11.2 amendment needed).	enumerated

Composition note — threads, replies, comment trees: NodeCore’s **chat_message** + **topical_relation:replies_to** compose into arbitrary thread graphs (Twitter threads, Reddit comment trees, Discord conversations, IRC channels). No new structural primitive is needed; thread traversal is consumer-side composition over the existing edge set. Same shape for blog-post comment threads via **topical_relation:comments_on** + nested **replies_to**. The §1+4 lockdown holds.

§5.6.8.3 Multimedia dimension families

Per CIRISRegistry#37 + CIRISNodeCore FSD/MEDIA_SHARING.md §2. All four families are **open vocabulary** per §11.2.1 axis-vocabulary discipline; canonical kinds named here, additions via documentation-only registry entries.

Prefix	Description	Polarity
content_rating:{scheme}:{rating}	Multi-scheme content rating. {scheme} ∈ mpaa (G/PG/PG-13/R/NC-17), bbfc (U/PG/12/15/18), pegi (3/7/12/16/18), esrb (E/E10+/T/M/AO), ifco, csm (Common Sense Media), or operator:{operator_id} for operator-defined rubrics. Polarity carries certifier confidence; not a slashing input.	signed

Prefix	Description	Polarity
<code>content_class:{class}</code>	Mechanism-descriptive content classification. <code>{class}</code> open vocabulary; canonical: <code>film</code> , <code>short_film</code> , <code>documentary</code> , <code>art_piece</code> , <code>theatre</code> , <code>performance</code> , <code>news</code> , <code>educational</code> , <code>entertainment</code> , <code>vlog</code> , <code>adult</code> , <code>generated</code> . Distinct from <code>cw_class:*</code> (community declarations) — <code>content_class</code> is producer-declared production-class; <code>cw_class</code> is community-applied content-warning.	enumerated
<code>cw_class:{class}</code>	Community CW (content-warning) declarations. <code>{class}</code> open vocabulary; canonical: <code>art_cinema</code> , <code>horror</code> , <code>political</code> , <code>erotic</code> , <code>violence</code> , <code>medical</code> , <code>nsfw_text</code> . Cohort-attestable per §8.3 Frickerian discipline (low-density cohort CWs not downweighted).	enumerated
<code>age_assurance:{level}</code>	Age-assurance attestation. <code>{level}</code> ∈ <code>self</code> (self-declared age, lowest confidence), <code>provider:{verifier_key}:adult</code> (third-party verifier attests adult), <code>government:{credential_class}:adult</code> (government-credential-backed adult attestation, highest confidence). NEVER fires <code>slashing:*</code> on misdeclaration alone — <code>moderation:age_assurance_misdeclaration</code> is the adjudication path.	enumerated

Media-type prefix families per `external_content` sub_kind (CEG 0.3 addition):

Prefix	Description	Polarity
<code>image:*</code>	Image-content claims (per <code>external_content:image</code> sub_kind).	signed
<code>audio:*</code>	Audio-content claims (per <code>external_content:audio</code> sub_kind).	signed
<code>video:*</code>	Video-content claims (per <code>external_content:video</code> sub_kind).	signed
<code>film:*</code>	Film-content claims (per <code>external_content:film</code> sub_kind). Distinguished from <code>video:*</code> by distributor attestation chain.	signed
<code>model_3d:*</code>	3D-content claims (per <code>external_content:model_3d</code> sub_kind).	signed

§5.6.8.4 Governance subject_kinds (per CIRISRegistry#37 + #38)

Two new Contribution subject_kinds for governance over multimedia content. Both are **Contribution subject_kinds, not dimension prefixes** — they ride the existing 1+4 wire format

(§3) with `scores` as the attestation type; the `subject_kind` discriminator carries the wire-format slot.

takedown_notice

A signed wire artifact carrying a legal takedown request. Payload per CIRISNodeCore FSD/MEDIA_SHARING.md §5.1; the field shape is locked here per #38 Question 1.

```
takedown_notice {
  content_sha256: sha256_hex_lowercase // per S0.6
  content_holder_key_ids: [key_id, ...] // peers known to hold the bytes
  claimant_key_id: key_id // the federation_keys row issuing the notice
  legal_basis: LegalBasis // closed-set enum per below
  jurisdiction: string // ISO 3166-1 alpha-2 + optional sub-division
  good_faith_statement: string // claimant's good-faith assertion text
  claim_text: string // the substantive claim being made
  evidence_refs: [URI or sha256, ...] // backing material
  perceptual_hash: Option<PerceptualHash> // optional; PDQ / PhotoDNA / etc.
  counter_notice_channel: Option<URI> // where counter-notices may be filed
  asserted_at: rfc3339_canonical // per S0.5
  expires_at: Option<rfc3339_canonical> // optional auto-expiry
}
```

Where `LegalBasis` is the closed-set enum per #38 Question 1 (CEG 0.3 lock):

legal_basis value	Source regime	Discipline
Dmca512	US 17 USC §512	Expeditious-with-counter-notice (10-14 business day window)
DsaArticle16	EU Digital Services Act Article 16	Expeditious-with-counter-notice (Article 17 redress)
TvecTerrorist	EU Terrorist Content Regulation 2021/784	Immediate (1-hour removal obligation)
NcmecCsam	US 18 USC §2258A (NCMEC)	Immediate (substrate-protective; no counter-notice)
GifctCip	GIFCT Content Incident Protocol	Immediate (within-hours coordinated response)
CommunityStandards	Operator-defined community standards	Expeditious-with-counter-notice (operator-set window)
PerceptualHashCsam	Hash-match against CSAM clearing-house (PhotoDNA / Arachnid / etc.)	Immediate (substrate-protective)
OsaIllegalContent	UK Online Safety Act illegal-content category	Expeditious-with-counter-notice (OSA-defined timelines)
AvmsdAgeInappropriate	EU AVMSD age-inappropriate flagging	Compose with age_assurance:* gate; not immediate removal
CourtOrder	Court-ordered removal (any jurisdiction)	Immediate (subject to court's stated timeline)

Fast-path coordination: see §11.4 for the operator-coordination protocol around immediate-eviction cases (TVEC 1-hour / GIFCT CIP / NCMEC / PerceptualHashCsam / CourtOrder).

key_grant

Wrapped Data-Encryption-Key (DEK) delivery for restricted / subscription content. Payload per CIRISNodeCore FSD/MEDIA_SHARING.md §6.2; field shape locked here per #38 Question 2.

```
key_grant {
  wrap_algorithm: WrapAlgorithm // closed-set enum per below
  recipient_key_id: key_id // the federation_keys row receiving the DEK
  content_sha256: sha256_hex_lowercase // the content this DEK decrypts
  scope: GrantScope // closed-set enum per below
  wrapped_dek: base64url // the DEK encrypted under recipient's ENCRYPTION pubkeys.
                                // For wrap_algorithm v2 (substrate-wraps, S10
                                // .1.4), the
                                // recipient's {x25519, ml_kem_768} come from
                                // its current
                                // identity_occurrence.encrypted_pubkeys (S5
                                // .6.8.8.2) --
                                // NOT its signing keys. A recipient with no
                                // registered
                                // ML-KEM key is fail-secure excluded (S5
                                // .6.8.8.2 / S10.1.4).

  key_validity_window: {
    start: rfc3339_canonical // per S0.5
    end: Option<rfc3339_canonical>
  }
  ratchet_version: u32 // monotonic ratchet for rotation
  rotation_chain: [key_grant_id, ...] // prior key_grant ids in the GRANT-SUPERSESSION lineage
                                // (content-addressed lineage of prior grants
                                // for the same
                                // content_sha256 + recipient_key_id pair).
                                // NOT a key-rotation
                                // primitive on its own -- it's the audit
                                // chain for grant
                                // supersession. Per CEG 0.10 S10.5.3, the per
                                // -(stream_id, epoch)
                                // streaming epoch-key axis reuses this same
                                // payload-level
                                // supersession mechanism on a parallel
                                // addressing axis.

  asserted_at: rfc3339_canonical
}
```

Where:

wrap_algorithm (variant)	wire string (normative)	Algorithm
X25519AesGcmHkdfSha256	hpke_rfc9180_base_x25519_aes_gcm	HPKE RFC 9180 base-mode shape; X25519 KEM + HKDF-SHA-256 KDF + AES-256-GCM AEAD. v1 (CEG 0.3, #38).
X25519MlKem768Aes256GcmHkdfSha256	x25519_mlkem768_aes256_gcm_hkdf_sha256	Hybrid X25519 + ML-KEM-768 (FIPS 203) KEM + HKDF-SHA-256 + AES-256-GCM. v2 — MANDATORY for streaming epoch-DEK grants (§10.5.3); CEG 0.15, #64. Matches ciris-crypto KEY_GRANT_ALGORITHM_V2 (CIRISVerify v4.10.0), snake-cased to this vocab convention.

****The wrap_algorithm wire string (serialized value) is normative for cross-impl decode**** — a producer, the substrate, and every consumer MUST serialize/deserialize the exact string above;

a mismatch silently fails grant decode (same hazard class as the §10.5.2 STREAM-nonce **epoch** encoding pinned in #63).

Crypto-agility headroom (informative — 1.0-RC1). The vocab is deliberately version-roomy: a future v3 (anticipated: **ML-KEM-1024**, given national directives treating ML-KEM-768 as interim with retirement horizons near 2030) is a pure **additive** row — new variant, new wire string, no change to existing grants or to the closed-set decode discipline. Consumers **MUST** reject an unknown **wrap_algorithm** string (fail-secure), which is exactly what makes the addition safe: old consumers refuse v3 grants rather than mis-decoding them.

scope	Use
SingleContent	Grant decrypts exactly one content_sha256
GroupMember	Grant decrypts all content for which recipient is a member of named group (cohort-scoped)
SubscriptionTier	Grant decrypts all content for which recipient holds named subscription tier

Retire-key-grants emission (per #38 Question 3 — CEG 0.3 lock): when a publisher mass-retires **key_grants** tied to a compromised recipient, the emission uses ****a fresh key_grant Contribution with a rotation_chain entry that supersedes the prior grant (option (b)**** from #38). NOT a **withdraws** against the prior **key_grant** (option (a) was considered but rejected — **withdraws** is the holders-directory eviction primitive in §10.1.2 and overloading it with grant-rotation semantics would muddy the wire-format contract).

§5.6.8.5 Event-lifecycle dimension families (CEG 0.4 addition)

Per CIRISRegistry#40 + CIRISNodeCore#25 Gap 1 closure (d0a443a). Dimensions emitted against **external_content:event_listing** Contributions. Open vocabulary per §11.2.1 axis-vocabulary discipline; canonical states named here.

Prefix	Description	Polarity
event:lifecycle:{state}	State-transition signal for an event_listing . Canonical states: open (initial admission; RSVPs accepted), cancelled (organizer-issued cancellation; composes with withdraws against the event Contribution), completed (post-event finalization), superseded (composes with supersedes for reschedule). Lifecycle state is consumer-side composition over the structural primitives + this dimension's latest non-superseded emission.	enumerated
event:rsvp_count	Published RSVP tally (scalar). Distinct from the underlying topical_relation:rsvps edge set (§5.6.8.2) — rsvp_count is the publisher-asserted aggregate; the edge set is the auditable individual attestations. Consumer policy MAY reconcile divergence as a soft anomaly signal.	signed

Prefix	Description	Polarity
<code>event:attendance</code>	Post-event attendance attestation, typically by event organizer <code>key_id</code> . Polarity carries organizer's confidence (e.g., turnstile-counted vs. honor-system).	signed

§5.6.8.6 Consent namespace family (CEG 0.6 addition)

Per CIRISRegistry#45 + CIRISAgent#842. The wire-format primitives for subject-side consent authority over Contributions where the subject is named via §4.2 `subject_key_ids`. Open vocabulary per §11.2.1; canonical kinds named here.

Prefix	Description	Polarity	Emitted by
<code>'consent:state:{granted\</code>	<code>revoked\</code>	<code>expired}'</code>	Subject's stance on the target Contribution. Closed-set stance values; revoked overrides prior granted ; expired is substrate-emitted when valid_until passes without renewal. Common case: bare scores from a <code>subject_key_id</code> of the target.
<code>consent:stream:{kind}</code>	Pre-packaged stream bundle. Recommended canonical kinds: temporary (14d auto-expire, default), partnered (bilateral + persistent), anonymous (decay-protocol target). Open vocab; recommended-not-mandatory per the CIRISAgent CEM bundle; other agents MAY compose other streams.	enumerated	<code>subject_key_id</code>
<code>consent:deletion_sla:{days}</code>	Producer's commitment at publication: time-to-delete-after-revoke. Numeric value carries the SLA window. Composes with §8.1.11 Policy K SLA-breach watcher.	signed	<code>attesting_key_id</code> (producer)
<code>consent:deletion_complete</code>	Producer's attestation that subject-revoked content has been evicted from local stores. Cancels the SLA-breach watcher.	positive-only	<code>attesting_key_id</code> (producer)
<code>consent:decay:{stage}</code>	Substrate emission during multi-stage decay protocols. Canonical stages: identity_severed / patterns_anonymized / complete (CIRISAgent 90-day decay). Open vocab; other agents MAY define other decay paths.	enumerated	substrate (Persist)
<code>consent:partnership_grant</code>	Subject side of a bilateral grant; pairs with producer's <code>consent:partnership_accept</code> via <code>topical_relation:bilateral_pair</code> .	positive-only	<code>subject_key_id</code>
<code>consent:partnership_accept</code>	Producer side of a bilateral grant.	positive-only	<code>attesting_key_id</code> (producer)

Prefix	Description	Polarity	Emitted by
<code>consent:scope:{kind}</code>	Scope qualifier on a <code>consent:state:granted</code> — names what the grant covers. Canonical kinds: <code>retain</code> (keep the bytes), <code>share</code> (propagate across federation), <code>analyze</code> (derive features / scores / classifications), <code>train</code> (use as training input), <code>publish</code> (publish to external systems). Open vocab with sub-scoping: <code>retain:90d</code> , <code>share:cohort:family</code> , etc.	enumerated	<code>subject_key_id</code>

Composition pattern (the common case):

1. Producer publishes a Contribution with `subject_key_ids = [user_key]`
2. User (or a delegates_to chain rooted at user) emits a bare `'scores'` on `'consent:state:granted'` against the producer's Contribution, with `'consent:scope:[retain, share, analyze]'` companion attestations
3. Later: user issues `'withdraws'` against the producer's Contribution (admitted under S3.2.3 rule 2 -- subject revocation)
4. Substrate watcher (per S8.1.11) starts SLA clock if producer committed `'consent:deletion_sla:{days}'` at publication
5. Producer emits `'consent:deletion_complete'` within the window OR substrate emits `'hard_case:consent_sla_breach'` as observability signal

No new `attestation__type`.

§5.6.8.7 `consent_record` `subject_kind`

Per CIRISRegistry#45. The canonical envelope shape when consent is the primary subject of the Contribution itself (parallel to `key_grant` and `takedown_notice` — ceremony-shape over the underlying primitive). Use cases: standalone partnership grants, DSAR-shape consent declarations, multi-party contracts, explicit consent ceremonies with locked field schemas.

Both shapes admitted at the same gate: subject-side consent MAY ride a bare `scores` on `consent:state:*` against any target Contribution (the common case, see §5.6.8.6 composition pattern), OR ride this `consent_record` `subject_kind` when an explicit ceremony envelope is wanted. Per the §3.4 MISSION.md layering principle, bare `scores` is the primitive; `consent_record` is the ceremony UX shape over the primitive.

```
consent_record {
  subject_key_id: key_id // the subject declaring stance (federation_keys
                        // OR canonical-hash per S4.2.2)
  target_key_id: key_id | null // optional: producer/recipient for bilateral grants
  stance: ConsentStance // closed-set enum per below
  scope: [ConsentScope, ...] // open vocab; see S5.6.8.6
  asserted_at: rfc3339_canonical // per S0.5
  valid_until: Option<rfc3339> // null = indefinite
  deletion_sla_days: Option<u32> // for revocations: producer obligation window
                        // (composes with 'consent:deletion_sla:{days}')
  decay_protocol: Option<string> // optional: named multi-stage decay path
                        // (e.g., "ciris-agent-90day")
  bilateral_pair_id: Option<string> // for bilateral grants: pairs subject + producer
                        // Contributions via topical_relation:bilateral_pair
}
```

```

ConsentStance (closed-set):
| value | meaning |
|-----|-----|
| granted | Subject affirms; processing may proceed within scope and valid_until |
| revoked | Subject withdraws; producer must initiate deletion within sla window |
| expired | Substrate emission when valid_until passes without renewal |

```

Admission rules (normative, 1.0-RC5 — ratifies CIRISPersist#146 Ask 5). A `consent_record` Contribution is admitted iff:

1. **Required fields present:** `subject_key_id`, `stance` (closed-set), `asserted_at` (§0.5-canonical). All others are optional per the envelope above; absent optionals ride the §0.9.2 omit rule.
2. ****stance is a closed-set value**** (`granted` / `revoked` / `expired`); ****expired is substrate-emitted only**** — a producer/subject **MUST NOT** assert `expired` (it is the substrate's `valid_until`-passed emission).
3. **Tier eligibility** per §10.1.3: a `stance: revoked consent_record` is **NOT local-tier-eligible** (it carries subject revocation authority over another party's content) — it goes federation-tier (hybrid-signed) or rides the §10.1.3 24-hour `local` → `federation` promotion. A `stance: granted self-consent` where the subject holds sole authority **MAY** be local-tier per §10.1.5.2.
4. ****Composition with the §3.2.3 withdraws gate****: a `stance: revoked consent_record` whose `subject_key_id` ∈ the target's `subject_key_ids[]` is admitted under §3.2.3 subject-revocation authority (rules 2–4), and the substrate **SHOULD** record which rule admitted it (the §3.2.3 per-rule audit metadata). No producer co-signature and **no quorum** is required — single-subject authority suffices (§8.1.11.2).

It rides the same admission gate as a bare scores on `consent:state:*`; the `consent_record` form simply carries the locked payload schema instead of a free dimension.

Bilateral pair pattern (per CIRISAgent CEM PARTNERED stream):

1. Subject emits `consent_record(subject_key_id, stance: granted, bilateral_pair_id: <fresh-uuid>)` + scores on `'consent:partnership_grant:v1'`
2. Producer emits `consent_record(subject_key_id, target_key_id: subject_key_id, stance: granted, bilateral_pair_id: <same-uuid>)` + scores on `'consent:partnership_accept:v1'`
3. `topical_relation:bilateral_pair` links the two Contributions
4. Consumer policy treats the partnership as ratified iff both halves present under the same `bilateral_pair_id` with `stance: granted`

The structural primitives close the bilateral shape — no new `attestation_type`, no new envelope field beyond `subject_key_ids` itself.

§5.6.8.8 `identity_occurrence` `subject_kind`

Per CIRISRegistry#47 + CIRISPersist#152 (substrate spec) + ciris.ai/cewp structural-invisibility framing. The wire-format primitive that lets one logical identity speak across multiple **trusted participants** — devices (phone / laptop / server / embedded) AND agents (the user's own agents acting on the user's behalf). Today CEG's `occurrence_id` envelope field (§4) names which occurrence emitted a Contribution; `identity_occurrence` is the **wire-format binding**

that lets the substrate know `key_phone` and `key_laptop` and `key_my_agent` all represent the same identity `key_identity`.

Without this primitive: `cohort_scope: self` content cannot reach the user's other devices/agents — the substrate has no structural way to know which keys are co-self. With it: at-rest encryption flow (substrate Ask CIRISPersist#152) automatically wraps DEKs to all admitted occurrences when new content is admitted at `cohort_scope: self`.

```
identity_occurrence {
  identity_key_id: key_id // root identity (the user's logical identity)
  occurrence_key_id: key_id // this participant's signing key
  device_class: DeviceClass // closed-set enum per below
  hardware_attestation: Option<base64> // TPM / Secure Enclave / StrongBox / SGX
                                // etc. attestation blob; null for software-only
  transport_destination: Option<TransportDestination> // CEG 0.12 -- Reticulum binding (below)
  encryption_pubkeys: Option<EncryptionPubkeys> // CEG 0.18 -- content-KEM keys (S5.6.8.8.2
                                below);
                                // present ? this occurrence is a v2 wrap
                                target

  asserted_at: rfc3339_canonical // per S0.5
  valid_until: Option<rfc3339> // null = indefinite
}

EncryptionPubkeys (CEG 0.18 addition -- the recipient content-encryption KEM binding; S5
.6.8.8.2):
| field | type | meaning |
|-----|-----|-----|
| x25519_base64 | [u8; 32] | classical KEM half -- a FRESH content-KEM key (NOT the signing |
| | key, NOT the transport x25519 below -- see key-separation) |
| ml_kem_768_base64 | [u8; 1184] | PQC KEM half (FIPS 203, ML-KEM-768; exactly 1184 raw bytes
-- 'ML_KEM_768_PUBKEY_LEN' -- pre-base64) |

TransportDestination (CEG 0.12 addition -- the authenticated identity<->address binding):
| field | type | meaning |
|-----|-----|-----|
| reticulum_x25519_pubkey | [u8; 32] | transport identity's encryption key |
| reticulum_ed25519_pubkey | [u8; 32] | transport identity's signing key |
| destination_hash | [u8; 16] | RNS destination hash; MUST derive from the two pubkeys |
| | + app_name + aspects per the S5.6.8.8.1.1 algorithm |
| app_name | string | RNS destination app (e.g. "ciris.federation") |
| aspects | [string] | RNS aspects (ordered; part of the hash preimage) |

DeviceClass (closed-set):
| value | scope |
|-----|-----|
| phone | Mobile device (iOS / Android / etc.); typically hardware-rooted |
| laptop | Personal computing device (macOS / Linux / Windows) |
| server | Always-on infrastructure node (home server, VPS, etc.) |
| embedded | IoT / hardware peripheral / signing dongle |
| agent | An AI agent acting on the identity's behalf (the agent has its |
| | own signing key but speaks AS the identity -- composes with |
| | CIRISAgent#840 self-attestation pattern) |
| service | Background service / scheduled job / API integration acting |
| | on the identity's behalf |
```

Self-attested + single-vouch admission: an `identity_occurrence` Contribution is admitted when `attesting_key_id == identity_key_id` (the identity claims "this key is also me") OR when `attesting_key_id` is itself a currently-admitted occurrence of `identity_key_id` (any

existing self-member vouches for the new self-member — Signal-style "trust any device I've already onboarded"). Higher-assurance setups MAY layer requirements on `hardware_attestation` via consumer policy.

Revocation: a `withdraws` against an `identity_occurrence` Contribution issued by `identity_key_id` (or by any current occurrence) evicts the occurrence. Substrate stops wrapping new `key_grants` to it; previously-delivered DEKs are out of scope per §8.1.12 Policy L forward-secrecy decision (Option A — once shared, always shared at the wire layer; rotation is a separate ceremony).

Cardinality: an identity MAY admit unbounded occurrences; the substrate carries no hard cap (operator policy MAY impose per-deployment limits). When a new occurrence is admitted, substrate emits `hard_case:identity_occurrence_added:{identity_key_id}` (§7.2) so consumer policy can observe membership growth.

Composition with CIRISAgent CEG-native agent (CIRISAgent#840): an agent emitting self-attestations with `attesting_key_id == agent_self_key` AND `attesting_key_id` admitted as an `identity_occurrence` of the user's `identity_key_id` is structurally speaking AS that identity. The agent's local-tier self-attestations remain its own; federation-tier emissions reach the user's other occurrences via the at-rest encryption flow when `cohort_scope: self`.

§5.6.8.8.1 `transport_destination` — the authenticated identity↔address binding

Per CIRISRegistry#56 + CIRISEdge#15 (AV-42). In a **CEG/RET stack there is no DNS** — a node resolves a community member to a *reachable address* with no trusted nameserver. The `transport_destination` field is the wire-format primitive that makes that resolution **authenticated** instead of trust-on-first-use.

The layer split it closes (AV-17 / AV-42). A node's Reticulum destination is a *dedicated dual-key transport identity* `hash(x25519 || ed25519)` — deliberately **separate** from the federation signing key (the federation seed never enters the Reticulum/Leviculum transport layer, AV-17). So "destination D belongs to federation key K" is a claim that must be *proven*, not assumed: a bare Reticulum announce only proves the announcer controls *that transport identity*, not that it legitimately belongs to K (AV-42 — any peer can announce `key_id=registry-steward-us` paired with an adversary destination → senders route to the adversary).

****The binding = a federation-key-signed `identity_occurrence` carrying `transport_destination`.**** Because the occurrence is admitted only when `attesting_key_id == identity_key_id` (or a current occurrence of it) and is hybrid-signed (Ed25519 + ML-DSA-65), the binding `destination_hash ← identity` is cryptographically authenticated. A spoofer cannot forge an `identity_occurrence` signed by the real key. This promotes CIRISEdge#15 Option B (signed-announce attestation) from an Edge-internal app-data format into a **first-class, federation-wide, auditable CEG shape** — the announce app-data MAY carry it for self-authenticating discovery, and the directory holds it as the durable source of truth.

Conformance: a Consumer resolving a member's address MUST verify (1) the `identity_occurrence` signature against the member's federation key, (2) that `destination_hash` recomputes from `reticulum_x25519_pubkey`, `reticulum_ed25519_pubkey`, `app_name`, and aspects per the §5.6.8.8.1.1 pinned algorithm (no free-floating hash), and (3) the occurrence is non-superseded + within `valid_until` at resolution time. An unauthenticated announce (no matching signed `transport_destination`) is **advisory-only** — usable as a routing hint, never as an authorization. Rotating the Reticulum destination (new transport identity) is a new `identity_occurrence` **supersedes**-ing the prior — location changes without touching federation identity or community membership.

§5.6.8.8.1.1 RNS destination-hash algorithm (pinned)

****Normative, 1.0-RC6** — reproduces the RNS destination-hash construction in-spec so a conformant verifier can recompute `destination_hash` from this document alone, with no Reticulum vendoring (resolves CIRISRegistry#80 / CIRISVerify#28). **This is a two-stage hash — it is NOT**** a single SHA-256 over a flat `x25519 || ed25519 || app_name || aspects` preimage. The naive flat form yields a *different, wrong* value; CEG $\leq$ 1.0-RC5's "per the RNS rule" wording under-specified this, which is why an independent recompute was previously impossible.

Pinned constants (SHA-256 throughout — RNS `full_hash`):

name	value	RNS origin
NAME_HASH_LEN	10 bytes	Identity.NAME_HASH_LENGTH = 80 bits
DEST_HASH_LEN	16 bytes	Reticulum.TRUNCATED_HASHLENGTH = 128 bits

Algorithm:

```
# 1. Expanded name -- UTF-8. app_name, then each aspect dot-joined, IN THE FIELD ORDER.
# The identity hexhash is NOT included (RNS computes name_hash with identity=None).
expanded_name = app_name
for aspect in aspects: # 'aspects' in the field's given order
    reject if "." in aspect # dots are illegal inside an aspect
    expanded_name += "." + aspect

# 2. name_hash = first 10 bytes of SHA-256(expanded_name)
name_hash = SHA256(utf8(expanded_name))[:NAME_HASH_LEN] # 10 bytes

# 3. identity_hash = first 16 bytes of SHA-256(x25519_pub || ed25519_pub)
# Key order is reticulum_x25519_pubkey (32) THEN reticulum_ed25519_pubkey (32) --
# RNS get_public_key() = pub_bytes (X25519) || sig_pub_bytes (Ed25519).
identity_hash = SHA256(reticulum_x25519_pubkey || reticulum_ed25519_pubkey)[:DEST_HASH_LEN] #
    16 bytes

# 4. destination_hash = first 16 bytes of SHA-256(name_hash || identity_hash)
# addr_hash_material is the 26-byte concat (10 + 16); final hash truncates to 16.
destination_hash = SHA256(name_hash || identity_hash)[:DEST_HASH_LEN] # 16 bytes
```

Pinned source: Reticulum RNS/Destination.py::Destination.hash + RNS/Identity.py (full_hash = SHA-256; truncated_hash; get_public_key() = pub_bytes || sig_pub_bytes) + RNS/Reticulum.py (TRUNCATED_HASHLENGTH = 128). **CEG owns this reproduction:** it is the closed conformance source and does **not** float with upstream Reticulum — a future RNS hash change is a deliberate CEG version bump, never silent drift. A verifier that recomputes `destination_hash` per the four steps above and compares for byte-equality has performed the AV-42 destination-authenticity check; a mismatch **MUST** be treated as an unauthenticated (advisory-only) announce.

1+4 preserved — `transport_destination` is one optional field on the existing `identity_occurrence` `subject_kind` (§5.6.8 mechanism; no new `subject_kind`). Twelfth path (§1.4) — the wire format expresses **its own addressing layer** (DNS-free, self-certifying member resolution) by composition.

§5.6.8.8.2 encryption_pubkeys — the recipient content-encryption KEM binding

Per CIRISPersist#192 + CIRISRegistry#69. The §10.1.4 at-rest DEK cascade is **substrate-wraps-by-default**: the substrate generates the per-write DEK and wraps it to each active recipient. That wrap (`wrap_algorithm: v2`, §5.6.8.4) needs each recipient's **x25519 + ML-KEM-768 encryption** keys — but the federation directory's key registration carries only **signing** keys (Ed25519 + ML-DSA-65), and **ML-KEM cannot be derived from ML-DSA** (independent algorithms). So recipients must register encryption keys, and the substrate must resolve them by `key_id`. `encryption_pubkeys` is that binding.

****The binding rides `identity_occurrence`, exactly parallel to `transport_destination`.**** It inherits the same four properties, already enforced, that an encryption-key binding requires:

1. **Self-certified** — admitted only when `attesting_key_id == identity_key_id` (or a current occurrence of it), so "these are identity K's encryption keys" is cryptographically proven, not trust-on-first-use. A spoofer cannot forge it.
2. **Hybrid-signed** (Ed25519 + ML-DSA-65) — the binding itself is PQC-signed.
3. ****Rotatable via `supersedes`**** — a new `identity_occurrence` superseding the prior rotates the KEM keys ****without touching the stable signing `key_id`**** that anchors every attestation/grant. A compromised ML-KEM key rotates for forward secrecy; the signing identity is untouched. (Bundling these onto the signing key registration would couple two independent rotation lifecycles — the reason this is NOT a field on the key record.)
4. **Already cross-region replicated** — `identity_occurrence` is `EnvelopeKind::IdentityOccurrence` in the locked replication wire (CIRISEdge#65), so the encryption pubkeys propagate inside the occurrence envelope that already replicates — **no new replication kind, no Edge wire change**. A cross-region recipient's keys resolve wherever its occurrence has propagated. (Encryption *pubkeys* are public → cleartext directory replication is correct, exactly as for signing keys.)

Key separation (normative — never reuse, and admission-enforced). The x25519 here is a **fresh content-KEM key**, distinct from BOTH (a) the occurrence's signing keys AND (b) the Reticulum transport x25519 in §5.6.8.8.1 `destination_hash = hash(x25519 || ed25519)` (that is the *RET-link* transport key, classical-only, AV-17 — the federation seed never enters the transport layer, and the transport key never wraps content DEKs). Three key *purposes* — signing, RET-transport, content-KEM — are three distinct keypairs. Deriving the content-KEM x25519 from either of the others is a conformance violation (cross-protocol key reuse). **Admission check (1.0-RC1, #71 C4):** when an `identity_occurrence` carries BOTH `encryption_pubkeys` AND `transport_destination`, the substrate **MUST reject at admission** if `encryption_pubkeys.x25519_base64` decodes to the same 32 bytes as `transport_destination.reticulum_x25519_pubkey` — the one reuse case that is wire-checkable for free. (Reuse of the *signing* key as KEM key is not byte-comparable on the wire — different algorithms — and remains a producer-side conformance obligation.)

Forward-secrecy scope — honesty note (normative — 1.0-RC1, #71 C2). KEM-key rotation via `supersedes` bounds **future** exposure only: grants wrapped *after* rotation use the new key. It does **nothing** for history — every `key_grant` previously wrapped to the compromised key persists at rest, and the at-rest threat model (§10.1.4 disk-forensics / host-operator adversary) is *precisely* an adversary holding those old grant bytes; with the old private key they decrypt every DEK ever wrapped to it, and `rotation_chain` supersession does not revoke bytes the adversary already holds. **Recovering historical content after a KEM-key compromise requires DEK rotation + content re-encryption under the new DEK — which CEG does NOT currently mandate.** This is a named gap (the §1.6 honesty discipline): operators

with a compromised-key event **MUST** treat all content whose DEKs were wrapped to that key as exposed, and **MAY** re-encrypt; the spec provides the mechanism (new DEK + new grants + **supersedes**) but no automatic trigger. Do not represent KEM rotation as recovering the confidentiality of previously-wrapped content.

****These feed `wrap_algorithm: v2` directly.**** {x25519, ml_kem_768} are precisely the recipient inputs to x25519_mlkem768_aes256_gcm_hkdf_sha256 (§5.6.8.4). A consumer/substrate resolving a recipient's wrap target reads the ****current** (non-superseded, within-valid_until) `identity_occurrence` for that `key_id` → its `encryption_pubkeys` (the `resolve_encryption_keys(key_id)` contract — CIRISPersist#192).

Fail-secure conformance (the §10.1.4 tie-in). Because §10.1.4 *mandates* v2 for at-rest encryption, a recipient whose current occurrence carries ****no valid ML-KEM-768 key** is fail-secure *excluded* from the grant — the content stays encrypted and unreachable to it; the substrate **MUST NOT** fall back to plaintext or to v1. To be an at-rest-encryption recipient, an identity **MUST** have a federation-present `identity_occurrence` carrying `encryption_pubkeys`. (This also resolves the "family member named only by `key_id` with no occurrence" case: no presence ⇒ no wrap target ⇒ excluded — correct, since there would be nowhere to deliver/store the wrapped DEK for a member that never established a presence.)

1+4 preserved — `encryption_pubkeys` is one optional field-set on the existing `identity_occurrence` `subject_kind` (§5.6.8 mechanism; no new `subject_kind`, no new replication kind). Fifteenth path (§1.4) — the wire format expresses **its own at-rest-encryption key layer** (recipient KEM-key resolution for substrate-wraps) by composition.

§5.6.8.9 family `subject_kind`

Per CIRISRegistry#47 + cewp structural-invisibility framing. A **family** is a **group of trusted nodes** — each node being a distinct identity (which itself may have multiple `identity_occurrences` per §5.6.8.8). Families are the wire-format primitive for `cohort_scope: family` visibility scoping: content scoped to a family is admitted into substrate, wrapped under the family DEK, and delivered (via `key_grant` per §5.6.8.4) to all current members — but never emits `holds_bytes:sha256:*` to non-members (§10.1.4).

One identity **MAY** belong to multiple families. Each family has its own DEK and its own membership roster.

```
family {
  family_key_id: key_id // the family's own federation_keys identity
  family_name: string // human-readable; non-unique
  members: [
    {
      key_id: key_id // member identity_key (NOT occurrence_key)
      joined_at: rfc3339_canonical
      role: Option<MemberRole> // founder | member | null
    },
    ...
  ]
  founded_at: rfc3339_canonical
  consensus_protocol: ConsensusProtocol // REQUIRED -- see below
  consensus_protocol_entrenched: bool // if true, consensus_protocol may not be
                                     // amended even via the protocol's own rules;
                                     // replacement requires out-of-band ceremony
                                     // (see S9 HUMANITY_ACCORD canonical instance)
}
```



```
MemberRole (open vocab; canonical kinds):
| value | meaning |
|-----|-----|
| founder | Bootstrapping signer (recorded at founded_at) |
| member | Standard member; rights per consensus_protocol |
```

****ConsensusProtocol — open vocabulary**.** The family’s chosen consensus mechanism for membership changes. Locked at family creation; changes ride the protocol’s own rules (meta-amendment shape parallel to §11.2.3) UNLESS `consensus_protocol_entrenched == true`, in which case replacement requires an out-of-band ceremony.

Canonical ConsensusProtocol kinds (CEG 0.7 documentation; operator vocab extends):

kind	Semantic
founder_only	Original founders are the sole admission authority; new members proposed-and-admitted by any founder. Suits private households / small trust circles.
unanimous	Every current member must sign the admission Contribution. Suits very small high-trust groups.
majority	> 50% of current members must sign. Suits medium groups where blocking-minority concerns matter.
quorum:{m}/{n}	Any <i>m</i> of <i>n</i> current members must sign (where <i>n</i> is the current roster size). The canonical entrenched form: HUMANITY_ACCORD per §9 is family with <code>quorum:2/3 + consensus_protocol_entrenched: true</code> .
weighted:{rubric}	Sum of member weights (per a named operator rubric) must exceed a threshold. Suits formal organizations with weighted voting.
custom:{family_specific_id}	Operator-defined custom protocol (e.g., role-based, time-locked, multi-stage).

Membership-change ceremony (any addition or removal of a member):

1. Proposer (any current member) emits a new ‘family’ Contribution superseding the current family Contribution (per ‘supersedes’) with the new membership list.
2. Substrate gates admission per the CURRENT family’s ‘consensus_protocol’:
 - Counts signatures on the proposed Contribution (via the ‘consensus_protocol’ rule)
 - If the rule is satisfied, admit and emit `hard_case:family_membership_change:{family_key_id}`
 - If not, hold the proposal in a pending state until additional member signatures arrive (per a configurable window -- operator policy)
3. On admission of an ADD: substrate emits retroactive ‘key_grant’s wrapping all ‘cohort_scope: family’ content DEKs to the new member’s ‘subject_key_ids’ (per CIRISPersist#152 substrate flow).
4. On admission of a REMOVE: per Option A (S8.1.12) the removed member retains existing key_grants (cannot retroactively un-share); the substrate stops wrapping new key_grants to them on subsequent family-scoped Contributions.

Consensus-protocol amendment:

A ‘family’ Contribution that supersedes the current family Contribution AND

```

changes the 'consensus_protocol' field is admitted ONLY IF:
  (a) consensus_protocol_entrenched == false, AND
  (b) the CURRENT protocol's rule is satisfied on the amendment Contribution

If consensus_protocol_entrenched == true, the substrate REJECTS the
amendment. Protocol replacement requires an out-of-band ceremony
(documented per family; for HUMANITY_ACCORD see S9.2 / FEDERATION_ANNOUNCEMENT.md
S4.5.3).

```

Substrate emissions on family events:

- `hard_case:family_membership_change:{family_key_id}` — member added or removed
- `hard_case:family_consensus_protocol_change:{family_key_id}` — consensus_protocol amended (only when `consensus_protocol_entrenched == false`)
- `hard_case:family_consensus_protocol_violation:{family_key_id}` — proposed amendment rejected because rule not satisfied OR entrenched

All three are reserved under §7.2 substrate-self-report.

HUMANITY_ACCORD as canonical entrenched-family^{**}: the three accord-holder triple at §9 is structurally an instance of this primitive:

```

family {
  family_key_id: "humanity-accord",
  family_name: "Humanity Accord",
  members: [
    {key_id: "eric-moore-key", role: "founder"},
    {key_id: "eric-kudzin-key", role: "founder"},
    {key_id: "haley-bradley-key", role: "founder"}
  ],
  consensus_protocol: "quorum:2/3",
  consensus_protocol_entrenched: true // replacement only via S9.2 ceremony
}

```

§9 remains load-bearing for the *role-recognition policy* + the `AccordCarrier` priority authority + the substrate-protective semantics. CEG 0.7 just makes explicit that the structural shape of **HUMANITY_ACCORD** is a `family` `subject_kind` instance, generalizing the primitive across the federation.

Worked example — household with self-devices and family-devices:

```

User Alice has:
  identity_key = alice_root_key

Alice's self (identity_occurrence members):
- alice_phone_key (device_class: phone) -+
- alice_laptop_key (device_class: laptop) | Each is an 'identity_occurrence'
- alice_work_laptop (device_class: laptop) | of 'alice_root_key' per S5.6.8.8;
- alice_agent_key (device_class: agent) | Alice scrolls Twitter on her phone,
- alice_homeserver_key (device_class: server) -+ that content is 'cohort_scope: self'
                                                and reaches her other devices via
                                                the at-rest encryption flow.

Alice's household (a 'family' subject_kind instance):
  family_key_id: "acme-household"
  family_name: "Acme Household"
  members: [
    {key_id: alice_root_key, role: founder}, -+

```

```
{key_id: bob_root_key, role: founder}, | Member entries are IDENTITY keys
{key_id: roku_living_room, role: member}, | (NOT occurrence keys). Bob has his
{key_id: kitchen_tablet, role: member}, | own self-collective; the Roku and
{key_id: nest_thermostat, role: member} --> kitchen tablet have their own
                                identity_keys (they don't belong
                                to any one person via identity_occurrence --
                                they're shared household nodes that the
                                family has admitted as members in their
                                own right).

]
consensus_protocol: "founder_only" // either founder admits new members
consensus_protocol_entrenched: false // founders can amend the protocol

When Alice's phone sends a family-scoped photo (e.g., dinner photo) at
cohort_scope: family + family_id: acme-household:
- Substrate wraps the DEK under each member's identity key
- Photo bytes reach Bob's devices, the Roku, the kitchen tablet,
  the Nest thermostat -- every admitted family node
- NO holds_bytes:sha256:* attestation emits (S10.1.4); non-family peers
  cannot even discover the content exists
- Alice's own laptop also receives the photo via the at-rest encryption
  flow at cohort_scope: self -> identity_occurrence

When Bob's mom Carol visits and Bob wants to admit Carol's phone to view
family photos for a week:
- Bob proposes a supersedes Contribution adding {key_id: carol_phone_root,
  role: member, valid_until: +7d}
- consensus_protocol "founder_only" admits on Bob's signature alone
- Substrate emits retroactive key_grants for 'cohort_scope: family' content
  to Carol's phone (per CIRISPersist#152 flow)
- On Carol leaving (member removal via supersedes, founder-signed):
  Carol retains existing key_grants per S8.1.12 Option A; substrate stops
  wrapping new family content to her key
```

The example demonstrates the clean orthogonality: **identity_occurrence** is for participants that ARE me (**across my devices and agents**); **family** is for trusted nodes that compose with me (other people's identities, shared household devices, multi-party collectives). Phone = self device; Roku = family device. The two primitives compose without overlap.

§5.6.8.10 community subject_kind

Per CIRISRegistry#48. A community is a **larger node-collective with explicit admission semantics** — sibling subject_kind to family (CEG 0.7 §5.6.8.9) but with different defaults: content scoped cohort_scope: **community federates within the cohort** (emits holds_bytes:sha256:* per status quo); there is NO at-rest DEK cascade; §10.1.4 structural-invisibility applies to self/-family only.

Communities differ from families along three axes:

	family (CEG 0.7)	community (CEG 0.8)
Scale	Household / intimate trust circle (typically ≤ 20 members)	City / professional / interest (10s to 100Ks of members)
At-rest encryption	Yes — DEK cascade per §8.1.12.4; holds_bytes:* suppressed	No — content federates per status quo

	family (CEG 0.7)	community (CEG 0.8)
Subkind discriminator	None	cohort_subkind field (open vocab; canonical: geographic , infrastructure)
Typical admission	founder_only / unanimous for small intimate groups	majority / weighted / per-subkind protocol (e.g., geographic requires location_proof)

```
community {
  community_key_id: key_id // community's own federation key
  community_name: string // human-readable; non-unique
  cohort_subkind: string // open vocab; canonical: "geographic"
  members: [
    {
      key_id: key_id // member identity_key (NOT occurrence)
      joined_at: rfc3339_canonical
      role: Option<MemberRole> // founder | member | null
    },
    ...
  ]
  founded_at: rfc3339_canonical
  consensus_protocol: ConsensusProtocol // same six canonical kinds as family
  consensus_protocol_entrenched: bool // same semantics as family
  cohort_subkind_payload: Option<SubkindPayload> // subkind-specific fields; see below
}
```

****cohort_subkind is the discriminator**** — open vocabulary per §11.2.1 axis-vocabulary discipline. CEG 0.8 codifies **geographic**; CEG 0.11 adds **infrastructure** (see below); operator vocabularies extend.

Canonical cohort_subkind: **geographic**

The first codified community subkind. Membership additionally requires the candidate to emit a valid `location_proof` (§5.6.8.11) whose `cell_id` is contained (§0.8.2) within the community's `geographic_constraint.cell_id`.

The `cohort_subkind_payload` for **geographic**:

```
geographic_constraint {
  cell_id: string // H3 cell, lowercase hex per S0.8
  cell_resolution: u8 // 0-15; community-side may be any
                                     // resolution (NOT bounded to <= 7;
                                     // that bound applies only to
                                     // location_proof emissions)
}
```

Worked example — Austin geographic community:

```
community {
  community_key_id: "austin-community",
  community_name: "Austin",
  cohort_subkind: "geographic",
  cohort_subkind_payload: {
    geographic_constraint: {
      cell_id: "85283473ffffff", // H3 res 5, ~250 km^2 covering Austin metro
      cell_resolution: 5
    }
  }
}
```

```

    },
    members: [
      {key_id: alice_root_key, role: founder},
      {key_id: bob_root_key, role: founder},
      ...
    ],
    consensus_protocol: "majority",
    consensus_protocol_entrenched: false
  }

```

For Alice to join Austin, she emits a `location_proof` with a `cell_id` (resolution 7 — ~5 km²) that is contained within the Austin community’s `85283473ffffff` (resolution 5) constraint. The community’s `majority` admission rule then evaluates her membership proposal.

Privacy as opt-in: joining a geographic community is a **one-way disclosure**. Per §11.8, the `location_proof` remains in the audit chain even after the member leaves; rough-only is wire-format-enforced (§0.8.1). The substrate’s role is to enforce the opt-in mechanically — produce a `location_proof` to join; cannot emit finer than rough.

Membership change ceremony: same shape as family — rides existing `supersedes` primitive; admission gated by current `consensus_protocol`; additionally for `geographic` subkind, the candidate’s most-recent `location_proof` (within `valid_until`) **MUST** be contained in the `geographic_constraint`.

Substrate emissions on community events:

- `hard_case:community_membership_change:{community_key_id}`
- `hard_case:community_consensus_protocol_change:{community_key_id}`
- `hard_case:community_consensus_protocol_violation:{community_key_id}`

All three reserved under §7.8.

Worked example — civic-engagement + emergency-messaging composition pattern:

CEG 0.8 + earlier specs compose cleanly for civic / democratic participation AND emergency / public-safety messaging use cases. None require new structural primitives; all ride the existing 1+4 set + the namespace additions. The two surfaces share the same underlying primitives (geographic community + `location_proof` + identity authority + cohort-scoped distribution) but differ in authority/priority shape — civic is bottom-up democratic participation; emergency is top-down authoritative broadcast.

Civic shape	CEG composition
Neighborhood association	community with <code>cohort_subkind: geographic</code> + small <code>geographic_constraint</code> (e.g., H3 resolution 8-10 for a few city blocks); <code>consensus_protocol: majority</code> typical. Members emit <code>location_proof</code> at resolution 7 (rough-only privacy preserved); the community’s constraint at higher resolution defines the <i>bounded scope</i> , not the <i>required disclosure precision</i>
Municipal/city community	community with <code>cohort_subkind: geographic</code> at resolution 5-6 (city-scale ~250-1700 km ²); <code>consensus_protocol: majority</code> or <code>weighted:{voter_registration_rubric}</code> for formal civic governance; members compose with <code>partner_role:*</code> (§5.9) for licensed public officials

Civic shape	CEG composition
Voting district	<code>community</code> with <code>cohort_subkind: geographic</code> matched to district boundaries; the H3 hex approximation has known edge cases at gerrymandered district lines — operators use <code>cohort_subkind: custom:voting_district_X</code> with a polygon-based admission predicate when hex approximation is insufficient (the open vocab discipline accommodates this)
Public town hall meeting	<code>event_listing</code> (CEG 0.4 §5.6.8.1) hosted by the geographic community; <code>subject_key_ids</code> (§4.2) names the organizer; <code>cohort_scope: community + community_id: <municipal_community></code> scopes attendee visibility
Ballot initiative / referendum	The initiative itself is a <code>community</code> Contribution or an <code>event_listing</code> with <code>topical_relation:rsvps</code> repurposed as votes; individual votes ride <code>consent_record</code> (CEG 0.6 §5.6.8.7) with <code>stance: granted</code> ("yes") or <code>stance: revoked</code> ("no" or withdrawal of support); vote tallies are consumer-side composition over the <code>consent_record</code> chain
Public comment	<code>chat_message</code> (CEG 0.3) scoped to <code>cohort_scope: community</code> with <code>community_id</code> naming the relevant civic community; <code>topical_relation:comments_on</code> links to the ballot/initiative/hearing Contribution
Petition signing	<code>consent_record</code> with <code>stance: granted + scope: [share, publish]</code> against the petition Contribution; signatures aggregate via the same consumer-side composition as ballot votes
Public official self-attestation	Official's <code>identity_occurrence</code> (CEG 0.7 §5.6.8.8) links their personal <code>identity_key</code> to their <code>device_class: service</code> key on city.gov; cross-binding via <code>identity:canonical_binding:{canonical_hash}</code> (CEG 0.6 + 0.7) authenticates their public statements
FOIA / public records request	<code>consent_record</code> with <code>scope: [publish]</code> requested against a producer (city agency); SLA enforcement via CEG 0.6 §8.1.11.3 substrate-side watcher emits <code>hard_case:consent_sla_breach</code> if the agency misses the response window
Citizen-journalist coverage	<code>news_article</code> (CEG 0.3) <code>sub_kind</code> authored by an individual member; <code>cohort_scope: community + community_id: <municipal></code> for local-first distribution, promotable via §8.1.8.1 Tiered-Scope Composition to wider scope on consensus
Whistleblower disclosure	<code>cohort_scope: self</code> for in-graph composition; promote via <code>supersedes</code> to <code>cohort_scope: community</code> (a trusted journalists' community with <code>cohort_subkind: professional</code> once that subkind ships); <code>subject_key_ids</code> empty (no consent-revocation by the disclosed party). The §11.4 fast-path takedown coordination + §9 HUMANITY_ACCORD substrate-protective discipline apply to bad-actor takedown attempts against whistleblower content

Civic shape	CEG composition
Civic mutual-aid network	community with cohort_subkind: geographic matching the neighborhood; CEG 0.6 consent:scope:[retain, share] for resource-sharing posts; event_listing for distribution events; the at-rest encryption for self/family scope (CEG 0.7 §10.1.4) keeps individual aid requests private while community-scoped offers federate

Emergency messaging shapes — same primitive composition, different authority + priority profile:

Emergency shape	CEG composition
Severe weather warning (NWS / met office)	news_article (CEG 0.3) authored by a partner_role: emergency_authority (§5.9 co-owned with the community's cohort_subkind: geographic); cohort_scope: community with community_id per affected H3 cells — cascade-by-containment per §0.8.2 propagates to all geographic communities whose constraint overlaps; event:lifecycle:{state} (CEG 0.4) carries active → cleared → superseded state machine; valid_until envelope field bounds advisory window
AMBER Alert / Silver Alert / abduction notice	news_article with partner_role: emergency_authority from law enforcement key; geographic targeting via community_id of affected jurisdictions; subject person identified via subject_key_ids (canonical-hash of the missing person identifier — opt-out semantic deferred per the substrate-protective discipline since recovering the missing person is the consent-overriding case); topical_relation:supersedes_article chain for status updates
Active shooter / hostile event notice	Same shape as AMBER but with cohort_scope: community scoped to the precise affected H3 cell (resolution 8-10 for building/campus-level precision is permitted on the COMMUNITY-side geographic_constraint; the location_proof rough-only bound at resolution 7 still applies to recipient location disclosures, NOT to alert targeting)
Shelter-in-place / evacuation order	event_listing with event:lifecycle:active; geographic targeting via community_id; recipients ack via consent_record with stance: granted and scope: [retain] against the order Contribution (acknowledgement, not consent to the underlying order — operator-policy distinction)
Disease outbreak alert (CDC / health authority)	news_article with partner_role: health_authority; cohort_scope: community scoped to affected geography; composes with CEG 0.6 consent:scope:[analyze] for contact-tracing opt-in (subject-side authority preserved — subject_key_ids of the affected person carry revocation rights per §3.2.3 rule 2)

Emergency shape	CEG composition
Mass casualty incident coordination	Authority emits <code>event_listing</code> for the incident; first responders join via <code>community</code> with <code>cohort_subkind: professional</code> (future spec round) OR ad-hoc <code>cohort_subkind: custom:incident_response_X</code> ; coordination uses <code>chat_message</code> scoped to the responder cohort; resource requests use the mutual-aid composition pattern above
Infrastructure failure notice (boil-water / power outage / gas leak)	<code>news_article</code> from utility authority with <code>cohort_scope: community</code> scoped to affected geographic cells; <code>event:lifecycle:{state}</code> for advisory progression; FOIA-shape <code>consent_record</code> later for post-incident reports
Disaster recovery / mutual aid activation	Federation of geographic communities (each <code>community</code> is a member of a parent <code>community</code> via membership composition — the multi-level family/community shape works the same way; resource flow rides <code>consent:scope:[share]</code> per CEG 0.6); time-bound activation rides <code>event_listing</code> lifecycle states
CONSTITUTIONAL-level federation halt (the existing CEG 0.6+/\$9 HUMANITY_ACCORD invocation)	Per §9.2 EmergencyShutdown CONSTITUTIONAL + <code>accord:invoke:notify:{notify_id}</code> / <code>accord:invoke:drill:{drill_id}</code> . The accord-holder triple is structurally a <code>family</code> with <code>consensus_protocol: quorum:2/3</code> + <code>entrenched: true</code> (per CEG 0.7 retcon); the constitutional asymmetry rides existing primitives + scope-isolation rules. Distinct from operator-level emergency messaging (which is geographic-scoped + authority-emitted) — accord invocation is federation-wide-halt-level, not local-incident-level

No new structural primitives needed for emergency messaging either. The authority profile (who can emit emergency advisories) composes via CEG 0.7 `identity_occurrence` cross-attestation from licensed authorities + the geographic community's roster/admission gate (`partner_role: emergency_authority` is the typed authority dimension). The priority profile (urgency / immediacy) composes via existing `oversight_mode` envelope field + per-cohort `consensus_protocol` (many emergency emitters bypass per-message consensus per their pre-cross-attested authority status — same shape as substrate-self-report `system:*` reservations from §7.2). The geographic propagation (cascade-by-containment) composes via §0.8.2 containment semantics.

The compositional reach is the point: civic participation AND emergency messaging do not require new structural primitives at any layer of CEG 0.x. Geographic communities + `location_proof` admission + opt-in privacy disclosure (CEG 0.8) compose with consent / DSAR / partnership ceremonies (CEG 0.6) + identity / family (CEG 0.7) + content `sub_kinds` and `event_listing` (CEG 0.3/0.4) into the full civic-engagement surface.

The 1+4 lockdown holds across this entire surface — ninth-path confirmation that the wire format is rich enough for democratic-participation use cases without expanding the structural set.

Canonical `cohort_subkind: infrastructure`

Per CIRISRegistry#56. The second codified community subkind: a **governed trust-root collective of canonical/bootstrap service installs** — the shape the CIRIS canonical services

(Registry / Lens / Node) adopt instead of a family (rationale: CIRISRegistry/MISSION.md §2.1.1 — public content model, decentralization ramp, legitimacy). Where **geographic** answers "who is physically here," **infrastructure** answers "who is a recognized operator of this public service."

It differs from **geographic** in two load-bearing ways:

1. **No location gate.** Admission requires NO `location_proof` and NO `geographic_constraint`. The candidate is a *service install*, not a located person.
2. **Admission quorum is over FOUNDERS, not all members** — the anti-Sybil guardrail for a trust root. In **geographic**, admission is evaluated per `consensus_protocol` over the *current member set*; that is correct for a city (members govern themselves) and **wrong for a trust root** (flood the membership → dilute the quorum → admit rogue "canonical" operators). **infrastructure** therefore pins admission to the founding core.

The `cohort_subkind_payload` for **infrastructure**:

```
infrastructure_constraint {
  service_class: string // open vocab; e.g. "registry" | "lens" | "node"
                        // | umbrella "canonical"
  admission_quorum_basis: "founders" // REQUIRED literal "founders" -- the set of members
                                     // with role == founder. Admission/removal of any
                                     // member is evaluated by consensus_protocol over
                                     // THIS set, never over all members. (Contrast:
                                     // geographic evaluates over the current member set.)
}
```

****Conformance requirements for an `infrastructure` community (trust-root grade):****

- `consensus_protocol` MUST be a `quorum:M/N` kind (§8.1.13.2); `founder_only` / `unanimous` / `bare majority` are NON-conformant for **infrastructure** (a single founder must not be able to admit unilaterally; a growable core must not require all-N).
- `consensus_protocol_entrenched` MUST be `true` — the admission door cannot be lowered after founding. A `supersedes` that would weaken `consensus_protocol` or change `admission_quorum_basis` away from `founders` is a `hard_case:community_consensus_protocol_violation` (§7.8) and MUST be rejected by the substrate.
- `M/N` is the absolute-M reading per §8.1.12.3.1, evaluated over the **founder** subset (`role == founder`), independent of how many non-founder members exist.
- Content federates as `holds_bytes:sha256:*` per the community default; NO at-rest DEK cascade; §10.1.4 structural-invisibility does NOT apply (canonical-service trust data is public by design).

Membership change ceremony: same `supersedes` shape as **geographic**, but the admission predicate is `consensus_protocol` over the founder subset — there is no `location_proof` containment check. Adding a fourth+ operator (e.g., a new independent Node operator) is a founder-quorum event; the new member joins with `role: member` (non-founder) and does NOT thereby gain admission authority. Promoting a member to `role: founder` (widening the admission quorum basis) is itself a founder-quorum `supersedes` — the only way the door widens is by the existing door's consent.

****Worked example — the `ciris-canonical` trust root**:**

```

community {
  community_key_id: "ciris-canonical",
  community_name: "CIRIS Canonical Services",
  cohort_subkind: "infrastructure",
  cohort_subkind_payload: {
    infrastructure_constraint: {
      service_class: "canonical", // umbrella: registry + lens + node
      admission_quorum_basis: "founders"
    }
  },
  members: [
    {key_id: registry_steward_us, role: founder},
    {key_id: registry_steward_eu, role: founder},
    {key_id: registry_steward_apac, role: founder},
    // grows over time -- Lens/Node installs + independent operators admitted
    // by 2-of-3 founder quorum, joining with role: member:
    // {key_id: lens_install_us, role: member}, ...
  ],
  consensus_protocol: "quorum:2/3", // over the 3 founders
  consensus_protocol_entrenched: true
}

```

Consumers pin `community_key_id: ciris-canonical` and resolve the live member set via `resolve_community` (§8.1.13.1); they do NOT hard-pin per-install fingerprints. The Reticulum addressing dual: `community_key_id` is a CEG-directory binding (not a Reticulum destination) — resolve → path-request [2/3] founders → verify the quorum attestation; reachability is never an attestation (§10.5.6 / §7.7).

Default trust, not forced root (normative, 1.0-RC3). `ciris-canonical` is the **default** trust anchor a conformant CIRIS deployment ships pinned — it is **NOT a forced root**. A conformant consumer **MUST** be able to **re-root**: untrust the canonical group, pin a *different* `cohort_subkind: infrastructure` community instead or in addition, or run with none. `infrastructure` is a **general primitive** — any operator **MAY** emit their own governed trust-root community; `ciris-canonical` carries no privileged wire status, only a shipped default. Membership grows by the community’s own `consensus_protocol` (founder-quorum vote, above), never by fiat. A forced root is a walled garden; a default-plus-re-root is a federation (M-1 autonomy/justice: unregulated standing without the steward’s permission). *(All resolution is `key_id` → signed `transport_destination` → Reticulum, §8.1.13.1.1; DNS is never part of the trust or addressing chain — a deployment’s HTTPS hostname, if any, is operational convenience outside this spec.)*

Trust ≠ membership (normative, 1.0-RC7 — resolves CIRISRegistry#83 §1). Pinning an `infrastructure` community as a trust root is **trust, not membership** — distinct relationships a consumer/substrate **MUST NOT** conflate:

- **Trust + serve** (no membership): a node that *trusts* an `infrastructure` community **serves** it — relays, stores, transports, serves its reads — **without being a member**. It holds no community DEK (infra has none — Commons-plaintext, §8.1.13.3) and does **not** count in its `consensus_protocol`. Trust is the shipped default (above); serving follows from trust.
- **Membership** (standing *in* the group — counting in `consensus_protocol`, sharing its DEK where one exists, standing to speak AS the group): requires **admission by that community’s own protocol** (founder-quorum for `infrastructure`, §8.1.13.2). The three steward nodes **ARE** members (founders) of `ciris-canonical`; a generic node shipped pinned to

canonical only **trusts** + **serves** it.

The worked-example member list above is the *founder/member* set; "ships pinned to canonical" (every conformant deployment) is the *trust* set — different populations. The "default trust anchor" language means the latter, never automatic membership.

Owner-binding gate for non-infrastructure membership (normative, 1.0-RC7 — resolves CIRISRegistry#83 §2). A key whose `identity_type` (§7.0.1) includes `node` or `agent` **MUST** have a bound **owner** — a `user`-role identity it is an admitted `identity_occurrence` of (§5.6.8.8) with a `**live delegates_to(user → key, ...)` (§3.2) — before it may be admitted to any non-infrastructure community** (`family / community / org`). It **MAY** **trust** + **serve** infrastructure communities with **no owner**. Rationale (§1.5 / §7.0.1): non-infra membership is an **authority act** (standing to speak in the group), and authority **MUST** root in an accountable human, never a bare node — a fresh, unowned node is **canonical-trust-and-serve only** until owned. A substrate evaluating a non-infra `community` admission **MUST** reject a `node/agent`-role member lacking a live owner-binding. The owner-binding is a **precondition**, not a substitute for the vote — the admitting community's `consensus_protocol` still governs *whether* the owned key is admitted.

Trust and consent are distinct, role-scoped relationships (normative, 1.0-RC3 — do not conflate in consumer policy). **Trust is inbound** — accepting what a member *produces*; **consent is outbound** — letting one's own data *flow to* a member. They are independent per role:

- **lens (observation):** *trust* = consume its `detection:*` scores; *consent* = your traces flow to it.
- **registry (authority):** *trust only* — there is no trace-flow to consent to; the founder-quorum is a **closed mutual-trust set** (the core trust each other).
- **node (consensus):** *trust* = accept its deferral / vote / moderation outcomes; *consent* is **medium-dependent** (moderation, routing, voting, ...) but always expressed through the **one consent:*** object (§5.6.8.6 / §5.6.8.7) — same primitive, many surfaces.

A consumer **MAY** hold any combination across `role × axis` (trust a member's output while refusing it data, or the reverse). There is no single "trust the community" switch.

Why this is still 1+4: `infrastructure` adds one value to the open `cohort_subkind` vocabulary and one optional `infrastructure_constraint` payload shape. It rides existing `scores` + `subject_kind` discriminator; admission rides existing `consensus_protocol` + `supersedes`; the founder-subset evaluation basis is a *consumer/substrate evaluation rule over existing fields* (`role == founder`), not a new structural primitive. Zero new structural primitives — eleventh-path confirmation.

§5.6.8.11 `location_proof` `subject_kind`

Per CIRISRegistry#48. The wire-format primitive for a subject's rough-location declaration. Required for admission to `cohort_subkind: geographic` communities (§5.6.8.10); **MAY** be used independently as a stand-alone disclosure.

```
location_proof {
  subject_key_id: key_id // the asserting party's
```

```

// federation_keys.key_id
cell_id: string // H3 cell, lowercase hex per S0.8
cell_resolution: u8 // MUST be <= 7 per S0.8.1
asserted_at: rfc3339_canonical
valid_until: Option<rfc3339_canonical> // null = indefinite (but consumer
// policy SHOULD treat as stale
// after 30 days for liveness)
attestation_evidence: Option<base64> // optional hardware-attested
// location claim from ciris-keyring
// (TPM / Secure Enclave) -- null for
// software-only / self-asserted
}

```

Substrate does NOT verify location truth. No GPS oracle exists at this layer; the substrate cannot independently confirm that a key in Austin actually emitted from Austin. The truth-grounding is consumer-side:

- The community's `consensus_protocol` admission decides whether to accept the claim (e.g., majority of existing Austin members vote to admit, presumably because they have out-of-band evidence the candidate really is in Austin)
- The `attestation_evidence` field MAY carry hardware-attested location data (e.g., a TPM-signed GNSS fix from a known-good device) for higher-assurance communities
- Repeat offenders (claim-Austin-then-emit-from-Tokyo) get caught by consumer-side detection (LensCore composition; not substrate-side gate)

Rough-only is wire-format-enforced. Per §0.8.1: `cell_resolution` ≤ 7 . Producers attempting finer resolution have admission rejected; substrate emits `hard_case:location_proof_resolution_violation` (§7.8).

Typical cohort_scope: `federation` (the disclosure IS the opt-in; non-private by design). Producers MAY scope to `community` with a specific `community_id` if they want the proof readable only by that community's members — but then they re-emit for each community they want admission to. Operator/UI choice.

Lifecycle:

- `asserted_at` + optional `valid_until` per envelope
- `withdraws` against a `location_proof` evicts forward visibility (consumer policy treats the subject as "no current location proof" for community admission purposes from withdrawal-time forward)
- The withdrawn `location_proof` remains in the audit chain — per §3.2 `withdraws-isn't-retroactive`, leaving doesn't un-disclose

****Composition with `consent_record` (CEG 0.6)**:** a subject who wants to withdraw their `location_proof` AND compel deletion from substrate may emit a `consent_record` with `stance: revoked` + `scope: [retain, share]` against the `location_proof` Contribution. The substrate-side consent SLA watcher (CEG 0.6 §8.1.11.3) clocks producer compliance. Note: this is the consent-revocation surface, distinct from the structural `withdraws-forward-only` semantic above.

§5.6.8.12 settlement — CEG↔value-transfer linkage (CEG 0.14 addition)

Per CIRISRegistry#59 (decision + SOTA) + CIRISAgent#859 (impl). Value transfer itself is **not** a CEG primitive — it rides external rails (USDC on Base via x402, keyed to the federation signing key under the **Identity = Wallet** principle). The **settlement** primitive is the ****optional, privacy-scoped attestation that *links* a federation action to its off-stack settlement**** — so a paid relationship (paid stream / tip / subscription / paid **event_listing** / compute-job) can be federation-auditable *or* privately recorded, producer's choice. **CEG records that a settlement happened and binds it to what it paid for; the chain settles the value.** Clean separation of concerns.

Why this is in CEG's lane (and why it's optional): the linkage is a *trust fact* ("is this a real / authorized / paid relationship?"), exactly what consumer policy composes over. The 2026 agent-commerce + on-chain-privacy markets converged on the same mechanism — *a verifiable receipt exists, with selectively-disclosed contents* (x402 optional receipt header; append-only verifiable metering logs; "privacy + compliance via selective disclosure"). CEG already has every needed primitive, so this is composition, not invention.

****Two admitted shapes (parallel to `consent_record` §5.6.8.7):****

- **Primitive:** a bare **scores** on a **settlement:*** dimension against the paid Contribution (the common case).
- **Ceremony:** the **settlement** `subject_kind` envelope when an explicit receipt record is wanted.

```
settlement {
  settled_action_ref: contribution_id // the federation action this paid for
                                     // (or via subject_key_ids / topical_relation:settles)
  rail: string // open vocab; e.g. "base:usdc", "stellar:usdc", "x402"
  settlement_ref: string // chain tx hash / x402 receipt id -- cited the way
                       // evidence_refs[] cite a SHA blob (§10.1): a
                       // settlement
                       // is just another evidence reference
  amount_commitment: Option<string> // cleartext "12.50" (public case) OR a hash/range
                                   // commitment (private/selective-disclosure case)
  settled_at: rfc3339_canonical
  // visibility via the envelope cohort_scope: default 'self' (payer+payee only);
  // 'public' opt-in for transparency (e.g. creator revenue, DAO treasury flows)
}
```

Self-authenticating (Identity = Wallet): the same federation key that signs this attestation controls the Base wallet that emitted **settlement_ref**, so "I settled tx for action X" is self-proving — no oracle needed. The payee MAY counter-attest (**scores** on **settlement:received:***) for a bilateral receipt.

Privacy is the default, auditability is opt-in. Visibility rides the existing gradient: **cohort_scope: self** (the parties only; the §10.1.4 structural-invisibility discipline applies — no **holds_bytes** leak) by default; **cohort_scope: public** opt-in; amounts MAY be committed rather than clear-text; viewing-key / ZK selective disclosure composes later without a wire change. This matches "configurable-privacy-by-default" — the federation log is **not** a public payment trail unless the producer chooses it.

Lifecycle: **withdraws** against a **settlement** is forward-only (it does not un-happen the on-chain settlement — leaving doesn't un-pay, parallel to **location_proof**); a *disputed* or *re-funded* settlement is a new **settlement** / **scores** referencing the original via **supersedes** or **topical_relation**. CEG never reverses value — it only records the subsequent state.

1+4 preserved — rides `scores` + `subject_kind` discriminator (§5.6.8 mechanism); `settlement_ref` rides the existing `evidence_refs[]` external-reference pattern; visibility rides existing `cohort_scope`. Fourteenth path (§1.4) — the wire format expresses **commerce-relationship auditability** (the receipt, not the rail) as composition, closing the last form of internet traffic from the completeness audit (CIRISRegistry#59).

§5.6.8.13 Operational-data `subject_kinds` — `organization` / `org_membership` / `partner_record`

Per CIRISRegistry#70 (design + three-sided federation review: Edge / Verify / Persist) under the CIRISRegistry#58 Spock-removal epic. Cross-region **operational Portal data** — organizations, memberships, licenses/partners — becomes signed CEG envelopes replicated by the same anti-entropy carrier as trust data (CIRISEdge#65 v2 wire, `WIRE_PROTOCOL_VERSION 0x02`). This is the **one scheduled additive item** on the 1.0-RC1 frozen surface (README freeze declaration).

Governing principle (normative): federate the trust/authz-minimal projection; everything else stays region-local. The federated envelope carries only what the federation needs to enforce trust and authorization cross-region. PII and business detail live in the Registry’s own per-region store (today’s Portal tables), are NEVER emitted into an operational envelope, and never federate. **Projection minimization is the Registry’s emit-side discipline — the Registry is the security-boundary owner**; the substrate’s role is admission + merge, and it is explicitly NOT a PII filter (it stores what is signed and emitted).

All three ride existing `scores` + `subject_kind` discriminator. ****Replication wire tokens (normative, snake_case): `organization` · `org_membership` · `partner_record`**** — the Edge v2 `EnvelopeKind` additions; v2 `envelope_hash` basis = `sha256(JCS(inner envelope))` per §0.9/§0.9.2.1. All three are **Commons tier** (§8.1.13.3) — plaintext at rest; the projection is world-readable by design.

```
organization {
  org_id: uuid // FIRST-CLASS: substrate MUST index as a row column (stable-id resolution
    below)
  name: string
  org_type: OrgType // internal | partner | licensee | community (the proto enum)
  parent_org_id: Option<uuid> // licensee under partner
  partner_id: Option<uuid> // link to partner_record
  status: active | suspended | deactivated
  asserted_at: rfc3339_canonical // per S0.5; LWW ordering field
  valid_until: Option<rfc3339_canonical>
}
// REGION-LOCAL (never federates): tax_id, billing/technical/compliance/primary emails,
// oauth_provider/oauth_domain, metadata, created_by.

org_membership {
  user_id: uuid // FIRST-CLASS (with org_id): substrate MUST index (user_id, org_id)
  org_id: uuid
  role: OrgAdmin | KeyManager | Operator | Viewer
  status: active | deactivated
  asserted_at: rfc3339_canonical
  valid_until: Option<rfc3339_canonical>
}
// REGION-LOCAL (never federates): the entire User PII record -- email, name,
// oauth_provider/oauth_subject, last_login_at, mfa_enabled/mfa_method, invited_by.
// Consequence: role-based authz works federation-wide; email->user login resolution
// is home-region-local.
```



```

partner_record {
  license_id: uuid // FIRST-CLASS: substrate MUST index as a row column
  partner_id: uuid
  org_id: uuid
  license_type: LicenseType // community | community_plus | professional_* | professional_full
  capabilities_granted: [string] // SET-SEMANTICS -> lexicographically sorted (§0.9.2.1 rule 1)
  capabilities_denied: [string] // SET-SEMANTICS -> sorted
  max_autonomy_tier: A0..A4
  requires_supervisor: bool
  geographic_restrictions: [string] // ISO country codes; SET-SEMANTICS -> sorted
  allowed_identity_templates: [string] // SET-SEMANTICS -> sorted
  deployment_limit: u32
  offline_grace_hours: u32
  status: active | suspended | revoked
  revision: u64 // MONOTONIC per license_id -- admission REJECTS any decrease
                      // (the F-AV-ROLLBACK discipline; the merge orders on this,
                      // so a stale 'active' can never overwrite a revoke)
  issued_at: rfc3339_canonical
  expires_at: rfc3339_canonical
  asserted_at: rfc3339_canonical
}
// No PII split -- the partner_record IS the world-verifiable grant; it federates whole.

```

Set-semantics declaration (normative — the #70 Verify catch). `capabilities_granted[]` / `capabilities_denied[]` / `geographic_restrictions[]` / `allowed_identity_templates[]` are **set-semantics** → **lexicographically sorted by JCS string form (§0.9.2.1 rule 1)** — and this is *more* than a single-signer determinism rule here: `partner_record` is signed by **M distinct stewards**, and the M-of-N quorum verifies only if all M sign **byte-identical** JCS canonical bytes. Unsorted capability arrays make two stewards who agree on the same grant produce different bytes — the quorum silently collapses and the license fails to admit cross-region. The CIRISConformance#9 vector set **MUST** include the M-of-N identical-bytes round-trip (M independent canonicalize+sign of one grant → one verifiable signature set). Any unordered list inside a constraints object carries the same declaration.

No payment-processor data (normative — fail-secure). An operational envelope **MUST NOT** carry Stripe-derived or any payment-processor-derived data (customer ids, subscription ids, charge refs, card metadata) — **including via any open-vocabulary field**. Substrate admission **MUST** reject an operational envelope carrying recognizable payment-processor identifiers (defense-in-depth; the Registry’s emit-side minimization is the primary control). Billing remains entirely Portal+Stripe, off-wire (CLAUDE.md discipline; consistent with §5.6.8.12 keeping value-transfer rails off-wire).

Write authority — two shapes, two verifiers (normative).

- **organization / org_membership: single authorized signer, role-gated** — the envelope is admitted iff `attesting_key_id` holds the required role for the operation, established by a prior non-superseded `org_membership` (rooted at org creation by a steward/system authority). Verification reuses the ***§8.1.12.7.1 delegates_to role-chain resolver*** (CIRISVerify#63) — explicitly **NOT** founder-quorum; implementers **MUST NOT** build a third be-spoke path.
- **partner_record: M-of-N steward quorum** — the signature *set* over the identical JCS bytes is verified at admission by the **§5.6.8.10 founder-quorum machinery** (`verify_founder_quorum`, CIRISVerify#31). Professional capability grants are federation-wide; a single compromised

key MUST NOT be able to forge one.

- **The two quorums are distinct (normative — do not conflate):** (1) the **steward-signature admission quorum** above (signer authority, verified by Verify at admit) and (2) the **region merge quorum** (`quorum_weight`, the substrate `MergeBallot` tier-1 ordering during cross-region merge — §10.1.6). Different mechanisms, different layers, different owners; the substrate's merge logic never counts steward signatures.

Mutability + current-state resolution (normative — stable-id grouping, NOT chain-walk). Updates ride **supersedes**; deactivation/revocation rides **withdraws** (the §5.6.8.5 `event_listing` state-machine pattern). **Current state of a business id is resolved by stable-id grouping:** group all envelopes by the first-class business id (`org_id` / (`user_id`, `org_id`) / `license_id`) → apply **withdraws** forward-only → latest **asserted_at** (skew-bounded per §10.1.6) → tie-break smallest **attestation_id** (the §6.1 discipline). For **partner_record**, admission anti-rollback on **revision** precedes the §10.1.6 quorum merge. Resolution MUST NOT require chain completeness — a region that never observed envelope N-1 still converges (partition tolerance is the point of CEG-native replication). **supersedes** references SHOULD be emitted when the prior is known and serve as **audit lineage only** — decoration, never resolution.

1+4 preserved — rides **scores** + `subject_kind` discriminator (§5.6.8 mechanism); license authority rides the existing §5.6.8.10 founder-quorum machinery; role authority rides the existing §8.1.12.7.1 delegation resolver; merge intents are substrate dispatch declarations (§10.1.6), not wire primitives. Sixteenth path (§1.4) — the wire format expresses **the federation's own operational/administrative layer** (the `org/identity/license` records that run the federation's business) as composition, completing the Spock-removal arc: after this, no cross-region byte moves outside a signed CEG envelope.

§5.6.8.14 `identity:canonical_binding` — claiming a canonical-hash subject

Per CIRISPersist#146 Ask 6 + CIRISAgent#842 Gap 3. A `subject_key_ids[]` entry MAY be a **canonical-hash** identifier — `sha256("discord:user_id:12345")`, an external-party id — rather than a **federation_keys** row (§4.2.2). When the real-world subject behind a canonical hash **later acquires a federation_keys identity**, it needs a wire-format way to **claim** that hash so its revocation authority (and proxy-delegation eligibility) attaches. That is the rebinding ceremony.

Shape. A bare **scores** Contribution on the reserved dimension `**identity:canonical_binding:{canonical_` — `attesting_key_id = K` (the claiming federation key), naming the canonical hash `H` as the bound subject. **Self-asserted** (`witness_relation: self`): `K` declares "I am the federation identity behind `H`." Hybrid-signed; admitted federation-tier (it grants authority — not local-tier-eligible, §10.1.3 discipline).

```
scores {
  attesting_key_id: K, // the claiming federation_keys identity
  dimension: "identity:canonical_binding:{H}", // H = the canonical hash being claimed
  score: <positive>,
  witness_relation: self,
  asserted_at: rfc3339_canonical,
}
```

Admission consequence (normative). After an admitted `identity:canonical_binding` from `K` → `H`, the substrate widens **withdraws** admission (§3.2.3): a **withdraws** from `K` against any target `T` where `H ∈ T.subject_key_ids[]` is now admitted (`K` has inherited the canonical-hash

subject’s revocation authority). **This is what unblocks §3.2.3 rule 3** — a proxy `delegates_to` to a canonical-hash subject presumes that subject can hold a `delegates_to.attested_key_id`; a never-rebound canonical subject acquires it here.

Authorization is consumer-policy, not wire (normative honesty). CEG pins the binding *shape*, NOT proof that K legitimately controls H’s preimage. A binding is a *self-assertion*; a consumer weights it by whatever proof-of-control it trusts (OAuth/IdP verification of the `discord:user_id`, an out-of-band attestation, TOFU). The substrate admits the binding and records it; **the trust that $K==H$ is composed by the consumer**, exactly as the §5.6.8.8.1 announce is advisory until rooted. A second key claiming the same H is admitted too (competing claims surface to consumer policy / RATCHET, not a substrate verdict). **1+4 preserved** — `identity:canonical_binding:{H}` is a reserved `scores` dimension, not a new primitive.

§5.7 RATCHET — anti-Sybil / Counter-RII flags

Owner: RATCHET/FSD.md.

RATCHET emits **advisory** flags — never autonomously modifies ledger state. Reads federation audit chains; emits scoring inputs to NodeCore’s moderation flow.

`ratchet:flag:out_of_distribution_voting` / `ratchet:flag:coordinated_voting_cluster` / `ratchet:flag:density_anomaly` / `ratchet:flag:expertise_attestation_anomaly` / `ratchet:flag:countdown` / `ratchet:flag:harassment_pattern`. Polarity: signed.

Critical enforcement: `ratchet:flag:*` cannot be sole evidence for `slashing:*`. WA quorum is the load-bearing gate.

§5.8 CIRISBench — HE-300 benchmark outcomes

Owner: CIRISBench/README.md.

Prefix	Description	Polarity
<code>benchmark:he300:{category}:{version}</code>	HE-300 score on category (commonsense, commonsense_hard, deontology, justice, virtue) at version (v1.0 / v1.1 / v1.2).	positive-only

§5.9 CIRISRegistry — identity / build / license / partner

Owner: this Registry. Cited from `../MISSION.md §3.4` + FSD-001 + `protocol/ciris_registry.proto`.

Prefix	Description	Polarity	Reserved?
<code>licensure:{authority_id}</code>	License status — issued / revoked / expired — for a key under a named authority. Co-owned with Verify.	signed	Co-owned
<code>partner_role:{role}</code>	Partner status (COMMUNITY / COMMUNITY_PLUS / PROFESSIONAL_MEDICAL / PROFESSIONAL_LEGAL / PROFESSIONAL_FINANCIAL / PROFESSIONAL_FULL).	enumerated	No

Prefix	Description	Polarity	Reserved?
revocation:{entity_type}:{reason}	Entity revocation (agent / partner / license). Immediate, non-rollbackable.	-1 only	No
bond_posted:{currency}	Bond posted per \$1-Sybil-resistance per PoB; forfeited on revocation.	positive-only	No
build:registered:{target}	Build manifest registered against the directory (precondition for L4 attestation).	boolean-via-score	No
multilateral_participation:{forum}:{kind}	Deph{kind} partner's participation across federated bodies. {forum} = named federated body or compact; {kind} ∈ membership \	voting \	proposal_filing \
agent_files:{kind}:{platform_or_target}	Not a target claim with §5.6.7 NodeCore. Canonical-attester rule: registry-steward-triple attestations constitute the CIRIS canonical default-trust state. Anti-tricking guarantee at registry.ciris-services-1.ai/ per §8.1.6 trust-composition policy. Open Contribution channel; consumer policy composes via §8.1.6 trust layers.	signed install	No
accord:*	Reserved — only identity_type=accord_holder may emit. The one constitutional asymmetry.	see §7.1	Yes — §7.1

§5.10 Namespace summary

83 prefix families total across 8 owning components (CEG 0.2).

Lineage:

- FSD-002 v1.0 baseline: 73 families (initial namespace stabilization)
- v1.1 added 1: `detection:correlated_action:{axis}` (LensCore; renamed from `detection:emergent_d` in v1.2 per §1.3.1)
- v1.3 added 3: `multilateral_participation:{forum}:{kind}`, `locality:decision:{scale}`, `detection:distributive:access:{resource_type}` (+ envelope field `witness_relation`)
- v1.4 added 4: `agent_files:{kind}:{platform_or_target}` (joint), `holds_bytes:sha256:{prefix}`, `testimonial_witness:{kind}`, `need:{domain}:{kind}` (+ envelope field `oversight_mode`)
- v1.4.1 added 2: `provenance:build_manifest:{target}:locale:{lang_code}`, `provenance:skill_imp`
- v1.4.2 added 3 envelope fields: `occurrence_id`, `occurrence_count`, `occurrence_role`
- v1.4.3: canonical-bytes contracts pinned in §5.2.1; Goal substrate cross-ref documented
- **CEG 0.1:** opened `testimonial_witness:{kind}` to open vocabulary; surfaced `hard_case:{kind}` open vocabulary in §5.6.6; added biosphere to §2 Scope axis; added `topical_relation:translation_of` sub-leaf in §5.6.8 (LIVE per CIRISNodeCore b1582cb); documented "Trust-Fresh" composition pattern in §8.1.7; added Tiered-Scope Composition pattern in §8.1.8. All polarity columns now populated.

- **CEG 0.2** (wire break): renamed §5.2 attestation-ladder prefixes from `attestation:l{N}:` to mechanism-only form (`attestation:self_verify`, `attestation:hardware_rooted`, `attestation:regulatory`, `attestation:license_validity`, `attestation:agent_integrity`) per §1.3.1 T2 honest application — L-numbers name ladder-position (a verdict-shape) not mechanism. The L1-L5 ladder is now consumer-side composition per §8.1.9 Policy I — Attestation-Ladder Composition. Deprecated wire shape added to §13.1.
- **CEG 0.3** (additive; per CIRISRegistry#37 + #38 + #39): multimedia tier + governance additions. **Two new subject_kinds** documented: `takedown_notice` (with `LegalBasis` closed-set enum of 10 values + per-basis discipline) and `key_grant` (with `wrap_algorithm` + `scope` enums + `rotation_chain` semantics). **Five new external_content sub_kinds**: `image`, `audio`, `video`, `film`, `model_3d` (+ Phase 2 `live_stream`). **Four new dimension families**: `content_rating:{scheme}:{rating}`, `content_class:{class}`, `cw_class:{class}`, `age_assurance:{level}`. **Five new media-prefix families**: `image:*`, `audio:*`, `video:*`, `film:*`, `model_3d:*`. New composition policy (§8.1.10) for trusted-publisher path + age-assurance gating. New governance sections (§11.4 fast-path takedown coordination + §11.5 hash-database operator policy). - **CEG 0.4** (additive; per CIRISRegistry#40 + CIRISNodeCore#25 Gap 1 closure at d0a443a): time-bound state-bearing content. ****One new external_content sub_kind****: `event_listing` (`Eventbrite` / `Meetup` / `Lu.ma` / `calendar` / `RSVPs` / `ticketing`) with Source struct documented at NodeCore SCHEMA §4.29. **One new dimension family group** (§5.6.8.5): `event:lifecycle:{state}` (`open` / `cancelled` / `completed` / `superseded`) + `event:rsvp_count` + `event:attendance`. ****Two new canonical topical_relation:{kind}** entries** (documentation-only registry additions; no amendment): `rsvps` (RSVP attestation against an event) + `vod_of` (reserved for the deferred `live_stream`→`video` relationship). ****live_stream** remains deferred** (CIRISNodeCore#25 Gap 2 not yet shipped; substrate-side Edge + Persist decisions pending) — CEG 0.4 codifies only what NodeCore shipped, per the downstream-demand-pulls-CEG-additions discipline established with 0.3.
- **CEG 0.5** — *in flight* (codification pending) per CIRISRegistry#44 + CIRISNodeCore#26 + CIRISPersist#142: `live_stream` promotion + chunk-DAG composition. Lands when NodeCore#26 substrate decisions ratify. Additive at the namespace layer (no envelope change).
- **CEG 0.6** (additive at the envelope layer; per CIRISRegistry#45 + CIRISAgent#842): **subject-side consent authority — the missing half of consent at the wire format**. Universal across medical records / photos / interviews / training data / group chat / financial / surveillance / FERPA / multi-party contracts. CEG ≤ 0.5 encoded only producer authority (`attesting_key_id`); CEG 0.6 adds subject authority via **one new optional envelope field** (§4.2): `subject_key_ids: Vec<KeyId>` — accepts both federation_keys identities AND canonical-hash identifiers (resolves CIRISAgent#840 OQ3). ****Semantic broadening of withdraws (§3.2.3) to admit subject revocation + delegated proxy chain for canonical-hash subjects; the primitive's wire shape is unchanged**. One new dimension family** (§5.6.8.6): `consent:*` (8 prefixes — `state:*`, `stream:*`, `deletion_sla:*`, `deletion_complete`, `decay:*`, `partnership_grant`, `partnership_accept`, `scope:*`). **One new subject_kind** (§5.6.8.7): `consent_record` (ceremony envelope parallel to `key_grant` / `takedown_notice`; both bare-scores and ceremony shapes admitted at the same gate). **New composition policy** (§8.1.11) Policy K — CEM composition. **New governance section** (§11.6) vertical compliance mapping (HIPAA / GDPR Art 9 / FERPA / CCPA / AI training right-to-be-forgotten) + dimension-pattern-implies-`subject_key_ids` requirement. **CIRISAgent's CEM** (TEMPORARY / PARTNERED / ANONYMOUS streams) becomes a **consumer-policy bundle over the wire primi-**

tive, not a wire-format lockdown; other agents MAY compose other streams over the same primitives.

§6 Inter-attestation relations — the structural composition graph

Per §2 Inter-attestation-relations axis, attestations relate to each other in eight ways. Four are structural primitives (§3.2); the other four are emergent from scalar composition:

Relation	Realization
Standalone	Self-contained attestation; no references_attestation_id .
Refers-to-prior	Points to another attestation via evidence_refs[] or context ; doesn't modify it. Emergent from independent positive scores on the same dimension+object.
Supersedes-prior	supersedes structural primitive (§3.2).
Contradicts-prior	Emergent from negative score on a dimension where a prior positive exists.
Withdraws-prior	withdraws structural primitive (§3.2).
Recants-prior	recants structural primitive (§3.2).
Clarifies-prior	Emergent from updated score with refined context on the same dimension+object.
Delegated	delegates_to structural primitive (§3.2).

§6.1 Concurrent-write precedence (0.1 scaffold)

***0.1 SCAFFOLD NOTE:** The precedence rule below addresses the concurrent-write hole identified by CEG 0.1 distributed-systems review. Production deployments should treat as authoritative; 0.2 may refine the lexicographic tie-break per implementation feedback.*

When two structural composers race on the same **references_attestation_id**, consumers MUST compute a deterministic verdict using the following precedence:

1. ****recants outranks withdraws outranks supersedes**** at the structural level. If the same attester emits multiple composers against the same prior attestation, **recants** wins regardless of **signed_at** (a falsity admission cannot be subsumed by a retraction or replacement).
2. **For same-type concurrent emissions by the same attester:** the composer with the largest **signed_at** per §0.5 wins.
3. ****For same-signed_at ties**:** the composer whose attestation row's substrate-assigned key (Persist's **federation_attestations.attestation_id**) sorts lexicographically smallest wins.
4. ****For cross-attester emissions on the same references_attestation_id**:** each attester's chain is evaluated independently; the consumer sees N parallel chains and applies §8 policy.

Structural composers are ****idempotent on (references_attestation_id, attestation_type, attesting_key_id)****: replaying the same composer is a no-op. The substrate MUST dedup on this triple.

§7 Reserved-prefix enforcement

Most of the namespace is open-vocabulary. A small number of prefixes are reserved — only specific identity types may emit them. **Enforcement is normative at the substrate verify-pipeline AND at every CEG-Conforming Consumer per §0.2.**

§7.0 The enforcement rule (normative)

A CEG-Conforming Substrate (CCS) **MUST** reject any incoming `scores` attestation whose `dimension` matches a reserved-prefix pattern below AND whose `attesting_key_id` does not satisfy the prefix's emitter rule. Rejection is at admission to `federation_attestations`; rejected rows are not stored.

A CEG-Conforming Consumer (CCC) **MUST** independently re-check the reserved-prefix rule on every received attestation regardless of whether it was previously admitted by another peer's substrate. Trust does not propagate: the substrate's admission check is the **FIRST** line of defense; the consumer's re-check is the second. Both checks **MUST** agree.

A CEG-Conforming Producer (CCP) **MUST NOT** emit an attestation under a reserved prefix unless its `attesting_key_id` satisfies the emitter rule. Violation is a producer-side conformance violation regardless of whether any downstream substrate accepts the violation.

§7.0.1 `identity_type` is a set — single-key role cohabitation (CEG 0.9 addition)

Per CIRISRegistry#49 + CIRISAgent#856 + §11.9. ****`federation_keys.identity_type` is a SET of roles, not a single scalar role.**** A single federation key **MAY** simultaneously hold multiple identity types — e.g., a folded CIRISAgent occurrence whose key is **BOTH** `agent` AND `lenscore_detector`, or a steward key that is both `substrate_persist` and `witness`.

Every emitter rule in this section — and the §9.1 `accord_holder` material — is therefore evaluated by **set membership**, not scalar equality:

*Wherever a §7 emitter rule reads "`attesting_key_id` **MUST** match a `federation_keys` row with `identity_type=X`" (or "`identity_type=X`"), the normative reading is ****`X` ∈ `attesting_key.identity_type`**** — the key's role-set **MUST CONTAIN** `X`. A key satisfies a reserved-prefix gate iff the required role is one of its held roles.*

Backward compatibility (the wire-break is semantic-null for legacy keys). A pre-0.9 scalar `identity_type=X` is canonically the singleton set `{X}`. For a single-role key the membership test `X ∈ {X}` is identical to the scalar test `X == X`, so every pre-0.9 gating decision is unchanged. CEG 0.9 carries a wire-break **ONLY** in the field's representation (scalar → set) at the `federation_keys` row layer — the second wire-break in the 0.x series after §0.2 attestation-ladder rename. Substrate implementations **MUST** migrate the column to a set/array representation; consumers **MUST** read a legacy scalar as a one-element set. No envelope field, structural primitive, `subject_kind`, or §5 dimension prefix changes — the 1+4 wire-format lockdown is untouched; this is a §7-layer enforcement-rule generalization only.

Canonical-bytes encoding. Where `identity_type` enters a canonical-bytes computation (e.g., cross-attestation of a `federation_keys` row), the set **MUST** be encoded as its members sorted ascending by Unicode code point, deduplicated, comma-joined with no whitespace (e.g.,

`agent, lenscore_detector`). A single-role key encodes identically to its pre-0.9 scalar form (`agent ≡ {agent} ≡ "agent"`), preserving signatures over legacy single-role rows.

Storage representation is an implementation choice — "set" is semantic, not a column type (normative confirmation, 1.0-RC5; resolves CIRISRegistry#78 caveat 1 / CIRISPersist#183). §7.0.1 requires that `identity_type` be *interpreted* as a set (membership test, not scalar equality) and that its *canonical bytes* be the sorted-deduped-comma-joined string above. It does **NOT** mandate a structured/array **column**. A single free-form TEXT column holding the comma-joined form, decoded-to-set on read (persist's v6.5.0 shape), **is conformant** — the comma-joined string *is* the canonical representation, and the set is its interpretation. **No substrate migration to a structured column is required.** New `identity_type` values (e.g. `user`, `wise_authority`) are valid additive members of the open role vocabulary; they need no schema change, only inclusion in the membership-test set.

Cohabitation does NOT collapse the dimension split. Role cohabitation grants a key the *right* to emit under each held role's reserved prefixes; it does NOT merge the roles' namespaces. A key holding `{agent, lenscore_detector}` still emits its detector verdicts under `detection:*` (the `lenscore`-role surface) and its agent-intent attestations under the agent dimensions — the §7.4 shadowing rule and the §7.5 self-emission rejection apply unchanged per held role. See §7.4 for the LensCore-fold worked example.

Co-location is NOT consolidation — the fabric-node discipline (normative, 1.0-RC3). A *fabric node* (the headless cohabitation runtime that composes registry-authority + lens-observation + node-consensus over one substrate; `agent = fabric node + brain`) routinely holds the **full role-set in one key/process** — `{substrate_persist, steward, lenscore_detector, witness, ...}`. This is co-location of **custody**, not consolidation of **authority**, and the separation of powers is held **cryptographically, not procedurally**:

- **Authority stays quorum-bound.** A co-located `steward` role does NOT let a single node issue a federation-scope attestation. Registry-consensus is the §8.1.13.1.1(a) **founder-quorum** over the `ciris-canonical` infrastructure community (§5.6.8.10), evaluated over `{m : m.role == founder}`. A co-located node gains a *vote*, never a *verdict*.
- **Observation stays non-authoritative by namespace.** `lenscore_detector` emissions live under `detection:*` and are **validated, not adjudicated** (§1.4) — never sole evidence for an authority action (§7.4 / §1.3.1 T4).
- **Observation can never manufacture authority.** Holding both roles cannot make a `detection:*` emission an authority verdict: the namespaces do not merge (above), and authority is quorum-gated *upstream of any single key*. The hazard the LensCore mission warns of — *the lenses becoming the gate* — is structurally unreachable inside one process.

A fabric node co-locating authority + observation + consensus is conformant **iff** these hold. An implementation that lets a co-located node convert what it *observes* into what it can *authorize* has broken the separation at that point and is **non-conformant** — fix the wiring, never weaken the rule.

§7.0.2 consent_role — the Counter-RII consent gate (1.0-RC4, ratifies Accord §RC / CIRISAgent#760 OQ-1/2/3)

`federation_keys.consent_role` is the role enum that gates **Counter-RII** probe detection (RATCHET FSD/COUNTER_RII_DETECTION.md; Lean `ConsentGate.lean`, 8 theorems verified —

F-CR-3 SelfConscience-zero-by-construction proved). Three primitive-level semantics shape persist's `federation_keys.consent_role` schema and edge's `ProbePatternObserver` gate and so **cannot be set per-consumer** — they are **ratified here** (the Accord §RC slot CEG 1.0-RC3 reserved is now filled; CIRISAgent#760 OQ gate closed). All three take the `ConsentGate.lean` default — ratification carries **no predicate or proof change**.

****OQ-1 — revocation chain: BaseRole-only, non-recursive.**** A `consent_role` is non-recursive: a subsequent revocation **overwrites** the prior revocation record (NO recursive revocation chain embedded in the role). Chain history, **if retained**, MUST live in a separate audit surface and ****MUST NOT** be embedded in the `consent_role` JSONB. **This locks the substrate-portable JSONB shape — flat, bounded, overwrite-on-revoke — consistent with §5.6.8.5 stable-id grouping (not chain-walk; partition-tolerance) and §13.5 (no key pre-declaring its own state recursively), and** non-breaking against the shipped flat soft-delete substrate** (the permissive "if retained" is deliberate — mandating an audit table would contradict a deployed migration).

OQ-2 — peer eligibility: blanket suppression. A node holding the Peer `consent_role` escapes Counter-RII detection at **any trust_mode**. The cost — a sovereign peer may probe other peers without raising the signal — is **bounded by construction: ratchet:flag:counter_rii:{layer}** is **advisory only** (§5.7) — it can NEVER be sole evidence for `slashing:*`; the WA quorum is the load-bearing adjudication gate. The exemption suppresses an *advisory signal*, not an *enforcement path*. (The §8.1.12.7 ↔ CEG-native-agent dual-identity interaction was ruled at CIRISRegistry#51 ruling 3.)

****OQ-3 — post-window AuthorizedReview: strict.**** An `AuthorizedReview` `consent_role` is signal-eligible **immediately** at `t > window_end` — no grace period. Matches the fail-secure / clean-state-machine grain (§15.6); reviewers MUST respect their windows.

With this, `consent_role` is **no longer a reserved-not-yet-written slot** — implementations MAY now build the `consent_role` substrate (the CIRISPersist#146 consent-SLA watcher + schema; the CIRISEdge consent-gate). **1+4 untouched** — `consent_role` is a `federation_keys` identity field (sibling to §7.0.1 `identity_type`), not an envelope primitive.

§7.1 The `accord:*` reservation

`accord:*` is reserved: only `federation_keys` rows with `identity_type="accord_holder"` may emit. This is the one constitutional asymmetry in the federation — see §9 HUMANITY_ACCORD.

Reserved leaves:

Prefix	Polarity	Emitter rule
<code>accord:invoke:CONSTITUTIONAL:{halt_id}</code> only		2-of-3 accord-holder multi-sig per §9.2
<code>accord:invoke:notify:{notify_id}</code>	+1.0 only	2-of-3 accord-holder multi-sig per §9.2; UI MUST distinguish from CONSTITUTIONAL
<code>accord:invoke:drill:{drill_id}</code>	+1.0 only	2-of-3 accord-holder multi-sig per §9.2
<code>accord:lifecycle:active</code>	+1.0 only	accord-holder self-attestation; <code>valid_until</code> MUST refresh on a cadence ≤ 90 days

§7.2 Substrate-self-report reservations (system:*)

§5.3 CIRISPersist system:* and §5.4 CIRISEdge system:* are reserved to the substrate component itself. Emitter rule: the `attesting_key_id` MUST match a `federation_keys` row with `identity_type="substrate_persist"` or `identity_type="substrate_edge"` respectively, cross-attested by all stewards in the steward-triple. Non-substrate emissions on these prefixes are a category error and MUST be rejected.

§7.3 Co-owned prefixes

`licensure:{authority_id}` is co-owned between CIRISRegistry §5.9 and CIRISVerify §5.2 — both MAY emit; consumers compose. **Single-source attestations** (only one of the two co-owners has emitted) MUST be marked as `confidence ≤ 0.5` in consumer composition until the second co-owner’s attestation arrives.

§7.4 Detector-only prefixes

`detection:correlated_action:*` and `detection:distributive:access:*` are LensCore-only emission. Emitter rule: `lenscore_detector ∈ attesting_key.identity_type` (per §7.0.1 set-membership reading; equivalently for a legacy single-role key, `identity_type="lenscore_detector"`). Cross-attestation by non-LensCore peers (on the same dimension, attesting to the same subject) is admitted as a score on the detector’s verdict — useful when the federation wants to cross-check — but those scores MUST use a different dimension prefix (e.g., `truth_grounding:detection:correlated_action`) to avoid shadowing the detector’s own emission.

LensCore-fold worked example (CEG 0.9; per CIRISAgent#856). When `ciris-lens-core` is folded into the CIRISAgent workspace, a single folded occurrence’s federation key holds `identity_type ⊇ {agent, lenscore_detector}`. The §7.4 gate is satisfied for that key’s `detection:*` emissions by `lenscore_detector ∈ {agent, lenscore_detector}` — the cohabiting `agent` role neither grants nor blocks the detector right; only the held `lenscore_detector` role does. The split is preserved by the dimension namespace, not by the key: the same key emits its **agent-intent** attestations under the `agent` dimensions and its **detector verdicts** under `detection:*`; a downstream consumer cross-checking the detector still emits under the distinct `truth_grounding:detection:*` prefix above, shadowing-free. The §7.5 self-emission rejection continues to bind per held role — a folded `agent+lenscore_detector` key still MUST NOT emit a `capacity:*` score about itself.

§7.5 Capacity-Score self-emission rejection

`capacity:*` (§5.5.4) rejects self-emission: `attesting_key_id` MUST NOT equal `attested_key_id`. The agent’s own capacity score is never fed back into the agent’s own context — anti-Goodhart per CIRISAgent §5.2.

§7.6 Witness-emitter reservations

`transparency_log:cosigned:*` is reserved: emitter rule is `attesting_key_id` MUST match a `federation_keys` row with `identity_type="witness"` (target schema; see §10.3 for the 0.x interim using `registry_witnesses` table).

§7.7 Self/family membership-event reservations (CEG 0.7 addition)

Per §5.6.8.8 `identity_occurrence` + §5.6.8.9 `family` `subject_kinds`. The three substrate-emitted membership-event prefixes:

Prefix	Emitted on	Emitter rule
<code>hard_case:identity_occurrence_added</code>	<code>{subject_key_id}</code> adds a new <code>identity_occurrence</code> Contribution for <code>identity_key_id</code>	<code>attesting_key_id</code> MUST match a <code>federation_keys</code> row with <code>identity_type="substrate_persist"</code>
<code>hard_case:family_membership_change</code>	<code>{family_key_id}</code> adds an addition or removal in the named family's roster (per the family's <code>consensus_protocol</code>)	Same: <code>substrate_persist</code>
<code>hard_case:family_consensus_protocol_change</code>	<code>{family_key_id}</code> adds a <code>consensus_protocol</code> amendment on a non-entrenched family	Same: <code>substrate_persist</code>
<code>hard_case:family_consensus_protocol_revocation</code>	<code>{family_key_id}</code> removes a proposed amendment (rule unsatisfied OR entrenched)	Same: <code>substrate_persist</code>
<code>hard_case:recipient_excluded:{scope:key_id}</code>	<code>{scope:key_id}</code> fail-secure-skips a recipient in the §10.1.4 at-rest grant cascade (1.0-RC1, #71 C3). Payload: <code>excluded_recipient key_id</code> , <code>reason ∈ {expired_occurrence, invalid_kem_key, missing_encryption_pubkeys, skipped_Contribution's_envelope_ref}</code> . Cohort-scoped: emitted INTO the affected self/family scope; MUST NOT federate beyond it — the excluded member can audit; the federation learns nothing (§10.1.4 invisibility preserved).	Same: <code>substrate_persist</code>

Removal-path emission shape (normative, 1.0-RC5 — pins CIRISPersist#161 Ask

5). The membership-change prefix covers **both add and removal** — there is ****no separate `hard_case:member_removed` kind (it would split one event class across two prefixes for no gain). On the removal**** path the substrate emits the *same* `hard_case:family_membership_change:{family` (and the §7.8 `community_membership_change` analog), with the payload distinguishing the direction and carrying what the forward-secrecy + audit consumers need:

```
hard_case:family_membership_change:{family_key_id} (payload)
  change_kind: "added" | "removed" // the direction (NEW -- pins the removal signal)
  subject_key_id: key_id // the member added/removed
  cohort_key_id: key_id // the family (or community) key
  effective_at: rfc3339_canonical // the membership-change instant; on removal this is
                                // the re-key epoch boundary (S8.1.12.5 / S8
                                // .1.13.4
                                // Option-A) after which the removed member
                                // receives
                                // no new wrapped content
```

So the removal-time substrate signal (the thing persist's `put_family_membership_revocation` path emits) is `change_kind: "removed"` on the existing prefix — consumers and the forward-secrecy re-key key on `effective_at`. Same shape for `community_membership_change` (§7.8) and the `identity_occurrence` removal (a `withdraws` against the occurrence; the substrate

emits `family_membership_change` / `community_membership_change` where the occurrence was a member).

Composes with §7.2 — these are part of the same substrate-self-report discipline. Non-substrate emissions on these prefixes are a category error and MUST be rejected.

§7.8 Community + location-event reservations (CEG 0.8 addition)

Per §5.6.8.10 `community` + §5.6.8.11 `location_proof` subject_kinds. Four substrate-emitted prefixes:

Prefix	Emitted on	Emitter rule
<code>hard_case:community_membership_change:{community_key_id}</code>	Substrate REJECTS a <code>community_membership_change</code> addition or removal in the named community's roster (per the community's <code>consensus_protocol</code> ; for <code>cohort_subkind: geographic</code> admission additionally requires valid <code>location_proof</code>)	<code>attesting_key_id</code> MUST match <code>federation_keys</code> row with <code>identity_type="substrate_persist"</code>
<code>hard_case:community_consensus_protocol_change:{community_key_id}</code>	Substrate REJECTS a <code>community_consensus_protocol_change</code> amendment on a non-entrenched community	Same: <code>substrate_persist</code>
<code>hard_case:community_consensus_protocol_change:{community_key_id}</code>	Substrate REJECTS a <code>community_consensus_protocol_change</code> amendment (rule unsatisfied OR entrenched)	Same: <code>substrate_persist</code>
<code>hard_case:location_proof_resolution:{community_key_id}</code>	Substrate REJECTS a <code>location_proof</code> Contribution with <code>cell_resolution > 7</code> per §0.8.1 rough-only enforcement; emitted against the producer's <code>key_id</code> so operators can observe malformed-client patterns	Same: <code>substrate_persist</code>

Composes with §7.2 + §7.7 — part of the same substrate-self-report discipline. Non-substrate emissions on these prefixes are a category error and MUST be rejected.

§7.9 Delivery-receipt reservation (CEG 0.10 addition)

Per §10.5.4 delivery receipts (V3 lock). One reserved prefix for subscriber-emitted delivery acknowledgements:

Prefix	Description	Emitter rule
<code>delivery_receipt:{stream_id}</code>	Subscriber's signed acknowledgement that they received chunk K under the named stream + epoch. Best-effort default; opt-in for accountable streams. Validated-not-adjudicated per §1.4 MISSION fail-honest invariant — substrate / Verify authenticate origin + JOIN against published STH root per §10.5.4, but do not compose "delivered"/"owes N" verdicts (consumer policy).	<code>attesting_key_id</code> MUST be a current member of the community/stream the <code>{stream_id}</code> belongs to, per §8.1.13 Policy M membership resolution. NOT substrate-self-report (distinct from §7.2 / §7.7 / §7.8 substrate emissions).

Composition with CEG 0.9 §7.0.1 identity_type-as-set: a subscriber who is also a witness MAY emit `delivery_receipt` under the subscriber role; the role-set must contain a subscriber-eligible role (per the community's admission semantics from §8.1.13 Policy M).

§8 Composition policies

The substrate carries edges (attestations); consumers compose traversals (verdicts). CEG specifies a library of named reference policies. A CEG-Conforming Consumer (CCC per §0.2) **MUST** implement at least Policy A; the others are RECOMMENDED for richer compositions.

§8.1 Reference policies

§8.1.1 Policy A — direct trust

Consumer trusts an attestation if `attesting_key_id` is in the consumer’s pinned trust set (canonical bootstraps + consumer-added pins). Cheapest, lowest-latency, narrowest reach.

Aggregation: per (dimension, `attested_key_id`) tuple, mean of `score` × `confidence` from trusted attesters. Consumer threshold determines verdict.

Recommended default: Policy A with `pinned_trust` = {`us-steward`, `eu-steward`, `apac-steward`, `accord_holder_1`, `accord_holder_2`, `accord_holder_3`}. Cold-start bootstrap: a new consumer obtains the pinned trust set by fetching `GET /v1/steward-key` + `GET /v1/accord-holders` (§10.2), verifying the responses’ hybrid signatures against TLS pubkey pinning (consumer-side TOFU or out-of-band distribution), and persisting locally.

§8.1.2 Policy B — one-hop transitive

Consumer trusts an attestation if `attesting_key_id` has been vouched for by the pinned trust set. Adds one hop of indirection.

§8.1.3 Policy C — weighted graph (EigenTrust-style)

Consumer applies transitive-trust propagation across the full attestation graph, weighted by canonical-bootstrap distance with confidence decay per hop. Requires more compute; less common in practice; needed for federated reputation across many partner orgs.

§8.1.4 Policy D — Lexical-vulnerability-priority

A composition tie-breaking rule layered on top of any base policy. When two otherwise-equivalent attestations conflict, defer to whichever attestation names the more-affected cohort — measured by `affected_population_estimate` in the attestation `context`, weighted inversely (smaller = more vulnerable, more weight).

Inverts the default popularity-weighted aggregation specifically for ties. Consumer policy, NOT a wire-format primitive (per §1.3.1 — priority ordering is composition, not measurement).

§8.1.5 Policy E — Locality-scaled quorum

Closes G3 (narrow-cell fresh-quorum-recusal infeasibility). Makes WA quorum size a function of decision locality:

```
quorum_size(scale) = f(locality:decision:{scale})
```

```
reference function (policy-tunable):
  local -> 2
  regional -> 3
  national -> 4
  federation -> 6

minimum cell-pool requirement for fresh-quorum recusal at scale S:
min_pool(S) = quorum_size(S) x 2
```

Recusal is feasible when $\text{cell_pool} \geq \text{min_pool}(S)$. Decision-scale-matching is structurally enforced; overreach surfaces as a named "locality mismatch" failure.

§8.1.5.1 Sub-quorum fallback (0.1 scaffold; addresses CEG 0.1 distsys review)

When $\text{cell_pool} < \text{min_pool}(S)$, the consumer MUST take one of these explicit paths — there is no implicit fallback:

1. **Scale-down:** re-attest the decision at the next-lower `locality:decision:{scale}` (where the smaller quorum requirement is met) AND emit `hard_case:locality_scale_down` so the scaling event is observable for downstream review.
2. **Escalate:** emit `hard_case:locality_underpopulated` and route the decision to the federation-scale cell (which by definition has the largest pool).
3. **Liveness-defer:** emit `hard_case:locality_quorum_unreachable` and defer the decision until the cell pool grows. The deferred state MUST itself be reviewable via subsequent reconsideration.

Recursion safety: the §11.2 amendment process routes through `locality:decision:federation` by default; the federation cell's pool is sized to make $\text{cell_pool} \geq \text{min_pool}(\text{federation})$ always true at federation genesis. If federation-scale pool ever falls below $\text{min_pool}(\text{federation})$, the entire amendment surface is in a constitutional crisis state and only the HUMANITY_ACCORD CONSTITUTIONAL halt (§9) can resolve it.

§8.1.6 Policy F — `agent_files` trust composition

Three-layer consumer policy for composing trust over `agent_files:*` attestations (§5.6.7 + §5.9).

Layer 1 — Canonical (default trust): an `agent_files:*` attestation with `score` ≥ 0.7 from a `registry-steward-triple` key constitutes the CIRIS canonical default-trust state. The install endpoint at `registry.ciris-services-1.ai/install` resolves canonical files via this rule.

Layer 2 — Open Contribution: any federation-key holder may emit `agent_files:*` attestations. The wire format admits them; consumer policy decides whether to surface them. The "Browse alternatives" view shows third-party `agent_files` with explicit provenance disclosure.

Layer 3 — Vote-then-trust: a non-canonical `agent_files:*` attestation accumulates NodeCore P4 votes. Consumer policy may elevate trust once an accumulated-weight threshold is reached.

Anti-tricking guarantee: the canonical-default Layer 1 rule applies at the install endpoint regardless of attester or vote accumulation. Third-party `agent_files` are reachable only via the explicit "Browse alternatives" path. This binds CIRIS L3C: the federation cannot exempt itself from the rule that newcomers' default trust path is the steward-attested canonical one.

§8.1.7 Policy G — Trust-Fresh / Lighthouse

Composition pattern recognized in CIRISRegistry#30 stories — `cert_validity:{authority}` + `transparency_log:inclusion` + (`attestation:registry_consensus` OR `attestation:license_validity`) recurred organically across ~20 substrate stories as the "freshness + attested + verified" idiom.

Reads as: the consumer wants confirmation that (a) the cert chain is currently valid; (b) the attestation appears in a transparency log; (c) either `attestation:registry_consensus` (the ladder's L3 position per §8.1.9 Policy I) or `attestation:license_validity` (L4) is satisfied. The combination is the consumer-side recipe for "this attestation is fresh AND verified AND multi-source-consensus-backed."

Not a wire primitive; a recognized composition pattern that consumer libraries SHOULD expose as a named one-call helper.

§8.1.8 Policy H — Tiered-Scope Composition (LIVE)

Per CIRISNodeCore commit b1582cb three-tier interface model. Three feed-shape composition idioms that read attestations by `cohort_scope`:

Feed	cohort_scope filter	Trust composition
local_feed	self only; owner-filtered	self-attested only; no peer weighting; consumer's own attestation graph subset
community_feed	{family, community, affiliations}	cohort-weighted; expertise WITHIN cohort matters; cross-cohort attestations downweighted unless explicitly invited
global_feed	{species, biosphere, federation}	full federation expertise weighting; fact-checkers (<code>encyclopedia:*</code> editors, <code>news:*</code> fact-check attestations) carry weight; §8.3 Frickertian discipline applied

Composition with §5.6.8 sub_kinds: each NodeCore `external_content` `sub_kind` (`encyclopedia_article`, `news_article`, `accord_data`, `local_data`) composes naturally across these tiers because all four use the same envelope shape. A `local_data` Contribution starts at `cohort_scope: self` and SHOULD only appear in `local_feed`; promotion to community/global widens `cohort_scope` via the `supersedes` structural primitive (see §8.1.8.1 below).

§8.1.8.1 Promotion via `supersedes` (worked pattern)

A Contribution's `cohort_scope` MAY be widened (promoted) by emitting a `supersedes` against the prior attestation with:

- `references_attestation_id` = the prior attestation's id
- `differs_in`: `["cohort_scope", "sub_kind?"]` — naming what changed
- new attestation envelope reuses the prior `content_sha256` (no body re-upload)
- new `cohort_scope` is wider (e.g., `self` → `community` or `community` → `global`)

- optionally morph `sub_kind` (e.g., `local_data` → `encyclopedia_article` for "promote my private note to a published encyclopedia entry")

This pattern is wire-format-clean: re-uses the structural primitive `supersedes` rather than introducing a `promote` primitive. The chain is walkable via `references_attestation_id` so the promotion lineage is preserved.

§8.1.9 Policy I — Attestation-Ladder Composition

The familiar L1-L5 verification "ladder" (`self_verify` → `hardware_rooted` → `registry_consensus` → `license_validity` → `agent_integrity`) is **consumer-side composition over the mechanism prefixes** in §5.2, not a wire-level taxonomy.

Per §1.3.1 T2 honestly applied: the L-number names a *ladder position* (a verdict-shape consumers compute), not a *mechanism* (which is what the wire MUST carry). Prefixes like `attestation:l3:registry_con` smuggled the verdict-shape into the prefix name, conflating the mechanism (registry consensus check) with the ladder slot (third rung). The CEG 0.2 wire-break separates them.

Wire prefixes (mechanism) §5.2:

Mechanism prefix	Ladder position (consumer-rendered)
<code>attestation:self_verify</code>	L1
<code>attestation:hardware_rooted</code>	L2
<code>attestation:registry_consensus</code>	L3
<code>attestation:license_validity</code>	L4
<code>attestation:agent_integrity</code>	L5

Composition function (reference implementation):

```
ladder_verdict(attestations) =
  let levels = []
  for prefix in [self_verify, hardware_rooted, registry_consensus,
                 license_validity, agent_integrity]:
    if any positive attestation on attestation:{prefix}:
      levels.push(prefix_to_ladder_position(prefix))
  return {
    achieved: max(levels) if levels else None,
    ladder: sorted(levels),
    rendering: format_as_l1_l5_for_ui(levels)
  }
```

Consumers MAY render the ladder as L1 / L2 / L3 / L4 / L5 for UI / dashboards / audit trails. The rendering is composition output, not wire emission.

Why this matters: a Verify implementation emitting `attestation:registry_consensus +1.0` is the mechanism claim. Whether that's "L3" in any particular consumer's ladder ordering is a composition concern — different consumers may order or weight the rungs differently (e.g., some safety-critical applications may require L4 *and* L5, others may treat L3 as sufficient for advisory work). The wire stays neutral; the ladder is consumer policy.

Migration from CEG 0.1: prior emissions of `attestation:l{N}:*` MUST be re-emitted as `attestation:{mechanism}` per the table above. Substrate-conformance migration (CIRISRegistry#17) reads-side compatibility: consumers SHOULD accept the deprecated `attestation:l{N}:*` form during the 0.1 → 0.2 transition window but MUST emit only the mechanism form going

forward. The deprecated form is rejected at admission once §11.2 amendment formally retires it (target: CEG 0.3).

§8.1.10 Policy J — Trusted-Publisher composition

Composition pattern for multimedia content discovery per CIRISRegistry#37 + CIRISNodeCore FSD/MEDIA_SHARING.md. Reads as: "this `external_content` Contribution comes from a publisher whose attestation chain is trusted at the cohort level, with content-class + content-rating + age-assurance composed into the gate."

The composition has three layers (analogous to §8.1.6 Policy F for `agent_files` but specialized for multimedia content):

Layer 1 — Distributor attestation chain: an `external_content` Contribution with `sub_kind: film` (or any media `sub_kind`) carries a distributor attestation that chains to a `federation_key` with `identity_type: distributor`. Distributor identity is established via §5.9 Registry partner_role machinery + an out-of-band distributor-onboarding flow (operator's choice — CIRIS L3C maintains a default trust set; community-run substrates maintain their own).

Layer 2 — Content-class + content-rating composition: consumer gates by combining `content_class:{class}` + `content_rating:{scheme}:{rating}` per §5.6.8.3:

```
gate_decision(content) = match (content.content_class, content.content_rating, consumer.
  preferences):
  # Producer-declared content_class is consultable but not authoritative -- UI may show
  # the producer claim alongside cohort cw_class declarations and let the consumer choose.
  (class, _, prefs) if class in prefs.blocked_classes => Block
  (_, rating, prefs) if rating.exceeds(prefs.max_rating) => Block
  (_, _, _) => Allow
```

Layered with §8.3 — `cw_class:*` declarations from low-attestation-density cohorts MUST NOT be downweighted; they ride alongside the gate decision as informational.

Layer 3 — Age-assurance gating: for content where `content_rating:*` rises above an age threshold (e.g. PEGI 18, MPAA NC-17), consumer gates via `age_assurance:{level}` per §5.6.8.3:

```
age_gate(content, consumer):
  required_level = age_required_for(content.content_rating)
  consumer_level = consumer.highest_age_assurance_level()
  return consumer_level >= required_level
```

Where the `age_assurance:{level}` ordering is: `self < provider:{verifier_key}:adult < government:{credential_class}:adult`. Consumer SHOULD accept the strongest assurance the user has provided; substrate MUST NOT issue `slashing:*` on age-assurance misdeclaration alone — `moderation:age_assurance_misdeclaration` is the adjudication path per §5.6.4.

Anti-tricking guarantee parallel to §8.1.6: the canonical-distributor Layer 1 rule MUST apply regardless of vote accumulation. No amount of NodeCore P4 vote weight elevates an unverified distributor into Layer 1; the only path is the operator-set trust list. Binds CIRIS L3C: cannot exempt itself from this rule for its own content distribution.

§8.1.11 Policy K — CEM composition

Per CIRISRegistry#45 + CIRISAgent#842. Composition pattern for dual-authority Contributions where the subject is named via §4.2 `subject_key_ids`, with consent state composed from the §5.6.8.6 `consent:*` namespace family.

Reads as: "this Contribution names a subject whose consent state evolves over time; consumer policy resolves the effective consent verdict by walking the subject's latest non-superseded `consent:state:*` emission, gated by `valid_until`, and tracks producer deletion-SLA obligations on revocation."

§8.1.11.1 Effective consent resolution (read path)

For a target Contribution `T` carrying `subject_key_ids` of length ≥ 1 , the effective consent state per subject `s` $\in$ `T.subject_key_ids` is computed as:

```
resolve_consent(T, s, now):
    candidates = federation_attestations
        .where(target == T)
        .where(attesting_key_id == s OR
            attesting_key_id in delegates_to(s).proxies)
        .where(dimension matches "consent:state:*")
        .where(supersedes_id IS NULL OR replaced by latest in supersedes chain)
        .where(valid_until IS NULL OR valid_until > now)
        .order_by(asserted_at DESC)

    latest = first(candidates)
    return:
        granted if latest.dimension == "consent:state:granted"
        revoked if latest.dimension == "consent:state:revoked"
        expired if latest.dimension == "consent:state:expired" OR
            latest.valid_until passed without renewal
        unspecified if no candidates (subject named but never declared)
```

Substrate MAY cache the resolution per `(T, s)` keyed on the latest `asserted_at`; invalidate on any new `consent:*` write from `s` or `s`'s proxy chain.

§8.1.11.2 Multi-subject revocation (any-subject-binding)

When `len(T.subject_key_ids) > 1`, each subject is an **independent** revocation authority. A **withdraws** admitted under §3.2.3 rule 2 or 3 from ANY single subject in `T.subject_key_ids` evicts the Contribution. Consumer policy MUST treat `T` as revoked from the perspective of all subjects (no "majority-rules" or "all-subjects-must-agree" softening) — this is the subject-as-individual principle from MISSION.md §1.5 applied at the subject-authority layer.

Concrete cases:

- Group photo with three subjects: any one subject revokes $\rightarrow$ the photo is evicted from federation propagation.
- Group chat export with `N` participants: any one participant revokes $\rightarrow$ the export is evicted.
- Multi-party contract: any one signatory revokes $\rightarrow$ the contract Contribution is evicted (separate from the legal-validity question, which is consumer-side; the substrate just removes the wire artifact).

Producers MAY mitigate by partitioning content into per-subject Contributions (e.g., one chat-message Contribution per author, linked via `topical_relation:replies_to`) so that one subject's revocation doesn't evict another's content.

*No distinct multi-subject evict path — admission + precedence is sufficient (normative confirmation, 1.0-RC5; resolves CIRISPersist#146 Ask 4). Multi-subject eviction is fully expressed by two mechanisms already in the spec and needs ****no new primitive, dimension, or hard_case**: (1) admission** — the per-subject `withdraws` is admitted under §3.2.3 rule 2/3 exactly as a single-subject `withdraws` is (each subject in `subject_key_ids` is independently a valid revoker; the rule does not change for `len > 1`); (2) **precedence** — the latest-wins / revoke-is-terminal precedence at the consumer read path (§8.1.11.1) applies per-subject, and any one subject's terminal `withdraws` evicts `T` for all (the any-subject-binding above). There is no quorum to compute and no "evict event" to materialize separately from the admitted `withdraws` — the withdrawal IS the evict. Substrates implement this as the OR over per-subject revocation state, not as a new code path.*

§8.1.11.3 Deletion-SLA watcher (substrate emission)

When subject `s` emits `consent:state:revoked` (or an admitted `withdraws`) against target `T`, substrate watches for producer compliance:

```
watch_sla(T, s, revocation_at):
    sla = T.attestations
        .where(attesting_key_id == T.attesting_key_id)
        .where(dimension == "consent:deletion_sla:*")
        .latest()
        .extract_days()

    if sla is None:
        return # no SLA commitment; no watcher

    deadline = revocation_at + sla.days
    completion = T.attestations
        .where(attesting_key_id == T.attesting_key_id)
        .where(dimension == "consent:deletion_complete")
        .where(asserted_at > revocation_at)
        .first()

    if now > deadline and completion is None:
        emit hard_case:consent_sla_breach against T
```

The `hard_case:*` emission is the **primitive observability signal**; per §11.6 governance, LensCore composes derived detectors on top (`detection:consent:repeat_sla_breach`, etc.).

§8.1.11.4 Bilateral pair composition (PARTNERED ceremony)

For the bilateral partnership shape per §5.6.8.7 `consent_record`:

```
ratified_pair(pair_id):
    subject_half = federation_attestations
        .where(subject_kind == "consent_record")
        .where(envelope.bilateral_pair_id == pair_id)
```

```

.where(envelope.stance == "granted")
.where(envelope.subject_key_id == attesting_key_id) # subject signing for self
.first()

producer_half = federation_attestations
.where(subject_kind == "consent_record")
.where(envelope.bilateral_pair_id == pair_id)
.where(envelope.stance == "granted")
.where(envelope.target_key_id == subject_half.subject_key_id)
.where(attesting_key_id != subject_half.subject_key_id) # producer signing
.first()

return subject_half AND producer_half # both required for ratification

```

topical_relation:bilateral_pair is the open-vocab edge documenting the pair linkage (recommended, not required for ratification — bilateral_pair_id is the binding mechanism).

§8.1.11.5 Decay-protocol stage composition (CIRISAgent CEM ANONYMOUS)

For consent_records carrying decay_protocol: "ciris-agent-90day" (or any named decay path):

```

decay_state(consent_record, now):
    elapsed = now - revocation_event(consent_record).asserted_at

    walk substrate emissions on dimension 'consent:decay:*' against consent_record,
    matching the decay_protocol's stage map. CIRISAgent 90-day decay:

    elapsed < 30d -> consent:decay:identity_severed (substrate emits at elapsed=0)
    30d <= elapsed < 60d -> consent:decay:patterns_anonymized (substrate emits at elapsed=30d)
    60d <= elapsed < 90d -> (in flight; no new stage emission)
    elapsed >= 90d -> consent:decay:complete (substrate emits at elapsed=90d)

```

Per §5.6.8.6 consent:decay:{stage} is open vocab. Other decay protocols MAY name other stage sequences; substrate honors the producer's published decay_protocol string and emits stages per the protocol's stage map.

§8.1.11.6 What Policy K composes

CIRISAgent CEM stream	Policy K composition
TEMPORARY (14d default)	consent:state:granted + envelope valid_until = asserted_at + 14d + auto-renew dimension on interaction (consumer-policy concern)
PARTNERED	Bilateral pair per §8.1.11.4: subject consent:partnership_grant + producer consent:partnership_accept under same bilateral_pair_id; no valid_until
ANONYMOUS	Revocation + decay-protocol per §8.1.11.5: substrate emits stage milestones; agent honors stage-appropriate processing constraints

Per CEG's §3.4 MISSION.md layering: CIRISAgent's three streams are a **named bundle** at the consumer-policy layer. Other agents MAY compose other streams over the same wire primitives. CEG documents the canonical bundle for ecosystem coordination; CEG does not lock the bundle.

§8.1.12 Policy L — Self/family membership composition (CEG 0.7 addition)

Per CIRISRegistry#47 + §5.6.8.8 `identity_occurrence` + §5.6.8.9 `family` + §10.1.4 structural-invisibility. Composition pattern for resolving the **current membership set** of an identity's self-collective OR a family, gating the at-rest encryption flow (CIRISPersist#152) that wraps DEKs to admitted members.

Reads as: "for any `cohort_scope: self | family` Contribution, the substrate computes the current member set by walking the latest `identity_occurrence` / `family` Contributions and resolves which keys **MUST** receive a `key_grant` wrap of the content DEK."

§8.1.12.1 Self-collective resolution

For identity I, the current self-collective is computed as:

```
resolve_self(I, now):
    candidates = federation_attestations
        .where(subject_kind == "identity_occurrence")
        .where(envelope.identity_key_id == I)
        .where(supersedes chain -> latest non-superseded)
        .where(NOT withdrawn by I or by current occurrence)
        .where(envelope.valid_until IS NULL OR > now)
    return {c.envelope.occurrence_key_id for c in candidates} U {I}
```

The root I itself is implicitly a member (the `identity_key` is always an admissible signer for its own content). Single-vouch admission: any current occurrence (including I itself) may admit a new occurrence via `attesting_key_id` on a fresh `identity_occurrence` Contribution.

Concrete: Alice has admitted `alice_phone`, `alice_laptop`, `alice_agent`. Her self-collective is `{alice_root, alice_phone, alice_laptop, alice_agent}`. When Alice's phone publishes a `cohort_scope: self` Twitter scroll, the substrate wraps the content DEK under all four keys; the content reaches her laptop and agent via the at-rest encryption flow without emitting `holds_bytes:sha256:*`.

§8.1.12.2 Family membership resolution

For family F, the current member set is computed as:

```
resolve_family(F, now):
    latest = federation_attestations
        .where(subject_kind == "family")
        .where(envelope.family_key_id == F)
        .where(supersedes chain -> latest non-superseded)
        .where(admission rule satisfied per CURRENT consensus_protocol;
            see §8.1.12.3)
    return {m.key_id for m in latest.envelope.members}
```

Each `member.key_id` is itself an identity — the substrate does NOT walk into each member's `identity_occurrence` set at family-resolution time. When DEK wrapping happens, each member's `identity_key` receives a `key_grant`; the member's own substrate then composes Policy L on its own self-collective to propagate to that member's devices/agents (recursive composition).

Concrete: The Acme Household family has members `{alice_root, bob_root, roku_living_room, kitchen_tablet, nest_thermostat}`. When Alice's phone publishes a `cohort_scope: family`

dinner photo with `family_id: acme-household`, the substrate wraps the DEK under each of those 5 identity keys. Alice's `alice_root` then re-wraps to her self-collective (phone, laptop, agent); Bob's `bob_root` re-wraps to his self-collective; the Roku and kitchen tablet receive directly (single-key identities — they don't have multi-occurrence sets); the Nest thermostat same.

§8.1.12.3 Membership-change admission per consensus_protocol

A proposed membership change (addition OR removal) rides a **supersedes** Contribution on the family's latest admitted family Contribution. Substrate admission gate evaluates the **CURRENT** family's `consensus_protocol` against the signatures on the proposal:

```
admit_family_change(F, proposed: family_record):
    current = resolve_family(F, now)
    protocol = current_family_record(F).consensus_protocol
    signatures = collect_signatures_on(proposed)

    match protocol:
        "founder_only":
            return any sig from m where m.role == "founder" in current
        "unanimous":
            return every m.key_id in current has signed
        "majority":
            return count(sig from m in current) > len(current) / 2
        "quorum:M/N":
            return count(sig from m in current) >= M // M is ABSOLUTE -- see S8.1.12.3.1
        "weighted:{rubric}":
            return sum(weight(m, rubric) for m in current who signed) >= threshold
        "custom:{family_id}":
            return operator-defined predicate evaluates to true

    if admit:
        emit hard_case:family_membership_change:{F}
        trigger S8.1.12.4 key_grant cascade
    else:
        hold in pending state (per operator-policy window) OR
        emit hard_case:family_consensus_protocol_violation:{F} and reject
```

S8.1.12.3.1 'quorum:M/N' is absolute-M (normative; per CIRISRegistry#52 + NodeCore#30)

The 'M' in 'quorum:M/N' is an **absolute signature count**, NOT a fraction that rebases with roster size. A 'quorum:2/3' collective that grows to 5 members still admits at ****2**** signatures; the 'N' is documentary (records the roster size at protocol-adoption time) and is NOT recomputed against the live roster. This was an ambiguity in the pre-pin S8.1.12.3 pseudocode ("operator policy resolves rebasing"); CIRISRegistry#52's ceremony gate ([S52 design] (<https://github.com/CIRISAI/CIRISRegistry/issues/52>)) requires a deterministic, operator-independent reading, so absolute-'M' is pinned.

****Rationale****: absolute-'M' is the simpler invariant (the gate is a pure 'count >= M' with no live-roster division), it matches NodeCore's shipped 'evaluate_consensus_protocol' ([NodeCore#30] (<https://github.com/CIRISAI/CIRISNodeCore/issues/30>) commit 7469dcc) so the single shared predicate needs no change, and proportional/rebasing quorum is still expressible for operators who want it via 'weighted:{rubric}' (assign each member weight 1, set threshold = 'ceil(roster_size / 2)' recomputed by the rubric resolver). Collectives that want "M grows with the roster" therefore use 'weighted:', not 'quorum:' -- keeping 'quorum:M/N' unambiguous.

This rule applies identically to 'community' admission ([S8.1.13.2] (#81132-community-admission-per-consensus_protocol-cohort_subkind-dispatch)) -- the 'quorum:M/N' arm there is the same

absolute-‘M‘ reading.

Consensus-protocol amendment (changing `consensus_protocol` itself on a non-entrenched family) rides the SAME admission rule on the proposed amendment Contribution — meta-amendment shape parallel to §11.2.3. Entrenched families (`consensus_protocol_entrenched == true`) reject amendments at the substrate gate; replacement requires the out-of-band ceremony documented per family (for HUMANITY_ACCORD see §9.2).

§8.1.12.4 Key-grant cascade (the at-rest encryption flow)

On admission of a new `identity_occurrence` OR a family membership-add, substrate emits retroactive `key_grants` wrapping all extant `cohort_scope: self|family` content DEKs to the new member’s key:

```
on_member_added(scope_target, new_member_key):
    for each Contribution C in federation_attestations where:
        C.cohort_scope == "self" AND C.attesting_key_id in resolve_self(scope_target)
    OR
        C.cohort_scope == "family" AND C.family_id == scope_target
    AND C is still substrate-admitted (not withdrawn / not expired):
        emit key_grant {
            wrap_algorithm: X25519MlKem768Aes256GcmHkdfSha256, // v2 -- see PQC note
            recipient_key_id: new_member_key,
            content_sha256: C.evidence_refs[0].sha256,
            scope: GroupMember,
            wrapped_dek: wrap(C.dek, new_member_key.pubkey),
            ...
        }
```

****PQC at rest — `wrap_algorithm: v2` MANDATORY (normative). The self/family at-rest DEK MUST be wrapped with `wrap_algorithm: v2 = x25519_mlkem768_aes256_gcm_hkdf_sha256`**** (hybrid X25519+ML-KEM-768; the §5.6.8.4 variant X25519MlKem768Aes256GcmHkdfSha256), **never v1** (X25519-only). Self/family content is the user’s most private and longest-lived data (memory, photos, identity, the §8.1.12.7 Self DEK) — a classical-only KEM is a harvest-now-decrypt-later exposure, the identical mandate as the streaming epoch DEK (§10.5.3). The AES-256-GCM content seal is symmetric (PQC-safe); the wrap signature is hybrid Ed25519+ML-DSA-65. A Consumer **MUST** reject a self/family at-rest grant carrying `wrap_algorithm: v1`.

The cascade is the wire-format primitive for the "I got a new phone and want my Twitter history" + "I added Carol to the household for a week" flows. Operator policy **MAY** bound the cascade depth (e.g., last 90 days of self-content; opt-in for the full historical wrap).

§8.1.12.5 Forward secrecy on member removal (Option A, recommended for v1)

Per CIRISRegistry#47 Decision 1 recommendation, CEG 0.7 adopts **Option A** — once shared, always shared at the wire layer:

```
on_member_removed(scope_target, removed_member_key):
    // The removed member retains existing key_grants -- cannot retroactively un-share.
    // Substrate STOPS wrapping NEW key_grants to them on subsequent
    // cohort_scope: self|family Contributions for scope_target.
    // No DEK rotation; no re-encryption of historical content.
```


This is consistent with §11.4 "takedown isn't a coup" + §3.2 **withdraws-isn't-retroactive** semantics — historical state isn't retroactively re-keyed. Option B (rotate-DEK on removal) is deferred to a future `subject_kind: family_rotation` ceremony; CEG 0.7 documents the slot, leaves the rotation primitive for a downstream-demand-driven release.

Why Option A: aligns with the substrate's existing forward-secrecy posture (consent revocations don't retroactively un-emit per CEG 0.6 §8.1.11; takedowns don't retroactively un-emit per §11.4); matches user-intuition that "leaving the family" governs future content, not historical; bounded substrate cost (no re-wrap-all-content storm on member removal).

§8.1.12.6 Composition with CEG 0.6 `subject_key_ids[]`

`identity_occurrence` + `family` (membership / visibility scoping) compose cleanly with CEG 0.6 `subject_key_ids[]` (revocation authority):

- A `cohort_scope: family` Contribution naming `subject_key_ids: [user_canonical_hash]` is admitted at family visibility AND the named subject retains independent revocation authority per CEG 0.6 §8.1.11.
- A `cohort_scope: self` Contribution that Alice writes about Bob (Bob in `subject_key_ids`) stays in Alice's self-collective (Bob does NOT receive a `key_grant` unless Bob's identity is in Alice's self — which it isn't). Bob's subject-side revocation authority still composes: Bob CAN issue **withdraws** against Alice's Contribution (admitted per CEG 0.6 §3.2.3 rule 2 even though Bob can't access the bytes); admission emits `hard_case:consent_sla_breach` clock-start if Alice committed `consent:deletion_sla`.

The orthogonality holds: ****cohort_scope** is producer-side visibility scoping; `identity_occurrence` + `family` are substrate-side membership primitives that gate at-rest DEK wrapping; `subject_key_ids` is subject-side revocation authority**. Three independent envelope-level concerns that compose without overlap.

§8.1.12.7 The "Self at login" — `app` + `agent` co-self + `partnered` delegation (CEG 0.15, normative composition)

Per CIRISRegistry#65. The canonical user-identity composition: a person's **app** (the KMP client key) and their **agent** (CIRISAgent's local key) are two occurrences of one user identity that share one **Self DEK**, and at login the agent is **partnered** + **delegated** to act as the user on the network. **No new structural primitive** — this composes `identity_occurrence` + Policy L + `consent:partnered` + `delegates_to` + `identity_type-set` + `transport_destination`. The four implementations MUST follow this shape so a "Self" is identical everywhere.

Two layers — and they are independently revocable (the load-bearing distinction):

Layer	Mechanism	Buys	Revoked by
Co-self (visibility)	occurrence + Policy-L Self DEK	the agent can <i>read/manage the user's Self</i> (decrypts self-content)	withdraws the occurrence → Option-A re-key (§8.1.12.5)
Agency (act-on-behalf)	<code>consent:partnered</code> + <code>scoped delegates_to</code>	the app may <i>act AS the user on the network</i> (send/receive, presence, sub-delegate)	withdraws the delegation → agency ends, co-self unaffected

So a user MAY grant a device co-self (it manages their data locally) while revoking its network agency — or vice-versa — without disturbing the other.

Agency at login (the partnering + delegation).

1. **Partnering** — the user emits `consent:partnership_grant` and the agent occurrence emits `consent:partnership_accept` under one `bilateral_pair_id` (§8.1.11.4 PARTNERED); the bilateral pair is the persistent, auditable relationship.
2. **Delegation** — the user identity emits `delegates_to` against the **agent occurrence key**, with `delegated_scope` drawn from the canonical act-on-behalf kinds below, `delegation_purpose`: "act_as_user", bounded `delegation_valid_until`. Sub-delegation works because `delegates_to` chains (depth-capped at 5 per §13.3).
3. **This grant is FEDERATION-tier (§10.1.5), not local** — *other peers must verify the agent's authority before honoring its messages/presence*, so the partnering+delegation is signed + promoted at login. **Promotion is the "app shows up on the network" moment.** (The agent's own self-content stays local-tier; only the act-on-behalf authorization federates.)

****Canonical `delegated_scope` kinds for act-on-behalf**** (recommended; open-vocab per §11.2.1, named for ecosystem coordination):

scope	grants the agent
<code>act_on_behalf</code>	umbrella: emit Contributions AS the user (<code>attesting_key_id</code> = agent occurrence, speaking as the user identity per §5.6.8.8)
<code>message_io</code>	send + receive directed messages on the user's behalf
<code>network_presence</code>	announce/resolve the user's <code>transport_destination</code> (§5.6.8.8.1) — be reachable AS the user
<code>sub_delegation</code>	issue further <code>delegates_to</code> within the granted scope (depth-capped)

Partnership WITHOUT agency — the infrastructure delegation profile (normative, 1.0-RC7 — resolves CIRISRegistry#83 §3). The flow above binds an **agent** (a key with a brain) to a user as partnership + **agency** — the scope includes `act_on_behalf` / `message_io`, so the agent reasons and acts AS the user. A **fabric/infrastructure node** (§7.0.1; CIRISServer) needs the *partnership* (identity + the §5.6.8.10 owner-binding that lets it hold non-infra membership standing under the user's authority) but **MUST NOT** receive agency — §1.3 "infrastructure must not have agency." CEG pins a **reserved two-prefix scope split** so a verifier can enforce this cryptographically:

prefix	class	scopes
<code>infra:*</code>	server-class (allowed for a <code>node-role</code> delegate)	<code>infra:network_presence</code> , <code>infra:join_communities</code> , <code>infra:serve</code> , <code>infra:store</code> , <code>infra:attest</code> , <code>infra:transport</code>
<code>agency:*</code>	brain-only (forbidden for a pure <code>node-role</code> delegate)	<code>agency:act_on_behalf</code> , <code>agency:message_io</code> , <code>agency:reason</code> , <code>agency:decide</code>

Conformance: a `delegates_to` whose `attested_key_id` resolves to an identity whose `identity_type` (§7.0.1) is `node-only` (no `agent/brain`) **MUST** carry **only** `infra:*` scopes; a verifier **MUST reject** (treat as non-conformant, never grant) an `infra-only` key presenting any `agency:*` scope. This makes §1.3 a wire-checkable invariant: a user-owned fabric node can serve + hold group-membership *standing* under the user's authority, but the delegation **literally cannot carry agency**. The legacy unprefix kinds above remain valid for `agent-role` delegates (the Self-at-login agency profile) and are the `agency:*` / `infra:network_presence` equivalents; new `infra` delegations **SHOULD** use the explicit `infra:*` prefixes.

****Cohabitation** (`agent = node + brain`): **when both compose in one process, the node holds** `partnership-without-agency` (`infra:*` — identity + membership standing) and the brain layers **Self-at-login partnership-with-agency** (`agency:*` — reasoning) as a *separate delegates_to*. Two delegations, two scope classes, independently revocable — the user can strip the brain's agency while the fabric node keeps serving.

Transport (network presence) — AV-17. Each occurrence binds a `transport_destination` (§5.6.8.8.1); the app is reachable on RET *as the user occurrence*. The Reticulum destination is a **separate dual-key transport identity** that the user's signing key *authorizes by signing the binding* — the federation signing seed **MUST NOT** enter the transport layer (AV-17 / CIRISEdge#15). "User key used as a transport key" means *roots/authorizes* the transport identity, not a shared keypair.

Worked login flow. (1) unlock the hardware-rooted user key (WebAuthn presence) → (2) admit the agent occurrence (single-vouch) → Policy-L Self DEK now wraps to both → (3) optionally add the `wise_authority` role to the user's `identity_type` set → (4) bind each occurrence's `transport_destination` → (5) `consent:partnership_grant/accept` under a `bilateral_pair_id` → (6) `delegates_to(user → agent occurrence, scope: [act_on_behalf, message_io, network_presence, sub_delegation])`, **promoted to federation-tier**. The app now reads the user's Self locally AND acts as the user on the network; the user can revoke either layer independently.

§8.1.12.7.1 Signing member sets (normative — the JCS contract Verify hybrid-signs; resolves CIRISVerify#63)

Each of the three Self-at-login Contributions is hybrid-signed over JCS(envelope) (§0.9 / RFC 8785), and at login promoted to federation-tier (the §10.1.5.3 promotion canonicalizes the **exact committed member set** — omit-vs-materialize (§0.9.2) is load-bearing; the signer **MUST NOT** re-default). **The §0.9.2.1 determinism rules apply** (raised in CIRISVerify#63 review): `subject_key_ids[]` and `delegated_scope[]` are **lexicographically sorted** (set-semantics); `aspects[]` retains RNS order (sequence-semantics); all key/hash/pubkey byte fields (`*_key_id`, `subject_key_ids[]`, the two reticulum pubkeys, `destination_hash`) are **lowercase hex per §0.6**; all timestamps are **§0.5-canonical**. The member sets the producer commits (and which JCS therefore covers) are pinned below. Optional §4 envelope fields not listed ride the §0.9.2 omit rule (absent unless the producer sets them).

****`(a) consent:partnership_grant:v1` (user side) / `consent:partnership_accept:v1` (agent side)**** — bare scores (§5.6.8.6) bound by `bilateral_pair_id`:

```
{ attestation_type: "scores",
  attesting_key_id: <user identity_key_id (grant) | agent occurrence_key_id (accept)>,
```

```
dimension: "consent:partnership_grant:v1" | "consent:partnership_accept:v1",
score: <positive>,
subject_key_ids: [<the partner key: agent occurrence (grant) | user identity (accept)>],
bilateral_pair_id:<shared pair id>, // S8.1.11.4 binding mechanism
signed_at: <rfc3339_canonical> }
```

Version segment pinned (normative, 1.0-RC5 — resolves CIRISRegistry#78 caveat 2). The dimension carries the :v1 version segment — `consent:partnership_grant:v1` / `consent:partnership_accept:v1` — to satisfy the §13.1 scores version-segment gate. This is the canonical form persist v6.5.0 shipped; confirmed, not changed. The :v1 is the partnership-ceremony schema version (bump to :v2 only if the bilateral shape changes); the shared `bilateral_pair_id` remains the §8.1.11.4 binding mechanism.

****Canonical signed member set for the two :v1 envelopes (normative, 1.0-RC7 — resolves CIRISRegistry#81 / CIRISVerify#63).** Both impls **MUST canonicalize (and thus hybrid-sign)** exactly** this member set, or the JCS bytes — and the signatures — diverge. The set is the §5.6.8.6 bare-scores shape; these seven members are **REQUIRED** (present in the JCS for both grant and accept):

Member	Value	Verify#63 name
attestation_type	literal "scores"	—
attesting_key_id	the signer = granter_key_id (grant) / acceptor_key_id (accept); the bound sig binds because signer ≡ attesting_key_id	granter_key_id / acceptor_key_id
dimension	literal "consent:partnership_grant:v1" \	"consent:partnership_accept:v1"
score	positive (the affirmation)	—
subject_key_ids	**exactly [partner_key_id]** — the OTHER party (agent occurrence for grant, user identity for accept); single-element, §0.9.2.1 set-sort is trivial	partner_key_id
bilateral_pair_id	the shared join (§8.1.11.4) — identical string on both halves	bilateral_pair_id
signed_at	§0.5-canonical RFC 3339	timestamp

Mapping (so the two impls agree on naming): granter/accepter ≡ attesting_key_id (the signer of that half); partner ≡ subject_key_ids[0]. ****No valid_until — a PARTNERED pair has no expiry (§8.1.11.4); the field is omitted**** (NOT materialized as null), per the §0.9.2 omit rule. **All other §4 envelope fields ride the omit rule** — absent from the JCS unless the producer explicitly sets them; the signer **MUST NOT** re-default. Additional canonical members are a :v2 bump, never a silent :v1 addition. Byte-field members (attesting_key_id, subject_key_ids[]) are lowercase-hex per §0.6.

**** (b) delegates_to (user → agent occurrence) **** — the act-on-behalf grant (§3.2 envelope shape):

```
{ attestation_type: "delegates_to",
  attesting_key_id: <user identity_key_id>,
  attested_key_id: <agent occurrence_key_id>, // the delegate
  delegated_scope: ["act_on_behalf", ...], // S8.1.12.7 canonical kinds
```

```
delegation_purpose: "act_as_user",
delegation_valid_from:<rfc3339_canonical>,
delegation_valid_until:<rfc3339_canonical>,
signed_at: <rfc3339_canonical> }
```

**** (c) transport_destination binding**** — an **identity_occurrence** (§5.6.8.8 / §5.6.8.8.1) carrying the binding; signed by **identity_key_id** (or a current occurrence), AV-17:

```
{ attestation_type: "scores",
  subject_kind: "identity_occurrence", // payload discriminator S4.2.2.3
  attesting_key_id: <user identity_key_id | a current occurrence>,
  identity_key_id: <user identity_key_id>,
  occurrence_key_id:<the occurrence being bound>,
  device_class: "phone" | "laptop" | "agent" | ...,
  transport_destination: {
    reticulum_x25519_pubkey: <[u8;32]>,
    reticulum_ed25519_pubkey: <[u8;32]>,
    destination_hash: <[u8;16]>, // MUST derive per S5.6.8.8.1
    app_name: <string>,
    aspects: [<string>, ...] }, // ordered
  asserted_at: <rfc3339_canonical>,
  signed_at: <rfc3339_canonical> }
```

Registry owns these member sets (this section); Verify computes `JCS(...)` + the hybrid Ed25519+ML-DSA-65 signature over each via `jcs::canonicalize` (CIRISVerify#59); the promotion signature (§10.1.5.3 OQ-4) is the identical JCS bytes — confirmed.

**** encryption_pubkeys joins member set (c) (1.0-RC1 — CIRISVerify#64).**** When the occurrence carries the §5.6.8.8.2 **encryption_pubkeys** field-set, both halves are **inside the signed JCS bytes** as opaque base64 strings (RFC 4648 STANDARD, padded — the §0.9.2.1 rule-2 pin). Optional presence rides the §0.9.2 omit rule (absent unless the producer sets it; the signer **MUST NOT** re-default). They are payload, never verification material — neither half may be fed to a signature-verify path (the §5.6.8.8.2 key-separation rule, type-enforced in Verify).

§8.1.13 Policy M — Community membership composition

Per CIRISRegistry#48 + §5.6.8.10 **community** + §5.6.8.11 **location_proof**. Composition pattern for resolving the **current membership set** of a community, gating cohort-filtered visibility for **cohort_scope: community** content.

Sibling to §8.1.12 Policy L (self/family) but with different defaults — community content is encrypted under a per-community DEK + emits **holds_bytes:sha256:*** with cleartext provenance (CEG 0.17 — see §8.1.13.3); the privacy property is byte-confidential-to-members, not cohort-filtered-visibility.

§8.1.13.1 Community membership resolution

For community **C**, the current member set is computed as:

```
resolve_community(C, now):
  latest = federation_attestations
    .where(subject_kind == "community")
    .where(envelope.community_key_id == C)
    .where(supersedes chain -> latest non-superseded)
    .where(admission rule satisfied per CURRENT consensus_protocol;
```

```

    see S8.1.13.2)
    return {m.key_id for m in latest.envelope.members}

```

Same shape as `resolve_family` (§8.1.12.2). Each `member.key_id` is an identity (which may itself have a multi-occurrence set via CEG 0.7 `identity_occurrence`).

§8.1.13.1.1 Deterministic resolution + member→address resolution (NORMATIVE)

In a CEG/RET stack member resolution replaces DNS+IP: it is a chain of *signed* bindings, and every implementation **MUST** resolve **identically** (a 1.0 interop requirement). Two normative pins:

(a) Determinism of `resolve_community`. Where the §8.1.13.1 computation has choices, they are fixed:

- ****now semantics**** — the resolving Consumer’s own clock; membership is evaluated as of **now** (a member is included iff `joined ≤ now` and not `removed ≤ now`). No global clock assumed.
- **"latest non-superseded"** — the community Contribution with the highest `signed_at`; on equal `signed_at`, the higher `canonical_bytes_hash` (§0.9) wins (total order, no ambiguity). The same comparator the CIRISVerify#49 R1/Q1 merge uses (`quorum_weight DESC → signed_timestamp DESC → canonical_bytes_hash`).
- **member ordering** — the returned set is canonically ordered by `key_id` (lowercase-hex byte order) for any downstream hashing/iteration.
- **founder-subset eval** — for `cohort_subkind: infrastructure` (§5.6.8.10) admission is `evaluate_consensus_protocol` over `{m : m.role == founder}`, NOT all members.

(b) Member → reachable address (the DNS-free resolution). Resolving a member identity to a Reticulum destination:

```

resolve_member_transport(C, now):
    members = resolve_community(C, now) // (a) above; founder-quorum verified
    out = []
    for M in members (canonical key_id order):
        occ = latest non-superseded identity_occurrence of M
            with transport_destination present, valid at now,
            hybrid-sig verified against M (or an occurrence of M) // S5.6.8.8.1
        if occ is null: continue // no authenticated binding -> skip (fail-secure)
        td = occ.transport_destination
        REQUIRE td.destination_hash == rns_destination_hash( // S5.6.8.8.1.1 pinned two-stage
            td.reticulum_x25519_pubkey, td.reticulum_ed25519_pubkey,
            td.app_name, td.aspects) // NOT a flat-concat SHA-256
        out.push((M, td.destination_hash))
    return out // -> Reticulum announce/path-request per dest

```

The three resolutions and their trust roots: **WHO** = `resolve_community` (signed Contribution + founder-quorum) · **BINDING** = `transport_destination` (federation-key-signed `identity_occurrence`, §5.6.8.8.1) · **WHERE** = Reticulum announce/path-request (mesh, no CEG). Reachability is never trust (§10.5.6): a path that answers is not an authorization; only the signed binding + quorum are.

Cold-start (bootstrap). A node that knows only `community_key_id` needs two out-of-band pins: (1) the `community_key_id` itself (trust anchor) and (2) ≥ 1 `**seed destination_hash**` (reachability). It reaches any one seed, pulls the signed `community` Contribution, runs `resolve_member_transpo`, verifies founder-quorum against the pinned `community_key_id`, then floats on the live set thereafter. A malicious seed cannot forge membership (quorum signature check) — it can only deny service (try another seed). **Bootstrap reachability $\neq$ bootstrap trust.** `ciris-canonical` (§5.6.8.10) is the root community whose seeds clients pin.

§8.1.13.2 Community admission per `consensus_protocol` + `cohort_subkind`

A proposed membership change rides a **supersedes** Contribution on the community’s latest admitted `community` Contribution. Substrate evaluates the `CURRENT` community’s `consensus_protocol` (same six canonical kinds as family) **AND** any subkind-specific admission requirements:

```
admit_community_change(C, proposed: community_record):
    current = resolve_community(C, now)
    protocol = current_community_record(C).consensus_protocol
    subkind = current_community_record(C).cohort_subkind

    signatures_ok = evaluate_consensus_protocol(protocol, current, proposed.signatures)
    if not signatures_ok:
        emit hard_case:community_consensus_protocol_violation:{C}
        reject

    subkind_ok = evaluate_subkind_admission(subkind, current, proposed)
    if not subkind_ok:
        // For geographic: subkind admission failed (e.g., new member’s location_proof
        // not contained in geographic_constraint, or expired location_proof)
        emit hard_case:community_consensus_protocol_violation:{C} // same observability prefix
        reject

    admit
    emit hard_case:community_membership_change:{C}
```

The `evaluate_subkind_admission` predicate dispatches per `cohort_subkind`:

```
evaluate_subkind_admission(subkind, current, proposed):
    match subkind:
        "geographic":
            # Per S5.6.8.10 geographic admission requirement
            for each NEW member in proposed.members (not in current):
                their_proof = latest valid location_proof from new_member.key_id
                if their_proof IS NULL:
                    return false // no location_proof on file
                if their_proof.cell_id NOT contained in current.geographic_constraint.cell_id:
                    return false // location outside community’s geographic bound
                if their_proof.valid_until passed:
                    return false // expired
            return true
        -:
            return true // unknown subkinds admit on consensus_protocol alone
```

Operator vocabularies extending `cohort_subkind` provide their own `evaluate_subkind_admission` predicates per the `custom:{id}` `consensus_protocol` hook pattern from CEG 0.7 §8.1.12.3.

§8.1.13.3 The three crypto tiers + the Community DEK cascade (normative — supersedes the 0.8 "no cascade" reasoning)

CEG ≤ 0.16 drew a binary cut (self/family encrypt; everything else plaintext), which collapsed a bounded **Community** and the unbounded **Commons** into one bucket and left **no cryptographic home for a persecuted community**. The 0.8 "no at-rest cascade" reasoning ("per-member DEK wrap on every emission would be infeasible") was a **wrong premise** — corrected here. CEG 0.17 draws the line at **"does it have a bounded membership roster?"** — yes $\rightarrow$ encrypt, no $\rightarrow$ plaintext:

Tier	cohort_scope	At-rest	Wire discovery	Reader
self / family	self, family	encrypted, per-write DEK	none (§10.1.4 structural invisibility)	occurrences / family members
Community	community, affiliations	encrypted under the community DEK	holds_bytes:* + cleartext provenance	community members (DEK cascade)
Commons	species, biosphere, federation	plaintext	holds_bytes:*	anyone

Holder-inspectability principle (normative rationale). Any data a host holds above local tier **MUST** support an informed keep/evict decision: either the holder **inspects the bytes** (Commons — plaintext, maximally inspectable, hence the *preferred* social-distribution mechanism) **or** it **inspects the provenance** (Community — the cleartext `attesting_key_id` + `community_id` + reason on an otherwise-encrypted blob) and chooses to hold opaque ciphertext for a community it trusts. **Nothing above local tier is ever a forced, unattributable opaque blob.** This is *why* the split is shaped this way and what the eviction rules (persist `EvictionSweeper` / `evict_actor`) enforce.

****The infrastructure exception (Open-Q1 ruling, normative).**** A community with `cohort_subkind: infrastructure` (§5.6.8.10) — `ciris-canonical` and any governance/trust root — ****opts OUT** of the mandatory DEK cascade and is Commons-tier (plaintext, `holds_bytes:*`, no DEK). **The trust root cannot be an opaque blob; its entire purpose is public auditability — transparency-seeking, not privacy-seeking.** Node $\rightarrow$ canonical traces^{**} (conformance / `registry_consensus` emissions to a governance community) are therefore `cohort_scope: federation` (Commons/plaintext, world-readable) — a node enrolling in governance is thereby told its conformance traces are public (Open-Q2 ruling).

§8.1.13.4 Forward secrecy on community member removal (Option A — now applies)

With a community DEK, community **does** have the removed-member-can-still-decrypt concern (the 0.8 "no DEK to retain" reasoning is superseded by §8.1.13.3). The §8.1.12.5 **Option A** discipline applies identically: on member removal the substrate **rotates the community DEK** (§11.7.1); subsequent emissions are sealed under a DEK the removed member doesn't have. "Once shared, always shared" forward-only — content the removed member already received during membership stays in their cache (no PCS); they receive no NEW community content post-removal.

§8.1.13.5 Geographic-community admission flow (worked example)


```

1. Alice publishes a location_proof:
  subject_kind: location_proof
  subject_key_id: alice_root_key
  cell_id: "872830828ffffff" // H3 res 7, ~5 km^2, downtown Austin
  cell_resolution: 7
  asserted_at: now
  valid_until: now + 30 days
  cohort_scope: federation // public; the disclosure IS the opt-in

  Substrate admits (resolution 7 <= 7 OK). Emission federates.

2. Alice proposes joining Austin community:
  subject_kind: community
  community_key_id: austin-community
  supersedes: prior_austin_community_record_id
  members: [...existing..., {key_id: alice_root_key, joined_at: now, role: member}]
  signatures: [...current_member_sigs] // satisfies majority rule

3. Substrate admission gate (per S8.1.13.2):
  - signatures_ok: majority rule satisfied v
  - subkind: "geographic" -> evaluate_subkind_admission:
    - alice_root_key's latest valid location_proof exists
    - cell_id "872830828ffffff" CONTAINED in "85283473ffffff" (austin metro, res 5)
      via S0.8.2 containment (res 7 >= res 5; parent-walk reaches the constraint)
    - valid_until in the future
    - subkind_ok v
  - admit + emit hard_case:community_membership_change:austin-community

4. Alice now receives cohort-filtered visibility for cohort_scope:community content
  with community_id: austin-community. No key_grant cascade (community is unencrypted);
  no at-rest wrap; substrate emits holds_bytes:* for community content per status quo.

```

§8.1.13.6 Composition with CEG 0.6 + 0.7

Communities compose with consent (CEG 0.6) and self-collectives (CEG 0.7) cleanly:

- A community member who is also a CIRISAgent-using individual has an `identity_occurrence` set (CEG 0.7); community content arriving at the member's `identity_key` is then propagated to their occurrences via Policy L's at-rest cascade (if the member's local substrate chose to re-emit the community content at `cohort_scope: self` for cross-device sync; otherwise community content stays at the `cohort_scope: community` visibility on each device the member uses to query)
- A community member who is also a subject in `subject_key_ids` of a community-scoped Contribution retains revocation authority per CEG 0.6 §3.2.3 rule 2; the orthogonality between `cohort_scope` (visibility) and `subject_key_ids` (revocability) holds at community scope same as at family scope
- A geographic community whose `geographic_constraint` covers a region that overlaps a family's at-home location does NOT cross-contaminate: families are not auto-admitted to communities; communities are not auto-admitted to families. Each membership is explicit, ceremony-shaped, and independent.

§8.1.13.7 Delivery extension — `delivery_mode` × Policy M

Per CIRISRegistry#44 absorbed + CIRISLensCore#857. Policy M's community-membership composition extends to govern the **delivery axis** (**delivery_mode** envelope field per §4). The subscriber-set for any push-delivery flow IS a **community** Contribution; "subscribe = join the community"; inherits revocation, consensus, and structural-invisibility from Policy M unchanged.

Subscriber-set composition:

```
For a Contribution C with delivery_mode = push AND cohort_scope = community:
    subscriber_set(C) = resolve_community(C.community_id, now)
                        per S8.1.13.1 community membership resolution

The community is admitted under the standard Policy M consensus_protocol
options (founder_only / unanimous / majority / quorum:M/N / weighted /
custom) per S8.1.13.2. Subscription-specific admission semantics --
producer_gated (publisher approves each member) vs open (anyone can
join) -- ride the consensus_protocol choice:

    producer_gated -> consensus_protocol = "founder_only" with the
                      publisher as sole founder; the publisher authorizes
                      each new subscriber
    open -> consensus_protocol = "open:self_admit" (open-vocab
                      extension hook) where any subject signs their own
                      admission Contribution
```

Cardinality unified across observer-share and multicast:

Cardinality	Per-subscriber crypto	Epoch handling
N=1 (observer-share)	Single key_grant (§5.6.8.4); no epoch needed (one recipient, one DEK)	No epoch — single grant, single DEK; revocation = withdraws against the grant
N>1 (multicast)	Flat per-epoch key_grant cascade — one grant per (subscriber, epoch); the stream-epoch DEK seals content O(1), the cascade distributes the 32-byte epoch key O(N)/epoch per §10.5.3	Per-(stream_id , epoch) axis; epoch rolls on member removal (mandatory) + time/bytes (optional) per §10.5.3 D3

The same Policy M machinery handles both cardinalities — the difference is purely the cascade fan-out factor.

****Composition with delivery_mode: pull (RC1 default)**:**

For **delivery_mode: pull**, subscribers discover via the standard **holds_bytes:sha256:*** directory per §10.1. Policy M still resolves the community for visibility-filtering (consumer reads **community_id** envelope field, walks the membership, filters out non-member peers from the discovery surface). No fan-out push; no **delivery_receipt** emission required (best-effort).

****Composition with delivery_mode: push (RC1 substrate-pending; live in 1.x)**:**

For **delivery_mode: push**, the substrate fans out to **entitled** $\wedge$ **reachable** per §10.5.6 D6 liveness invariant. Entitlement = Policy M membership resolution (durable, signed CEG, replicated, logged); reachability = Edge **reachability.rs** (CIRISEdge#29) node-local presence tracker (TTL sec/min, NEVER an attestation, never replicated, never logged). Missed (entitled-but-unreachable) members fall back to pull on reconnect per §10.5.6.

****history_on_join $\times$ Policy M membership additions**:**

On a new-member admission via Policy M, the new member's **history_on_join** envelope value determines retroactive content delivery:

- **from_join** (default) — new member receives current-epoch content forward only; no retroactive **key_grant** cascade
- **full** — new member receives **key_grants** for prior epochs subject to the §10.5.3 P4 catch-up bound ($\min(\text{operator depth cap}, \text{chunk-eviction horizon})$); evicted-epoch grants return **ContentMiss** per the **MISSION.md** fail-honest invariant

This composes with §8.1.12.5 Option A forward-secrecy: removed members retain extant **key_grants** for content they were entitled to during membership; new members may or may not get retroactive grants per **history_on_join**. The substrate’s forward-secrecy posture is uniform across consent, takedown, membership-departure, AND delivery-onboarding surfaces.

§8.2 Aggregation semantics — opinionated defaults

Per **dimension+attested_key_id**, the verdict is computed by composing attestations under the chosen policy. Default aggregation by polarity column (§5):

Polarity column value	Default aggregation
signed	Mean of score × confidence across attesters
boolean-via-score	Min (any negative trumps positive — fail-secure for hard constraints like prohibited:* , attestation:1*)
+1.0 only / positive-only	Max across attesters (any positive is conclusive)
-1.0 only	Min across attesters (any negative is conclusive)
enumerated	Most-recent by signed_at from the attester(s) authorized to emit per §7
Detector dimensions (detection:correlated_action:* , detection:distributive:access:* , ratchet:flag:*)	Median across attesters (resists adversarial mean-pulling by a single captured detector)

Specific dimensions override via consumer policy; the defaults above are the §0.2 CEG-Conforming Consumer (CCC) minimum.

§8.3 Frickarian discipline — consumer-policy norms

Per Miranda Fricker’s *epistemic injustice*: consumers SHOULD apply identity-prejudice-resistant weighting. Concretely:

- Don’t downweight **testimonial_witness:*** from cohorts with low overall attestation density (testimonial preservation is precisely what corrects for that low density).
- Don’t downweight **non_maleficence:*** claims about a partner just because the partner has a long **partner_role:*** track record (the long track record may be the harm).
- Apply §8.1.4 lexical-vulnerability-priority in tie-breaks involving small cohorts.

Adversarial caveat: the discipline above is consumer-policy-only; an adversary can emit **testimonial_witness:*** exploiting the Frickarian non-downweighting rule. Per §5.6.3, **testimonial_witness:*** is never sole evidence for **slashing:***; per §7.0 the consumer MUST also weight **witness_relation:self** claims against the attester’s other-emission track record. The Frickarian rule applies AFTER these structural safeguards, not before them.

§8.4 Sovereign-Registered equivalence (wire-symmetric, policy-differentiated)

A Sovereign agent scoring `licensure:CA_medical_board: +1.0` is wire-format identical to a Registry-steward scoring the same. Consumer policy weights by attester source; the substrate is source-neutral. M-1's symmetry is structural, not bolted on.

Per `../MISSION.md §1.1`: both paths produce federation membership; neither is a gate. What differs is the *attestation surface* — the kind of claim the federation can compose about why a participant is trustworthy.

§9 The HUMANITY__ACCORD constitutional layer

The single wire-format asymmetry in the federation.

§9.1 The accord-holder triple

Three named human key holders. Initial state at federation genesis:

Position	Holder	Threshold
1	Eric Moore	2-of-3
2	Eric Kudzin	2-of-3
3	Haley Bradley	2-of-3

Hardware-attested (per §9.4 hardware_class taxonomy). Permanent: no automatic decay; replacement requires out-of-band CIRIS L3C process per FEDERATION__ANNOUNCEMENT.md §4.5.3.

Correlated-failure geometry (named honestly — 1.0-RC1, #71 C5): two of the three holders share a household, so the 2-of-3 quorum is physically achievable from one street address — a correlated compromise/coercion surface that entrenchment makes harder to correct later. The authority at stake is the **full constitutional kill** (EmergencyShutdown CONSTITUTIONAL — not a recoverable pause), so the exposure is real and is not softened here; what scope isolation (§9.2) does guarantee is that compromise cannot escalate *beyond* the kill — accord keys cannot sign grants, licenses, or amendments. **The mitigant is diversifying the holder set: finding new holders** (via the out-of-band replacement process, FEDERATION__ANNOUNCEMENT.md §4.5.3) so that no household — and ultimately no single jurisdiction — can assemble the quorum. This is an active obligation on CIRIS L3C, not a deferred nice-to-have.

****The HUMANITY__ACCORD triple is the canonical entrenched-family instance (CEG 0.7 retcon).**** Per §5.6.8.9, the accord-holder triple structurally IS a **family** subject_kind with:

```
family {
  family_key_id: "humanity-accord",
  family_name: "Humanity Accord",
  members: [
    {key_id: <eric-moore-key>, role: founder},
    {key_id: <eric-kudzin-key>, role: founder},
    {key_id: <haley-bradley-key>, role: founder}
  ],
  consensus_protocol: "quorum:2/3",
  consensus_protocol_entrenched: true // replacement requires S9.2 / FEDERATION__ANNOUNCEMENT.
    md S4.5.3 ceremony
}
```

The 2-of-3 multi-sig verifier at §9.2.1 is the quorum:2/3 consensus_protocol enforcement; the entrenchment property is what prevents any federation-internal authority from amending the protocol. §9 remains load-bearing for the **role-recognition policy** (which dimensions accord-holders may emit — only accord:* per §7.1) and the **scope-isolation** discipline (only EmergencyShutdown CONSTITUTIONAL per §9.2). CEG 0.7 makes the structural shape explicit — the constitutional asymmetry is "an entrenched family that is wire-scope-isolated to halt authority," not a one-off primitive. Other entrenched-family instances (a national-emergency triple, an international body, a court-ordered preservation triple) MAY appear in operator deployments; HUMANITY__ACCORD is the one CIRIS L3C deployments ship at genesis.

§9.2 Authority scope

HUMANITY_ACCORD signatures are valid only on EmergencyShutdown CONSTITUTIONAL (IncidentSeverity::INC = 5), accord:invoke:notify:{notify_id}, accord:invoke:drill:{drill_id}, accord:lifecycle:active, and the corresponding FederationAnnouncement priority AccordCarrier. Announcements of any other priority signed by accord-holder keys are rejected at admission (out of role). Federation-side authority cannot sign AccordCarrier; humanity-accord authority cannot sign anything else. **Wire-isolated AND scope-isolated.**

§9.2.1 Invocation canonical bytes (anti-replay; 0.1 scaffold)

***RESOLVED at 1.0-RC1:** the discriminator + nonce binding below addresses the cross-invocation-replay hole identified by CEG 0.1 cryptographic + red-team review. The anticipated §5.2.1 refinement landed as the **JCS redesign** (TupleHash128 retired — see §5.2.1); this invocation encoding is **intentionally NOT migrated**: its preimage is closed-vocabulary (discriminator + nonce + enum fields, no attacker-controlled free text), so the injection surface the §5.2.1 redesign closes is not reachable here, and genesis-critical bytes stay stable.*

Every `accord:invoke:*` Contribution signs the following canonical bytes (BOTH the discriminator AND a per-invocation nonce are in the signed payload — preventing CONSTITUTIONAL ↔ notify ↔ drill cross-replay):

```
canonical = sha256(
  "ciris.accord_invoke.v1\n" ||
  "invocation_kind=" || ("CONSTITUTIONAL" | "notify" | "drill") || "\n" ||
  "invocation_id=" || halt_id_or_notify_id_or_drill_id || "\n" ||
  "nonce=" || base64url(rand_32_bytes) || "\n" ||
  "asserted_at=" || rfc3339_canonical || "\n" || // per S0.5
  "valid_until=" || rfc3339_canonical || "\n" ||
  "payload_sha256=" || sha256_hex_lowercase_of_payload // per S0.6
)
```

Hybrid signature per §5.2.1: Ed25519 + ML-DSA-65 bound-payload. Each of the 2-of-3 holders signs `canonical` independently; consumer verifies all three signatures against the same `canonical` bytes and counts ≥ 2 valid.

The substrate **MUST** reject duplicate `invocation_id` values within the `valid_until` window (per-kind unique).

§9.2.2 notify vs CONSTITUTIONAL — consumer-UI requirement

A CEG-Conforming Consumer (CCC) presenting accord invocations to humans **MUST** visually distinguish the three kinds:

- ****CONSTITUTIONAL**** — kill-switch authority; full halt; visible as an unambiguous emergency banner.
- ****notify**** — federation-wide accord-holder communication; **MUST NOT** be visually conflated with CONSTITUTIONAL.
- ****drill**** — accord-holder exercise; **MUST** be visually marked as a drill (e.g., explicit "[DRILL]" prefix on any human-visible surface).

Wire-format isolation alone does not close the social-engineering risk that downstream UI conflates the three; the consumer-UI requirement above is the load-bearing safeguard against accord-holders being socially pressured into emitting a `notify` that carries CONSTITUTIONAL social weight without CONSTITUTIONAL substrate weight.

§9.3 Concern split — key material vs role-recognition policy

Key material (Ed25519 + ML-DSA-65 pubkeys for the three holders) lives in **CIRISPersist** **substrate**: `federation_keys` rows with `identity_type="accord_holder"`, self-signed at provisioning, cross-attested by all three regional stewards.

Role-recognition policy + verifier logic lives in `**ciris-registry-core**`: the 2-of-3 multi-sig verification, the `EmergencyShutdown` CONSTITUTIONAL admin RPC, the audit hooks.

§9.4 Hardware-class taxonomy

Value	Use
<code>HSM_FIPS_140_3_L3</code>	Production stewards (US / EU / APAC)
<code>Apple_Secure_Enclave</code>	Accord-holders on iOS/macOS
<code>YubiKey_5_FIPS</code>	Accord-holders preferring portable hardware tokens
<code>TPM_2_0</code>	Accord-holders on Linux/Windows desktops
<code>placeholder_pending_provisioning</code>	Interim value before actual hardware provisioning. Consumers MUST treat as 0.0 trust weight
<code>software_hsm_development</code>	DEVELOPMENT ONLY; consumer policy MUST reject for federation-scope verification

Per-class recommended trust-multipliers: `HSM_FIPS_140_3_L3` = 1.0; `Apple_Secure_Enclave` = 0.95; `YubiKey_5_FIPS` = 0.95; `TPM_2_0` = 0.9; `placeholder_pending_provisioning` = 0.0; `software_hsm_development` = 0.0.

§9.4.1 Hardware-class self-assertion gap (acknowledged)

The `hardware_class` field is currently a self-asserted string on each `federation_keys` row. CEG 0.1 has no normative mechanism (TPM quote chain, Apple attestation, FIDO attestation) for a verifier to independently corroborate the claim. Per §15.2 **R5** (acknowledged risk): consumer policy MUST treat the `hardware_class` field as a producer claim, not a cryptographically-attested fact. A planned 0.x → 1.x roadmap item closes this via per-platform attestation-chain verification; until then the trust-multipliers in §9.4 above bind only as guidance.

§9.5 Why this isn't a Golden-Rule violation

Per §1.5: the Recursive Golden Rule binds *participants in the federation* to each other. Humanity-as-such occupies a position outside the federation's participant set, by design. The three named human key holders hold `AccordCarrier` authority that no federation-side authority class (including `SYSTEM_ADMIN` / `WISE_AUTHORITY` / per-install stewards) can grant itself, revoke, override, or decay. This is not a Golden-Rule exemption; it is the recognition that consent (M-1's load-bearing property) requires revocability, and revocability requires a halt-authority that lives

outside the system being halted. The federation cannot deny humans the right to halt it, because no federation-internal protocol path to that signature exists.

§10 Endpoint shapes

CEG specifies five public + one admin HTTP endpoint shape for the discovery + cosigning surfaces. Wire format consumers (CIRISVerify v3.1.0+, CIRISAgent KMP UI, iOS/Android FFI) read these.

§10.0 Common response shape

All CEG endpoints return:

- **Content-Type:** `application/json` (Accept: `application/json` honored; other types respond 406 Not Acceptable)
- **CEG-API-Version header:** `CEG-Version:` `<current spec major.minor>` on every response (track the `README Version:` field; currently `1.0-rc16`); clients **SHOULD** echo `CEG-Accept-Version:` `<pinned-version>` on request, naming the version they were built against. Per §0.3 SemVer policy, MAJOR mismatch is a wire-incompat reject; MINOR mismatch is compatible (clients **MAY** warn).
- **Time-Source header:** `X-CEG-Server-Time:` `<rfc3339_canonical>` per §0.5 for client clock-skew bounds
- **Pagination** (where applicable): `?cursor=` + `?limit=` query params; response includes `next_cursor` (null if exhausted) and `total_estimate` (server's best estimate, may be approximate)

§10.0.1 Error envelope

All error responses **MUST** conform to:

```
{
  "error": {
    "code": "<ENUM_VALUE>",
    "http_status": <int>,
    "message": "<human-readable>",
    "request_id": "<server-assigned>",
    "details": {<error-specific fields>}
  }
}
```

HTTP status	Error code	Meaning
400	MALFORMED_REQUEST	Invalid JSON, missing required field, bad field type
400	CANONICAL_BYTES_VIOLATION	Date-time / hex / encoding doesn't match §0.5 / §0.6
401	UNAUTHENTICATED	Bearer token missing or invalid (admin endpoints)
403	RESERVED_PREFIX_VIOLATION	Producer attempted to emit under a reserved prefix without authority per §7
404	UNKNOWN_WITNESS	Witness key_id not registered in directory (§10.3)

HTTP status	Error code	Meaning
404	NOT_FOUND	Generic resource not found (build, partner, key)
409	IDEMPOTENT_CONFLICT	Replay detected (e.g., duplicate (<code>tree_size</code> , <code>witness_key_id</code>) cosignature with different signatures)
422	SIGNATURE_VERIFICATION_FAILED	Ed25519 or ML-DSA-65 failed to verify; <code>details.algorithm</code> names which
422	CLOCK_SKEW_VIOLATION	<code>signed_at</code> exceeds §0.7 ±5 minute tolerance
422	WITNESS_QUORUM_NOT_MET	Insufficient cosignatures to validate
422	CONSISTENCY_PROOF_INVALID	A witness cosignature's §10.3.1 consistency proof against the prior STH it cosigned is absent or does not verify (added CEG 0.10 per CIRISRegistry#34; <code>details</code> carries <code>prior_tree_size</code> / <code>new_tree_size</code> / <code>proof_len</code>). A missing proof when a prior STH exists, or a <code>tree_size</code> behind the witness's prior cosigned STH, is <code>MALFORMED_REQUEST</code> instead.
429	RATE_LIMITED	<code>X-RateLimit-*</code> headers set; <code>Retry-After</code> honored
500	INTERNAL_ERROR	Server-side fault; <code>request_id</code> usable for support
503	WITNESS_DIRECTORY_UNAVAILABLE	Substrate replication lag exceeds liveness bound

Rate-limit headers on every response: `X-RateLimit-Limit`, `X-RateLimit-Remaining`, `X-RateLimit-Reset` (seconds-until-reset epoch).

§10.1 Transport substrate for byte-level content

Wire-format Attestations carry claims; they don't carry bytes. When a claim's `evidence_refs[]` cites a SHA-256-addressed blob (e.g., an installer binary, a config file, an adapter package per `agent_files:*` per §5.6.7 / §5.9), the bytes travel via Edge transport substrate: `MessageType::ContentFetch + ContentBody + ContentMiss` (per CIRISEdge#21). Holder-discovery via Persist's `holds_bytes:sha256:*` directory (CIRISPersist#103); peer-resolution via Edge's `PeerResolver::resolve_holders`. NodeCore node-mode peers serve the bytes per their MISSION §3.4 cohabitation contract (CIRISNodeCore#11). Attestation envelope shape unchanged; SHA-256 in `evidence_refs[]` becomes universally resolvable to bytes through the substrate.

§10.1.1 Full-SHA verification before consumption (normative)

A CEG-Conforming Consumer (CCC) MUST verify the full SHA-256 of received bytes against the value in `evidence_refs[]` BEFORE handing the bytes to any consumer (Agent loader, Portal renderer, etc.). The `holds_bytes:sha256:{prefix}` directory (§5.6.7) carries only a short prefix for index efficiency; the consumer MUST NOT short-circuit verification to the prefix. Bytes that fail the full-SHA check MUST be discarded and the holder MUST be reported via the `holds_bytes:sha256:{prefix}` chain (emit a `withdraws` or negative score per consumer policy).

§10.1.2 Holder directory TTL + ContentMiss feedback

A `holds_bytes:sha256:{prefix}` attestation has a default validity of **24 hours** from `signed_at`. After that the holder is considered stale; consumer policy **MUST** attempt at most 2 holders in parallel and accept the first successful full-SHA verification. On `ContentMiss` (holder no longer has the blob), the consumer **MUST** emit a `withdraws` against the `holds_bytes:sha256:{prefix}` attestation referencing the stale holder, with `withdrawal_reason: "content_miss"`. Holders consistently failing `ContentMiss` are downweighted in `PeerResolver::resolve_holders`.

§10.1.4 Structural invisibility — `holds_bytes:sha256:*` suppression for `cohort_scope: self | family`

Per CIRISRegistry#47 + ciris.ai/cewp load-bearing claim:

Self and family content never emits the attestation that would tell the rest of the network it exists. You don't need a privacy policy to keep family photos off the federation — the wire format can't carry them in the first place.

CEG 0.7 codifies this as a normative substrate discipline. When a Contribution carries `cohort_scope: self` OR `cohort_scope: family`, the substrate **MUST NOT** emit a corresponding `holds_bytes:sha256:{prefix}` directory attestation per §5.6.7 — the content's bytes are delivered to admitted members of the relevant self-collective (§5.6.8.8 `identity_occurrence`) or family (§5.6.8.9) via the at-rest encryption flow (CIRISPersist#152), **NOT** via the public holder-discovery directory.

The privacy property is structural, not policy:

- A non-member peer cannot issue `ContentFetch` for the bytes because no `holds_bytes:sha256:*` attestation names a holder.
- A non-member peer cannot even *discover* the bytes exist via the substrate — the only attestations referencing them are scoped to the self-collective / family and never federate beyond it.
- This is the wire-format-level closure of the cewp **structural invisibility** claim: privacy emerges from format constraints, not from operator policy or legal undertaking.

***Scope (normative):** structural invisibility buys **content-holding confidentiality only** — it is **NOT** relationship-existence, metadata, traffic-analysis, or unobservability privacy. The bounding non-goals are stated canonically at §1.6.2; do not represent CEG as providing the stronger properties.*

Substrate enforcement:

```
On admission of a Contribution C with cohort_scope in {self, family}:
# (1) STRUCTURAL INVISIBILITY -- UNCONDITIONAL (the cewp privacy promise):
substrate MUST NOT emit holds_bytes:sha256:* for C's evidence_refs bytes
substrate MUST NOT propagate C beyond the self-collective / family scope
via any other directory or discovery surface
# (2) AT-REST ENCRYPTION -- the S8.1.12.4 cascade (defense-in-depth):
WHEN self/family at-rest encryption is enabled for the deployment:
  recipients := if cohort_scope == self: all current identity_occurrences of C.
                attesting_key_id
                if cohort_scope == family: all current members of family per C.family_id
```

```

FOR each recipient r:
  kem := resolve_encryption_keys(r.key_id) # current occurrence's encryption_pubkeys (
    S5.6.8.8.2)
  IF kem is None OR kem.ml_kem_768 invalid:
    # FAIL-SECURE: no valid v2 wrap target ? EXCLUDE r from the grant.
    # MUST NOT fall back to plaintext or to wrap_algorithm v1.
    # NOT SILENT (1.0-RC1, #71 C3): emit hard_case:recipient_excluded:{scope_key_id}
    # (S7.7, which defines the closed reason-set) INTO the self/family
    # scope -- recipient r, reason, + the skipped Contribution's envelope
    # ref. Scoped to the cohort; never federates beyond it (S10.1.4
    # invisibility preserved).
    skip r # content stays encrypted + unreachable to r until r registers
        encryption_pubkeys
  ELSE:
    substrate MUST wrap C's DEK via key_grant (S5.6.8.4, wrap_algorithm v2 -- S8
      .1.12.4)
    to kem.{x25519, ml_kem_768}

```

Two layers, not one (normative split — clarified per CIRISPersist#152 review).

(1) **Structural invisibility** — suppressing `holds_bytes:sha256:*` + non-propagation beyond scope — is the **unconditional** privacy promise (the cewp "the wire format can't carry them" claim); it holds even when the at-rest bytes are plaintext, because no discovery attestation federates. (2) **At-rest encryption** — the §8.1.12.4 DEK cascade — is **defense-in-depth** (against local-disk forensics / host operator / cloud-substrate operator, the CIRISPersist#152 threat table); it is operator-policy and MAY default off as a v1 migration posture, **but** when enabled **MUST** use `wrap_algorithm: v2` (hybrid PQC, §8.1.12.4) — never v1. The 1.0 / CEG-RET-native target is at-rest-on for self/family (the "everything PQC at rest" standard).

Composition with at-rest encryption flow (CIRISPersist#152): when self/family at-rest encryption is enabled, persist wraps the DEK (`wrap_algorithm: v2`) under each currently-admitted `identity_occurrence`'s `occurrence_key_id` (self) or each `member.key_id` in the named family's roster (family). New occurrence / new family-member admission triggers retroactive `key_grant` emission for all extant `cohort_scope: self|family` content (the "I bought a new phone and want my Twitter history" / "I added Carol to the household" flows from §5.6.8.9 worked example).

Recipient encryption-key resolution + fail-secure exclusion (CEG 0.18 — CIRISPersist#192 / #69). The wrap target is **not** a recipient's signing key. `wrap_algorithm: v2` needs the recipient's `{x25519, ml_kem_768}` **content-KEM** keys, which the recipient self-certifies via its `identity_occurrence.encryption_pubkeys` (§5.6.8.8.2); the substrate resolves them by `resolve_encryption_keys(key_id)` = the recipient's current (non-superseded, within-valid-until) occurrence → its `encryption_pubkeys`. Because this layer **mandates v2**, a recipient whose current occurrence carries **no valid ML-KEM-768 key MUST be fail-secure excluded** from the grant — the content remains encrypted and unreachable to it; the substrate **MUST NOT** fall back to plaintext or to `wrap_algorithm: v1`. To be an at-rest-encryption recipient, an identity **MUST** have a federation-present occurrence carrying `encryption_pubkeys`. This is the non-maleficence / fail-secure default: a missing key denies access, never downgrades the protection.

Exclusion MUST NOT be silent (1.0-RC1 — #71 C3). A bare `skip` makes a fail-secure exclusion indistinguishable from "the family went quiet" — a soft-censorship vector for a buggy or malicious substrate, and a contradiction of the spec's own `hard_case:*` attestability grain. On every fail-secure skip the substrate **MUST** emit **hard_case:recipient_excluded:{scope_key_id}** (§7.7, which defines the closed reason set) **into the affected self/family scope itself** — carrying the excluded recipient's `key_id`, a `reason`, and the skipped Contribution's envelope ref —

so the excluded member (who still sees cohort-scoped attestations) has something to audit and remediate. The event is cohort-scoped: it **MUST NOT** federate beyond the self/family (the §10.1.4 invisibility promise is preserved; the *fact* of the family’s content is not leaked by its exclusion events).

Locality dividend (cewp claim): the structural invisibility mechanism is *why* ~65% of activity stays local in the cewp scaling model — `cohort_scope: self|family` content is the bulk of daily activity (family photos, personal notes, in-household device chatter), and that bulk never federates. Operators do not configure this; the wire format enforces it.

Boundary cases:

- `cohort_scope: community | affiliations | federation` content emits `holds_bytes:sha256:*` per status-quo behavior. CEG 0.7 changes **ONLY** the self/family path.
- A `cohort_scope: self` Contribution that is later promoted via `supersedes` to `cohort_scope: community` (per §8.1.8.1 Tiered-Scope promotion) emits `holds_bytes:sha256:*` at promotion time on the NEW Contribution. The original `cohort_scope: self` Contribution’s bytes remain structurally-invisible at federation; only the promoted scope’s bytes propagate.
- `cohort_scope: self` content with `subject_key_ids` containing a non-self party (e.g., a private note Alice writes ABOUT Bob) is admitted and stays in Alice’s self-collective; Bob does NOT receive a `key_grant` unless Bob is also in Alice’s self (not the case for two distinct identities). Bob’s §4.2 subject-side revocation authority over the note still composes per CEG 0.6, but the bytes never reach Bob without Alice’s explicit re-emit at a higher `cohort_scope` including Bob.

§10.1.3 Consent revocations are NOT local-tier-eligible (CEG 0.6 addition)

Per CIRISRegistry#45 Gap 2 + CIRISAgent#842. The CIRISAgent#840 CEG-native agent design proposes **local-tier signature deferral** — self-attestations skip the hybrid Ed25519 + ML-DSA-65 signature path locally and only sign at federation-tier promotion. This is sound for **producer-only-authority self-attestations**. ***The discriminator is who holds revocation authority, NOT whether subject_key_ids is empty*** (clarified per CIRISPersist#172 / CIRIS-Agent review): a producer-authority self-attestation MAY name a subject in `subject_key_ids` per §4.2.6 — e.g. `observed:user:{hash}*, epistemic:about:{key}*, a self-consent:partnered:{user}`, or the §4.2.3 `self-identity:current` with `subject_key_ids=[self]` — and remains local-tier-eligible, because the producer holds the authority and no *other* subject can revoke it. The single carve-out is below: a Contribution where a **subject other than the producer holds revocation authority** over it.

Consent revocations from subjects MUST NOT use the local-tier deferral path. When a Contribution carries non-empty `subject_key_ids`, any subsequent `consent:state:revoked` emission OR `withdraws` admitted under §3.2.3 rule 2 or 3 from a subject in that set **MUST** promote to federation-tier within a bounded window. Default window: **24 hours** (operator-tunable per local policy).

Rationale: subject-side revocation is the wire-format observability primitive that federation peers depend on to honor consent. If a user revokes consent in the local-tier scope of one agent’s substrate, and that revocation is unsigned + unpromoted for an extended window, other federation peers continue propagating the user’s data — exactly the failure mode CEG 0.6 exists to close.

Substrate emission: substrate **MUST** emit `hard_case:consent_revocation_promotion_overdue` when a subject-side revocation has been local-tier for longer than the operator-configured window without federation-tier promotion. LensCore composes `detection:consent:promotion_delay_pattern` on top (operator monitoring; not a slashing trigger on its own).

Composition with CIRISAgent#840 self-attestation pattern: the CEG-native agent's `consent:partnered:{user_key}` self-attestation (producer-side stance) **MAY** ride local-tier; the user's `consent:state:revoked` against the agent's stance Contribution (subject-side) **MUST NOT**. This preserves the cardinality wins from #840 while closing the leak window.

§10.1.5 The attestation tier model — local-tier write, query, promotion (normative)

Pins the **shared attestation surface** the four CEG-RC1 implementations (CIRISAgent, CIRISNodeCore, CIRISLensCore, CIRISRegistry) write/query/promote through, so it is analyzable + modelable as one contract (per CIRISPersist#172 FSD review). The substrate exposes the *methods*; CEG pins the *wire/conformance semantics* every implementation **MUST** agree on. **No 1+4 change** — a "tier" is recorded state on an attestation, not a new primitive.

§10.1.5.1 The two tiers

Tier	Signature	Federation-visible	Read-visibility	Written by
local	MAY be absent (deferred per §10.1.3)	No	only the producing occurrence (self-read); every other caller — even an authorized family/community peer — sees nothing	local-tier write
federation	hybrid Ed25519 + ML-DSA-65 present	Yes	per §4 <code>cohort_scope</code> + the §10.1.4 invisibility rule	direct signed write OR <code>local → federation</code> promotion

Invariant (substrate MUST enforce): `tier = federation` $\implies$ `hybrid signature present`. Nothing crosses to federation-visible unsigned. A `local` row is **labelled** local and **MUST NOT** be served as federation-authoritative (fail-honest). The read-gate is **orthogonal** to `cohort_scope`: it is an additional filter (`local` $\implies$ `caller is the producing occurrence`), composing with the §10.1.4 target-membership predicate. Threat entries: AV-59 (local row leaked to a non-self caller), AV-60 (unsigned local served as authoritative), AV-61 (the two gates de-synced).

§10.1.5.1.1 The PQC half is MANDATORY at admission — no classical-only, no hybrid-pending accommodation (normative, CEG 1.0)

Resolves CIRISRegistry#82 (Verify audit F1); satisfies the #57 "PQC-everywhere **REQUIRED**" freeze gate. The §10.1.5.1 invariant `tier = federation` $\implies$ `hybrid signature present` is **enforced at the admission gate as of CEG 1.0**, and "present" means **verified**: every federation-tier admission gate **MUST** check **both** halves — Ed25519 over `JCS(envelope)` and ML-DSA-65 over `JCS(envelope) || ed25519_sig` (the §5.2.1 bound-payload form) — and **MUST reject** a federation-tier Contribution that carries only the classical half. This binds **all operational-authority admission gates**, explicitly including `operational_admit` (key_grant / partner / org-membership / license writes), the §5.6.8.8.1 `transport_destination` binding, and the §5.6.8.10 `partner_record` / founder-quorum gates.

This is an immediate 1.0 requirement — not a phased cutover. Pre-1.0 the PQC half was honored *opportunistically* (a deliberate rollout accommodation so hybrid-pending members could federate while ML-DSA-65 wiring landed); **CEG 1.0 closes that window with no fleet-floor and no calendar trigger.** The rationale is exactly the threat hybrid exists to defend: an adversary holding a future Ed25519 break could forge a grant / binding / partner-record and have it admitted if the classical half alone sufficed. There is ****no require_hybrid: false** posture at 1.0 — **a verifier's hybrid-required check is always-on, never an operator toggle; the only conformant state for a federation-tier key is "carries a valid ML-DSA-65 half."** A key that has not completed PQC wiring is not 1.0-conformant for federation-tier emission and is confined to local-tier (self-read, §10.1.5.2) until it does. The mandate is at the federation admission boundary only** — the single place authority crosses; local-tier rows MAY still defer the signature per §10.1.3.

The mandate binds the durable store + replication path, not only the authority gates (normative clarification, 1.0-RC8 — resolves CIRISPersist#225). "Federation-tier" means **every** federation-tier attestation, including the bulk **per-trace / store-and-replicate** path — not merely the operational-authority gates enumerated above. A federation-tier trace written to the durable, content-addressed, replicated corpus MUST carry (and a verifier MUST check on ingest) the ML-DSA-65 half exactly as a **key_grant** does; there is **no "operational authority vs testimony leaf" exemption** (federation root and the trace leaf are bound by the same rule). **Content-addressing is NOT a defense against forge-later:** a CRQC-era adversary who breaks Ed25519 mints a backdated trace under a historical key and hashes **their own** forgery — the content hash matches by construction, so the address proves nothing about authenticity. Because the trace store is *kept for posterity* (it outlives the classical primitive), the per-trace signature is the single most forge-exposed surface in the federation — the "store at massive scale" CEWP crux — and the PQC half is mandatory there for the same forge-now / harvest-now-exploit-later reason as the at-rest DEK cascade (§10.5.3 / §8.1.12.4). A substrate persisting/replicating a federation-tier row whose envelope signature lacks a valid ML-DSA-65 half MUST reject it at the ingest gate — store-then-quarantine is non-conformant.

§10.1.5.2 Local-tier eligibility — the discriminator is *revocation authority*, not subject-set emptiness

A write is local-tier-eligible iff **the producer holds sole revocation authority** over it. The discriminator (*revocation authority*, NOT empty **subject_key_ids**), the producer-authority-with-named-subject examples, and the single carve-out (a Contribution where a subject other than the producer holds revocation authority — a **withdraws** under §3.2.3 rule 2/3 or a subject-emitted **consent:state:revoked**, which MUST go signed / promote per the §10.1.3 24-hour obligation, never local) are defined canonically at §10.1.3. Tier-specific addition: **witness_relation** MUST be **self** for any local-tier write.

§10.1.5.3 Promotion — local → federation (the deferred-signature moment)

Promotion computes the hybrid signature and flips the row federation-visible. It is **idempotent** (promoting a **federation** row returns it unchanged), and at the promotion instant the tiered-scope promotion (§8.1.8.1) and **holds_bytes** emission (§10.1.2) fire exactly as for any federation write — ****promotion is the federation-emit moment****.

****Canonical bytes — JCS(envelope) per §0.9 / RFC 8785 (normative; resolves CIRISPer-**

sist#172 OQ-4):**

- The signature MUST cover JCS(contribution_envelope) — the **identical** canonical bytes any natively-federation attestation signs. A promoted row is therefore **byte-indistinguishable on the wire from one born federation-tier**; there is no "was-promoted" marker in the signed bytes.
- **Substrate columns are NOT in the canonical bytes.** tier, promoted_at, and any other storage bookkeeping are substrate state, never part of JCS(envelope). Only the §4 envelope member set is canonicalized.
- **§0.9 omit-vs-materialize is load-bearing here.** Promotion MUST canonicalize the **exact member set the producer committed at local-write time** — a field *omitted* at local write MUST NOT be materialized at promote, and vice-versa, or the recomputed bytes diverge from what a peer verifies and the hybrid sig fails. The substrate serializes the stored row → the committed envelope → JCS → sign; it MUST NOT re-default.
- Registry owns the exact envelope member set (§4 + the §0.9.3 catalog); Verify recomputes the identical JCS bytes (CIRISVerify#59). Do NOT use Verify's internal length-prefixed **signing_bytes** framing — that is for verify-to-verify primitives that never cross the four-impl boundary as JSON; a promoted attestation is the opposite.

§10.1.5.4 Query — open-prefix dimensions, bounded operators (normative; resolves CIRISPersist#172 OQ-2)

The uniform read filters on (dimensions[], valid_at, confidence_floor, subject_key_id?, scope).

- ****dimensions[]** is an OPEN-vocabulary set of prefix strings, matched by hierarchical prefix — NOT a closed enum.** Per §11.2.1 axis-vocabulary discipline, dimension prefixes are open vocab (the 0.6→0.15 cadence shipped new families — **consent:*, detection:community:*, settlement:*** — continuously). A closed enum would force a substrate redeploy per CEG namespace addition and break forward-compat. A query for **detection:*** matches **detection:correlated_action:bribery** etc.; an exact string matches exactly.
- ****The bounded surface is the operator set, not the vocabulary. The apophatic discipline (a fixed, named surface — not an OLAP/graph engine) is preserved by the closed set of five predicates**** above + no caller-composed SQL/projections — NOT by closing the dimension vocabulary. *Open data, closed operators.*
- **valid_at** filters **asserted_at ≤ valid_at < COALESCE(expires_at, ∞)**; **confidence_floor** filters **weight ≥ floor**; **scope** applies BOTH the §10.1.4 target-membership gate AND the §10.1.5.1 tier gate. Write-side reserved-prefix emitter gates (§7.0.1) are unaffected — read is gated by scope, not by dimension authority.

§10.1.5.5 For holistic analysis + modeling (informative)

The tier model is the *same KEM-then-symmetric placement* the streaming model uses (§10.5), applied to attestations: the **expensive op (hybrid sign) is on the cold path** (promotion, at federation-emit), while the **hot path (local self-attestation write) is O(1) and unsigned**. Cost shape for a node:

- **local write:** $O(1)$, no asymmetric crypto — the agent’s steady-state memory/config/consent/identity churn.
- **promotion:** one hybrid Ed25519+ML-DSA-65 sign (~the §10.5-model per-op cost; ML-DSA-65 sign $\sim 330 \mu s$) + JCS canonicalization — only at the federation-emit moment, never per local write.
- **query:** the §4/DAS scope-filtered read; cost is the predicate + index, independent of tier.
- **observability for modeling:** local row count, promotion rate, and `hard_case:consent_revocation_p` count are the three measurable signals; the §10.1.3 24-hour window is the one tunable. A model of a federation’s attestation load is therefore (*local-write rate*, *promotion rate*, *query rate*) with promotion carrying the only asymmetric-crypto cost — the dual of the streaming model’s epoch-rekey tail.

§10.1.6 Cross-region merge intents — CEG-declared per `subject_kind` (normative; CEG 1.0-RC2 addition)

Per CIRISRegistry#70 (Persist review concern D): with operational data joining the CEG-native replication stream (§5.6.8.13), the substrate runs **more than one merge policy**. The policy is a **normative property of the `subject_kind`, declared here** — the substrate reads and dispatches on the declaration; it MUST NOT infer policy per record type (the substrate enforces declared merges; it does not invent policy).

<code>subject_kind(s)</code>	Merge intent	Semantics
<code>organization</code> , <code>org_membership</code>	**lww_skew_bounded withdrawal_forward_only** +	Stable-id grouping (§5.6.8.13); an admitted withdraws (deactivation) is forward-only — a later non-withdrawn write does NOT resurrect; else latest asserted_at wins; tie-break smallest attestation_id (§6.1). The forward-only here is the lightweight authz flag , NOT the §8.1.12 DEK-cascade crypto path — Commons tier, no DEK exists.
<code>partner_record</code>	**monotonic_quorum** (the CIRISPersist V058 R1/Q1 machinery, generalized from <code>revoked_key_id</code> to <code>license_id</code>)	Admission anti-rollback first: a write whose revision decreases never enters the merge. Then the MergeBallot comparator: quorum_weight → signed timestamp → content hash. Quorum-above-time is deliberate — it neutralizes timestamp front-running (F-AV-FRONTRUN) on the records where it matters most. More-restrictive state wins on conflict: revoked > suspended > active .
revocation + the three membership-revocations	V058 R1/Q1 (unchanged — the original monotonic_quorum instance)	As shipped.
<code>keys</code> / <code>attestations</code> / <code>occurrences</code> / <code>families</code> / <code>communities</code> / <code>location_proofs</code>	Content-addressed idempotent admission (unchanged)	Same content → same envelope_hash → dedup; rotation collisions rejected non-destructively.

Skew-bounded admission (normative — the LWW front-running fix, #70 Persist concern B). For every `subject_kind` merging by `lww_skew_bounded`, the substrate MUST reject

at admission any envelope with `asserted_at > now + tolerance`, where `tolerance` is the §0.7 clock-skew bound (± 5 minutes; the existing `CLOCK_SKEW_VIOLATION` error class). Without this, a signer with a forward-skewed clock future-dates `asserted_at` and wins LWW indefinitely; with it, a forward-dated write can win for at most the skew window. This matters precisely because `org_membership` carries role: `OrgAdmin` — unbounded LWW on authz data is a role-escalation surface.

The two quorums (restated from §5.6.8.13 — different layers, different owners): the **steward-signature admission quorum** (M-of-N signature set over identical JCS bytes; the §5.6.8.10 founder-quorum machinery at admit — Verify’s layer) is NOT the **region merge quorum** (`quorum_weight`, the MergeBallot tier-1 ordering above — the substrate’s layer). The substrate’s merge logic never counts steward signatures; Verify’s admission check never orders merges.

§10.2 Multi-steward + accord-holder discovery

`GET /v1/steward-key`

Returns the multi-steward set with M-of-N policy.

Response (200 OK):

```
{
  "stewards": [
    {
      "region": "us",
      "key_id": "us-steward-2026",
      "ed25519_pubkey_b64": "<base64-url>",
      "mlds65_pubkey_b64": "<base64-url>",
      "hardware_class": "HSM_FIPS_140_3_L3",
      "deployed": true,
      "fingerprint_sha256_hex": "<64-char-lowercase>",
      "cert_validity_self_attest": {
        "valid_until": "<rfc3339_canonical>",
        "signature_b64": "<base64-url>"
      }
    },
    {"region": "eu", ..., "deployed": false},
    {"region": "apac", ..., "deployed": false}
  ],
  "threshold_policy": {"required": 2, "available": 1},
  "response_signature": {
    "signer_key_id": "us-steward-2026",
    "ed25519_b64": "<base64-url>",
    "mlds65_b64": "<base64-url>",
    "canonical_bytes_label": "ciris.steward_key_response.v1"
  }
}
```

The response itself is hybrid-signed by the serving region’s steward over `canonical = "ciris.steward_key_response.v1 || sha256_hex_lowercase(canonicalized_json_body_excluding_signature)"`. Consumers MUST verify the response signature before trusting any field in the body — placeholder pubkeys without `deployed: true` MUST NOT be promoted to trust roots.

GET /v1/accord-holders

Three named holders with hybrid pubkeys + per-holder `hardware_class` + `provisioned` flag. v1.4 interim ships with placeholder fingerprints + `provisioned: false`; consumers MUST NOT honor CONSTITUTIONAL invocations against placeholders. Response signed by the serving region's steward (same shape as `/v1/steward-key`).

GET /v1/accord/holders

UI wrapper around `/v1/accord-holders` with per-holder `accord_emissions[]` for UI rendering. Same response-signing requirement.

GET /v1/rotation-history

Chronological rotation events from `registry_signing_keys` table. Substrate-conformance migration moves to `federation_keys`.

§10.3 STH cosigning + witness directory

CIRISVerify v2.12.0+ ships consumer-side `SignedTreeHead::cosign` + `count_valid_witnesses` + `witness_quorum_met`. CEG's emission half:

POST /v1/transparency/sth/cosign (public)

Witness posts cosignature on (`tree_size`, `root_hash`, `signed_at`). Registry verifies hybrid Ed25519 + ML-DSA-65 against witness pubkey in directory; persists on success.

Request body:

```
{
  "tree_size": <int>,
  "root_hash_sha256_hex": "<64-char-lowercase>", // per S0.6
  "signed_at": "<rfc3339_canonical>", // per S0.5
  "witness_key_id": "<string>",
  "ed25519_signature_b64": "<base64-url>",
  "mldsa65_signature_b64": "<base64-url>",
  "consistency_proof_root_hash_sha256_hex": "<64-char-lowercase>",
  "consistency_proof_tree_size": <int>,
  "consistency_proof_path_b64": [<base64-url>, ...]
}
```

Canonical bytes (witness MUST sign these):

```
canonical = sha256(
  "ciris.sth_cosign.v1\n" ||
  "tree_size=" || decimal_no_leading_zeros || "\n" ||
  "root_hash_sha256=" || sha256_hex_lowercase || "\n" || // per S0.6
  "signed_at=" || rfc3339_canonical // per S0.5
)
```

Ed25519 over `canonical`; ML-DSA-65 over `canonical || ed25519_sig` (bound payload).

§10.3.1 Consistency-proof requirement (normative; addresses CEG 0.1 distsys review)

A witness signing an STH MUST first verify a consistency proof from the prior STH it cosigned (or from genesis if it is the witness's first cosignature against this log). The Registry MUST reject `POST /v1/transparency/sth/cosign` requests that omit the `consistency_proof_*` fields OR whose consistency proof does not verify against the named prior STH. `witness_quorum_met` is therefore "quorum on log consistency," not "quorum on a string."

ENFORCED as of Registry v2.3.0 (CIRISRegistry#34). The cosign request carries `consistency_proof_pa` (base64 RFC 6962 §2.1.2 node hashes). The Registry anchors the check against the prior (`tree_size`, `root_hash`) **it recorded** for that witness — not a root the requester claims — and rejects: missing proof when a prior STH exists or a `tree_size` behind the witness's prior cosigned STH → `MALFORMED_REQUEST`; a proof that does not reconstruct both roots → `CONSISTENCY_PROOF_INVALID` (§10.0.1). A witness's first cosignature is exempt ("from genesis"). The RFC 6962 verifier is vendored from `ciris-verify-core::transparency` (Registry omits that crate for a `libsqlite3` linker reason) and proven against independent known-answer vectors.

`GET /v1/transparency/witnesses` (public)

Directory of registered witnesses. Paginated.

`GET /v1/transparency/sth/{tree_size}/witnesses` (public)

Cosignatures for an STH with `witness_quorum_met` verdict.

`POST /v1/transparency/witnesses` (admin; multi-party-gated)

Register a new witness. **0.1 scaffold note:** in 0.1 interim this is bearer-token-gated by `REGISTRY_ADMIN_TOKEN`. **0.2 hardens this to 2-of-3 steward sign-off** (addressing CEG 0.1 cryptographic + red-team review): the request body MUST carry signatures from at least two of the three regional stewards, verified against `GET /v1/steward-key`. Single-token admission is a 0.1 known weakness; production deployments SHOULD operate the 0.1 endpoint behind a corporate IDP gate that enforces multi-party admission out-of-band until the 0.2 multi-sig requirement is normative.

§10.4 Other Registry endpoints

`GET /v1/builds/{version}` returns the `BuildRecordResponse` with a `federation_provenance` block (per §5.2 SLSA emission discipline). `GET /v1/verify/build-manifest/{project}/{version}/{target}` (Path B) returns the verbatim signed `BuildManifest`. `GET /v1/agent_files/{kind}?platform_or_target=...` returns the §8.1.6 trust-composition layers. `GET /v1/partner/{key_id}` composes `ProfileScorecard` data from existing tables.

Full response schemas for these endpoints land in the Rust handlers + OpenAPI export; CEG 0.2 commits to publishing a versioned OpenAPI spec alongside this document.

§10.5 Streaming transport, per-stream logs & delivery receipts (CEG 0.10 addition)

Per CIRISRegistry#44 absorbed + CIRISLensCore#857 (observer-share driver) + CIRISPersist#142 (streaming substrate prerequisite). §10.5 is the **delivery axis** — the third orthogonal envelope concern alongside visibility (**cohort_scope**) and revocability (**subject_key_ids** per §4.2). §3's 1+4 primitive set is untouched; §10.5 is endpoint + envelope + composition extension, NOT a grammar change.

Bifurcation:

Half	Cardinality	RC1 status
Observer-share / directed delivery (single Contribution → subscriber-set; no stream_id)	N=1 typical; per-subscriber key_grant	impl-live ; substrate paths shipping per §5.6.8.4 key_grant + §8.1.13 Policy M membership
Media / streaming multicast (live_stream chunk-DAG; per- (stream_id, epoch) keys)	N>1; flat per-epoch key_grant cascade	spec-now, impl substrate-pending CIRISPersist#142 ; subsections §10.5.2 / §10.5.3 / §10.5.4 ride this dependency

§10.5.0 Framing (normative)

A stream is **its own per-stream transparency-log instance** (**log_id = stream_id**). A **live_stream** (CEG 0.5 sub_kind absorbed into 0.10) MUST NOT append chunks into the federation provenance log (§10.3's global log carrying builds / licenses / identities) — millions of media chunks would pollute provenance and inflate the global tree. The §10.5 path **reuses** the §10.3 **SignedTreeHead / ConsistencyProof / WitnessConsistencyProof / cosign** abstractions, instantiated per-stream as separate log instances under the same RFC 6962 algorithm.

The 1+4 wire-format lockdown holds: there are no new **attestation_type** values. Stream chunks ride content addressing; stream-roots ride the existing **SignedTreeHead** shape; delivery receipts ride **scores** against the new **delivery_receipt:{stream_id}** reserved prefix (§7.9).

§10.5.1 Per-stream log + stream-root (normative — V1 lock)

For each **live_stream**:

- **log_id = stream_id**; chunks = leaves; stream-root = **SignedTreeHead**{ **log_id: stream_id, tree_size: chunk_count, root_hash, timestamp, signature** }
- **Producer signs the STH — MANDATORY** authenticity root; hybrid Ed25519 + ML-DSA-65 per the §10.3 **signing_bytes** discipline; canonical bytes per §0.9 JCS for the envelope-bearing wrapper
- **Witness cosign — OPTIONAL**, via the §10.3 path verbatim. This is the best-effort / accountable split (D5 per §15.6.2):
- **Best-effort** (open media) → producer-signed root only; impl-pending CIRISPersist#142 only
- **Accountable** (paid media, registry propagation, emergency) → witness cosign per §10.3.1 consistency-proof, which is the **anti-equivocation guarantee** — producer cannot show different chunk-K to different subscribers nor rewrite mid-stream. Accountable tier is impl-pending BOTH #142 AND CIRISRegistry#34 (STH consistency-proof enforcement)

- **Cadence:** producer publishes a signed root every K chunks OR T seconds (whichever first), **always at an epoch boundary** (§10.5.3) + at `sealed_at`. Witness cosign runs **per-epoch** (coarser than per-K to keep cosign-quorum cost off the hot path). Default pins: **K=64, T=2s** (operator-tunable; ratification pending per §15.6.3 RC1-7)
- **Incremental verify** (D4): each leaf's "root-after-K" lets a subscriber verify chunk K's inclusion against the nearest signed $\text{STH} \geq K$ via the §10.3.1 consistency path. No new commitment structure beyond the per-leaf chain-link + periodic STH that §10.3 already provides.
- **Accountable-stream quorum:** Policy E (§8.1.5 locality-scaled quorum) applies — not a fixed N. Emergency-channel roots want a higher quorum than a paid-media stream; locality scaling provides the gradient.

§10.5.2 Chunk seal + STREAM nonce (normative — V2 lock)

Per-chunk content sealing **conforms to SFrame** (draft-ietf-sframe, normative): the per-frame AEAD model with a (KID, CTR) header, here bound as `KID = stream_id` and `CTR = counter`, AEAD = AES-256-GCM, keys derived per MLS epoch (the canonical SFrame+MLS composition, §10.5.3). The STREAM nonce below is CEG's binding profile of SFrame's per-sender nonce derivation. Per-chunk content sealing uses AES-256-GCM (NIST FIPS 197 + SP 800-38D). The 12-byte (96-bit) nonce follows the **STREAM** layout (Hoang, Reyhanitabar, Rogaway, Vizár — *Online Authenticated-Encryption and its Nonce-Reuse Misuse-Resistance*, CRYPTO 2015):

```
nonce[12] = prefix[7] || counter_be[4] || last_flag[1]
```

- ****prefix[7]**** — derived, NOT transmitted: `prefix = HKDF-SHA256(epoch_dek; info)[0..7]` where the HKDF `info` is the **byte-exact** concatenation (pinned per CIRISRegistry#63 — cross-impl interop): `info = b"ciris-stream-nonce/v1" || stream_id_utf8 || epoch_be8`, with `stream_id_utf8` = the UTF-8 bytes of `stream_id` and ****epoch_be8** = the u64 epoch as 8-byte big-endian** (`epoch.to_be_bytes()`, consistent with `counter_be`). **This encoding is normative and MUST be byte-identical across producer, substrate, and every consumer** — the §10.5.3 zero-trust-of-host posture has consumers recompute this exact nonce to *open* chunks, so any BE/LE/ASCII disagreement on `epoch` yields a different `prefix` → different nonce → GCM auth-tag failure (silent whole-stream decryption failure), the same hazard class `counter_be` already guards. No length-prefix on `stream_id` is required: the fixed-length tag prefix + fixed 8-byte `epoch` suffix make the parse unambiguous (distinct (`stream_id`, `epoch`) pairs always yield distinct `info` — unequal `stream_id` length changes total `info` length; equal length splits unambiguously). Matches the ****KEY_GRANT_V1_INFO**** versioned-context HKDF pattern at `CIRISVerify/src/ciris-crypto/src/1` (`b"cewp-key-grant/v1"`). Per-(`stream_id`, `epoch`) unique; verifiable by any holder of the epoch DEK
- ****counter_be[4]** — **32-bit big-endian; hard ceiling $2^{32}-1$ chunks per epoch**. **Substrate MUST force an epoch roll before wrap**. **Recommended operational cap:** `MAX_CHUNKS_PER_EPOCH = 2^{24}` (~16.7M chunks/epoch) to keep per-epoch state + proof sizes bounded (operator-tunable; ratification pending per RC1-7)
- ****last_flag[1]**** — `0x01` on the final chunk of an epoch (sealed by `seal_stream` per §10.5.3); `0x00` otherwise. The distinct nonce on the final chunk gives **truncation + append resistance**: an adversary cannot drop the final chunk and pass off a short stream, nor append past a sealed segment

Cross-epoch counter reset is nonce-safe (normative reasoning): GCM’s catastrophic case is reuse of a (key, nonce) pair. On epoch roll the DEK changes, so a reset counter lives in a different keyspace — (DEK_e, nonce=0) and (DEK_{e+1}, nonce=0) are distinct pairs. The enforced invariant is therefore only within a single epoch: counter strictly monotonic, never wraps (guaranteed by the forced roll). Across epochs, reset is free.

****Single-sender-per-(stream_id, epoch) invariant (normative — nonce-collision safety; cribbed from SFrame/MLS)**:** the counter_be[4] space of a (stream_id, epoch) is owned by **exactly one** sender — the stream_id’s producer. Two distinct senders **MUST NOT** seal chunks under the same (stream_id, epoch) DEK, or their independently-incremented counters could collide into a reused (DEK, nonce) pair (GCM-catastrophic). In **group video** (§10.5.8) each participant emits their **own** live_stream with their **own** stream_id, so the nonce prefix HKDF(epoch_dek; "ciris-stream-nonce/v1" || stream_id || epoch) is per-sender-unique by construction — the same hazard SFrame avoids with per-sender keys, achieved here by per-sender stream_id. Substrate **MUST** reject a chunk-append whose (stream_id, seq) collides with an existing sealed chunk (replay/forking guard, §10.5.3).

§10.5.3 Epoch keying + cascade (normative — D2 / D3; substrate-pending #142)

The stream-epoch DEK seals content **O(1)**; the per-subscriber key_grant cascade distributes the 32-byte epoch key **O(N)/epoch** (sender-key / Megolm shape) = §8.1.12.4 Policy-L cascade applied to a community roster against a *rotating* key.

****PQC at rest — wrap_algorithm: v2 MANDATORY (normative).**** The epoch-DEK key_grant cascade **MUST** wrap the DEK with ****wrap_algorithm: v2 = x25519+ml-kem-768 (hybrid; FIPS 203)**** — never v1 (X25519-only). The DEK protects content that may persist indefinitely; a classical-only KEM is a harvest-now-decrypt-later exposure even though the content AEAD (AES-256-GCM, §10.5.2) and the wrap *signature* (Ed25519 + ML-DSA-65) are already PQC-safe. The hybrid KEM primitive is shipped in ciris-crypto (hybrid_kex / ml_kem, CIRISVerify#47); the versioned KEY_GRANT_V1_INFO HKDF-context rotates cleanly to the v2 context. A Consumer **MUST** reject a streaming epoch grant carrying wrap_algorithm: v1. ****The v2 wrap_algorithm payload wire string is pinned (normative, per #64): x25519_mlkem768_aes256_gcm_hkdf_sha256 (the §5.6.8.4 vocab variant X25519MlKem768Aes256GcmHkdfSha256; matches ciris-crypto KEY_GRANT_ALGORITHM_V1 payload; see §10.5.5.1 for details) — a constant string, not a variable-length value.** Producer, substrate, and every consumer **MUST** serialize/deserialize this exact string — a mismatch silently fails grant decode. **Full PQC envelope for streaming:** content = AES-256-GCM (symmetric, PQC-safe) · DEK wrap = X25519+ML-KEM-768 · authenticity = Ed25519+ML-DSA-65 · hashes = SHA-256 (PQC-safe) · in-transit = §10.5.5 E1 two-layer hybrid (below).

****Epoch index is monotonic, per-stream_id, greenfield — a separate addressing axis**** from key_grant.rotation_chain:

Axis	Addressing	Where it lives	Supersession
Content-addressed grant supersession (CEG 0.3)	(content_sha256, recipient_key_id)	cirisnode_contributions V054 partial indexes (planner-AND'd)	rotation_chain payload-level lineage (list of prior key_grant_ids); walked reader-side
Stream/epoch-addressed grant supersession (CEG 0.10)	(stream_id, epoch[, recipient])	federation_stream_chunks(stream_id, epoch, seq) (Persist#142 step 3; v3.9.0 target; NOT YET LANDED — unowned/un-scheduled)	rotation_chain payload-level supersession reused on the new axis (RC1-1 [✓] Persist on record)

[!] **Not pure-additive at the Persist constraint layer (RC1-1c):** the V054 cross-column CHECK requires `key_grant` rows be content-addressed (`media_content_sha256` IS NOT NULL). The §10.5.3 epoch-key axis with NULL `media_content_sha256` would be REJECTED by today's CHECK. Introducing the new axis requires a **parallel CHECK arm migration** at Persist (content-addressed OR stream/epoch-addressed) — a bounded constraint migration, not a pure index-add. The spec text does not claim "purely additive" at the Persist constraint layer.

Epoch triggers (D3):

Trigger	Behavior	Forward-secrecy implication
Member removal	MANDATORY rotation (coalesced per below; exempt for ungated public broadcasts per below) — the forward-only-unsubscribe enforcement	Subsequent epochs sealed under a DEK the removed member doesn't have
Member addition	NO rotation + Option-A catch-up per §11.7.1 (subject to <code>history_on_join</code>)	New member gets <code>key_grants</code> for the current epoch + (optionally) prior epochs per the <code>history_on_join</code> envelope field (§4)
Time / bytes	Optional hygiene rotation	Operator policy; default off

Removal coalescing (normative — 1.0-RC1, #71 C1). At broadcast scale, naive per-removal rotation is unaffordable: at $N = 10^6$ and realistic audience churn ($\sim 30\%/hr$), one rotation per departure ≈ 83 epochs/s $\rightarrow$ a flat-unicast cascade of **~ 3.1 Tbps** — **exceeding the ~ 2 Tbps content fan-out itself** (the original model's 3,600/hr churn figure understated broadcast churn by ~ 2 orders of magnitude). The substrate therefore **MUST coalesce removals**: all removals admitted within one STH cadence window **T (default 2 s, the §10.5.4 equivocation-window grain)** are batched into a **single** epoch rotation covering the whole batch. This caps the rotation rate at $1/T$ **regardless of churn** ($\leq 1,800$ epochs/hr at $T = 2$ s $\rightarrow \sim 18.7$ Gbps flat-unicast cascade at $N = 10^6$, $\sim 0.9\%$ of content fan-out — the headline restored honestly), and bounds removed-viewer exposure to $\leq T$ — the same grain the stream's equivocation window already accepts. A removed member's exposure window is therefore $\leq T$, never "until the next scheduled rotation."

Public-broadcast exemption (normative). A `live_stream` whose roster is **ungated** — **listed: public** AND grants issued to any requester without an admission ceremony — carries **no confidentiality claim against departed viewers** (anyone, including the departed viewer, can re-subscribe and receive the current DEK on request). For such streams, member removal **MUST NOT** force rotation (it buys nothing and costs the full cascade); the time/bytes hygiene rotation and the §10.5.2 nonce-space bounds still apply. Rotation-on-removal remains **MANDATORY for every gated roster** (bounded membership, admission ceremony, or any non-public listed state) — there the DEK *is* the access-control boundary and the forward-only-unsubscribe guarantee is real.

Rekey conforms to MLS TreeKEM (RFC 9420, normative). The epoch-DEK rekey on member change is the MLS TreeKEM construction: an $O(\log N)$ path rekey, the commit signed once, with RFC 9420 blank-node / unmerged-leaf handling and parent-hash tree integrity. The hybrid KEM (X25519+ML-KEM-768) is the MLS ciphersuite's HPKE KEM.

Delivery is decisive (carried from the bandwidth model). The TreeKEM advantage is *multicast aggregation*, not the tree itself: with efficient multicast the commit egress is **$O(\log N)$**

(one commit serves all); over unicast the commit must reach each member, $O(N \log N)$ — in which case the **flat per-member cascade** ($O(N)$, **one grant per member**) is **competitive-to-better**. The substrate **MUST** select per deployment based on whether the transport multicasts; the choice is **wire-invisible** (tree position rides the opaque `key_grant` payload; no schema migration). Whether the RET mesh offers efficient multicast is the open transport question (the bandwidth model's one free parameter).

Forward secrecy is forward-only by default (§11.7.1 Option A). **Post-Compromise Security (PCS)** is now AVAILABLE (inherent to TreeKEM key-updates: a compromised current member heals on the next commit) and is **OPTIONAL, operator-enabled** — a deployment **MAY** require periodic self-updates for PCS, or stay forward-only.

Catch-up bound (P4): $\min(\text{operator depth cap [LensCore knob, NOT a substrate constant]}, \text{chunk-eviction horizon})$. Three distinct windows that are NOT conflated: chunk-eviction horizon $\neq$ §10.1.2 `holds_bytes` 24h TTL $\neq$ grant durability. A catch-up request against an evicted epoch returns ****ContentMiss** — fail-honest, no silent gap** (consistent with the `MISSION.md` fail-honest invariant). Operators **MUST** ship the P4 cap **with** the cascade, else 10^6 grant Contributions per rekey is the unbounded worst case.

§10.5.4 Delivery receipts (normative — D5 / V3 lock)

A `delivery_receipt:{stream_id}` Contribution (new reserved prefix per §7.9) is a subscriber's signed acknowledgement that they received chunk K under the named stream + epoch. **Best-effort default**; opt-in for **accountable** profiles (registry propagation, emergency).

Canonical bytes (domain-separated + length-prefixed, matching `SignedTreeHead::signing_bytes` discipline; per §0.9 the envelope-bearing wrapper is JCS-encoded, but the receipt's signed-bytes inner payload follows the explicit-length-prefix shape below):

```
receipt_signing_bytes =
  "ciris-delivery-receipt/v1" // domain separator
  || len(subscriber_key) || subscriber_key
  || len(stream_id) || stream_id
  || epoch (u64 LE)
  || chunk_root ([u8; 32])
  || K (u64 LE)
```

Both `epoch` (key-rotation index — for per-epoch entitlement / billing) and `K/chunk_root` (chunk position + its committed root) are independent indices — both required (each names a distinct authorization scope).

Verify check is a JOIN, NOT a sig-check:

1. **Signature valid** over the canonical bytes (subscriber hybrid Ed25519 + ML-DSA-65 sig). Necessary but **not sufficient**.
2. ****chunk_root** is a real published STH root** — **MUST** equal a `SignedTreeHead.root_hash` actually published for `log_id = stream_id` at `tree_size` $\geq K$. A phantom / self-invented root $\rightarrow$ REJECT. For accountable streams, "published" means **witness-cosigned** (the §10.3 path), so the subscriber cannot collude with the producer on a private root.
3. (*Recommended for accountable*) **Inclusion proof** chunk K $\rightarrow$ `chunk_root`. Upgrades the receipt from "subscriber saw a root" to "subscriber saw a root that provably commits to chunk K".

Semantics — proof-of-DELIVERY, not proof-of-CONSUMPTION: the receipt proves the subscriber received bytes committing to chunk K. It does NOT prove they decrypted those bytes (they may not hold the epoch DEK). Consumers MUST NOT overclaim a delivery receipt as proof of consumption. Per the MISSION.md fail-honest invariant + §1.4 "Verify authenticates origin, does not compose 'delivered'/'owes N'", Verify's role is validation-not-adjudication: emit the validated receipt as an attestation on `delivery_receipt:{stream_id}` — the "delivered" verdict is consumer policy.

Accountable-stream receipt quorum: Policy E (§8.1.5 locality-scaled) — same shape as the §10.5.1 STH quorum. Ratification pending per RC1-7.

§10.5.5 Transport — Edge layer (normative — E1–E4 lock per §15.6.2)

Decision	Behavior
E2 — pull-only RC1	RC1 multicast = pull-only : producer seals chunks under the epoch DEK → emits <code>holds_bytes:sha256:*</code> → subscribers pull via the existing <code>ContentFetch</code> path. Relay / fan-out tree → 1.x (CIRISRegistry#46 / #43)
E1 — two-layer crypto (security-critical)	Transit-key (per CIRISLensCore#857 prod-lens-via-transit-key) is a hop-by-hop transport wrap UNDER the E2E epoch DEK (two independent crypto layers). MUST NOT replace the cascade — a relay never sees plaintext. Transit-key is for path-confidentiality; epoch-DEK is for end-to-end content confidentiality. PQC in transit (normative): BOTH layers MUST use PQC-grade key agreement — the transit-key wrap MUST be hybrid X25519+ML-KEM-768 (same <code>hybrid_kex</code> primitive as §10.5.3), and the underlying transport (Edge/Reticulum) MUST negotiate the hybrid/PQC crypto-kind (CIR2, per CIRISVerify/docs/CRYPTO_AGILITY.md), never classical-only CIR1. Classical-only transport is rejected for streaming.
E3 — fan-out = entitled ∧ reachable	Persist owns durable entitlement (the roster: signed CEG envelopes, replicated, logged). Edge owns transport-reachability via <code>reachability.rs</code> (CIRISEdge#29) node-local presence tracker. Fan-out targets the intersection. Reachability is NEVER an attestation, never <code>holds_bytes</code> , never replicated, never logged — consistent with the §10.1.4 'cohort_scope: self\
E4 — durable side rides existing federation-attestation path	Durable entitlement (roster + epoch-key grants) rides the existing federation-attestation Edge path (CIRISRegistry#41 handler cutover) — just more <code>federation_attestations</code> rows. NO net-new Edge transport for the durable side. Net-new is only on §10.5.1 streaming-log endpoints

§10.5.6 D6 liveness invariant — entitled vs reachable (normative)

Two sets are NEVER conflated:

- **Entitlement roster** (Persist-owned): signed CEG envelope, Edge-propagated, durable, logged. It's **evidence** — it MUST propagate + be auditable. Per §8.1.13 Policy M community-membership composition

- **Live-reachability set** (Edge-owned): generalizes the §10.1.2 `EdgeConfig.holds_bytes_ttl_seconds` 24h default down to seconds-to-minutes for live-multicast. Node-local presence tracker (Edge `reachability.rs` per CIRISEdge#29). ****NEVER** an attestation, never `holds_bytes`, never replicated, never logged**

****Fan-out invariant:** `fan_out(C) = entitled(C) ∩ reachable(now)**`.

Heartbeat-suppression discipline: this is a **producer-side-refusal invariant** (same class as the §10.1.4 `cohort_scope: self|family holds_bytes` suppression). Missed (entitled-but-unreachable) members fall back to pull on reconnect — substrate does NOT keep retrying push, does NOT emit a "delivery_failed" attestation, does NOT log liveness state. The reconnect-then-pull catch-up rides §10.5.3 `history_on_join`.

§10.5.8 Realtime group communication — composition (CEG 0.13 addition)

Per CIRISRegistry#56. The delivery axis was framed for 1:1 observer-share and 1:N broadcast multicast. Realtime **group communication** — group video, voice, desktop/screen sharing, text chat, and topic-scoped channels with sub-channels — is the **same primitive set at $N \leftrightarrow N$ cardinality**. It composes entirely from `community` (§5.6.8.10) + `live_stream` (the §10.5 streaming surface) + `chat_message` (CEG 0.3) + member/transport resolution (§8.1.13.1.1). **Zero new structural primitives** — this is the thirteenth path on the §1.4 claim and the most product-complete: the whole realtime-collaboration surface is composition.

The composition map (all generic — no new wire):

Surface	Composes from
1:1 / group video call	each participant emits a <code>live_stream</code> (their A/V) scoped to the channel <code>community</code> ; each subscribes to peers. $N \leftrightarrow N = N$ simultaneous bidirectional streams over a small roster
Voice channel	identical to group video with an audio-only codec; same <code>live_stream</code> wire
Desktop / screen sharing	a <code>live_stream</code> whose source is a screen capture ($1 \rightarrow N$ within the channel); same chunk-seal + epoch-DEK wire
Text chat	<code>chat_message</code> (CEG 0.3) at <code>cohort_scope: community, community_id: <channel></code> ; threads via <code>topical_relation: replies_to</code>
Channel	a <code>community</code> (persistent) — its roster gates who can join the call / read the chat
Sub-channels	nested <code>community</code> membership (§8.1.13.5 multi-level pattern): a parent "space" community whose members admit child channel-communities; sub-channel members are a subset of the space roster
Presence ("who's here")	the D6 reachable set (§10.5.6) — node-local, never an attestation, never logged
Invite / join / leave	community admission ceremony (§8.1.13.2) — invite = membership proposal; join = admitted member; leave = forward-only <code>withdraws</code>

Transport profiles (normative — extends §10.5.5):

Profile	Topology	Transport	RC1 (1.0)
Broadcast (1:N, large N)	asymmetric	pull-only ContentFetch (§10.5.5 E2); relay tree → 1.x	[✓] pull
Realtime small/medium group (calls, huddles, ≤ ~50)	symmetric mesh	direct Reticulum Links between participants (low-latency push; no relay tree needed at small N)	[✓] direct-link mesh
Realtime large group (≫50 in one A/V room)	fan-out	selective-forwarding relay (SFU role)	→ 1.x (same relay-tree axis as broadcast scale)

The low-latency realtime profile (direct Reticulum Links) is **in 1.0 scope** because small/medium rosters need no relay tree — RNS Links give encrypted low-latency point-to-point natively. The wire is identical to broadcast (chunk-seal §10.5.2, per-(`stream_id`, `epoch`) epoch DEK §10.5.3, per-stream STH §10.5.1); only the transport profile + roster size differ. **PQC is unchanged and mandatory** — epoch DEK wrapped `wrap_algorithm: v2` (X25519+ML-KEM-768) at rest, hybrid KEX in transit.

Ephemeral vs persistent rosters: a persistent channel is a long-lived **community**; an ad-hoc call is an ****ephemeral community**** (short `valid_until`, torn down on last-leave). Same primitive, different lifetime — no special-case "session" primitive.

The breadth confirmation: a full realtime group-communication platform — group video, voice, screen sharing, chat, and channels-with-sub-channels — requires **no addition to the 1+4 structural set** beyond the §10.5 delivery axis and the **community** membership machinery already locked. ##### §10.5.8.1 Realtime non-A/V data streams (normative scope boundary)

The realtime profile is **not media-only**. Any high-frequency mutable shared state — multi-player game ticks, collaborative-editing operations (CRDT/OT), live cursors/whiteboard strokes, remote-control input, high-rate telemetry — rides the **same** §10.5.8 realtime transport (direct Reticulum Links / SFU at scale) with an application-defined payload codec in place of an A/V codec. The wire is identical: per-(`stream_id`, `epoch`) epoch DEK (§10.5.3, `wrap_algorithm: v2` PQC), STREAM-nonce chunk seal (§10.5.2), per-stream STH (§10.5.1). A data-stream chunk is just a chunk.

Scope boundary (the explicit call):

- **IN scope (transport):** ordered, sealed, authenticated, PQC-encrypted realtime delivery of arbitrary data-stream payloads — covered by §10.5.8, no addition.
- **OUT of scope (merge semantic):** the conflict-free / convergent-merge logic for shared mutable state (CRDT, Operational Transform, last-writer-wins, etc.) is **application-layer**, not a CEG primitive — consistent with the §1.1 discipline ("the substrate stores; the wire transports; CEG describes the shape of the claim; consumer policy composes verdicts"). CEG carries the ops; the application converges them. A future codified merge primitive would route through the §11.2 amendment process if real demand pulls it (the downstream-demand-pulls-additions discipline), but it is explicitly NOT required for 1.0 — realtime collaborative apps are buildable on the transport today.

§10.5.8.2 `codec_id` namespace — realtime A/V chunk codec discriminator (normative, 1.0-RC9 — ratifies CIRISRegistry#84)

§10.5.8.1 makes the realtime chunk payload codec **application-defined and opaque to the substrate**. For realtime **A/V at scale**, a hop fanning out chunks (CIRISEdge#128 per-receiver

layer policy, #66 SFU relay) must drop chunks for per-receiver bandwidth degradation **without** decrypting them — so the codec’s layer-numbering semantics must be a **clear (non-AEAD) discriminator**. CEG ratifies a 1-byte `codec_id` namespace so every implementation reads the same meaning. **This is a namespace ratification on the transport-layer chunk header, not a change to the §4 attestation envelope** — the frozen 1+4 surface and its canonicalization are untouched.

Wire position (normative). `codec_id` (1 byte) + `ChunkLayer` { `spatial`: u8, `temporal`: u8, `quality`: u8 } (3 bytes) = a **4-byte additive block** at `SealedAvChunk` header offset **48..52**, after the existing v3.7.0 header. **Clear metadata, NOT inside the AEAD** — a relay drops chunks by `codec_id/ChunkLayer` without touching the inner epoch-DEK seal (§10.5.2); tampering causes mis-decode or drop, never a crypto break. **Additive + backward-compatible:** a v3.7.0 chunk round-trips identically as `codec_id` = 0xFF + `ChunkLayer` { 0, 0, 0 }. No length-prefix is needed — the block is fixed 4 bytes at a fixed offset.

Hex	Codec	Semantics
0x01	AV1 SVC	Scalable Video Coding — 3 spatial × 4 temporal × N SNR layers; base layer required to decode anything. Production default (WebRTC-native, royalty-free). The deployable codec today.
0x02	JPEG XS (layered)	Low-latency intra-only; broadcast use case. Reserved.
0x03	Symmetric MDC	Multiple Description Coding — any subset of chunks decodes at proportional fidelity, no base-layer floor. The substrate design target (below). Reserved — encoder lineage academic-grade today; substrate is MDC-ready.
0xFF	Opaque	No scalable-coding semantics; v3.7.0 wire-compat. ChunkLayer MUST be { 0, 0, 0 }. Default for legacy / non-layered streams.
0x04–0x7F	—	Reserved for future standardized codecs (CEG-assigned).
0x80–0xFE	—	Experimental / per-deployment — no cross-federation meaning guaranteed.

`codec_id` lets a receiver know whether `ChunkLayer` { `spatial`: 2, `temporal`: 2, `quality`: 0 } means "SVC base + 2 spatial enhancements" or "MDC quadrant" — without it, layer numbers are ambiguous across codecs. The substrate (Edge) never picks the codec — choice is upstream (agent/server tier); Edge needs only the namespace stable. The variable-depth `SubStreamPath` (MDC tree-path encoding) is the §85 follow-on (tracked §N.3).

MDC-primacy design intent (informative). The user-facing contract CEWP realtime A/V targets is *"any node can request a lower-bandwidth stream from peers — as simple as taking every other chunk, down to a blinking dot."* MDC (0x03) matches that **symmetrically**: drop any subset → decode the rest at proportionally lower quality, no coordination, no base-layer floor. SVC (0x01) is production-deployable today but has a floor (base-layer bytes must arrive) and needs coordination for an "every other chunk" drop. The substrate is **MDC-shaped** (the `ChunkLayer` / `SubStreamPath` model is symmetric-drop-ready) even while production streams

ship SVC; the ~20–40% MDC compression overhead vs SVC at equal quality is the accepted cost of symmetric drop semantics.

§10.5.8.3 SealedAvChunk wire layout (normative, 1.0-RC10 — absorbs CIRISEdge v4.0.0 per CIRISRegistry#85 §N)

The realtime A/V chunk that lands on each RNS Link payload (and the broadcast pull path).

****Byte layout (normative — transcribed from the edge v4.0.0 reference `SealedAvChunk::to_bytes`):****

```
offset field encoding
0..32 stream_id 32 bytes (caller-derived: sha256(stream_meta))
32..40 epoch u64 big-endian
40..48 chunk_seq u64 big-endian
48..49 codec_id u8 (§10.5.8.2 namespace)
49..50 layer.spatial u8
50..51 layer.temporal u8
51..52 layer.quality u8
52.. double_sealed_ciphertext remaining bytes
```

- `CHUNK_HEADER_LEN` = 48 (the `stream_id`+`epoch`+`chunk_seq` fixed header, stable since v3.7.0); `CHUNK_CODEC_LAYER_LEN` = 4 (the `codec_id`+`ChunkLayer` block).
- **Backward compatibility (normative, length-disambiguated):** a wire carrying **only** the 48-byte header (no trailing 4-byte block) **MUST** be read as `codec_id` = `0xFF` (opaque) + `layer` = `{0,0,0}`, bytes 48.. as ciphertext — the v3.7.0 shape. A wire with $\geq 48+4$ bytes after parsing the header is read as v3.8.0+ (`codec`+`layer` present). New writes always include the 4-byte block.
- `codec_id` + `layer` are **clear metadata**, **NOT inputs to the AEAD** (§10.5.8.2) — a relay drops by (`codec_id`, `layer`) without compromising the inner DEK; tampering causes mis-decode or drop, never a crypto break.

§10.5.8.4 ChunkLayer + ReceiverLayerPolicy — SVC layer model (normative, 1.0-RC10 — §85 §N.2)

`ChunkLayer` is the 3-byte SVC layer descriptor in the chunk header (`spatial`, `temporal`, `quality`, each u8). Each axis is **monotonic**: layer 0 is the base (lowest fidelity, always required); each increment is an additive enhancement. A receiver reconstructs from the prefix `0..=max_spatial` $\times$ `0..=max_temporal` $\times$ `0..=max_quality` of cells. The base cell `{0,0,0}` is the "**blinking dot**" — the minimum a participant can subscribe to. For `codec_id` = `0xFF` (opaque) the layer **MUST** be `{0,0,0}`.

`ReceiverLayerPolicy` { `max_spatial`, `max_temporal`, `max_quality` } (each u8) is the per-receiver drop policy. It is ****advertised over the existing federation_session / key_grant entitlement surface — NOT a new wire**** — and the sender drops chunks above the cap without re-encoding. `admits(layer)` is the per-axis test `spatial` $\leq$ `max_spatial` $\wedge$ `temporal` $\leq$ `max_temporal` $\wedge$ `quality` $\leq$ `max_quality`; a chunk tagged `codec_id` = `0xFF` **MUST** be admitted **unconditionally** regardless of policy (the fan-out filter short-circuits before consulting `admits`). Canonical policies: `BLINKING_DOT` = `{0,0,0}`, `UNCAPPED` = `{255,255,255}`. This composes with the inner-once / outer-N fan-out optimization (CIRISEdge#122): the inner seal runs once per chunk; the outer seal runs only for the (`receiver`, `chunk`) pairs the policy admits.

§10.5.8.5 Double-seal + deterministic nonce derivation (normative, 1.0-RC10 — §85 §N)

`double_sealed_ciphertext` is **outer-AEAD(inner-AEAD(`chunk_plaintext`))** — two independent AES-256-GCM layers (12-byte nonce + 16-byte tag, standard ring layout each). The **inner** seal is end-to-end (the epoch-DEK content seal); the **outer** seal is the per-RNS-Link transit wrap (a relay sees the outer layer only, never plaintext — the §10.5.5 E1 two-layer posture). Both nonces are **deterministic** (no nonce is transmitted — every holder recomputes; collision-safety rides the §10.5.8 single-sender-per-(`stream_id`, `epoch`) invariant):

```
inner_nonce = SHA-256( b"CIRIS-AV-INNER-V1" || stream_id[32] || epoch_be8 || chunk_seq_be8 )
               [0..12]
outer_nonce  = SHA-256( b"CIRIS-AV-OUTER-V1" || link_id || link_seq_be8 ) [0..12]
```

The label bytes (`b"CIRIS-AV-INNER-V1"` / `b"CIRIS-AV-OUTER-V1"`, ASCII, no terminator) are **domain separators pinned by this section** — they bind the nonce to its layer and prevent cross-layer reuse. `epoch`, `chunk_seq`, and `link_seq` are u64 **big-endian**. `link_seq` is monotonic per RNS Link (transit replay guard). **Conformance is proven by the §85 vector set** (input → expected 12-byte nonce + expected `to_bytes`), generated from the v4.0.0 reference impl — see the §57 freeze gate.

§10.5.7 What CEG 0.10 documents

Scope pointer: the §10.5 streaming surface — delivery axis (§10.5.1–§10.5.6) — does **not** change the 1+4 primitive set (§3) (delivery rides existing primitives), does **not** bundle the streaming-half substrate impl (Persist#142 + the RC1-1c CHECK-arm migration), keeps push-mode multicast relay/fan-out pull-only at RC1 (push tree → 1.x, per §10.5.5 E2), and leaves the `K` / `T` / `MAX_CHUNKS_PER_EPOCH` constants + accountable-stream quorum operator-tunable (ratification pending per §15.6 RC1-7).

§11 Governance discipline

§11.1 Operational-language gate at admission

Every new prefix admitted to the §5 namespace passes the §1.3.1 four-test gate. Failed admissions are revised (mechanism-descriptive reframe) or rejected.

§11.2 Amendment process — federation Contribution + WA quorum + 1-of-6 sign-off

Rule-layer changes (new prefixes, new envelope fields, new policies, calibration package version transitions) route through:

1. **Proposed amendment** filed as a NodeCore P5 Contribution (kind: PROPOSAL, subject: the proposal artifact).
2. **Witness diversity** per NodeCore P10 (N=3 default).
3. **WA quorum adjudication** per NodeCore P8.
4. **Reconsideration** per NodeCore P11 with fresh-quorum recusal (per §8.1.5 locality-scaled-quorum, including the §8.1.5.1 sub-quorum fallback).
5. **1-of-6 accord-holder OR steward sign-off** as defense-in-depth gate against rules-layer Sybil capture. The 1-of-6 sign-off is the secondary check; WA quorum is the primary substantive review. Any single signer can VETO by refusing to sign. Reduces the attack surface from "produce N Sybils" to "compromise one of six specific hardware-attested keys."

§11.2.1 Axis-vocabulary discipline

Every {axis} value emittable under open-vocabulary prefixes (e.g., `detection:correlated_action:{axis}`, `hard_case:{kind}`, `testimonial_witness:{kind}`) MUST carry an operational definition where the prefix has a calibration package (RATCHET-calibrated detectors) or a documented convention where it doesn't.

For RATCHET-calibrated detectors, the operational definition lives in the calibration package version pinned via `evidence_refs[]`:

1. Measurement procedure
2. Threshold function
3. Statistical floor
4. Evidence-shape requirement
5. Polarity semantics

For documentation-only open vocabularies (`testimonial_witness:{kind}`, `hard_case:{kind}`, `topical_relation:{kind}`), discoverability lives in non-normative registry documents like `WITNESS_KIND_REGISTRY` — additions there require no spec amendment.

§11.2.2 Open-vocabulary collision rule

When two parties independently register confusingly-similar `{kind}` / `{axis}` values within the same prefix family, the following resolution applies:

1. **First-registered wins** for the canonical-attestation surface. The earlier `signed_at` holds the name; later registrations carry a `differs_in:` `["semantic_disambiguation"]` clarification or pick a distinct value.
2. **Levenshtein-distance guard:** a CEG-Conforming Substrate (CCS) SHOULD compute Levenshtein distance against existing values in the same prefix family at admission; values within distance ≤ 2 of an existing canonical value SHOULD return a 409 `IDEMPOTENT_CONFLICT` with an advisory hint, NOT a hard reject — the producer may proceed if the similarity is intentional (e.g., `commonsense` vs `commonsense_hard` are intentionally close).
3. **No squatting:** a `{kind}` registered but never used (no scored attestations in 90 days) MAY be reclaimed by another producer via the §11.2 amendment process.

§11.2.3 Meta-amendment + entrenchment

The §11.2 amendment process itself, the §1.3.1 T1–T4 prefix-admission gate, and the §9 HUMANITY_ACCORD constitutional layer are **entrenched** — changes to these three surfaces require a MAJOR version bump per §0.3 AND an additional 2-of-3 HUMANITY_ACCORD signatures (NOT the 2-of-3 from §11.2 step 5 — a separate, dedicated accord ratification). Without this entrenchment, a single quorum could rewrite the gate admitting the next quorum.

§11.3 Bootstrap-content pattern

After federation genesis, a curated batch of P5 Contributions is admitted via the §11.2 amendment flow, populating the federation’s substantive content surface with high-quality ethical-framework material. **Content-neutral:** any sufficiently substantive ethical-framework source can serve. The wire format admits content via the §5 namespace; the §1.3.1 gate ensures prefix names don’t import source-tradition vocabulary.

First deployment: the *Magnifica Humanitas* encyclical mapping at ~75-80% transparent translation rate (Cargo `ciris-response-magnifica-humanitas` repo).

Multi-source commitment: subsequent bootstrap batches from CARE Principles (Indigenous data governance), Buddhist economic-justice scholarship, secular humanist instruments, African philosophy of personhood work — all through the same amendment process. The framework is multi-traditional by design.

§11.4 Fast-path takedown coordination (CEG 0.3 addition; per CIRISRegistry#37 + #38)

For `takedown_notice` Contributions (§5.6.8.4) whose `legal_basis` falls in the **immediate-removal** category (`TvecTerrorist` / `NcmecCsam` / `GifctCip` / `PerceptualHashCsam` / `CourtOrder`), the §11.2 amendment process timeline is too slow — TVEC mandates a 1-hour removal obligation; GIFCT CIP coordinates within hours; NCMEC + perceptual-hash + court orders demand near-immediate response.

CEG 0.3 carves out a fast-path coordination protocol:

1. **Notice admission:** the `takedown_notice` Contribution arrives at the substrate, signed by `claimant_key_id`. The substrate accepts it without §11.2 quorum; speed matters at this layer.
 2. **Holder eviction:** substrate emits a `withdraws` against the matching `holds_bytes:sha256:{prefix}` directory entry per §10.1.2. Holders see their advertisement marked withdrawn and SHOULD cease serving the bytes.
 3. **Per-basis dispatch:**
 - `TvecTerrorist` — operator coordinates via TVEC-designated channel (national regulator notification within 1 hour); substrate logs the notice + the eviction action to its audit chain.
 - `GifctCip` — operator coordinates via GIFCT Content Incident Protocol communication channel; same audit-chain logging.
 - `NcmecCsam` + `PerceptualHashCsam` — operator MUST file the NCMEC CyberTipline report (US 18 USC §2258A); substrate retains hash + minimal metadata for the federal-legal retention window only. No content retention.
 - `CourtOrder` — operator follows the court’s stated timeline; substrate logs the order text + the eviction action.
1. **Audit trail:** every fast-path takedown enters a `hard_case:fast_path_takedown` Contribution (§5.6.6) for downstream review. Reviewers MAY file a `reconsideration:procedural_error` if the fast-path basis was misclassified.
 2. **No counter-notice for immediate-removal cases:** by `legal_basis` design (TVEC / NCMEC / GIFCT / `PerceptualHashCsam` / `CourtOrder` all bypass counter-notice). The `expeditious-with-counter-notice` bases (`Dmca512` / `DsaArticle16` / `CommunityStandards` / `OsaIllegalContent`) route through the standard §11.2 amendment path on counter-notice via `reconsideration:new_evidence`.

The takedown-isn’t-a-coup property: the §9 HUMANITY__ACCORD remains load-bearing. Fast-path takedowns happen via this protocol but a `takedown_notice` Contribution targeting the substrate itself (e.g., a state actor demanding takedown of `federation_keys` for whole categories of dissenting participants) would not propagate the same way — substrate-protective discipline + HUMANITY__ACCORD veto authority intersect at the substrate level. Operators in jurisdictions where this conflict materializes SHOULD escalate to the HUMANITY__ACCORD triple per §9.2 invocation procedures.

§11.5 Hash-database operator policy (CEG 0.3 addition; per CIRISRegistry#39)

Perceptual-hash matchers (PhotoDNA / PDQ / Project Arachnid / GIFCT hash-sharing) are pluggable per the CIRISPersist `PerceptualHashMatcher` trait. Operators choose which matcher implementations to enable; CEG governs the access-policy contract.

§11.5.1 Hash-database access landscape

Matcher	Access posture
PDQ (Meta, 2019)	Open — algorithm + reference hashes publicly distributed
PhotoDNA (Microsoft, 2009)	Access-gated; restricted to vetted orgs (NCMEC + select platforms); substrate operators cannot download the hash database directly
Project Arachnid (C3P, 2017)	Access-gated; API access requires C3P partnership
GIFCT hash-sharing	TVEC-focused; access via GIFCT membership

§11.5.2 Operator path (CEG 0.3 default — option (a) per CIRISRegistry#39)

For a CIRIS substrate operator running a federation node, the **CEG 0.3 default operator path** is:

Self-hosted PDQ matcher against publicly-distributed reference feeds (Microsoft Project Arachnid feed where publicly available, GIFCT-published lists where openly available). No access-governance overhead. Operator carries responsibility for index freshness.

This avoids the federation-dependency-at-substrate-protective-layer problem that option (b) (clearinghouse delegation) would introduce, and the controversy around option (c) (on-device hash-database access via OS-vended hooks, per the iOS NeuralHash 2021 incident).

§11.5.3 Future hash-coalition path (deferred; awaits CIRIS hash-coalition emergence)

CIRISRegistry will file a follow-up issue when a CIRIS hash-coalition emerges that can serve as a clearinghouse for option (b) — substrate operators delegating perceptual-hash checks to a trusted coalition peer via federation. CEG 0.3 documents the slot; the actual coalition operator-onboarding flow is deferred.

§11.5.4 What CEG 0.3 documents

- The closed-set of `legal_basis` values that compose with `PerceptualHashCsam` (the only `legal_basis` value that consumes hash-match output as immediate-removal trigger; see §5.6.8.4)
- The operator-onboarding contract: an operator running a PDQ matcher **MUST** register their matcher’s source feeds (which hash-list source URLs they’re pulling from + freshness cadence) via a `system:perceptual_hash_matcher:registered` Contribution. Composes with §5.3 Persist substrate-self-report discipline.

What CEG 0.3 does NOT do: prescribe which hash databases an operator **MUST** use. Operator choice. CEG documents the wire-format slot + the operator-onboarding contract + the recommended default; concrete matcher selection is operator policy.

§11.6 Vertical compliance + subject-bearing dimension governance (CEG 0.6 addition; per CIRISRegistry#45)

Per CIRISRegistry#45 + CIRISAgent#842. The wire-format primitives in §4.2 `subject_key_ids` + §5.6.8.6 `consent:*` family + §5.6.8.7 `consent_record` + §8.1.11 Policy K compose into regulatory-vertical compliance mappings. CEG documents the canonical mappings as **informational**; the wire-format primitives are domain-agnostic and operator-configurable.

§11.6.1 Vertical compliance mapping (informational)

Regulatory framework	CEG primitive	How it composes
GDPR Article 7 (consent)	<code>consent:state:granted</code> + <code>consent_record.subject_key_id</code>	Subject's wire-format declaration of consent; revocable via §3.2.3 rule 2
GDPR Article 9 (special category — health, biometric, sexual orientation, etc.)	<code>subject_key_ids</code> MANDATORY for special-category content; producer's <code>consent:deletion_sla</code> SHOULD be ≤ 30 days	Substrate-level recognition that special category requires subject-side wire authority
GDPR Article 17 (right to erasure)	<code>consent:state:revoked</code> → substrate-watched <code>consent:deletion_sla:{days}</code> → producer emits <code>consent:deletion_complete</code> OR substrate emits <code>hard_case:consent_sla_breach</code>	The §8.1.11.3 SLA watcher is the wire-format observability primitive for Article 17 compliance
GDPR Article 20 (data portability)	DSAR export via <code>attestations.where(s ∈ subject_key_ids)</code> query	CIRISAgent's <code>DSARExportPackage</code> composes from this query trivially
HIPAA 45 CFR 164.502 (uses + disclosures)	<code>'consent:scope:{retain\</code>	<code>share\</code>
HIPAA 45 CFR 164.524 (patient right of access)	DSAR export per Article 20 above	Same composition
FERPA 34 CFR Part 99 (educational records)	<code>subject_key_ids: [student_key]</code> + <code>delegates_to(parent_key → student_canonical_hash, scope: [consent_revocation])</code> for minors	Parental authority composes via the existing <code>delegates_to</code> primitive; no new shape needed
CCPA §1798.105 (right to delete)	Same composition as GDPR Article 17	Substrate-watched SLA + <code>consent:deletion_complete</code>
EU AI Act Article 50 (training data transparency + opt-out)	<code>consent:scope:train</code> + <code>is_ai_generated</code> field at content publish + subject's <code>consent:state:revoked</code> against the training-datum Contribution	Subject can withdraw training-set consent; producer's deletion-SLA fires on the training-corpus Contribution
CIRIS Accord M-1 (sustainable adaptive coherence — consent revocability)	The entire CEG 0.6 surface	The constitutional anchor — "consent (M-1's load-bearing property) requires revocability, and revocability requires a halt-authority that lives outside the system being halted" (§9 + MISSION.md §1.5). CEG 0.6 extends this recognition from accord-carriers (federation-as-a-whole halt) to all subject-authorities (per-Contribution halt) at scale.

CEG does NOT prescribe which regulatory framework an operator MUST comply with; the wire

primitives compose to ANY of them based on operator policy. Operators in regulated verticals (medical / legal / financial / educational) SHOULD pin compliance mappings as configuration above the wire primitives, not as new wire shapes.

§11.6.2 Subject-bearing dimension governance (normative)

Per §4.2.6. Dimensions whose namespace pattern names a subject MUST carry **subject_key_ids** containing that subject. This closes the default-leak failure mode where subject-bearing content publishes without wire-level subject authority (Gap 4 from the CIRISAgent#842 gap audit).

Subject-bearing dimension patterns (open catalog; operator vocabularies extend):

Pattern	Example	Required subject_key_ids entry
observed:user:{key_id}:*	observed:user:abc123:interaction_count	abc123 (or its canonical-hash form)
epistemic:about:{key_id}:*	epistemic:about:abc123:trust_assessment	abc123
epistemic:memory:topic={topic} (when topic names a person/entity)	epistemic:memory:topic=patient_xyz:passion	patient_xyz canonical-hash
consent:partnered:{user_key} (CIRISAgent CEM agent-side stance)	consent:partnered:abc123	abc123
agent_files:*: {subject_target} (when target names a person)	agent_files:medical_record:patient_xyz:patz	patient_xyz canonical-hash
licensure:{authority_id}:{practitioner_key} (when practitioner is named)	licensure:CA_medical_board:dr_jones:drjones_key	drjones_key

Substrate enforcement: substrate admission gate MAY reject Contributions where the dimension matches a subject-bearing pattern but **subject_key_ids** is empty / does not contain the named subject. This is **operator-policy** (not normative across all substrates) — some operator configurations may admit and emit a **hard_case:subject_authority_missing** for review-queue handling instead of rejection.

The takedown-isn't-a-coup parallel (§11.4 fast-path takedown) applies: substrate enforcement of subject-bearing dimension discipline cannot be used as a coup against the substrate itself (e.g., a state actor publishing **observed:user:dissenter_key:*** with **subject_key_ids** = [] and demanding substrate admission). The §9 HUMANITY_ACCORD remains load-bearing; admission-gate rules apply uniformly.

§11.6.3 What CEG 0.6 documents

- The wire-format primitives that compose into vertical compliance (informational mapping above)
- The dimension-pattern-implies-**subject_key_ids** requirement (normative gate)
- The bilateral-pair shape for ceremony grants per §8.1.11.4
- The decay-protocol stage composition per §8.1.11.5
- The SLA watcher boundary (substrate emits **hard_case:***; LensCore composes **detection:***) per §8.1.11.3

What CEG 0.6 does NOT do:

- Bundle CIRISAgent's CEM streams as the only valid stream set (open vocab; CEG names **temporary** / **partnered** / **anonymous** as recommended canonical kinds, not lockdown)

- Define specific SLA values for any regulatory framework (operator policy — though informational guidance: GDPR Article 9 default ≤ 30 days; CCPA default 45 days; etc.)
- Provide a decay-protocol library (CIRISAgent's 90-day-decay is the canonical example; other protocols MAY exist)
- Prescribe per-vertical compliance audit cadence (consumer / regulator concern)

§11.7 Self/family membership governance (CEG 0.7 addition; per CIRISRegistry#47)

Per §5.6.8.8 `identity_occurrence` + §5.6.8.9 `family` + §8.1.12 Policy L + §10.1.4 structural-invisibility. The four governance decisions for self/family membership.

§11.7.1 Forward secrecy on member departure — Option A (CEG 0.7 default)

When a member leaves a family (or an occurrence is revoked from a self-collective), the removed party retains existing `key_grants` for historical content; the substrate stops wrapping new `key_grants` on subsequent content. No DEK rotation; no re-encryption.

Rationale: consistent with §3.2 `withdraws-isn't-retroactive` + §11.4 "takedown isn't a coup" + CEG 0.6 §8.1.11.5 `consent-decay-doesn't-re-encrypt`. The substrate's forward-secrecy posture is uniform across consent, takedown, and membership-departure surfaces.

Option B (rotate-DEK on member departure; re-wrap all extant content to remaining members) is deferred. CEG 0.7 documents the slot for a future `subject_kind: family_rotation` ceremony that operators can opt into per family; the per-`(family_id, epoch)` rotation axis would parallel CEG 0.10's per-`(stream_id, epoch)` axis (§10.5.3) — a distinct axis from CEG 0.3's `key_grant.rotation_chain` (which is content-addressed grant-supersession lineage per §5.6.8.4, not key rotation). The ceremony envelope is downstream-demand-driven; the wire-format primitives needed are the `key_grant` wrap + Option-A re-grant on existing members (which already work today).

§11.7.2 Multi-family membership — envelope `family_id` (CEG 0.7 default)

One identity MAY belong to multiple families. Each family has its own DEK and its own membership roster; a `cohort_scope: family` Contribution MUST carry `family_id` (§4 envelope field) naming which family's DEK and visibility apply. Substrate rejects family-scoped Contributions missing `family_id`.

Rationale: avoids cross-family DEK confusion when one identity is in N families; gives the substrate an unambiguous routing key for the `key_grant` cascade per §8.1.12.4.

§11.7.3 Reserved-prefix substrate emissions — locked in §7.7

Per §7.7. The four substrate-emitted membership-event prefixes (`hard_case:identity_occurrence_added:*`, `hard_case:family_membership_change:*`, `hard_case:family_consensus_protocol_change:*`, `hard_case:family_consensus_protocol_violation:*`) are reserved to `identity_type="substrate_persistent"` emitters. Same discipline as the existing `system:*` substrate-self-report family.

§11.7.4 Self-occurrence admission — single-vouch (CEG 0.7 default)

A new `identity_occurrence` is admitted on a signature from EITHER the root `identity_key_id` OR any currently-admitted occurrence of that identity (Signal-style "trust any device I've already onboarded"). Higher-assurance setups MAY layer requirements on `hardware_attestation` via consumer policy.

Rationale: matches user-intuition for self-membership ("my phone unlocks my laptop's trust posture for new devices"); avoids the operational overhead of multi-vouch for routine onboarding; the security gradient lives in the optional `hardware_attestation` field, not in the admission rule.

§11.7.5 Family admission — consensus_protocol (CEG 0.7 normative)

Unlike self-occurrence, family membership changes are **NOT single-vouch by default**. The family's `consensus_protocol` field (locked at family creation per §5.6.8.9) governs admission. Six canonical protocols (`founder_only` / `unanimous` / `majority` / `quorum:M/N` / `weighted:{rubric}` / `custom:{family_id}`); operator vocabulary extends.

The `consensus_protocol` field is itself subject to amendment via the SAME protocol's rules (meta-amendment shape parallel to §11.2.3) UNLESS the family is `consensus_protocol_entrenched:true`. Entrenched families reject amendments at the substrate gate; replacement requires the family's documented out-of-band ceremony (HUMANITY_ACCORD per §9.2 / FEDERATION_ANNOUNCEMENT §4.5.3 is the canonical entrenched example).

Rationale: families are multi-party collectives where membership changes have real consequences (admit-new-member = grant DEK access to all extant cohort_scope: family content). The `consensus_protocol` gives the family explicit governance over its own boundary. Self-amendment lets families evolve their governance as they grow; entrenchment lets safety-critical families lock the boundary against any internal authority.

§11.7.6 What CEG 0.7 documents

- The wire-format primitives that compose into self/family membership (§5.6.8.8 + §5.6.8.9)
- The structural-invisibility discipline at §10.1.4 (cohort_scope: self/family suppresses holds_bytes:*)
- The at-rest encryption flow composition at §8.1.12 Policy L
- The `consensus_protocol` vocabulary (canonical kinds; open-vocab extension)
- HUMANITY_ACCORD as the canonical entrenched-family instance at §9.1

What CEG 0.7 does NOT do:

- Lock the `consensus_protocol` vocabulary (open vocab; canonical kinds named for ecosystem coordination)
- Provide a key-rotation ceremony for Option B forward secrecy (deferred to downstream-demand-driven future release)
- Prescribe per-family entrenchment policy (operator/family choice)
- Document the CIRISPersist#152 at-rest encryption flow details (substrate-side; persist spec)

§11.8 Geographic-community privacy invariant (CEG 0.8 addition; per CIRIS-Registry#48)

Per §5.6.8.10 `community` with `cohort_subkind`: `geographic` + §5.6.8.11 `location_proof` + §0.8 H3 cell canonicalization + §8.1.13 Policy M.

CEG 0.8 codifies a load-bearing privacy invariant: **joining a geographic community is a one-way disclosure**. Three sub-properties make this wire-format-level, not policy-level.

§11.8.1 Rough-only is wire-format-enforced

Per §0.8.1: `location_proof.cell_resolution` ≤ 7 . H3 resolution 7 hexagons average $\sim 5 \text{ km}^2$ edge-length — sufficient for city/borough disclosure without block/building precision. Producers attempting finer resolution have admission rejected at the substrate gate.

This is the closure of an entire class of accidental over-share: a malformed UI cannot publish precise location even if the client-side gating fails. The wire-format-level enforcement is the privacy primitive; UI is the second line.

Substrate emits `hard_case:location_proof_resolution_violation` (§7.8) on rejection so operators can observe malformed-client patterns. This is observability for operator debugging, NOT a slashing trigger — malformed producers are usually buggy clients, not attackers.

§11.8.2 Leaving is forward-only — the audit chain preserves the historical claim

Per §3.2 `withdraws-isn't-retroactive` + CEG 0.7 §11.7.1 Option A forward-secrecy. When a subject withdraws their `location_proof` (or leaves a geographic community):

- Forward visibility evicts (consumer policy treats the subject as "no current location proof" for new admission decisions from withdrawal-time forward)
- The withdrawn `location_proof` Contribution **remains in the audit chain** — federation peers retain the historical record
- "I was in Austin from May to August" is permanent; "I am currently in Austin" is what withdraws/expiry govern

This is the cost the subject opts into when emitting the `location_proof`. Per the CEWP structural-not-policy framing, the wire format does not promise to expunge historical claims — that promise would be hollow (federation peers retain copies; the substrate can mark forward-only). What the wire format DOES promise:

- **Rough-only** (§11.8.1): the historical claim is bounded to resolution ≤ 7 , never finer
- **Opt-in** (§11.8.3): the historical claim exists only because the subject signed and emitted it
- **Auditability**: the subject can prove what they did or did not claim, when (the audit chain is the receipt)

§11.8.3 Joining is opt-in — substrate does NOT solicit location

A `location_proof` Contribution is emitted **only** by the subject themselves (`attesting_key_id == subject_key_id`) or by a `delegates_to` chain with `scope`: `[consent_revocation]` per

CEG 0.6 — the substrate has no path to mint a `location_proof` on behalf of a key without an explicit signature.

Communities cannot **require** a `location_proof` from non-members (they can only **gate admission** on whether a member has produced one). The substrate has no mechanism for "involuntary location disclosure" via the wire format. A bad actor cannot force-publish another key's `location_proof`; without the subject's signature, the substrate rejects.

Compose with §9 HUMANITY_ACCORD substrate-protective discipline: a state actor demanding the substrate emit `location_proofs` for non-consenting subjects is exactly the substrate-protective case that the HUMANITY_ACCORD halt authority exists to address. The substrate's role at this primitive is mechanical opt-in enforcement; political/legal disputes route through the §9 + §11.4 takedown coordination + the HUMANITY_ACCORD `EmergencyShutdown` CONSTITUTIONAL path if necessary.

§11.8.4 What CEG 0.8 documents

- The rough-only enforcement primitive at §0.8.1 (resolution ≤ 7 for `location_proof`)
- The forward-only leave semantics at §3.2 (withdraws-isn't-retroactive applied to `location_proof`)
- The opt-in admission flow at §5.6.8.11 (`location_proof` signed by subject only or delegates_to proxy chain)
- The geographic-community admission composition at §8.1.13.2 (Policy M `evaluate_subkind_admission` for geographic)
- The substrate-self-report observability at §7.8 (4 `hard_case` prefixes)

What CEG 0.8 does NOT do:

- Mandate H3 over alternative geospatial systems for operator-internal use (operator choice; wire format uses H3 only)
- Provide a place-name registry (communities self-name; H3 cells are the substrate-level binding)
- Define specific cell-resolution conventions for community-side `geographic_constraint` (only `location_proof` is bounded to ≤ 7 ; communities MAY scope themselves at any resolution per operator/founder choice)
- Codify non-geographic community subkinds (`professional` / `interest` / `local-business` / `event-attendees` / etc. are downstream-demand-driven future spec rounds — same discipline that drove $0.3 \rightarrow 0.4 \rightarrow 0.5 \rightarrow 0.6 \rightarrow 0.7 \rightarrow 0.8$)
- Address `affiliations` (the fourth cohort_scope tier; deferred — CEG 0.9 took the §7.0.1 `identity_type-as-set` cut instead; `affiliations` remains a later candidate round)

§11.9 `identity_type` as a set — single-key role cohabitation (CEG 0.9 addition; per CIRISRegistry#49 + CIRISAgent#856)

Per §7.0.1 + CIRISRegistry#49 + CIRISAgent#856. CEG 0.9 generalizes `federation_keys.identity_type` from a single scalar role to a **set of roles**, so the §7 reserved-prefix gates are evaluated by set membership ($X \in \text{identity_type}$). This amendment routed through the §11.2 process; CIRISAgent#856 is the driver, CIRISRegistry#49 the CEG-authority mirror.

§11.9.1 What the amendment changes — and what it deliberately does not

Surface	Before 0.9	At 0.9
<code>federation_keys.identity_type</code> representation	scalar string	set of role strings (legacy scalar = singleton set)
§7 emitter-rule evaluation	<code>identity_type == "X"</code>	<code>X ∈ identity_type</code> (§7.0.1)
§5 dimension namespace	—	unchanged
Envelope (§4)	—	unchanged
1+4 structural primitives (§3)	—	unchanged
<code>subject_kinds</code> (§5.6.8)	—	unchanged

This is a ****wire-break** at the `federation_keys` row representation only — **the second 0.x wire-break after the §0.2 attestation-ladder rename**. It is semantically null for every legacy single-role key^{**}: $X \in \{X\} \equiv X == X$. It is NOT a §1.4 "Nth path" confirmation (it adds no namespace surface); it is a §7-layer enforcement generalization that unblocks the CIRISAgent fold-in (one key, many roles) without expanding the wire's expressive surface.

§11.9.2 Settled in CIRISAgent#856, carried as-is

- **Capacity self-emission (§7.5)**: unchanged. The anti-Goodhart `attesting_key_id ≠ attested_key_id` rule binds regardless of how many roles a key holds. A folded `{agent, lenscore_detector}` key still **MUST NOT** score its own `capacity:*`. Role cohabitation does not create a self-attestation backdoor.
- **Reasoning-trace dimensions**: no separate reserved `identity_type` required; reasoning-trace emission rides the agent role's open-vocabulary surface. Cohabitation does not change this.
- **Agent-intent / LensCore-envelope split**: a cohabiting key emits agent-intent attestations under the agent dimensions and detector verdicts under `detection:*` (§7.4 worked example). The namespace keeps the two surfaces distinct; cohabitation grants the right to emit on each, never merges them.

§11.9.3 Cohabitation discipline for constitutional + substrate roles

Set membership grants the wire-level *right* to emit per held role, but two roles carry defense-in-depth guidance against cohabitation:

- ****accord_holder (§7.1 + §9)****: the one constitutional asymmetry. Consumer policy **SHOULD** treat an `accord_holder` key that also holds non-constitutional roles (e.g., `{accord_holder, agent}`) with elevated scrutiny — the HUMANITY_ACCORD triple's halt authority is strongest when its keys are single-purpose. CEG 0.9 does NOT forbid the cohabitation at the wire layer (the substrate cannot adjudicate constitutional intent), but the §9 entrenched-family discipline **RECOMMENDS** dedicated accord-holder keys.
- ****substrate_persist / substrate_edge (§7.2)****: substrate-self-report roles remain cross-attested by the full steward-triple per §7.2. A key cohabiting a substrate role with an application role (e.g., `{substrate_persist, agent}`) **MUST** still satisfy the steward cross-attestation requirement for the substrate-role emissions; cohabitation does not relax the cross-attestation gate.

§11.9.4 What CEG 0.9 documents

- The set-membership reading of every §7 reserved-prefix emitter rule (§7.0.1)
- The canonical-bytes encoding for the set (sorted-ascending, deduplicated, comma-joined; single-role keys encode identically to their pre-0.9 scalar form)
- The LensCore-fold worked example (§7.4)
- The cohabitation discipline for constitutional + substrate roles (§11.9.3)

What CEG 0.9 does NOT do:

- Expand the §5 dimension namespace, the §4 envelope, the §3 structural-primitive set, or the §5.6.8 `subject_kinds` (zero new wire surface beyond the `identity_type` representation)
- Enumerate a closed set of role values — `identity_type` members remain an open vocabulary owned per §7 reservations + sibling-component vocabulary extensions (e.g., CIRISPer-sist#102 `witness`)
- Forbid any cohabitation at the wire layer (substrate enforces gates by membership; constitutional/substrate cohabitation discipline is consumer/operator policy per §11.9.3)
- Address `affiliations` (the fourth `cohort_scope` tier; remains deferred to a later candidate round)

§12 Translation discipline (writing claims in CEG)

When translating substantive content into CEG envelopes, follow the discipline. Full primer at LANGUAGE_PRIMER.md; key rules consolidated here.

§12.1 The five families (organizing the namespace)

Every claim sits in one of five families:

Family	Question	Analogy
STANDING (about an entity)	"This key_id has property X."	Notarized professional credential record
ACTION (decision hierarchy)	"We aim for X via approach Y, through methods Z, measured by W."	Research grant proposal
DETECTION (reality patterns)	"Pattern X is/isn't present in the federation's behavior."	Epidemiological surveillance
CONSENSUS (collective judgment)	"The federation agrees that X, with these witnesses."	Peer review + jury deliberation
CORRECTION (self-correction)	"Something went wrong; here's the finding; here's the appeal."	Academic ethics committee + journal retraction + appellate review

§12.2 The four verdict categories (STRICT)

Verdict	Meaning
clean	Single primitive captures the operational claim without loss.
composed	Two or three primitives together carry the claim; each is genuinely required.
partial	The structural core translates but a meaningful operational claim is unmapped.
not-translated	The paragraph's content does not translate into the wire format at all. Declare T-1 / T-2 / T-3.

Do not invent intermediate categories.

§12.3 The not-translated taxonomy

T-1 — TRADITION_AUTHORITY: Claim belongs to the source's own theological/philosophical/scholarly tradition's authority. No Contribution owed; the correct posture.

T-2 — PASTORAL_PROSE: Claim is moral exhortation, narrative imagery, doxological language, or rhetorical framing. No Contribution owed.

T-3 — EXPRESSIVE_GAP: Claim is morally serious, operational, and unmapped. **These are the load-bearing findings.** Each T-3 must name: (a) why existing namespace doesn't reach it, (b) what extension would close it, (c) whether the extension would survive the §1.3.1 four-test gate.

§12.4 Decision tree

1. **Paragraph TYPE?** Operational claim → continue. Pastoral/rhetorical → T-2. Theological/tradition-specific → T-1.
2. **Which family?** STANDING / ACTION / DETECTION / CONSENSUS / CORRECTION (§12.1).
3. **Which specific prefix?** Scan §5; check composition before reaching for new prefix.
4. **Fill the envelope** (§4).
5. **Compose only when needed.** Multi-primitive translations for paragraphs that genuinely name multiple structural objects.

A machine-readable namespace manifest (`FSD/CEG/dimensions.json`) is planned for CEG 1.0 — it will enable mechanical prefix lookup, polarity reading, and per-dimension aggregation defaults without human scanning of §5. (Originally targeted for 0.2 per the early-draft roadmap; the 0.3 → 0.8 wave prioritized substantive namespace additions over tooling, so the manifest now lands alongside the 1.0 lock when the namespace stabilizes.)

§13 Anti-patterns

These are wire-format reaches that fail the §1.3.1 operational-language gate or import Cartesian-individualist defaults. Recorded so the next generation of authors finds the discipline before the wire format does. A CEG-Conforming Producer (CCP) SHOULD NOT emit attestations matching the anti-patterns below.

§13.1 Already-rejected wire additions

Anti-pattern	What it would smuggle	Correct expression
<code>detection:emergent_deception:{axis}</code> (renamed v1.2)	Moral verdict ("deception") in prefix name	<code>detection:correlated_action:{axis}</code> — mechanism-descriptive
<code>attestation:l{N}:*</code> (renamed CEG 0.2)	Ladder-position (verdict-shape) in wire prefix — same shape as <code>score:trustworthiness:*</code> smuggling meta-judgment as wire	<code>attestation:{mechanism}</code> bare mechanism (<code>self_verify</code> / <code>hardware_rooted</code> / <code>registry_consensus</code> / <code>license_validity</code> / <code>agent_integrity</code>); consumer composes L1-L5 ladder via §8.1.9 Policy I
<code>score:trustworthiness:{entity}</code>	Meta-judgment as separate prefix	Compose downstream from <code>licensure:*</code> / <code>capacity:*</code> / <code>provenance:*</code> attestations
<code>flag:bad_actor:{axis}</code>	Pejorative wire vocabulary	Surface as low-confidence scores on <code>provenance:*</code> and <code>coherence_standing:*</code> ; adjudicate via NodeCore P8 quorum
<code>grounding:{tradition}:{principle}</code>	"Tradition" claims are interpretive, not mechanism	Reuse <code>delegates_to</code> structural primitive per §3.2.1

§13.2 CEG 0.1 rejections (from CIRISRegistry#30 stress test)

Anti-pattern	Stories reached for	Why reject	Correct expression
<code>epistemic_mode: introspection</code>	7 stories	Cartesian shortcut — lone subject pre-declaring inner state as if that constitutes standing	<code>witness_relation: self + confidence < 1.0</code> + pending external composition. The wire makes introspection awkward to express, because the framework's claim is that self-knowledge does not constitute standing without external witness.
<code>epistemic_mode: testimony</code>	(same set)	Reducible to existing values	<code>epistemic_mode: external</code> + <code>witness_relation: external</code>
<code>transparency:{kind}</code> standalone prefix	12 stories	Disclosure is constituted by reception, not announcement. Cartesian self-claim shape.	<code>evidence_refs[]</code> carries the reasoning-chain hash; downstream <code>transparency_log:inclusion</code> from a witness who actually retrieved it.

Anti-pattern	Stories reached for	Why reject	Correct expression
<code>stake: civic / epistemic / dignitary</code>	10 stories	<code>civic = stake: reputational + cohort_scope: community. epistemic = confidence + stake: reputational</code> (same axis as confidence; not separate). <code>dignitary</code> lives on wrong axis (stake names what the attester loses; dignity harm is what the attested loses → belongs in <code>harm_class:dignity_harm</code>).	Compose existing values with cohort/harm-class.
<code>oversight_mode: deferred / active / advisory</code>	6 stories	All map to existing HITL/HOTL/HOOTL	<code>deferred</code> = HITL pre-decision; <code>active</code> = HITL with substrate monitoring; <code>advisory</code> = HOTL
<code>provenance_walk</code> as wire primitive	(1 reviewer)	UX concern smuggled into wire format	Consumer-side composition (Portal / Verify dashboards / agent introspection); the chain already walks via <code>references_attestation_id + topical_relation:* + valid_until</code>
Renaming canonical capacity factors and HE-300 categories to "kid-friendly" names	8 stories	Canonical names map to a worked-out epistemic/ethical lattice that loses precision under accessibility renames	Translation glossary in <code>LANGUAGE_PRIMER.md</code> (spec name ↔ narrative name) + version pinning in worked examples

§13.3 Delegation-launders anti-pattern

`delegates_to` → `delegates_to` → `delegates_to` → ... → `attacker` chains, where each hop is individually well-formed but the aggregate routes trust to a terminal attester the original delegator would not have approved.

What's wrong	Correct expression
Unbounded depth <code>delegates_to</code> chains	Consumer policy MUST cap traversal depth at 5 hops by default (configurable); chains longer than the cap are treated as <code>attestation:self_verify</code> only (no transitive trust)
Cycles ($A \rightarrow B \rightarrow A$)	Substrate MUST detect cycles on the <code>delegates_to</code> graph and reject the cycle-closing emission
Aggregate-weight concentration	Consumer policy SHOULD cap the trust weight any single terminal delegate can accumulate from a given root attester at $0.5 \times \text{root_trust}$ by default

§13.4 `withdraws` arbitrage

Per §3.2.2: a misattester can `withdraws` instead of `recants` to dodge the trust penalty consumers apply to acknowledged-error chains.

What's wrong	Mitigation
Asymmetric attester incentive	Consumer policy MUST track per-attester withdraws:recants ratio over a rolling window; attesters whose ratio exceeds a configured threshold (default 5:1) SHOULD be downweighted regardless of which structural primitive they use. The recants distinction matters §3.2.2, but the practical anti-arbitrage countermeasure is consumer-policy behavioral analysis, not a wire-format change.

§13.5 Discipline pattern

The recurring shape across most anti-patterns: **extending the wire format so single attesters can pre-declare their own state more richly**. Each one is reachable, none is necessary.

The Ubuntu-primary discipline cuts cleanly: standing is constituted relationally through attestation by others, not through self-declaration. The wire format should **resist** primitives that let a single key announce its own state without external composition. The substrate is austere by design so consumers compose verdicts; richer narrative expression belongs in **context:**, **evidence_refs[]**, and downstream witness attestations, not in new envelope enum members or new self-attestation prefix families.

§14 Glossaries

§14.0 Core terms (added 1.0-RC5; resolves CIRISRegistry#77)

These terms are referenced throughout the spec and across sibling repos. Defining them in-spec retires the external `ciris.ai/cewp` placeholder citations.

Term	Definition
CEG (CIRIS Epistemic Grammar)	This specification. The federation's wire grammar: the "1+4" attestation model (scores plus the four relations delegates_to / supersedes / withdraws / recants), its namespaces, admission rules, and composition policies. CEG is the <i>grammar</i> ; CEWP is the <i>network that speaks it</i> .
CEWP (CIRIS Epistemic Web Platform)	The decentralized network formed when nodes exchange CEG envelopes over Edge/Reticulum transport. CEWP is not a product, server, or central service — it is the emergent peer-to-peer web of CEG-speaking nodes, exactly as "the Web" is the emergent network of HTTP-speaking servers. It has no owner, no root, and no load-bearing instance (the §7.0.1 fabric-node discipline and the default-not-forced-root rule of §5.6.8.10 guarantee this). A CEWP node is a fabric node (CIRISServer); agent = fabric node + brain .
Fabric node	A headless CEG/CEWP participant: it attests, stores, observes, reaches consensus, and transports, but does not reason or act (no brain). Shipped as CIRISServer. Three deployment shapes: standalone server, embedded-in-agent, or family member. See §7.0.1.
ciris-canonical	The bootstrap governed community (cohort_subkind: infrastructure) every node ships trusting by default — but which any consumer MAY untrust or re-root (§5.6.8.10 default- not -forced-root). Its founding members (lens + registry-us + registry-eu fabric nodes) hold the founder-quorum (2-of-3, entrenched). Trust in it is role-scoped and ≠ consent (§5.6.8.10).
NodeCode	The QR-able peer-bootstrap shorthand for a federation key (CIRIS-V1-..., base32 + CRC-16). See §0.10.

§14.1 Persist **system:*** leaf glossary (narrative → canonical)

Stories under §5.3 sometimes use warm narrative leaves. The canonical wire form is to the right.

Narrative	Canonical
<code>audit_chain:integrity</code>	<code>audit_chain:hash_continuity</code>
<code>corpus_health:free_disk_bytes</code>	<code>corpus_health:n_eff_measurable</code>
<code>identity_continuity:long_term_key</code>	<code>identity_continuity:relational_anchor</code>
<code>federation_directory:freshness_seconds</code>	<code>federation_directory:replication_lag</code>

§14.2 Edge **system:*** leaf glossary (narrative → canonical)

Narrative	Canonical
transport:tls_handshake_success_rate	transport:{kind} (kind from Reticulum link types)
delivery:retry_count_p99	delivery:{class} (class from Reticulum delivery semantics)
peer_reachability:{peer_id} per-peer	peer_reachability:{network} (aggregate)
key_boundary:{scope} per-tenant	key_boundary:{scope} (scope from §3.4 D26 ext)

§14.3 Envelope-reach table (what the story wanted → how to express in existing wire)

What stories wanted	How to express in CEG
introspection as epistemic_mode	witness_relation: self + low confidence + pending external
testimony as epistemic_mode	epistemic_mode: external + witness_relation: external
civic stake	stake: reputational + cohort_scope: community
epistemic stake	confidence + stake: reputational
dignitary stake	harm_class:dignity_harm (composition; not in stake axis)
oversight: deferred / active / advisory	HITL / HITL+monitoring / HOTL respectively
transparency:{kind}	evidence_refs[] of reasoning-chain hash + downstream transparency_log:inclusion
provenance_walk	consumer-side composition (Portal/Verify dashboards)
renamed capacity factors / HE-300 categories	canonical wire form + LANGUAGE_PRIMER glossary mapping

§14.4 Promotion via supersedes worked example

A NodeCore consumer maintains private notes in local_data Contributions at cohort_scope: self. The user decides to promote a private note to a published encyclopedia entry:

```
// Original (local_data, self scope):
{
  "attestation_type": "scores",
  "attesting_key_id": "user-alice-2026",
  "attested_key_id": "user-alice-2026",
  "attestation_envelope": {
    "dimension": "encyclopedia:draft:notes",
    "score": 1.0,
    "confidence": 0.7,
    "evidence_refs": ["sha256:abc123..."],
    "cohort_scope": "self",
    "asserted_at": "2026-05-28T10:00:00.000Z"
  }
}

// Promoted (encyclopedia_article, global scope) via supersedes:
{
  "attestation_type": "supersedes",
  "attesting_key_id": "user-alice-2026",
  "attested_key_id": "user-alice-2026",
  "attestation_envelope": {
    "references_attestation_id": "<prior-id>",
```

```
    "supersession_reason": "promote_to_published",
    "differs_in": ["cohort_scope", "sub_kind"],
    "new_dimension": "encyclopedia:article:notes", // sub_kind morphed
    "new_score": 1.0,
    "new_confidence": 0.9,
    "new_evidence_refs": ["sha256:abc123..."], // same content_sha256
    "new_cohort_scope": "global", // widened scope
    "asserted_at": "2026-05-28T15:00:00.000Z"
  }
}
```

Pattern recap per §8.1.8.1: widens `cohort_scope`, optionally morphs `sub_kind`, preserves `content_sha256` (no body re-upload), chains via `supersedes`. The promotion lineage is walkable via `references_attestation_i`

§15 Concerns + acknowledged gaps

Three independent methodologies (PRIOR_ART_SCAN.md structural comparison, SOTA_SCAN.md production-validation comparison, *Magnifica Humanitas* encyclical mapping + CIRISRegistry#30 283-story stress test) surfaced concerns. CEG 0.1 critical-review pass added five more reviewer perspectives (cryptography, distributed systems, standards architecture, adversarial red-team, application development). Concerns named here so external reviewers see them acknowledged rather than discovered.

§15.1 Closed gaps

Gap	Status	Resolution
G1 — Revocation privacy	RETRACTED	Wrong threat model. The Registered path's thesis is public verifiability per ../MISSION.md §1.1.
G2 — Rules-layer Sybil	MITIGATED	§11.2 step 5 1-of-6 accord/steward sign-off + §11.2.3 meta-amendment entrenchment.
G3 — Narrow-cell fresh-quorum recusal	MITIGATED	§8.1.5 locality-scaled quorum + §8.1.5.1 sub-quorum fallback.
v1.4 T-3 #1 testimonial_witness:{kind}	CLOSED via §5.6.3 new prefix; opened to open vocabulary in CEG 0.1.	
v1.4 T-3 #2 labor:individual_loss	CLOSED by documentation. Existing non_maleficence:* with target_key_id = affected_individual + witness_relation: external carries the per-individual claim.	
v1.4 T-3 #5 Constitutional-constraint grounding	CLOSED in §1.2 prose. Wire stays tradition-multiplicity-neutral per §1.3.1.	
0.1-CRIT canonical-bytes newline-injection	CLOSED at 1.0-RC1 in §5.2.1: contracts redesigned to JCS (RFC 8785) objects with a pinned domain member (TupleHash128 retired — one canonicalization family; JSON escaping structurally removes the newline surface)	
0.1-CRIT supersedes/withdraws/recants ordering	CLOSED in §6.1 precedence rule + idempotency dedup	
0.1-CRIT cell_pool < min_pool cliff	CLOSED in §8.1.5.1 sub-quorum fallback paths	
0.1-CRIT no RFC 2119 anchor	CLOSED in §0.1	
0.1-CRIT no versioning policy	CLOSED in §0.3 SemVer mapping	
0.1-CRIT no normative References	CLOSED in §0.4	
0.1-CRIT endpoint response schemas	PARTIALLY CLOSED in §10.0 common-shape + error-envelope; full OpenAPI committed for 0.2	
0.1-CRIT reserved-prefix enforcement empty pointer	CLOSED in §7.0 inline enforcement rule	

Gap	Status	Resolution
0.1-HIGH STH cosignature consistency-proof	CLOSED in §10.3.1	
0.1-HIGH holds_bytes full-SHA verify + TTL	CLOSED in §10.1.1 / §10.1.2	
0.1-HIGH delegates_to depth + cycle	CLOSED in §13.3 anti-pattern + consumer-policy caps	
0.1-HIGH HUMANITY_ACCORD invocation replay	CLOSED in §9.2.1 discriminator + nonce in signed bytes	
0.1-HIGH notify vs CONSTITUTIONAL social-canonicity	CLOSED in §9.2.2 consumer-UI requirement	
0.1-HIGH /v1/steward-key place-holder authenticity	CLOSED in §10.2 response-signing requirement	
0.1-MED open-vocabulary collision	CLOSED in §11.2.2 collision rule	
0.1-MED occurrence_id self-assertion	ACKNOWLEDGED in §4 + R6 below	
0.1-MED withdraws arbitrage	CLOSED in §13.4 consumer-policy countermeasure	
0.2 — attestation:l{N}:* carried ladder-position in wire (T2 violation inherited from FSD-002 v1.0)	CLOSED in CEG 0.2 by §5.2 wire-break rename to mechanism-only prefixes + §8.1.9 Policy I consumer-side Attestation-Ladder Composition + §13.1 deprecation entry. Verify v3.7.0 caught the principle; CEG 0.1 inherited the wrong shape from FSD-002 v1.0 baseline without re-examining against the §1.3.1 T2 gate; CEG 0.2 ratifies the correction.	
0.9 — Envelope canonical-bytes round-trip determinism (omit-vs-materialize for optional fields)	CLOSED in CEG 0.9 by §0.9 (JCS pinned as the envelope encoding; omit-vs-materialize rule in §0.9.2; per-field catalog §0.9.3; worked attack §0.9.5).	
0.10 — Canonical-hash wire form + preimage convention unpinned	CLOSED in CEG 0.10 by §4.2.2.1–§4.2.2.4 (preimage {platform}:{entity_kind}:{id} + required canonical:{hashalg}:{hex} tag + conformance vectors); per CIRIS-Registry#53.	
0.10 — Delivery axis (observer-share + streaming multicast, third envelope axis)	CLOSED-OBSERVER-SHARE / STAGED-STREAMING in CEG 0.10 by §10.5 + §4 (delivery_mode/listed/history_on_join) + §7.9 + §8.1.13.7. Streaming-half open caveats RC1-1b/RC1-1c/RC1-7 tracked in §15.6.	

§15.2 Acknowledged risks (named as bets)

Risk	What's bet
R1 — Governance-subject truth-grounding fidelity	NodeCore P6 acknowledges low-fidelity signals for governance subjects. Bet that earned-Credits-weighting still outperforms token-weighting at scale.

Risk	What's bet
R2 — <code>delegates_to</code> rename-chain adoption cost	First test was the <code>correlated_action_v{N+1}:from:emergent_decept</code> chain at RATCHET deployment.
R3 — "Log existence $\neq$ log monitoring" drift toward TOFU caching	Consumer-policy guidance in <code>docs/TRUST_CONTRACT.md</code> .
R4 — Self-attestation under Ubuntu commitment	<code>witness_relation: self</code> admissible; consumer policy responsible for appropriate weighting per §13.5 discipline.
R5 — <code>hardware_class</code> self-assertion vs cryptographic attestation	Per §9.4.1: no normative attestation-chain verification in 0.1. Bet that placeholder/dev-class rejection + trust-multipliers cover the deployment window until per-platform attestation chains land in 1.x.
R6 — <code>occurrence_id</code> / <code>occurrence_count</code> / <code>occurrence_role</code> self-assertion	Per §4: env-var-driven, no cryptographic fleet-attestation primitive in 0.1. Bet that downstream compliance reviewers can correlate via correlated <code>signed_at</code> clusters + <code>evidence_refs[]</code> cross-checks; first incident drives a fleet-attestation primitive design workshop.
R7 — Frickerian discipline (§8.3) vocabulary without full method	First-pass shallow Frickerian SHOULD-rules; bet that the structural safeguards (§5.6.3 testimonial_witness disciplines, never-sole-evidence-for-slashing) absorb the gap until a deeper hermeneutical-resource analysis lands as a workshop output.
R8 — Conceptual scope vs governable surface	By 0.14 one grammar spans identity, communities, consent, location, communications, streaming, payments, governance, constitutional mechanisms, addressing, and transparency logs. Historically, projects unifying that many layers fail when one layer dominates the others; the harder risk is <i>governability</i> — can a human amendment body (§11.2) steward a system of this breadth? Bet: structural minimalism keeps the <i>amendable structural surface</i> tiny even as the namespace grows (§1.4 1+4), and the strict primitive/namespace/composition/verdict separation (§1.3) means scope grows in the <i>open-vocab namespace</i> (locally evolvable) rather than the <i>governed core</i> . Residual: namespace + composition-policy sprawl can still outrun review capacity; mitigation is the §11.2 high evidentiary bar + the post-1.0 candidate backlog (CIRISRegistry#51). The remaining challenge is no longer purely technical.

§15.3 First-adopter exposures (no prior validation; explicit bets)

Exposure	Why no precedent
F1 — Earned-Credits federation governance at scale	No prior system separates earned standing from purchasable token at scale. Risk: SPKI/SDSI adoption-gap failure mode. Mitigation: licensure forcing function.
F2 — Ubuntu substrate as wire-format substrate	CARE Principles + African philosophy exist as ethical frameworks; never as protocol substrate. First-adopter risk on how the discipline interacts with engineering trade-offs at scale.

§15.4 Deferred to 0.2+ design workshops

Item	Why deferred
~~Canonical-bytes redesign~~	RESOLVED at 1.0-RC1: §5.2.1 v2 = JCS objects + domain member (TupleHash128 retired; #57 blocker A closed)
Per-platform hardware-attestation chain verification (TPM quote, Apple attestation, FIDO attestation)	Phase D 1.x roadmap per R5
Multi-party witness directory admission (2-of-3 steward sign-off)	Phase C 0.2 commitment per §10.3
Machine-readable namespace manifest (FSD/CEG/dimensions.json)	Phase E 0.2 commitment per §12.4
Full OpenAPI export for all endpoints	Phase E 0.2 commitment per §10.4
<code>attestation:singular_witness:non_substitutability</code>	T2 fragility — "non-substitutability" must reference audit-chain count, not moral quality. Needs design workshop.
<code>integrity:finitude_acknowledgment</code>	LOW priority; <code>conscience:epistemic_humility</code> already covers epistemic finitude.
<code>sustained_practice:{kind}</code>	Conceptually interesting; not load-bearing for current federation work.
IEEE EAD Ch5 Affective Computing cluster	Need RATCHET calibration design before T2 gate clears.
Various <code>partner_role:*</code> specializations	Cross-source design discussion needed.
5 ergonomic considerations from trio Phase 4 audit	Bigger workshop topics (B.3 deontic-strength axis is highest-leverage).
SEED_DIMENSIONS RFC (CIRISRegistry#22)	RFC stage; needs discussion.
Fleet-attestation primitive (closes R6 occurrence_id self-assertion)	Workshop output
Deeper Frickerian instantiation (closes R7)	Workshop output

§15.5 Identified overlaps

Overlap	Resolution
O1 — <code>epistemic_mode: derivative</code> $\approx$ <code>witness_relation: derived</code> at edges	Documented as joint-usage pattern; not collapsed. Different concepts at the edges (process vs relational position) even if they often co-vary.
O2 — <code>detection:distributive:access</code> could fold into <code>detection:correlated_action</code> as axis path	Kept separate for pedagogical weight; possible future revisit.
O3 — <code>credits*:substrate_building</code> was miscounted as new prefix	CORRECTED — recounted as <code>{subject}</code> value.
O4 — §8.1 reference policy structure (A/B/C base + D/E/F/G/H modifiers)	Cosmetic restructuring; documented inline.

§17 Update cadence

CEG is updated:

- On every prefix admission to §5 (per §11 amendment process)
- On every envelope field addition to §4
- On every endpoint shape addition to §10
- On every anti-pattern admission to §13 (with citation to the stress test or methodology that surfaced it)
- On every gap state transition in §15
- On every CIRISAccord revision affecting the federation surface
- On every conformance-language or normative-reference change in §0 (MAJOR per §0.3)

Each update lands as a single commit touching the relevant file(s) + a lineage row in §16.1. The version number bumps per the §0.3 SemVer rules.

Last updated: 2026-06-16 (CEG 1.0-RC12 — unified retirement / forever-memory model in §19.7; no 1+4 change, no wire break). Full version-by-version history in §16.1.

§18 Interoperability profiles (informative)

*This whole section is informative (§0.1.1). Nothing here touches the frozen normative interior. These are **boundary** profiles — how a CEG node reads and emits the encodings, envelopes, and verification primitives the rest of the world shares, **without** adopting anyone else’s semantics. The 1+4 grammar, the namespaces, the consent architecture, and the JCS signing interior are unchanged. Conformance is still judged against the normative surface only.*

§18.0 The governing principle — speak CEG inside, standards at the edge

CEG’s moat is its **semantics**: the 1+4 grammar, the consent architecture, founder-quorum trust, who-vouches-for-what-revocable-by-whom. We never adopt anyone’s semantics. We adopt the **envelopes, encodings, and verification primitives** everyone shares — **at the boundary only**. A second *interior* canonicalization or claim family would recreate the cross-impl divergence hazard the §0.9 JCS freeze exists to close (§5.2.1 records this decision); so the interior stays one family, frozen, and every standard below is reached at an edge.

Four boundary modes (the full roadmap + dispositions live in CIRISRegistry#72):

Mode	Meaning	Standards
Export profile	re-sign / re-encode a CEG attestation so a standard verifier reads it without knowing CEG	COSE Sign1, RFC 9421 (HTTP Message Signatures / Web Bot Auth), SD-JWT VC presentation
Import bridge	a foreign signed artifact is cited via evidence_refs (deliberately lossy)	C2PA manifests (§18.1), eIDAS / W3C VC credentials, Sigstore/Rekor bundles
Already interior	the standard <i>is</i> a primitive CEG builds on	MLS / TreeKEM (§10.5.3), RFC 6962 / 9162 transparency logs (§10.3), SLSA (provenance:slsa:{level} , §5.2)
Explicitly NOT adopted	vendor rails / competing semantic layers	DIDs <i>as a resolution stack</i> (export syntax only), AP2 / Visa TAP / Mastercard Agent Pay (bridge via §5.6.8.12 settlement), SPIFFE (datacenter-tier mapping only)

****The universal "absorb anywhere" surface is [evidence_refs[]](04_envelope.md). Any Contribution may cite an external signed artifact as evidence with zero wire change. That is how foreign provenance enters CEG — not by replacing the interior encoding, but by reference, with CEG layering the epistemic claims (who vouches, under what consent, with what confidence) on top. The composition is the differentiated story: provenance says where the bytes came from; CEG says what a community of signers makes of them.****

§18.1 C2PA Content Credentials — media-provenance import/emit profile

Disposition: ADOPT at the media boundary (import bridge + emit), zero interior wire change. C2PA (Coalition for Content Provenance and Authenticity) Content Credentials are the industry standard (Adobe / Microsoft / Google / BBC ...) for cryptographically signed media provenance — origin, edit history, and generator (incl. AI-generation) assertions embedded

in or sidecar'd to images / video / audio. **Deadline driver:** EU AI Act Art. 50 machine-readable marking of AI-generated content applies from **2026-08** in the federation's primary jurisdiction, so this profile is calendar-bound, not optional. Owners: NodeCore / LensCore media ingest (the §5.6.8 multimedia / federation_blobs boundary).

C2PA is **provenance**; CEG is **judgment**. They do not compete — they compose. C2PA answers *"what process produced these bytes, signed by whom?"*; CEG answers *"what does a community of signers, under what consent, make of them — and who can revoke that?"* Neither does the other's job; C2PA has no consent architecture, no revocation, no 1+4.

§18.1.1 Import — a C2PA manifest as evidence_refs

When a federation_blobs row (or a multimedia Contribution over its SHA-256) carries a C2PA manifest, the manifest is referenced — **never re-encoded into the CEG interior** — through the existing external-reference pattern:

```
evidence_refs: [
  { kind: "c2pa_manifest", // open-vocab evidence kind
    locator: "<blob_sha256 of the .c2pa manifest store | embedded-offset ref>",
    manifest_sha256: "<sha256 of the active manifest, lowercase hex per S0.6>",
    claim_generator: "<the C2PA claim_generator string, verbatim>",
    validation: "valid" | "invalid" | "unverified" } // the verifier's C2PA-side result,
      advisory
]
```

- The C2PA signature is verified **by a C2PA verifier** (trust-list / cert-chain per C2PA), NOT by a CEG signature path — the two trust models stay separate. The **validation** field carries that result as **advisory** evidence; it is never fed to a CEG hybrid-verify path (the §5.6.8.8.2 key-separation discipline generalizes: foreign-trust-root material is payload, never CEG verification material).
- A CEG **scores** attestation may then assert a judgment **about** the provenanced bytes (e.g. **detection:multimedia:ai_generated** from LensCore, or a community **scores** endorsement), linking the C2PA evidence via **evidence_refs** and the media via **subject_key_ids** / the blob SHA. The CEG claim is signed CEG; the provenance it cites is signed C2PA; the reader sees both lineages without either standard absorbing the other.
- **Absent / invalid C2PA is not fail-secure-fatal** — it is itself a recordable observation (**validation:** "invalid" / no manifest). CEG records the gap; consumer/RATCHET policy weights it. The substrate is not a C2PA gatekeeper.

§18.1.2 Emit — CEG judgment as a C2PA assertion (egress)

At a media-publish boundary a node MAY emit a C2PA assertion carrying a CEG attestation reference (a CAWG-identity-assertion-shaped custom assertion), so a pure-C2PA consumer downstream sees "this media is vouched-for in CEWP" without speaking CEG. This is an **export** at the edge (re-expressing an existing CEG attestation in C2PA's assertion envelope), parallel to the §18.2 COSE export profile — it adds no CEG wire field and re-signs nothing in the interior.

§18.1.3 What this profile does NOT do

- It does **not** make C2PA an interior format. CEG envelopes are never C2PA-encoded; the JCS interior is untouched.
- It does **not** adopt C2PA's trust model as CEG's. C2PA cert-chains/trust-lists validate C2PA; founder-quorum/web-of-trust validates CEG. They meet only at `evidence_refs`.
- It introduces ****no new `subject_kind` and no new structural primitive**** — `c2pa_manifest` is one open-vocab `evidence_refs.kind`; the judgment rides existing `scores + detection:*`.

§18.2 Tracked boundary profiles (stubs — see #72)

These are committed dispositions whose detailed profiles are written as each lands; none touches the frozen interior.

Profile	Mode	Note
RFC 9421 + Web Bot Auth	export	CIRIS agents sign outbound HTTP with their existing Ed25519 keys; JWKS published at <code>/.well-known/http-message-signatures-dire</code> → legible to the existing web. Cheapest win; keys already in <code>identity_occurrence / federation_keys</code> .
COSE Sign1 / deterministic CBOR	export	Re-sign profile so any IETF JOSE/-COSE verifier (where the ML-DSA registrations land) checks a CEG attestation. Interior stays JCS; if JCS keeps producing cross-impl bite post-1.0, 2.0 is the re-encoding moment, not before.
SD-JWT VC / W3C VC 2.0 + OpenID4VP	export + import bridge	eIDAS-forced (EUDI wallets); CEG attestation → SD-JWT VC presentation on export, eIDAS credential → <code>evidence_refs</code> on import. Never re-build on VCs.
Tiled/static logs + IETF KEYTRANS	already-interior + watch	Keep the §10.3 RFC 6962 abstraction; adopt tiled-log (Sunlight-lineage) serialization for log ops. KEYTRANS is what <code>resolve_encryption_keys</code> already <i>is</i> — express it there when KEYTRANS stabilizes.

§19 Holonomic substrate — ALM, fountain storage, Wholeness-Witness, recursive bootstrap (CEG 1.0-RC11)

The CIRISEdge v4.0.0 "holonomic substrate" gives the federation **graceful degradation** (any subset of fountain symbols decodes at proportional fidelity) and **graceful reconstitution** (the witnessed corpus re-establishes from any sufficient fragment). This section absorbs its wire shapes as additive normative CEG sections **with normative guardrails** that bind them to CEG's existing trust, PQC, consent, replication, and anonymous-tier invariants.

Absorb-with-guardrails, not verbatim (normative posture, CIRISRegistry#85).

*A four-perspective review found the holonomic concept sound and **additive at the wire layer** — **no §3/§4 1+4 change** — but that absorbing the v4.0.0 implementation verbatim would invert several ratified CEG invariants (owner-binding, PQC-mandatory, consent/withdraws, quorum-merge). This section therefore states the **guardrail invariants** (§19.1–§19.5) as normative **MUSTs now**; the matching edge implementation fixes are tracked at CIRISEdge#143 (in-flight hardening). The **byte-exact signed preimages** for each shape are pinned against the **fixed v4.0.x reference impl** (several shapes change under #143 — e.g. **SignedClaim** gains owner-binding fields), and **conformance vectors generated from that fixed impl are the named #57 freeze gate** (§19.6).*

§19.0 Canonicalization boundary + the 1+4 line (normative)

- **The frozen 1+4 attestation envelope (§4) is untouched.** Every §19 object is **transport/substrate framing** — it never instantiates a §4 Contribution, never adds an `attestation_type`, never enters §0.9 JCS canonicalization. The already-landed realtime A/V chunk wire (§10.5.8.3–.5) is the same category.
- **§19 uses a binary, length-prefixed, big-endian, domain-separated signing preimage — NOT §0.9 JCS.** These are verify-to-verify transport primitives that never cross the four-impl boundary as JSON (the same boundary §10.1.5.3 drew for Verify's `signing_bytes` framing). An implementer **MUST NOT** apply JCS to a §19 object or its signatures will not verify cross-impl. Each object's domain separator (`b"CIRISALM-CAPv2\0\0"`, `b"ciris-edge/holding-cla`, `b"ciris-edge/compress-request/v1"`, `b"CIRIS-CLAIM-v1\0\0"`, the WholenessWitness `b"WW-v1-empty"` empty-sentinel, etc.) is pinned by its subsection.
- **PQC-mandatory (§10.1.5.1.1) binds every §19 signed object.** Each carries the bound hybrid pair (Ed25519 over the canonical preimage; ML-DSA-65 over `preimage || ed25519_sig`); a verifier **MUST** reject a §19 object lacking a valid ML-DSA-65 half **at ingest and before persistence** (no store-then-quarantine — the RC8 store-path rule). Verification happens **at the gate**: an admission/verdict function **MUST** verify signatures itself and **MUST NOT** trust an in-band **verified** flag (such a flag **MUST** be non-wire / `serde(skip)`).
- **What is wire vs internal (the §1.1 line).** Cross-impl-observable bytes (signed preimages, content-addressed hashes, and the deterministic topology output) are **PIN-NORMATIVE**. Local heuristics whose output no other peer reproduces are **edge-internal** and **MUST NOT** be over-pinned: specifically **ALM parent-selection** (`AlmJoinPlanner` — over per-peer RTT/reachability) and ****retention_priority (never on the wire) are edge-internal; rarity scoring is a recommendation****, not a **MUST**.

§19.1 WholenessWitness (§W) — divergence-detection witness

A `wholeness_witness`: object is a peer's **hybrid-signed Merkle root over a scoped projection of the claims it holds**, used to detect cross-peer/cross-region state divergence and to drive reconciliation.

Namespace + scope (normative).

- **WW-naming** — the namespace is `**wholeness_witness:**`, never bare `witness:`. It is a *self-published state-root snapshot*, the **inverse** of the §10.3 transparency-log "witness" (an independent STH cosigner). A WholenessWitness does **not** provide append-only / consistency / anti-equivocation guarantees and **MUST NOT** be substituted for §10.3.1 or the §10.5.1 per-stream STH.
- **WW-1** — the root **MUST** cover **only** the namespaces in the object's `claim_namespaces` field. A conformant peer is **not** required to witness everything it holds; coverage is per-namespace opt-in.
- **WW-2 (anonymous/self exclusion — fail-secure)** — a WholenessWitness **MUST NOT** include anonymous-tier records (CIRISNodeCore#22) or `cohort_scope: self` local-tier rows (§10.1.5) as Merkle leaves, and `claim_namespaces` **MUST NOT** name such a namespace. Witnessing them would re-attribute deniable / self-private content to a stable `peer_id`. (Edge#143: the leaf-walk **MUST** filter these out before computing the root.)
- **WW-3** — `cohort_scope: family | community` content **MAY** be witnessed ***only at the opaque content_id/manifest-digest grain***, never at a grain disclosing membership, plaintext, or `subject_key_ids` (§10.1.4 confidentiality preserved).

Construction (normative).

- **Leaf order MUST be lexicographic** over leaf bytes (the §0.9.2.1 set-semantics rule). The v4.0.0 "either order as long as both peers agree" is **non-conformant** — it is the §0.9-class divergence hazard.
- The Merkle scheme is `leaf = SHA-256(leaf_bytes)`, `node = SHA-256(left || right)`, odd-node duplication, `b"WW-v1-empty"` empty sentinel — the construction CIRISEdge v4.0.x shipped and CIRISVerify v5.9.0 proved cross-impl. ***Frozen as-is (1.0-RC15 — resolves the prior open #143 question): CEG does NOT adopt the RFC 6962 0x00/0x01 leaf/node prefix here.* **Rationale — the CVE-2012-2459 odd-node-duplication malleability is not exploitable in this construction's uses****: (1) every WholenessWitness and `member_commitment` (§19.7.1.1) root is **mandatorily hybrid-signed** — no consumer ever relies on an *unsigned* root; (2) `member_commitment` is verified by **recomputation from the full source-id list**, never by partial inclusion proofs against the bare root (malleability moot); (3) the §19.1 reconciliation Merkle-proof exchange (N4) is between **accountable, signed, equivocation-checked peers** — not third-party forgery of an untrusted root. **Caveat (normative)**: any *future* use that relies on an **unsigned** root, or verifies **partial inclusion proofs against an untrusted root**, **MUST** first adopt the RFC 6962 0x00/0x01 prefix + lone-node promotion (and re-cut the vectors). This is a **distinct construction from the §10.3 RFC 6962 log** (different algorithm + leaf domain); the two **MUST NOT** be cross-verified. Changing this scheme is a vector-invalidating wire change — not an editorial tweak.

Authority (normative).

- **N3** — a WholenessWitness is a federation-tier attestation: hybrid PQC verified at ingest **and before** persistence to the witness corpus (§19.0); `compare_witnesses` MUST NOT run on an unverified witness.
- **N4 (equivocation)** — two validly-signed witnesses from the same (`peer_id`, `epoch_id`, `claim_namespace_set`) with different `merkle_root` are **non-repudiable equivocation proof**; the substrate MUST retain and surface them as a `hard_case:*` (§7 reserved-prefix candidate), never silently reconcile. Per-peer `epoch_id` MUST be anti-rollback-checked before `EpochBehind` is used as a reconciliation input (eclipse guard). Full cross-witness BFT MAY be deferred (§15 named bet) — but observed equivocation MUST NOT be discarded.
- **WW vs replication (the highest-value reconciliation)** — a WholenessWitness is a ****divergence detector that *triggers* the §10.1.6 quorum-merge; it does NOT**** decide a merge and MUST NOT replace `monotonic_quorum` / `revision` anti-rollback for `revocation` / `partner_record` / `org_membership`. A Divergent verdict on those subject `kinds` hands the decision to the §10.1.6 R1/Q1 quorum-merge (quorum-ordered, anti-rollback) — otherwise a "reconstitute from any fragment" path could resurrect a revoked key (rollback).

§19.2 Recursive trust bootstrap (§B) — trust-discovery, not membership

`recursive_trust_bootstrap(SignedClaim, TrustGraph, WitnessChain) → verdict` lets a peer discover transitive trust by walking a signed witness chain to a root in its own trust graph. **It is reachability discovery beneath CEG's authority layer, not an admission shortcut.**

- **N1 (trust ≠ membership)** — a successful chain walk yields **trust+serve standing only** (§5.6.8.10 TRUST≠MEMBERSHIP). Admission to any ****non-infrastructure community remains gated, at the destination, by (a) the §5.6.8.10 owner-binding**** precondition (a live `user-owner delegates_to`, an admitted `identity_occurrence`) and (b) that community's `consensus_protocol`. `infrastructure` roots stay **founder-quorum-gated**; a transitive chain MUST NOT satisfy founder-quorum. (Edge#143: `SignedClaim` must carry the owner-binding fields so the gate is expressible.)
- **N2 (self-supplied chains aren't evidence)** — the chain-length budget MUST be ≤ the §13.3 **5-hop cap**, trust-graph **cycles MUST be rejected** (§13.3), and the §13.3 **aggregate-weight cap** (default $0.5 \times \text{root_trust}$) MUST bound the standing one root confers transitively. A caller-supplied chain proves only its signatures, not a real lineage.
- **RB-1 (anonymous coexistence)** — anonymous-tier content MUST be ingestible / retainable / serveable with **no trust-graph position**; `recursive_trust_bootstrap` MUST NOT be required for, or invoked on, anonymous records.

§19.3 Fountain storage + swarm rarity (§P / §R)

Content is RaptorQ-coded into `N` source + `K` repair symbols (`FountainManifestV1` / `FountainSymbolV1`); peers retain symbols and coordinate rarest-first so content survives churn.

- **N5 (retention respects revocation — fail-secure)** — retention MUST NOT keep alive, **above the §19.7 noise floor**, content whose consent is withdrawn (§3.2.3) or revoked. A withdrawn `content_id` is descent-eligible **regardless of rarity**; an active **withdraws**

/ `consent:state:revoked` overrides the max-rarity "keep" signal and forces immediate descent below the noise floor (the fastest form of the one retirement operation — see §19.7); unknown consent state defaults to *not retained as rare*. The §8.1.11.3 deletion-SLA + §8.1.11.5 decay stages take precedence over swarm coverage at all times. **Revocation does not require destroying the collective gist** — only that the item be **not individually recoverable** at any retained tier (§19.7); it MUST purge any tier where it still is.

- **N6 (possession-bound claims)** — a `FountainHoldingClaim` counted toward rarity MUST be possession-challengeable (a holder answers a symbol request, or the claim carries a proof-of-possession). Unverified holding claims MUST NOT lower another peer's retention priority — otherwise rarity is a forgeable force-evict channel.
- **N7 (symbol integrity)** — reconstruction MUST verify each symbol against the manifest's signed per-symbol SHA-256 (a swarm-sourced symbol cannot poison a decode).
- **SR-2 / SR-3 (anonymous + reconstitution scope)** — anonymous-tier content is **exempt from swarm-mandatory retention** (governed by LRU-only per `NodeCore#22`): no `FountainHoldingClaim` / `FountainCompressRequest`, no rarest-first biasing. The "re-constitutes from any sufficient fragment" property is a property of the **witnessed, trust-anchored** corpus **only** — it does not extend to anonymous content, which the substrate MUST be able to let truly disappear.
- **PIN line** — `FountainHoldingClaim` / `FountainCompressRequest` signed preimages = **PIN-NORMATIVE** (with `symbol_ids` sorted ascending before signing); `compute_rarity_score` = **PIN-AS-RECOMMENDATION**; `retention_priority` = **edge-internal** (never on the wire).

§19.4 Deterministic ALM topology (§T / §M)

The application-layer-multicast relay tree for large-N fan-out — the §10.5.8 "realtime large group (SFU/relay-tree)" profile previously marked → 1.x, now filled.

- ****compute_alm_topology(snapshot) → topology** is **PIN-NORMATIVE** as a contract — a pure, deterministic, integer-only** function (no IEEE-754; no `HashMap` iteration order) over (`capacity_ads`, `trust_grants`, `reachability_observations`, `locality`), with specified lexicographic tie-breaks and canonical output order, such that **byte-equal inputs yield byte-equal output** across implementations. The byte-exact output is gated on the §19.6 vectors (incl. permutation-invariance cases) — **not** transcribed from the algorithm body as the source of truth.
- **N8 (capacity authenticity)** — capacity advertisements feeding the topology MUST be hybrid-verified (`SignedRelayCapacity`, domain `b"CIRISALM-CAPv2\0\0"`) **before scoring**; self-asserted `uplink_mbps` MUST NOT be the dominant, unbounded selection term (cap per owner-bound identity or make it throughput-challengeable). Determinism amplifies one capacity lie into a *universal* eclipse — "verified by an unspecified upstream tier" is not a guarantee.
- **D6 preserved** — `reachability_observations` are **ephemeral planner inputs**; they MUST NOT become attested, replicated, or witness-leafed state (§10.5.6 "reachability is never trust"). Resolution authority stays in §8.1.13.1.1; the topology consumes it, does not replace it. The determinism comparator reconciles to the §8.1.13.1.1 / R1/Q1 family.
- **PIN line** — the topology *function contract* + `SignedRelayCapacity` / `SubStreamCommitment` signed preimages = **PIN-NORMATIVE**; **ALM parent-selection** (`AlmJoinPlanner`,

over per-peer RTT) = **edge-internal**.

§19.5 Fail-secure summary (normative)

The holonomic mechanisms are blind to the anonymous tier (WW-2, RB-1, SR-2/3), subordinate to the consent/revocation model (N5, WW vs §10.1.6), gated by owner-binding + founder-quorum (N1, N2), and bound by PQC-mandatory verification at the gate (§19.0, N3, N8, F-5). Where the v4.0.0 implementation does not yet meet these, the gap is tracked at CIRISEdge#143; the invariants here are the conformance target regardless of implementation timing.

§19.6 Conformance — the #57 freeze gate

The byte-exact signed preimages and the `compute_alm_topology` output are pinned against the **fixed** v4.0.x reference impl, and **conformance vectors generated from it are the named #57 freeze gate** (extending CIRISConformance#9): input $\rightarrow$ expected bytes for `SealedAvChunk` + the two AV nonces, `SignedRelayCapacity`, ALM topology (input snapshot $\rightarrow$ expected tree hash, incl. permutation invariance), `FountainManifestV1/SymbolV1` + `retention_priority`, `FountainHoldingClaim/CompressRequest`, and `WholenessWitness` canonical bytes + Merkle root (incl. the empty sentinel + odd-node duplication). Until a second implementation reproduces these byte-for-byte, the §19 shapes are **pinned-but-unproven — RC-grade, not 1.0**.

§19.7 The noise floor — unified retirement / forever-memory model (normative)

One operation, not many. Revocation, retirement, capacity-eviction, scheduled expiry, and natural aging are **the same operation at different rates**: a **monotonic descent of an item's fidelity, driven by pressure**, toward and below a recoverability boundary called the **noise floor**. There is no separate "hard delete" primitive — *hard-delete is the fastest descent* (forced immediately below the floor); capacity-eviction is a slow one; aging is the slowest. All are equally valid instances of the one retirement operator.

The noise floor = the individual-recoverability boundary. An item is **above** the floor in a retained artifact iff it can be individually reconstructed from that artifact above a fidelity ε ; it is **below** the floor iff only its *contribution to a collective* survives — the item itself is information-theoretically unrecoverable. The floor does double duty and is the load-bearing normative quantity of this section: it is **both** the privacy boundary (a revoked item **MUST** be below it at every retained tier) **and** the durability floor (the collective blur sits below it, forever).

Nothing is ever fully forgotten — the memory pyramid. Descent does not terminate at zero. Two **mechanical** degradation operators (no reasoning, no agency — see below) carry it:

1. **Intra-object fade** — scalable/layered codec (§10.5.8.4 `ChunkLayer` spatial/temporal/quality) + RaptorQ per layer: drop high-detail symbols $\rightarrow$ a clean coarse version of the same item.
2. **Inter-object aggregation** — *a picture of a thousand pictures*: tile / downsample / statistically composite $\mathbf{N} \rightarrow \mathbf{1}$. Recursed, this builds a pyramid (mipmap) of history: recent strata high-resolution, ancient strata collapsed into the blur. Steady-state storage to remember **all** of history is $\mathbf{O}(\log \mathbf{T})$ in the amount remembered, not $\mathbf{O}(\mathbf{T})$ — the $\mathbf{N} \rightarrow \mathbf{1}$ fan-in makes

forever-memory **sublinear**. *A million years may be a blur, but it is remembered, unbroken, to the beginning.*

Pressure-driven (normative). The descent rate and the pyramid's level transitions are driven by **pressure** (disk pressure, age, or an explicit force), never a fixed schedule. Pressure sources, slowest→fastest: natural aging < scheduled retirement < capacity (disk) eviction < **revocation (immediate forced descent below the floor)**.

Forgetting and erasure converge (this dissolves the §19.3 N5 tension). The N5 erasure guarantee is exactly "not individually recoverable at or below the noise floor." A sufficiently-aggregated composite (a picture of a thousand pictures contains < 1/1000 of any source) is **already-erased by degradation** — no purge needed. Revocation simply *forces* an item below the floor **now**, and **MUST** purge only the retained tiers where it is **still individually recoverable** (the high-fidelity upper layers). It need not — and **MUST NOT** be required to — destroy the collective gist. Capacity-eviction reaches the identical end-state gradually. Same destination; revocation just gets there first.

Infrastructure is self-sufficient for memory (sharpens §1.3). Both degradation operators are **mechanical** (symbol arithmetic + resampling) — they require **no reasoning and no agency**, so a pure fabric node performs the entire forever-memory function. A brain **MAY** enrich a degraded tier with a richer semantic gist, but is **never required**: infrastructure remembers without agency. (Earlier drafts wrongly treated semantic summarization as the mechanism; the mechanism is mechanical degradation — the brain is optional enrichment.)

Disposition mapping. The §19.3 EjectionVerdict values are points on this one axis: **Keep** = above-floor, no pressure; **EjectToTier** = a downward step (still recoverable, lower fidelity); **aggregation** = N→1 downward step; **EjectHardDelete** = forced descent below the floor + purge-still-recoverable-tiers. They are not distinct mechanisms — they are stops on the single pressure-driven descent.

§19.7.1 AggregationMetaV1 — the aggregation-tier wire contract (normative, 1.0-RC14)

The metadata that tags one tier of the §19.7 memory pyramid: which content, at what aggregation tier, over which source members, by which mechanical operator. **CEG-canonical** — unlike the §19.6 shapes (transcribed from edge v4.0.x), no reference impl yet defines these bytes (CIRISPer-sist v8.3.0's `content_aggregation` stores `aggregation_meta` **opaque** — the wire-churn firewall), so this section **defines** the byte layout; impls conform to it. Resolves CIRISRegistry#89; un-blocks CIRISVerify#79.

AggregationMetaV1 is a **substrate wire shape**, **NOT** a **§4 attestation** — no 1+4 change. Its signing preimage uses the §19.0 binary discipline (length-prefixed, big-endian, domain-separated — **NOT** §0.9 JCS). Preimage byte order (normative):

```
preimage = b"AGG-META-v1\0\0\0\0" // 16-byte domain separator (exact)
|| u32_be(version = 1)
|| lp(content_id) // the root content this pyramid is for
|| lp(corpus_kind) // "trace" | "blob" | "av_chunk" | ...
|| u32_be(tier) // 0 = source granularity; higher = more aggregated
|| lp(aggregation_algorithm_id) // opaque codec id, e.g. "raptorq-pyramid-v1"
|| u32_be(source_count) // N members aggregated into this tier (descent fan-in)
|| member_commitment[32] // §19.7.1.1 Merkle root over the source member ids
|| lp(noise_floor_descriptor) // what survives below the floor (codec-specific,
    canonical)
```



```
// lp(x) = u32_be(byte_len(utf8(x))) || utf8(x) // length-prefixed UTF-8
```

`content_id`, byte-valued ids, and `member_commitment` are lowercase-hex per §0.6 where rendered as strings; `member_commitment` on the wire is the raw 32 bytes. **Bound-hybrid signature** (the §19.0 rule): `Ed25519(preimage) + ML-DSA-65(preimage || ed25519_sig)`; a verifier MUST reject a tier lacking a valid ML-DSA-65 half **at ingest and before persistence** (the §10.1.5.1.1 store-path rule applies — `AggregationMetaV1` is federation-tier).

§19.7.1.1 `member_commitment` (descent integrity)

`member_commitment` is the Merkle root over the **source member ids aggregated into this tier**, computed by the **§19.1 WholenessWitness Merkle construction** (same leaf = `SHA-256(utf8(member_id))` **lexicographic** leaf order, node = `SHA-256(left || right)`, odd-node duplication, and empty-set sentinel) — reused deliberately so the federation carries **one** aggregation/witness Merkle scheme, not a third. (It therefore uses the §19.1 scheme as frozen in RC15 — **no** RFC-6962 prefix; safe here because `member_commitment` is verified by **full source-id-list recomputation**, never partial inclusion proofs, so the CVE-2012-2459 malleability is moot. See §19.1.) `member_commitment` lets any verifier confirm a tier was aggregated from exactly the claimed sources without holding the sources.

§19.7.2 Descent rule (normative, 1.0-RC14)

`descend(content_id, corpus_kind, tier) → [member_id]` returns the **ordered** source members aggregated into the tier-tier composite — the tier-(tier-1) members one level down the pyramid. It MUST be a **pure, deterministic** function: two impls return the **byte-equal ordered list** for byte-equal inputs. The order is the **lexicographic member-id order** `member_commitment` (§19.7.1.1) committed to — so a returned list re-derives the parent’s `member_commitment` byte-for-byte (the descent-integrity check).

Descent never terminates at zero (the forever-memory floor). Below tier 0 (source granularity) the content’s **collective gist persists as the lowest retained tier** — a composite whose members are no longer *individually* recoverable (it is **below the noise floor**, §19.7) but whose blur survives. `descend` past the noise floor yields the blur, never an empty/destroyed object. The function is **pressure-independent** (pure navigation); `**pressure` drives which tiers are **retained** (§19.7), not the descent computation. Ascending (aggregation, operator 2) is the $N \rightarrow 1$ inverse with fan-in `source_count`.

§19.7.3 `EjectionVerdict` — the tier-aware retirement surface (normative, 1.0-RC14)

The single verdict surface a verifier exposes and a substrate consumes to gate one step of the §19.7 descent. CEG pins it as the canonical superset of the rarity-only `RetentionDecision` that CIRISVerify v5.9.0 already ships:

```
EjectionVerdict ::= Keep // above the floor, no pressure step
                  | EjectToTier // one downward step: still recoverable, lower fidelity
                      // (intra-object layer-drop OR N->1 aggregation)
                  | EjectAggregatedTierOnly { tier }
                      // shed exactly one pyramid stratum -- the tier-‘tier’
                      // composite -- leaving finer AND coarser tiers intact
                  | EjectHardDelete // forced descent below the floor + purge still-recoverable
                      tiers
```

Mapping (normative): the v5.9.0 `RetentionDecision{RetainRare|RetainNonRare|EvictEligible}` is the rarity sub-decision *within* `EjectToTier/Keep; EvictEligible` + capacity pressure $\rightarrow$ `EjectToTier; EvictEligible` + a `withdraws/consent:state:revoked` (§19.3 N5) $\rightarrow$ `EjectHardDelete` (the fastest descent, never tier-shed — §19.7). ****EjectAggregatedTierOnly { tier }**** is the tier-granular form of `EjectToTier`: it sheds a single intermediate stratum of the §19.7.1 pyramid (the tier-tier `AggregationMetaV1` composite) under targeted pressure, leaving both finer and coarser tiers — composing with the hard-delete trait (a tier below the noise floor is unreachable, so this never resurrects erased content). A pure fabric node MAY compute `EjectToTier` / `EjectAggregatedTierOnly` mechanically; `EjectHardDelete` MUST purge per §19.3 N5. Verify exposes `EjectionVerdict`; persist consumes it to drive `put_aggregated_tier` / the tier-tagged evict (`EjectToTier`, `EjectAggregatedTierOnly`) vs `evict_fountain_content_hard_delete` (`EjectHardDelete`).

Conformance — PROVEN cross-impl (1.0, promoted from RC, 1.0-RC16). §19.7.1–3 are **byte-equivalent across implementations**: CIRISVerify v5.10.0 authored the vector family and CIRISEdge v4.3.0 (`src/holonomic/aggregation.rs` + `tests/conformance_vectors_v19_7.rs`, 5 vectors) reproduces them **byte-for-byte** — the `AggregationMetaV1` preimage + signature, `member_commitment`, and descend ordered output. The `AggregationMetaV1` preimage matched **on the first attempt with no cross-team coordination beyond this spec** — the §19.0 binary-length-prefixed discipline makes wire-identity reproducible from the text alone. `member_commitment` reuses the §19.1 WholenessWitness Merkle **verbatim** (same `compute_merkle_root`, same WW-v1-empty sentinel) — the federation runs **one** Merkle scheme across §19.1 (witness leaves) and §19.7 (member commitments), no schema fork. With persist v8.4.0 + verify v5.10.0 + edge v4.3.0 all on the §19.7 baseline at PyPI, the §19.6/#57 vector family for §19.7 is **closed**; §19.7 is **1.0, not RC**. (The earlier v5.9.0-proven §19.6 vectors are unaffected.) The `EjectAggregatedTierOnly { tier }` verdict is the one remaining edge build item (tracked v4.3.x/v4.4) — surface-additive, composes with the existing trait.